Course Notes

Abstract Algebra III

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Chapter 1

Lecture 1: Modules

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1.1 Basic Definitions and Examples

Definition 1.1.1. *Let R be a ring with 1, and let M be an abelian group. We say that M is a left R-module if there is a map*

$$R \times M \to M, \quad (r, m) \mapsto rm,$$

such that

$$\begin{aligned} r(m+n) &= rm + rn, \\ (r+s)m &= rm + sm, \\ (rs)m &= r(sm), \\ 1m &= m \end{aligned}$$

for all $r, s \in R$ and $m, n \in M$.

*A **right R-module** is defined similarly, using a map $(m, r) \mapsto mr$ and the axiom $m(rs) = (mr)s$. If R is commutative, left R-modules and right R-modules coincide. In general, a right R-module may be viewed as a left R^{op}-module.*

For each $r \in R$, the map $\varphi_r \colon M \to M$ given by $\varphi_r(m) = rm$ is a homomorphism of abelian groups. Let

$$\mathrm{End}(M) = \{\, \varphi \colon M \to M \mid \varphi \text{ is a homomorphism of abelian groups} \,\}.$$

Then $\mathrm{End}(M)$ is a ring under composition.

Theorem 1.1.2. *Let M be an abelian group. Then M is an R-module if and only if there is a ring homomorphism*

$$\rho \colon R \to \mathrm{End}(M).$$

Proof. (⇒) Suppose M is an R-module. Define $\rho\colon R \to \mathrm{End}(M)$ by

$$\rho(r)(m) = rm \qquad (r \in R,\ m \in M).$$

For each $r \in R$, the map $\rho(r)\colon M \to M$ is a homomorphism of abelian groups since

$$\rho(r)(m_1 + m_2) = r(m_1 + m_2) = rm_1 + rm_2.$$

Moreover, ρ is a ring homomorphism:

- $\rho(r+s)(m) = (r+s)m = rm + sm = \rho(r)(m) + \rho(s)(m)$, so $\rho(r+s) = \rho(r) + \rho(s)$;
- $\rho(rs)(m) = (rs)m = r(sm) = \rho(r)(\rho(s)(m))$, so $\rho(rs) = \rho(r) \circ \rho(s)$;
- $\rho(1)(m) = 1m = m$, so $\rho(1) = \mathrm{id}_M$.

(⇐) Suppose $\rho\colon R \to \mathrm{End}(M)$ is a ring homomorphism. Define scalar multiplication by

$$(r, m) \mapsto \rho(r)(m).$$

Then

$$\begin{aligned} r(m_1 + m_2) &= \rho(r)(m_1 + m_2) = \rho(r)(m_1) + \rho(r)(m_2), \\ (r+s)m &= \rho(r+s)(m) = \rho(r)(m) + \rho(s)(m), \\ (rs)m &= \rho(rs)(m) = \rho(r)(\rho(s)(m)) = r(sm), \\ 1m &= \rho(1)(m) = \mathrm{id}_M(m) = m. \end{aligned}$$

Hence M is an R-module. □

Example 1.1.3. *(1) A vector space over a field is a module over that field.*

(2) An abelian group is a $\mathbb{Z}$-module.

(3) Any ideal of a ring R is naturally an R-module.

(4) Let M be an abelian group. Then M is an $\mathrm{End}(M)$*-module via*

$$\mathrm{End}(M) \times M \to M, \quad (\varphi, m) \mapsto \varphi(m).$$

(5) Let V be a vector space over a field K, and let $A\colon V \to V$ be a linear transformation. Then V is a $K[x]$-module via

$$K[x] \times V \to V, \quad (f(x), v) \mapsto f(A)v.$$

(6) If $|M| = 1$, then M is called the zero module.

(7) Let R be a ring with 1 and let $A \in M_n(R)$. Then the solution set of $AX = 0$, where $X \in R^n$, is a right R-module.

Definition 1.1.4 (R-submodule)**.** *Let R be a ring with 1, and let M be a left R-module. A subset $N \subseteq M$ is called an* R-submodule *of M if*

1. $0 \in N$*;*

2. $n_1, n_2 \in N$ *implies* $n_1 + n_2 \in N$*;*

3. $n \in N$ *implies* $-n \in N$*;*

4. $r \in R$ *and* $n \in N$ *imply* $rn \in N$*.*

Equivalently, N *is a subgroup of the abelian group* $(M, +)$ *that is closed under the* R*-action.*

Example 1.1.5. *(1) If* V *is a vector space over a field* K*, then every subspace of* V *is a* K*-submodule of* V*.*

(2) If G *is an abelian group, then every subgroup of* G *is a* $\mathbb{Z}$*-submodule of* G*.*

(3) The left regular module ${}_RR$ *has precisely the left ideals of* R *as its submodules.*

(4) Let G *be an abelian group, viewed as an* $\mathrm{End}(G)$*-module. Then the* $\mathrm{End}(G)$*-submodules of* G *are exactly the fully invariant subgroups of* G*.*

Remark 1.1.6. *A natural question is the following: are submodules of finitely generated modules again finitely generated? In general the answer is no. This property holds for all submodules if and only if the ring is Noetherian.*

Example 1.1.7 (Submodule of a finitely generated module that is not finitely generated)**.** *Let* k *be a field, and consider the polynomial ring*

$$R = k[x_1, x_2, x_3, \dots]$$

in infinitely many variables.

- *The ring* R *is a cyclic left* R*-module, generated by* 1*.*
- *Let* $I = \langle x_1, x_2, x_3, \dots \rangle$*. Then* I *is a submodule of the left* R*-module* R*.*
- *The submodule* I *is not finitely generated. Indeed, suppose*

  $$I = \langle f_1, \dots, f_n \rangle$$

  for some polynomials $f_1, \dots, f_n \in R$*. Let* N *be the largest index of any variable appearing in any of the* f_i*. Then every element of* $\langle f_1, \dots, f_n \rangle$ *lies in the subring* $k[x_1, \dots, x_N]$*, but* $x_{N+1} \in I$ *does not. This is a contradiction.*

Thus I *is a submodule of the finitely generated* R*-module* R *which is not finitely generated.*

Definition 1.1.8. *Let* M *be an* R*-module, and let* $(M_i)_{i \in I}$ *be a family of submodules of* M*. Then*

$$\bigcap_{i \in I} M_i$$

is an R-submodule of M. Moreover,

$$\sum_{i\in I} M_i = \left\{ \sum_{i\in I}^{\text{finite}} m_i \,\middle|\, m_i \in M_i \right\}$$

is also an R-submodule of M.

Definition 1.1.9. *Let M be an R-module, and let $S \subseteq M$. The submodule generated by S is*

$$\langle S \rangle = \left\{ \sum_{j=1}^{n} a_j s_j \,\middle|\, n \geq 0,\ a_j \in R,\ s_j \in S \right\}.$$

Equivalently,

$$\langle S \rangle = \bigcap_{\substack{N \leq M \\ S \subseteq N}} N.$$

If $M = \langle S \rangle$, then we say that M is generated by S. In particular, if $|S| = 1$, then M is called cyclic*; if $|S| < \infty$, then M is called* finitely generated.

Definition 1.1.10. *Let R be a ring, and let M be a left R-module. We say that M is* Noetherian *if every ascending chain of submodules*

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$

stabilizes; that is, there exists $n \geq 1$ such that

$$M_n = M_{n+1} = M_{n+2} = \cdots .$$

Equivalently, M is Noetherian if every submodule of M is finitely generated.

1.2 Quotient Modules and Homomorphism Theorems

Definition 1.2.1. *Let R be a ring, let M be a left R-module, and let N be a submodule of M. The* quotient module *(or* factor module*) of M by N is the quotient abelian group*

$$M/N = \{\, x + N \mid x \in M \,\},$$

with scalar multiplication defined by

$$r(x+N) = rx + N \qquad (r \in R,\ x \in M).$$

This is well defined, and with this operation M/N becomes a left R-module.

Example 1.2.2. *(1) If I is a left ideal of a ring R, then R/I is a quotient R-module of the left regular module ${}_RR$.*

(2) For $n \in \mathbb{Z}$, the subgroup $n\mathbb{Z}$ is a submodule of the $\mathbb{Z}$-module $\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ is a cyclic quotient module.

Definition 1.2.3. *Let R be a ring, and let M and N be left R-modules. A group homomorphism*

$$f\colon M \to N$$

*is called an R-*module homomorphism *(or R-*linear map*) if*

$$f(rx) = rf(x)$$

for all $r \in R$ and $x \in M$.

Definition 1.2.4. *Let $f\colon M \to N$ be an R-module homomorphism. The* kernel *of f is the set*

$$\ker f = \{\, x \in M \mid f(x) = 0 \,\}.$$

The kernel is a submodule of M. The image *of f is the set*

$$\operatorname{Im} f = \{\, f(x) \mid x \in M \,\}.$$

The image is a submodule of N.

If $\ker f = 0$, then f is injective. If $\operatorname{Im} f = N$, then f is surjective. A bijective R-module homomorphism is called an isomorphism.

Example 1.2.5. *Let R be a commutative ring, and let M and N be R-modules. Then $\mathrm{Hom}_R(M, N)$ is naturally an R-module.*

Theorem 1.2.6. *If $f\colon M \to N$ is an R-module homomorphism, then there is a canonical R-module isomorphism*

$$M/\ker f \to \operatorname{Im} f.$$

Theorem 1.2.7. *Let N be an R-submodule of M, and let $\pi_N\colon M \to M/N$ be the quotient map. Then π_N induces a natural bijection*

$$\{\, R\text{-submodules of } M \text{ containing } N \,\} \to \{\, R\text{-submodules of } M/N \,\}.$$

Theorem 1.2.8. *If $N \subseteq H$ are two R-submodules of M, then*

$$M/H \cong (M/N)/(H/N).$$

Theorem 1.2.9. *If N and H are R-submodules of M, then*

$$\frac{H+N}{N} \cong \frac{H}{H \cap N}.$$

Proposition 1.2.10. *Let M be an R-module, let N be a submodule of M, let M' be an R-module, and let $\varphi\colon M \to M'$ be a morphism such that $N \subseteq \ker\varphi$. Then there exists a unique morphism*

$$\varphi_N\colon M/N \to M'$$

such that the following diagram commutes. Moreover,

$$\ker\varphi_N = (\ker\varphi)/N.$$

$$\begin{tikzcd} M \arrow[r, "\varphi"] \arrow[d, "\pi_N"'] & M' \\ M/N \arrow[ur, dashed, "\varphi_N"'] & \end{tikzcd}$$

Remark 1.2.11. *The pair $(M/N, \pi_N)$ is unique in the following sense: if (S, θ) is a pair such that*

- *S is an R-module,*
- *$\theta : M \to S$ is a surjective morphism satisfying the property of $(M/N, \pi_N)$ described in the above proposition,*

then there exists a unique isomorphism $\psi : M/N \to S$ such that the following diagram commutes.

$$\begin{tikzcd} M \arrow[r, two heads, "\theta"] \arrow[d, two heads, "\pi_N"'] & S \\ M/N \arrow[ur, dashed, "\exists!\,\psi"'] & \end{tikzcd}$$

Proof. Since (S, θ) satisfies the same universal property as $(M/N, \pi_N)$, we may apply that property to the canonical projection

$$\pi_N : M \to M/N.$$

Because $N \subseteq \ker(\pi_N)$, there exists a unique morphism

$$u : S \to M/N$$

such that

$$u \circ \theta = \pi_N.$$

On the other hand, since $(M/N, \pi_N)$ has the universal property of the quotient by N, and since $N \subseteq \ker(\theta)$, there exists a unique morphism

$$\psi : M/N \to S$$

such that

$$\psi \circ \pi_N = \theta.$$

We now show that ψ and u are inverse to each other.

First, compute

$$(u \circ \psi) \circ \pi_N = u \circ (\psi \circ \pi_N) = u \circ \theta = \pi_N.$$

But the identity morphism $\mathrm{id}_{M/N} : M/N \to M/N$ also satisfies

$$\mathrm{id}_{M/N} \circ \pi_N = \pi_N.$$

By the uniqueness part of the universal property of $(M/N, \pi_N)$, it follows that

$$u \circ \psi = \mathrm{id}_{M/N}.$$

Similarly,

$$(\psi \circ u) \circ \theta = \psi \circ (u \circ \theta) = \psi \circ \pi_N = \theta.$$

But the identity morphism $\mathrm{id}_S : S \to S$ also satisfies

$$\mathrm{id}_S \circ \theta = \theta.$$

By the uniqueness part of the universal property of (S, θ), we obtain

$$\psi \circ u = \mathrm{id}_S.$$

Therefore ψ is an isomorphism, with inverse u.

Finally, the uniqueness of ψ follows directly from the universal property of $(M/N, \pi_N)$: indeed, ψ is the unique morphism

$$\psi : M/N \to S$$

such that

$$\psi \circ \pi_N = \theta.$$

Hence there exists a unique isomorphism

$$\psi : M/N \to S$$

for which the diagram commutes. $\square$

Proposition 1.2.12. *There is a unique isomorphism $\bar{\varphi} : \operatorname{coim}\varphi \to \operatorname{Im}\varphi$ such that the following diagram commutes:*

$$\begin{tikzcd}
M \arrow[r, "\varphi"] \arrow[d, "\pi_{\ker(\varphi)}"'] & M' \\
\operatorname{coim}(\varphi) \arrow[r, "\overline{\varphi}"] & \operatorname{Im}(\varphi) \arrow[u, hook]
\end{tikzcd}$$

Recall:

$$\begin{aligned}
&\varphi : M \to M', \quad R\text{-morphism} \\
&\operatorname{coker}(\varphi) = M'/\operatorname{Im}(\varphi) \\
&\operatorname{coim}(\varphi) = M/\ker\varphi
\end{aligned}$$

Definition 1.2.13. *Let M be an R-module and I an ideal of R. Denote by IM the submodule of M generated by all elements am where $a \in I$, $m \in M$.*

$$IM = \left\{ \sum_{\textit{finite}} a_i m_i \mid a_i \in I,\ m_i \in M \right\}$$

Proposition 1.2.14. *Let M be an R-module. Then $R \to \operatorname{End}(M)$ is a ring homomorphism.*

1. *$R \to \operatorname{End}(M)$ factors through the natural surjection $R \to R/I$ if and only if $IM = 0$:*

$$\begin{tikzcd}
R \arrow[rr, "\rho"] \arrow[dr, two heads, "\pi_I"'] & & \operatorname{End}(M) \\
& R/I \arrow[ur, dashed, "\exists\,\rho_I"'] &
\end{tikzcd}$$

   Where ρ factors through π_I means $\exists \rho_I : R/I \to \operatorname{End}(M)$ such that the diagram above commutes.

2. *The module M/IM is naturally endowed with a structure of R/I-module.*

3. *Let $\varphi : M \to M'$ be an R-homomorphism. Then $\varphi(IM) \subseteq IM'$, so φ induces a morphism of R/I-modules*

$$M/IM \to M'/IM'.$$

4. *If $M = M_1 \oplus \cdots \oplus M_n$, then $M/IM = M_1/IM_1 \oplus \cdots \oplus M_n/IM_n$.*

5. *$R^n/IR^n \cong (R/IR)^n$.*

Proof. **(1)** Suppose first that ρ factors through π_I. Then there exists a ring homomorphism

$$\rho_I : R/I \to \operatorname{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

If $a \in I$, then $\pi_I(a) = 0$, so

$$\rho(a) = \rho_I(0) = 0.$$

Therefore $am = 0$ for every $m \in M$. Since IM is generated by elements of the form am

with $a \in I$ and $m \in M$, it follows that

$$IM = 0.$$

Conversely, assume that $IM = 0$. Then for every $a \in I$ and every $m \in M$ we have

$$\rho(a)(m) = am = 0.$$

Thus $\rho(a) = 0$, so $I \subseteq \ker \rho$. Hence ρ factors uniquely through the quotient ring R/I: there exists a unique ring homomorphism

$$\rho_I : R/I \to \mathrm{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

(2) Define an action of R/I on M/IM by

$$(r + I) \cdot (m + IM) := rm + IM.$$

We must show that this is well-defined.

If $r + I = r' + I$, then $r - r' \in I$, so

$$(r - r')m \in IM,$$

hence

$$rm + IM = r'm + IM.$$

If $m + IM = m' + IM$, then $m - m' \in IM$. Since IM is a submodule of M, we have

$$r(m - m') \in IM,$$

so

$$rm + IM = rm' + IM.$$

Thus the action is well-defined. The module axioms follow immediately from the module axioms on M. Therefore M/IM is naturally an R/I-module.

(3) Let $\varphi : M \to M'$ be an R-module homomorphism. Take any element of IM, say

$$x = \sum_{j=1}^{t} a_j m_j \qquad (a_j \in I,\ m_j \in M).$$

Then

$$\varphi(x) = \sum_{j=1}^{t} \varphi(a_j m_j) = \sum_{j=1}^{t} a_j \varphi(m_j).$$

Since each $a_j \in I$ and each $\varphi(m_j) \in M'$, it follows that

$$a_j\varphi(m_j) \in IM'.$$

Hence $\varphi(x) \in IM'$, and therefore

$$\varphi(IM) \subseteq IM'.$$

Thus we may define

$$\overline{\varphi} : M/IM \to M'/IM', \qquad \overline{\varphi}(m + IM) = \varphi(m) + IM'.$$

This is well-defined: if $m + IM = m' + IM$, then $m - m' \in IM$, so

$$\varphi(m) - \varphi(m') = \varphi(m - m') \in IM',$$

hence

$$\varphi(m) + IM' = \varphi(m') + IM'.$$

Finally, $\overline{\varphi}$ is R/I-linear, since

$$\overline{\varphi}\big((r+I)\cdot(m+IM)\big) = \overline{\varphi}(rm+IM) = \varphi(rm)+IM' = r\varphi(m)+IM' = (r+I)\cdot\overline{\varphi}(m+IM).$$

(4) Assume that

$$M = M_1 \oplus \cdots \oplus M_n.$$

We first show that

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

If $x \in IM$, then x is a finite sum of elements of the form am with $a \in I$ and $m \in M$. Writing

$$m = m_1 + \cdots + m_n \qquad (m_i \in M_i),$$

we get

$$am = am_1 + \cdots + am_n,$$

and each $am_i \in IM_i$. Hence $x \in IM_1 + \cdots + IM_n$, so

$$IM \subseteq IM_1 + \cdots + IM_n.$$

The reverse inclusion is immediate because each $M_i \subseteq M$, hence each $IM_i \subseteq IM$. Therefore

$$IM = IM_1 + \cdots + IM_n.$$

To see that the sum is direct, suppose

$$x_1 + \cdots + x_n = 0 \qquad \text{with } x_i \in IM_i \subseteq M_i.$$

Since

$$M = M_1 \oplus \cdots \oplus M_n$$

is a direct sum, we must have $x_i = 0$ for all i. Thus

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

Now define

$$\Psi : M_1/IM_1 \oplus \cdots \oplus M_n/IM_n \to M/IM$$

by

$$\Psi(m_1 + IM_1, \ldots, m_n + IM_n) = (m_1 + \cdots + m_n) + IM.$$

This map is well-defined and R/I-linear. It is surjective because every element of M can be written as $m_1 + \cdots + m_n$. It is injective because

$$(m_1 + \cdots + m_n) + IM = IM$$

means that

$$m_1 + \cdots + m_n \in IM = IM_1 \oplus \cdots \oplus IM_n,$$

so each $m_i \in IM_i$. Hence

$$m_i + IM_i = IM_i$$

for all i, and the kernel is trivial. Therefore Ψ is an isomorphism:

$$M/IM \cong M_1/IM_1 \oplus \cdots \oplus M_n/IM_n.$$

(5) Since

$$R^n = R \oplus \cdots \oplus R$$

(n copies), part (4) yields

$$R^n/IR^n \cong R/IR \oplus \cdots \oplus R/IR.$$

Therefore

$$R^n/IR^n \cong (R/IR)^n.$$

This completes the proof. □

1.3 Direct Sums and Free Modules

Definition 1.3.1. *Let $(M_i)_{i\in I}$ be a family of R-modules. The* direct product *of $(M_i)_{i\in I}$ is the module*

$$\prod_{i\in I} M_i = \left\{ \varphi\colon I \to \bigcup_{i\in I} M_i \;\middle|\; \varphi(i) \in M_i \text{ for all } i \in I \right\},$$

with componentwise addition and scalar multiplication:

$$R \times \prod_{i\in I} M_i \to \prod_{i\in I} M_i, \quad (r,\varphi) \mapsto (r\varphi(i))_{i\in I}$$

The direct sum *of* $(M_i)_{i\in I}$ *is the submodule*

$$\bigoplus_{i\in I} M_i = \left\{ \varphi \in \prod_{i\in I} M_i \,\middle|\, \#\{\, i \in I \mid \varphi(i) \neq 0 \,\} < \infty \right\}.$$

These two modules agree when $|I| < \infty$. *In the category of* R*-modules, the direct product is the categorical product and the direct sum is the categorical coproduct.*

Theorem 1.3.2. *Suppose that* $M_1, \ldots, M_n$ *are* R*-submodules of* M *such that* $M = M_1 + \cdots + M_n$. *Let* $I = [n]$. *Then the following conditions are equivalent:*

1. *The map* $\varphi\colon \bigoplus_{i\in I} M_i \to M$ *given by* $\varphi((m_i)_{i\in I}) = \sum_{i\in I} m_i$ *is an isomorphism.*
2. *Any element of* M *can be uniquely written as a sum of elements of* M_i.
3. *The zero element can be uniquely written as a sum of elements of the* M_i.
4. *For each* $i \in I$, *one has*
$$M_i \cap \sum_{j\neq i} M_j = 0.$$

Proof. We prove

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (1).$$

(1) ⇒ (2)**.** If φ is an isomorphism, then for every $m \in M$ there exists a unique

$$(m_1, \ldots, m_n) \in \bigoplus_{i=1}^{n} M_i$$

such that

$$m = \varphi(m_1, \ldots, m_n) = m_1 + \cdots + m_n.$$

Hence every element of M can be written uniquely as a sum of elements of the M_i.

(2) ⇒ (3)**.** Apply (2) to the element $0 \in M$.

(3) ⇒ (4)**.** Fix $i \in I$, and let

$$x \in M_i \cap \sum_{j\neq i} M_j.$$

Then $x \in M_i$, and there exist $m_j \in M_j$ for $j \neq i$ such that

$$x = \sum_{j\neq i} m_j.$$

Therefore

$$0 = x - \sum_{j \neq i} m_j$$

is a decomposition of 0 as a sum of elements from the submodules $M_1, \ldots, M_n$. By the uniqueness in (3), this decomposition must be the trivial one. In particular,

$$x = 0.$$

Hence

$$M_i \cap \sum_{j \neq i} M_j = 0.$$

(4) ⇒ (1). Since

$$M = M_1 + \cdots + M_n,$$

the map φ is surjective. It remains to show that φ is injective.

Suppose that

$$\varphi(m_1, \ldots, m_n) = 0,$$

that is,

$$m_1 + \cdots + m_n = 0 \qquad \text{with } m_i \in M_i.$$

For each $i \in I$, we have

$$m_i = -\sum_{j \neq i} m_j \in \sum_{j \neq i} M_j.$$

Since also $m_i \in M_i$, it follows that

$$m_i \in M_i \cap \sum_{j \neq i} M_j = 0.$$

Thus $m_i = 0$ for all i, so

$$\ker \varphi = 0.$$

Therefore φ is injective, and hence an isomorphism.

This proves that the four conditions are equivalent. □

Definition 1.3.3 (free module)**.** *Let M be an R-module.*

1. $x_1, \ldots, x_n \in M$ *are linearly independent if*

$$a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = 0 \to a_1 = a_2 = \cdots = a_n = 0.$$

2. *M is called free if it is generated by a linearly independent subset. This subset is called a basis of M.*

3. *If M is an R-module with a basis $x_1, \ldots, x_n$, then $M = \langle x_1 \rangle \oplus \cdots \oplus \langle x_n \rangle$.*

Example 1.3.4. *$R \oplus \cdots \oplus R = R^n$.*

Theorem 1.3.5. *Let R be a commutative ring with 1. Let M be a finitely generated free R-module. Then any basis of M has the same cardinality. The number of elements in a basis is called the rank of the free module.*

Proof. Let $B = \{x_i\}_{i\in I}$ be a basis of M. We first show that any basis is finite. Since M is finitely generated, choose generators $y_1, \dots, y_n$ of M. Each y_k is a finite R–linear combination of elements of B, so there exists a finite subset $I_0 \subset I$ such that

$$y_1, \dots, y_n \in \sum_{i\in I_0} Rx_i.$$

Hence

$$M = \sum_{k=1}^{n} Ry_k \subseteq \sum_{i\in I_0} Rx_i \subseteq M,$$

so $M = \sum_{i\in I_0} Rx_i$. If $I \setminus I_0 \neq \varnothing$, pick $j \in I \setminus I_0$. Then

$$x_j \in M = \sum_{i\in I_0} Rx_i,$$

contradicting that B is linearly independent. Thus $I = I_0$ is finite.

Now let $x_1, \dots, x_r$ and $y_1, \dots, y_s$ be two bases of M. Let $J \subset R$ be a maximal ideal and set $F := R/J$, a field. Then M/JM is naturally an F-vector space. Write $\overline{m}$ for the class of $m \in M$ in M/JM, and $\overline{a}$ for the class of $a \in R$ in F. We claim that $\overline{x_1}, \dots, \overline{x_r}$ form an F–basis of M/JM.

Spanning. For any $m \in M$, write $m = \sum_{i=1}^{r} a_i x_i$. Then

$$\overline{m} = \sum_{i=1}^{r} \overline{a_i}\,\overline{x_i},$$

so $\overline{x_1}, \dots, \overline{x_r}$ span M/JM.

Linear independence. Suppose

$$\sum_{i=1}^{r} \overline{a_i}\,\overline{x_i} = 0 \quad \text{in } M/JM.$$

Then $\sum_{i=1}^{r} a_i x_i \in JM$, so there exist $a'_1, \dots, a'_r \in J$ such that

$$\sum_{i=1}^{r} a_i x_i = \sum_{i=1}^{r} a'_i x_i.$$

Hence

$$\sum_{i=1}^{r} (a_i - a'_i)\, x_i = 0.$$

By linear independence of $x_1, \dots, x_r$, we get $a_i - a'_i = 0$ for all i, so $a_i \in J$, thus $\overline{a_i} = 0$ for all i. Therefore $\overline{x_1}, \dots, \overline{x_r}$ are F-linearly independent, and form an F–basis of M/JM.

By the same argument, $\overline{y_1}, \ldots, \overline{y_s}$ also form an F–basis of M/JM. Hence

$$r = \dim_F(M/JM) = s.$$

Therefore any two bases of M have the same cardinality. $\square$

Chapter 2

Lecture 2: Tensor Product

2.1 Annihilator, Torsion and Projectivity of Free Modules

Definition 2.1.1. *Let M be an R-module, and let $m \in M$. The cyclic submodule generated by m is*

$$Rm = \langle m \rangle = \{\, rm \mid r \in R \,\}.$$

The annihilator *of m in R is*

$$\operatorname{Ann}_R(m) = \{\, r \in R \mid rm = 0 \,\}.$$

It is a left ideal of R.

Proposition 2.1.2. *For every $m \in M$, there is an isomorphism of left R-modules*

$$R/\operatorname{Ann}_R(m) \cong Rm.$$

In particular, every cyclic module is isomorphic to a quotient of R.

Definition 2.1.3. *An element $m \in M$ is called*

- ***torsion-free*** *if* $\operatorname{Ann}_R(m) = 0$*;*
- ***torsion*** *if* $\operatorname{Ann}_R(m) \neq 0$.

A module is called torsion-free *if every nonzero element is torsion-free.*

Definition 2.1.4. *The* torsion submodule *of M is*

$$\operatorname{Tor}(M) = \{\, m \in M \mid \operatorname{Ann}_R(m) \neq 0 \,\}.$$

Lemma 2.1.5. *If R is an integral domain, then the torsion part of M is a submodule of M.*

Proof. We verify the submodule criteria.

Non-empty. Since $r \cdot 0_M = 0_M$ for every $r \in R$ and $R \neq 0$, we have $\mathrm{Ann}_R(0_M) = R \neq 0$, so $0_M \in \mathrm{Tor}(M)$.

Closed under addition. Let $m, n \in \mathrm{Tor}(M)$. Then there exist $\lambda, \mu \in R \setminus \{0\}$ such that $\lambda m = 0$ and $\mu n = 0$. Hence

$$(\lambda\mu)(m + n) = \mu(\lambda m) + \lambda(\mu n) = 0.$$

Since R is an integral domain and $\lambda, \mu \neq 0$, we have $\lambda\mu \neq 0$, so $\mathrm{Ann}_R(m + n) \neq 0$ and $m + n \in \mathrm{Tor}(M)$.

Closed under scalar multiplication. Let $m \in \mathrm{Tor}(M)$ and $r \in R$. Choose $\lambda \in R \setminus \{0\}$ with $\lambda m = 0$. Then

$$\lambda(rm) = r(\lambda m) = 0,$$

so $\mathrm{Ann}_R(rm) \neq 0$ and $rm \in \mathrm{Tor}(M)$.

Therefore $\mathrm{Tor}(M)$ is a submodule of M. □

Example 2.1.6. *(1) The element $0 \in M$ is torsion.*

(2) Let R be a commutative ring with 1. Then R is an integral domain if and only if the left regular module ${}_RR$ is torsion-free. In this case $\mathrm{Tor}({}_RR) = 0$ and $\mathrm{Tor}(R^n) = 0$ for all $n \geq 1$.

(3) If I is a nonzero ideal of R, then $\mathrm{Tor}(R/I) = R/I$ when R/I is regarded as an R-module. If we regard R/I as an R/I-module, then it is torsion-free if and only if I is a prime ideal.

(4) More generally, let I be any nonzero ideal of R, and let M be an R-module. Then $\mathrm{Tor}(M/IM) = M/IM$ when M/IM is regarded as an R-module.

(5) Even when I is prime, M/IM need not be torsion-free as an R/I-module. For $\overline{m} \in M/IM$, one has $\mathrm{Ann}_{R/I}(\overline{m}) \neq 0$ if and only if there exists $r \in R \setminus I$ with $rm \in IM$. Thus primeness of I does not rule out nonzero operators in R/I annihilating nonzero classes in M/IM.

For instance, let $R = k[x, y]$, let $I = (y)$, and let $M = R/(x) \cong k[y]$. Then $IM = (y)$ and $M/IM \cong k$, while $x \cdot M = 0$ but $x \notin I$; hence $x + I \in R/I$ is nonzero and annihilates all of M/IM. If instead I is maximal, then R/I is a field, nonzero scalars act invertibly, and M/IM is always torsion-free over R/I. Item (3) is the case $M = R$.

Remark 2.1.7. *Let R be a commutative ring with 1, let M be an R-module, and fix $r \in R$. Then the map*

$$f \colon M \to M, \qquad f(m) = rm,$$

is an R-module homomorphism. If moreover M is torsion-free and $r \neq 0$, then f is a monomorphism.

Lemma 2.1.8. *If R is an integral domain and M is an R-module, then*

1. $\mathrm{Tor}(\mathrm{Tor}(M)) = \mathrm{Tor}(M)$.

2. $\operatorname{Tor}(M/\operatorname{Tor}(M)) = 0$.

Proof.

1. Since $\operatorname{Tor}(M)$ is a submodule of M, its torsion submodule is

$$\operatorname{Tor}(\operatorname{Tor}(M)) = \{\, m \in \operatorname{Tor}(M) \mid \operatorname{Ann}_R(m) \neq 0 \,\}.$$

Every $m \in \operatorname{Tor}(M)$ already satisfies $\operatorname{Ann}_R(m) \neq 0$, so $\operatorname{Tor}(\operatorname{Tor}(M)) \subseteq \operatorname{Tor}(M)$. Conversely, $\operatorname{Tor}(M) \subseteq \operatorname{Tor}(\operatorname{Tor}(M))$ because the annihilator of m in R does not depend on the ambient module: for any $m \in \operatorname{Tor}(M)$, its annihilator computed in the submodule $\operatorname{Tor}(M)$ is the same as $\operatorname{Ann}_R(m) \neq 0$, so $m \in \operatorname{Tor}(\operatorname{Tor}(M))$. Hence $\operatorname{Tor}(\operatorname{Tor}(M)) = \operatorname{Tor}(M)$.

2. Let $\pi \colon M \to M/\operatorname{Tor}(M)$ be the canonical projection. Write $\overline{m} = \pi(m) = m + \operatorname{Tor}(M)$. Suppose $\overline{m} \in \operatorname{Tor}(M/\operatorname{Tor}(M))$, i.e. there exists $\lambda \in R \setminus \{0\}$ such that

$$\lambda\,\overline{m} = \overline{0} \quad \text{in } M/\operatorname{Tor}(M).$$

This means $\lambda m \in \operatorname{Tor}(M)$, so there exists $\mu \in R \setminus \{0\}$ with $\mu(\lambda m) = 0$, i.e.

$$(\mu\lambda)\, m = 0.$$

Since R is an integral domain and $\mu, \lambda \neq 0$, we have $\mu\lambda \neq 0$, hence $m \in \operatorname{Tor}(M)$, so $\overline{m} = \overline{0}$. Therefore $\operatorname{Tor}(M/\operatorname{Tor}(M)) = 0$.

□

Definition 2.1.9. *Let J be a subscript set, denote $R^J = \prod_{j\in J} R.R^{(J)} = \bigoplus_{j\in J} R$.*

Let M be an R-module. Let $x_j \in M$ where $j \in J$ be a family of elements of M. Then the submodule generated by $(x_j)_{j\in J}$ is denoted by $\sum_{j\in J} Rx_j$.

Corollary 2.1.10. *We have a morphism of R-modules $\phi : R^{(J)} \to M$ such that $\phi(r_j)_{j\in J} = \sum_{j\in J} r_j x_j$ for all $(r_j)_{j\in J} \in R^{(J)}$.*

- *ϕ is injective if and only if $\{x_j\}_{j\in J}$ are linearly independent.*
- *ϕ is surjective if and only if $\{x_j\}_{j\in J}$ generate M.*
- *ϕ is an isomorphism if and only if M is free with basis $\{x_j\}_{j\in J}$.*

Lemma 2.1.11. *Let L be a free R-module with basis $\{x_j\}_{j\in J}$. For any R-module M, the map*

$$\Phi : \operatorname{Hom}_R(L, M) \to M^J, \qquad f \mapsto (f(x_j))_{j\in J}$$

is a bijection.

Proof. We prove that Φ is injective and surjective.

Injectivity. Let $f, g \in \mathrm{Hom}_R(L, M)$ and suppose that

$$\Phi(f) = \Phi(g).$$

Then for every $j \in J$ we have

$$f(x_j) = g(x_j).$$

Since $\{x_j\}_{j\in J}$ is a basis of L, every element $x \in L$ can be written uniquely as a finite sum

$$x = \sum_{j\in J} a_j x_j$$

with $a_j \in R$ and all but finitely many a_j equal to 0. Hence

$$f(x) = f\left(\sum_{j\in J} a_j x_j\right) = \sum_{j\in J} a_j f(x_j) = \sum_{j\in J} a_j g(x_j) = g\left(\sum_{j\in J} a_j x_j\right) = g(x).$$

Thus $f = g$, so Φ is injective.

Surjectivity. Let $(m_j)_{j\in J} \in M^J$ be given. We define a map

$$f : L \to M$$

by declaring that for

$$x = \sum_{j\in J} a_j x_j \qquad (a_j \in R,\ \text{all but finitely many } a_j = 0),$$

we set

$$f(x) := \sum_{j\in J} a_j m_j.$$

This is well defined because each element of L has a unique expression as a finite linear combination of the basis elements.

Now let

$$x = \sum_{j\in J} a_j x_j, \qquad y = \sum_{j\in J} b_j x_j, \qquad r \in R.$$

Then

$$f(x+y) = f\left(\sum_{j\in J}(a_j + b_j)x_j\right) = \sum_{j\in J}(a_j + b_j)m_j = \sum_{j\in J} a_j m_j + \sum_{j\in J} b_j m_j = f(x) + f(y),$$

and

$$f(rx) = f\left(\sum_{j\in J}(ra_j)x_j\right) = \sum_{j\in J}(ra_j)m_j = r\sum_{j\in J} a_j m_j = rf(x).$$

So f is an R-module homomorphism. Moreover, for every $j \in J$,

$$f(x_j) = m_j.$$

Therefore
$$\Phi(f) = (m_j)_{j\in J}.$$
Thus Φ is surjective.

Since Φ is both injective and surjective, it is a bijection. $\square$

Proposition 2.1.12. *Let L be a free R-module.*

1. *Let $\pi : X \to Y$ be an epimorphism of R-modules. For any R-module homomorphism $\varphi : L \to Y$ there exists a morphism $\tilde{\varphi} : L \to X$ such that the following diagram commutes:*

$$\begin{tikzcd} & L \arrow[dl, "\tilde{\varphi}"'] \arrow[d, "\varphi"] & \\ X \arrow[r, "\pi"'] & Y \arrow[r] & 0 \end{tikzcd}$$

2. *In particular, for any R-module M, any epimorphism $\pi : M \to L$ has a section, that is, there exists a morphism $\sigma : L \to M$ such that $\pi \circ \sigma = \mathrm{id}_L$. In that case σ is injective, $\mathrm{Im}(\sigma) \cong L$, and*

$$M = \ker \pi \oplus \mathrm{Im}(\sigma).$$

$$\begin{tikzcd} 0 \arrow[r] & \ker \pi \arrow[r] & M \arrow[r, two heads, shift left, "\pi"] & L \arrow[l, dashed, shift left, "\sigma"] \arrow[r] & 0 \end{tikzcd}$$

Proof. 1. Since L is free, let $\{x_j\}_{j\in J}$ be a basis of L.

For each $j \in J$, since $\pi : X \to Y$ is surjective, there exists an element
$$\tilde{x}_j \in X$$
such that
$$\pi(\tilde{x}_j) = \varphi(x_j).$$
Because L is free with basis $\{x_j\}_{j\in J}$, there exists a unique R-module homomorphism
$$\tilde{\varphi} : L \to X$$
satisfying
$$\tilde{\varphi}(x_j) = \tilde{x}_j \qquad \text{for all } j \in J.$$
We now check that $\pi \circ \tilde{\varphi} = \varphi$. For each basis element x_j,
$$(\pi \circ \tilde{\varphi})(x_j) = \pi(\tilde{\varphi}(x_j)) = \pi(\tilde{x}_j) = \varphi(x_j).$$
Hence $\pi\circ\tilde{\varphi}$ and φ agree on a basis of L, so they are equal as R-module homomorphisms. Therefore the diagram commutes.

2. Apply part (1) with
$$X = M, \qquad Y = L, \qquad \varphi = \mathrm{id}_L\,.$$
Since $\pi : M \to L$ is an epimorphism, there exists an R-module homomorphism
$$\sigma : L \to M$$
such that
$$\pi \circ \sigma = \mathrm{id}_L\,.$$
Thus σ is a section of π.

We first show that σ is injective. If $\sigma(\ell) = 0$ for some $\ell \in L$, then
$$\ell = (\pi \circ \sigma)(\ell) = \pi(0) = 0.$$
Hence $\ker \sigma = 0$, so σ is injective.

Next, since σ is injective, it induces an isomorphism
$$L \xrightarrow{\sim} \mathrm{Im}(\sigma), \qquad \ell \mapsto \sigma(\ell).$$
Thus $\mathrm{Im}(\sigma) \cong L$.

It remains to prove that
$$M = \ker \pi \oplus \mathrm{Im}(\sigma).$$
First, let $m \in M$. Then
$$m = \big(m - \sigma(\pi(m))\big) + \sigma(\pi(m)).$$
Now
$$\pi\big(m - \sigma(\pi(m))\big) = \pi(m) - (\pi \circ \sigma)(\pi(m)) = \pi(m) - \pi(m) = 0,$$
so
$$m - \sigma(\pi(m)) \in \ker \pi,$$
while clearly
$$\sigma(\pi(m)) \in \mathrm{Im}(\sigma).$$
Hence
$$M = \ker \pi + \mathrm{Im}(\sigma).$$
Finally, suppose
$$z \in \ker \pi \cap \mathrm{Im}(\sigma).$$
Then $z = \sigma(\ell)$ for some $\ell \in L$, and also $\pi(z) = 0$. Therefore
$$0 = \pi(z) = \pi(\sigma(\ell)) = (\pi \circ \sigma)(\ell) = \ell.$$

So $z = \sigma(\ell) = 0$. Thus

$$\ker \pi \cap \operatorname{Im}(\sigma) = 0.$$

Therefore

$$M = \ker \pi \oplus \operatorname{Im}(\sigma).$$

□

Exercise 2.1.13. *Let L be a free R-module. Let P be a direct summand of L. Show that whenever*

$$\pi : X \to Y$$

is an epimorphism and

$$\varphi : P \to Y$$

is a morphism, there exists a morphism

$$\tilde{\varphi} : P \to X$$

such that

$$\varphi = \pi \circ \tilde{\varphi}.$$

Proof. Since P is a direct summand of L, there exist R-module homomorphisms

$$i : P \to L \qquad \text{and} \qquad p : L \to P$$

such that

$$p \circ i = \mathrm{id}_P\,.$$

(Here i is the inclusion and p is the projection.)

Consider the composite

$$\psi := \varphi \circ p : L \to Y.$$

Since L is free and $\pi : X \to Y$ is an epimorphism, by the lifting property of free modules there exists an R-module homomorphism

$$\widetilde{\psi} : L \to X$$

such that

$$\pi \circ \widetilde{\psi} = \psi.$$

Now define

$$\widetilde{\varphi} := \widetilde{\psi} \circ i : P \to X.$$

Then

$$\pi \circ \widetilde{\varphi} = \pi \circ \widetilde{\psi} \circ i = \psi \circ i = \varphi \circ p \circ i = \varphi \circ \mathrm{id}_P = \varphi.$$

Therefore there exists a morphism $\widetilde{\varphi} : P \to X$ such that

$$\varphi = \pi \circ \widetilde{\varphi}.$$

□

2.2 Tensor product

Let R be a commutative ring, and let M and N be R-modules. Roughly speaking, the tensor product $M \otimes_R N$ linearizes bilinear maps out of $M \times N$.

Definition 2.2.1. *Let R be a commutative ring, and let M, N, and K be R-modules. A map*

$$f\colon M \times N \to K$$

is called bilinear *if for all $r \in R$, $m, m_1, m_2 \in M$, and $n, n_1, n_2 \in N$, one has*

- $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$;
- $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$;
- $f(rm, n) = rf(m, n)$;
- $f(m, rn) = rf(m, n)$.

Example 2.2.2. *If $f\colon M \times N \to K$ is bilinear, then*

$$f(x, 0) = 0 = f(0, y)$$

for all $x \in M$ and $y \in N$. Moreover,

$$f(-x, y) = -f(x, y), \qquad f(x, -y) = -f(x, y)$$

for all $x \in M$ and $y \in N$.

Definition 2.2.3. *The* tensor product *of M and N is a pair $(M \otimes_R N, t_{M,N})$, where $M \otimes_R N$ is an R-module and*

$$t_{M,N}\colon M \times N \to M \otimes_R N$$

is a bilinear map, usually written $t_{M,N}(m, n) = m \otimes n$, such that the following universal property holds:

whenever X is an R-module and $f\colon M \times N \to X$ is bilinear, there exists a unique R-module homomorphism

$$\tilde{f}\colon M \otimes_R N \to X$$

such that the following diagram commutes:

$$\begin{tikzcd} M \times N \arrow[r, "f"] \arrow[d, "t_{M,N}"'] & X \\ M \otimes_R N \arrow[ur, dashed, "\tilde{f}"'] & \end{tikzcd}$$

Theorem 2.2.4. *The tensor product of M and N exists and is unique up to isomorphism.*

Proof. We first prove uniqueness.

Suppose (T_1, t_1) and (T_2, t_2) are two tensor products of M and N. By the universal property, there exist unique R-module homomorphisms

$$\varphi\colon T_1 \to T_2, \qquad \psi\colon T_2 \to T_1$$

such that

$$t_2 = \varphi \circ t_1, \qquad t_1 = \psi \circ t_2.$$

Hence

$$\varphi \circ \psi \circ t_2 = \varphi \circ t_1 = t_2, \qquad \psi \circ \varphi \circ t_1 = \psi \circ t_2 = t_1.$$

By uniqueness in the universal property, it follows that

$$\varphi \circ \psi = \mathrm{id}_{T_2}, \qquad \psi \circ \varphi = \mathrm{id}_{T_1}.$$

Therefore φ and ψ are inverse isomorphisms, so $T_1 \cong T_2$.

Now we prove existence.

Let F be the free R-module with basis $M \times N$, say

$$F = \bigoplus_{(m,n)\in M\times N} Re_{m,n}.$$

Let B be the submodule of F generated by the following elements:

- $e_{m_1+m_2,n} - e_{m_1,n} - e_{m_2,n}$;
- $e_{m,n_1+n_2} - e_{m,n_1} - e_{m,n_2}$;
- $e_{rm,n} - re_{m,n}$;
- $e_{m,rn} - re_{m,n}$,

where $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, and $r \in R$.

Define

$$M \otimes_R N = F/B, \qquad t_{M,N}(m,n) = e_{m,n} + B.$$

Then $t_{M,N}\colon M \times N \to M \otimes_R N$ is a well-defined bilinear map.

It remains to verify the universal property.

Let X be an R-module, and let $f\colon M \times N \to X$ be bilinear. Since F is free on the set $M \times N$, there exists a unique R-module homomorphism

$$\hat{f}\colon F \to X$$

such that $\hat{f}(e_{m,n}) = f(m,n)$ for all $(m,n) \in M \times N$.

We claim that $B \subseteq \ker \hat{f}$. Indeed, since f is bilinear,

$$\begin{aligned}
\hat{f}(e_{m_1+m_2,n} - e_{m_1,n} - e_{m_2,n}) &= f(m_1+m_2,n) - f(m_1,n) - f(m_2,n) = 0, \\
\hat{f}(e_{m,n_1+n_2} - e_{m,n_1} - e_{m,n_2}) &= f(m,n_1+n_2) - f(m,n_1) - f(m,n_2) = 0, \\
\hat{f}(e_{rm,n} - re_{m,n}) &= f(rm,n) - rf(m,n) = 0, \\
\hat{f}(e_{m,rn} - re_{m,n}) &= f(m,rn) - rf(m,n) = 0.
\end{aligned}$$

Since B is generated by these elements and $\ker \hat{f}$ is a submodule, we obtain $B \subseteq \ker \hat{f}$.

By the universal property of quotient modules, $\hat{f}$ induces a unique R-module homomorphism

$$\tilde{f} \colon M \otimes_R N = F/B \to X$$

such that $\tilde{f} \circ \pi = \hat{f}$, where $\pi \colon F \to F/B$ is the canonical projection. In particular,

$$\tilde{f}(m \otimes n) = \tilde{f}(e_{m,n} + B) = \hat{f}(e_{m,n}) = f(m,n),$$

so $\tilde{f} \circ t_{M,N} = f$.

For uniqueness, suppose $g \colon M \otimes_R N \to X$ is another R-module homomorphism such that $g \circ t_{M,N} = f$. Then

$$g(m \otimes n) = f(m,n) = \tilde{f}(m \otimes n)$$

for all $m \in M$ and $n \in N$. Since the elementary tensors generate $M \otimes_R N$ as an R-module, it follows that $g = \tilde{f}$.

Therefore $(M \otimes_R N, t_{M,N})$ is a tensor product of M and N. $\square$

Definition 2.2.5. *The set of elements of the form $m \otimes n$ for $m \in M$ and $n \in N$ is called the set of elementary tensors. This set generates $M \otimes_R N$ as an R-module.*

Example 2.2.6. *Fix $m \in M$. The map $N \to M \otimes_R N, n \mapsto m \otimes n$ is a morphism of R-modules. Its image $m \otimes_R N = \{m \otimes n \mid n \in N\}$ is a submodule of $M \otimes_R N$ which is isomorphic to a quotient $N/\operatorname{Ann}_R(m)N$.*

Proposition 2.2.7. *We have the following isomorphisms:*

1. $(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$ *where* $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$.

2. $M_1 \otimes_R M_2 \cong M_2 \otimes_R M_1$ *where* $m_1 \otimes m_2 \mapsto m_2 \otimes m_1$.

3. $M_1 \otimes_R (M_2 \oplus M_3) \cong (M_1 \otimes_R M_2) \oplus (M_1 \otimes_R M_3)$ *where* $m_1 \otimes (m_2 \oplus m_3) \mapsto (m_1 \otimes m_2) + (m_1 \otimes m_3)$. *And more generally,*

$$M_1 \otimes_R (\bigoplus_{i=1}^{n} M_i) \cong \bigoplus_{i=1}^{n} (M_1 \otimes_R M_i)$$

where $m_1 \otimes (m_2 \oplus \cdots \oplus m_n) \mapsto (m_1 \otimes m_2) + \cdots + (m_1 \otimes m_n)$.

4. $R \otimes_R M \cong M$ *where* $r \otimes m \mapsto rm$. *And more generally,* $R^{(I)} \otimes_R M \cong M^{(I)}$ *where* $(r_i)_{i\in I} \otimes m \mapsto (r_i m)_{i\in I}$.

Proof. We prove (1):$(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$.

Step 1. We construct $\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$.

Fix $m_3 \in M_3$. For each m_3, define $f_{m_3} : M_1 \times M_2 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ by $f_{m_3}(m_1, m_2) = m_1 \otimes (m_2 \otimes m_3)$. This is bilinear in (m_1, m_2), so by the universal property of $M_1 \otimes_R M_2$, there exists a unique R-module homomorphism $\tilde{f}_{m_3} : M_1 \otimes_R M_2 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ such that $\tilde{f}_{m_3}(m_1 \otimes m_2) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{tikzcd} M_1 \times M_2 \arrow[r, "f_{m_3}"] \arrow[d, "t_{M_1,M_2}"'] & M_1 \otimes_R (M_2 \otimes_R M_3) \\ M_1 \otimes_R M_2 \arrow[ur, dashed, "\tilde{f}_{m_3}"'] & \end{tikzcd}$$

Now define $g : (M_1 \otimes_R M_2) \times M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ by $g(\alpha, m_3) = \tilde{f}_{m_3}(\alpha)$. We verify that g is bilinear. Linearity in the first argument follows because each $\tilde{f}_{m_3}$ is an R-module homomorphism. For the second argument, on elementary tensors:

$$\begin{aligned} g(m_1 \otimes m_2, m_3 + m_3') &= m_1 \otimes (m_2 \otimes (m_3 + m_3')) \\ &= m_1 \otimes (m_2 \otimes m_3 + m_2 \otimes m_3') \\ &= m_1 \otimes (m_2 \otimes m_3) + m_1 \otimes (m_2 \otimes m_3') \\ &= g(m_1 \otimes m_2, m_3) + g(m_1 \otimes m_2, m_3'), \\ g(m_1 \otimes m_2, rm_3) &= m_1 \otimes (m_2 \otimes rm_3) \\ &= m_1 \otimes (r(m_2 \otimes m_3)) \\ &= r(m_1 \otimes (m_2 \otimes m_3)) \\ &= r\, g(m_1 \otimes m_2, m_3). \end{aligned}$$

Since elementary tensors generate $M_1 \otimes_R M_2$, bilinearity holds on all of $(M_1 \otimes_R M_2) \times M_3$.

By the universal property of $(M_1 \otimes_R M_2) \otimes_R M_3$, there exists a unique R-module homomorphism

$$\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$$

such that $\varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{tikzcd} (M_1 \otimes_R M_2) \times M_3 \arrow[r, "g"] \arrow[d, "t"'] & M_1 \otimes_R (M_2 \otimes_R M_3) \\ (M_1 \otimes_R M_2) \otimes_R M_3 \arrow[ur, dashed, "\varphi"'] & \end{tikzcd}$$

Step 2. By an entirely symmetric argument (fixing m_1 first, then extending), we obtain an R-module homomorphism

$$\psi : M_1 \otimes_R (M_2 \otimes_R M_3) \to (M_1 \otimes_R M_2) \otimes_R M_3$$

such that $\psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3$.

Step 3. We show φ and ψ are inverses. On elementary tensors:

$$(\psi \circ \varphi)((m_1 \otimes m_2) \otimes m_3) = \psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3,$$

$$(\varphi \circ \psi)(m_1 \otimes (m_2 \otimes m_3)) = \varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3).$$

Since elementary tensors generate both modules, $\psi \circ \varphi = \mathrm{id}$ and $\varphi \circ \psi = \mathrm{id}$. The overall situation is summarized by the following commutative diagram:

$$\begin{tikzcd}
& M_1 \times M_2 \times M_3 \arrow[dl] \arrow[dr] & \\
(M_1 \otimes_R M_2) \otimes_R M_3 \arrow[rr, shift left, "\varphi"] & & M_1 \otimes_R (M_2 \otimes_R M_3) \arrow[ll, shift left, "\psi"]
\end{tikzcd}$$

where the diagonal arrows send (m_1, m_2, m_3) to $(m_1 \otimes m_2) \otimes m_3$ and $m_1 \otimes (m_2 \otimes m_3)$ respectively. Hence φ is an isomorphism with $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$. $\square$

Theorem 2.2.8. *Let M, N, P be R-modules. We have a canonical isomorphism:*

$$\phi_{M,N,P} : \mathrm{Hom}_R(M \otimes_R N, P) \to \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$$

such that $\phi_{M,N,P}(f)(m)(n) = f(m \otimes n)$. Furthermore, if R is commutative, then $\phi_{M,N,P}$ is an R-module isomorphism. If R is not commutative, then $\phi_{M,N,P}$ is only a $\mathbb{Z}$-module isomorphism.

Proof. Let $f \in \mathrm{Hom}_R(M \otimes_R N, P)$. For each $m \in M$, define $\phi_{M,N,P}(f)(m) : N \to P$ by

$$\phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

We verify that $\phi_{M,N,P}(f)(m) \in \mathrm{Hom}_R(N, P)$: for $n_1, n_2 \in N$ and $r \in R$,

$$\begin{aligned}
\phi_{M,N,P}(f)(m)(n_1 + n_2) &= f(m \otimes (n_1 + n_2)) = f(m \otimes n_1 + m \otimes n_2) \\
&= f(m \otimes n_1) + f(m \otimes n_2), \\
\phi_{M,N,P}(f)(m)(rn) &= f(m \otimes rn) = f(r(m \otimes n)) \\
&= rf(m \otimes n) = r\,\phi_{M,N,P}(f)(m)(n).
\end{aligned}$$

Next, we verify that $\phi_{M,N,P}(f) \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$: for $m_1, m_2 \in M$, $r \in R$, and any $n \in N$,

$$\begin{aligned}
\phi_{M,N,P}(f)(m_1 + m_2)(n) &= f((m_1 + m_2) \otimes n) \\
&= f(m_1 \otimes n) + f(m_2 \otimes n) \\
\phi_{M,N,P}(f)(rm)(n) &= f(rm \otimes n) \\
&= f(r(m \otimes n)) \\
&= rf(m \otimes n) \\
&= (r\,\phi_{M,N,P}(f)(m))(n).
\end{aligned}$$

So

$$\phi_{M,N,P}(f)(m_1 + m_2) = \phi_{M,N,P}(f)(m_1) + \phi_{M,N,P}(f)(m_2)$$

and $\phi_{M,N,P}(f)(rm) = r\,\phi_{M,N,P}(f)(m)$, confirming R-linearity.

Let $g \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$. Define $\hat{g} : M \times N \to P$ by $\hat{g}(m, n) = g(m)(n)$. We check that $\hat{g}$ is R-bilinear:

$$\begin{aligned}
\hat{g}(m_1 + m_2, n) &= g(m_1 + m_2)(n) = g(m_1)(n) + g(m_2)(n) = \hat{g}(m_1, n) + \hat{g}(m_2, n), \\
\hat{g}(m, n_1 + n_2) &= g(m)(n_1 + n_2) = g(m)(n_1) + g(m)(n_2) = \hat{g}(m, n_1) + \hat{g}(m, n_2), \\
\hat{g}(rm, n) &= g(rm)(n) = (rg(m))(n) = rg(m)(n) = r\,\hat{g}(m, n), \\
\hat{g}(m, rn) &= g(m)(rn) = rg(m)(n) = r\,\hat{g}(m, n).
\end{aligned}$$

By the universal property of $M \otimes_R N$, there exists a unique R-module homomorphism $\psi_{M,N,P}(g) : M \otimes_R N \to P$ such that $\psi_{M,N,P}(g)(m \otimes n) = g(m)(n)$:

$$\begin{tikzcd}
M \times N \arrow[r, "\hat{g}"] \arrow[d, "t_{M,N}"'] & P \\
M \otimes_R N \arrow[ur, dashed, "\psi_{M,N,P}(g)"'] &
\end{tikzcd}$$

For any $f \in \mathrm{Hom}_R(M \otimes_R N, P)$:

$$\psi_{M,N,P}(\phi_{M,N,P}(f))(m \otimes n) = \phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

Since elementary tensors generate $M \otimes_R N$, we have $\psi_{M,N,P}(\phi_{M,N,P}(f)) = f$.

For any $g \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$, for all $m \in M$ and $n \in N$:

$$\phi_{M,N,P}(\psi_{M,N,P}(g))(m)(n) = \psi_{M,N,P}(g)(m \otimes n) = g(m)(n).$$

So $\phi_{M,N,P}(\psi_{M,N,P}(g))(m) = g(m)$ for all m, hence $\phi_{M,N,P}(\psi_{M,N,P}(g)) = g$.

Therefore $\phi_{M,N,P}$ is a bijection of sets (in fact, of abelian groups).

It remains to show that $\phi_{M,N,P}$ respects the appropriate module structure.

Both $\mathrm{Hom}_R(M \otimes_R N, P)$ and $\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$ are abelian groups under pointwise addition. We verify $\phi_{M,N,P}$ is a group homomorphism: for $f_1, f_2 \in \mathrm{Hom}_R(M \otimes_R N, P)$,

$$\begin{aligned}
\phi_{M,N,P}(f_1 + f_2)(m)(n) &= (f_1 + f_2)(m \otimes n) \\
&= f_1(m \otimes n) + f_2(m \otimes n) \\
&= \phi_{M,N,P}(f_1)(m)(n) + \phi_{M,N,P}(f_2)(m)(n)
\end{aligned}$$

so $\phi_{M,N,P}(f_1 + f_2) = \phi_{M,N,P}(f_1) + \phi_{M,N,P}(f_2)$. Hence $\phi_{M,N,P}$ is a $\mathbb{Z}$-module isomorphism.

If R is commutative, $\mathrm{Hom}_R(M \otimes_R N, P)$ carries an R-module structure via $(r \cdot f)(x) =$

$rf(x)$, and similarly for $\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$. Then

$$\begin{aligned}\phi_{M,N,P}(rf)(m)(n) &= (rf)(m \otimes n)\\ &= rf(m \otimes n)\\ &= r\,\phi_{M,N,P}(f)(m)(n)\\ &= (r\,\phi_{M,N,P}(f))(m)(n)\end{aligned}$$

so $\phi_{M,N,P}(rf) = r\,\phi_{M,N,P}(f)$, and $\phi_{M,N,P}$ is an R-module isomorphism.

If R is not commutative, $\mathrm{Hom}_R(M \otimes_R N, P)$ does not naturally carry an R-module structure (scalar multiplication by r need not commute with R-linearity), so $\phi_{M,N,P}$ is only a $\mathbb{Z}$-module isomorphism. □

Definition 2.2.9. *Let $f : M \to M'$ and $g : N \to N'$ be R-module homomorphisms. The tensor product of f and g is the R-module homomorphism:*

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

defined by $(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$.

Definition 2.2.10. *Let $M_i, i \in I$ be a family of R-modules. Use the following notation:*

$$L^n(M_1, \dots, M_n; N) = \{f : M_1 \times \cdots \times M_n \to N \mid f \text{ is } n\text{-linear } \}$$

In particular,

$$L^2(M_1, M_2; N) \cong L(M_1 \otimes M_2, N) \cong \mathrm{Hom}_R(M_1 \otimes M_2, N)$$

or equivalently,

$$L^2(M_1 \otimes M_2, N) \cong L(M_1, L(M_2, N))$$

Proposition 2.2.11. *Let M, N be two R-modules where M is finitely generated and free. Denote*

$$M^\vee = \mathrm{Hom}_R(M, R)$$

Then we have a canonical isomorphism:

$$M^\vee \otimes N \cong \mathrm{Hom}_R(M, N)$$

Proof. Since M is a finitely generated free R-module, we have $M \cong R^m$ for some $m \geq 1$. We construct an explicit isomorphism

$$\Phi : M^\vee \otimes_R N \to \mathrm{Hom}_R(M, N).$$

Define a bilinear map $\beta : M^\vee \times N \to \mathrm{Hom}_R(M, N)$ by

$$\beta(f, n)(m) = f(m)\,n, \qquad f \in M^\vee,\ n \in N,\ m \in M.$$

We check R-bilinearity. For $f_1, f_2 \in M^\vee$, $n, n_1, n_2 \in N$, $m \in M$, and $r \in R$:

$$\begin{aligned}
\beta(f_1 + f_2, n)(m) &= (f_1 + f_2)(m)\, n = f_1(m)\, n + f_2(m)\, n \\
&= \beta(f_1, n)(m) + \beta(f_2, n)(m), \\
\beta(f, n_1 + n_2)(m) &= f(m)(n_1 + n_2) = f(m)\, n_1 + f(m)\, n_2 \\
&= \beta(f, n_1)(m) + \beta(f, n_2)(m), \\
\beta(rf, n)(m) &= (rf)(m)\, n = r\, f(m)\, n = r\, \beta(f, n)(m), \\
\beta(f, rn)(m) &= f(m)\, (rn) = r\, f(m)\, n = r\, \beta(f, n)(m).
\end{aligned}$$

By the universal property of the tensor product, there exists a unique R-module homomorphism

$$\Phi : M^\vee \otimes_R N \to \mathrm{Hom}_R(M, N), \qquad \Phi(f \otimes n)(m) = f(m)\, n.$$

Now let $\{e_1, \ldots, e_m\}$ be a basis of $M \cong R^m$, and let $\{e_1^*, \ldots, e_m^*\}$ be the dual basis; that is,

$$e_i^* \in M^\vee, \qquad e_i^*(e_j) = \delta_{ij}.$$

Define

$$\Psi : \mathrm{Hom}_R(M, N) \to M^\vee \otimes_R N, \qquad \Psi(g) = \sum_{i=1}^{m} e_i^* \otimes g(e_i).$$

We verify Ψ is an R-module homomorphism: for $g_1, g_2 \in \mathrm{Hom}_R(M, N)$ and $r \in R$,

$$\begin{aligned}
\Psi(g_1 + g_2) &= \sum_{i=1}^{m} e_i^* \otimes (g_1 + g_2)(e_i) = \sum_{i=1}^{m} e_i^* \otimes (g_1(e_i) + g_2(e_i)) \\
&= \sum_{i=1}^{m} e_i^* \otimes g_1(e_i) + \sum_{i=1}^{m} e_i^* \otimes g_2(e_i) = \Psi(g_1) + \Psi(g_2), \\
\Psi(rg) &= \sum_{i=1}^{m} e_i^* \otimes (rg)(e_i) = \sum_{i=1}^{m} e_i^* \otimes r\, g(e_i) \\
&= r \sum_{i=1}^{m} e_i^* \otimes g(e_i) = r\, \Psi(g).
\end{aligned}$$

We show Φ and Ψ are mutual inverses. For any $g \in \mathrm{Hom}_R(M, N)$ and $m = \sum_j r_j e_j \in M$:

$$\begin{aligned}
\Phi(\Psi(g))(m) &= \Phi\Big(\sum_{i=1}^{m} e_i^* \otimes g(e_i)\Big)\Big(\sum_j r_j e_j\Big) \\
&= \sum_{i=1}^{m} e_i^*\Big(\sum_j r_j e_j\Big) g(e_i) = \sum_{i=1}^{m} r_i\, g(e_i) \\
&= g\Big(\sum_{i=1}^{m} r_i e_i\Big) = g(m).
\end{aligned}$$

So $\Phi \circ \Psi = \mathrm{id}$.

For any elementary tensor $f \otimes n \in M^\vee \otimes_R N$:

$$\begin{aligned}\Psi(\Phi(f\otimes n)) &= \sum_{i=1}^m e_i^* \otimes \Phi(f\otimes n)(e_i) = \sum_{i=1}^m e_i^* \otimes f(e_i)\, n\\ &= \sum_{i=1}^m f(e_i)\,(e_i^*\otimes n) = \Big(\sum_{i=1}^m f(e_i)\, e_i^*\Big)\otimes n = f\otimes n,\end{aligned}$$

where the last equality uses $f = \sum_{i=1}^m f(e_i)\, e_i^*$(expansion of f in the dual basis). Since elementary tensors generate $M^\vee \otimes_R N$ and $\Psi\circ\Phi$ is R-linear, we have $\Psi\circ\Phi = \mathrm{id}$.

Therefore Φ is an R-module isomorphism, giving $M^\vee \otimes_R N \cong \mathrm{Hom}_R(M,N)$. □

Lemma 2.2.12. *Let P be a finitely generated projective R-module. Then there exists an R-module Q and an integer $t \geq 0$ such that*

$$P \oplus Q \cong R^t.$$

Proof. Since P is finitely generated, there exist elements

$$p_1, \dots, p_t \in P$$

such that

$$P = Rp_1 + \cdots + Rp_t.$$

Therefore we have a surjective R-module homomorphism

$$\pi : R^t \longrightarrow P$$

defined by

$$\pi(a_1, \dots, a_t) = a_1p_1 + \cdots + a_tp_t.$$

Equivalently, if $e_1, \dots, e_t$ denotes the standard basis of R^t, then

$$\pi(e_i) = p_i \qquad (1 \leq i \leq t).$$

Since $p_1, \dots, p_t$ generate P, the map π is surjective.

Now we use that P is projective. Since $\pi : R^t \to P$ is an epimorphism and P is projective, the map π splits. That is, there exists an R-module homomorphism

$$s : P \longrightarrow R^t$$

such that

$$\pi \circ s = \mathrm{id}_P\,.$$

In particular, s is injective, because if $s(p) = 0$, then

$$p = (\pi\circ s)(p) = \pi(0) = 0.$$

We claim that

$$R^t = \mathrm{Im}(s) \oplus \ker \pi.$$

First, let $x \in R^t$. Then

$$x = s(\pi(x)) + \bigl(x - s(\pi(x))\bigr).$$

Clearly,

$$s(\pi(x)) \in \mathrm{Im}(s).$$

Moreover,

$$\pi\bigl(x - s(\pi(x))\bigr) = \pi(x) - (\pi \circ s)(\pi(x)) = \pi(x) - \pi(x) = 0.$$

Hence

$$x - s(\pi(x)) \in \ker \pi.$$

Therefore

$$R^t = \mathrm{Im}(s) + \ker \pi.$$

It remains to show that the sum is direct. Suppose

$$y \in \mathrm{Im}(s) \cap \ker \pi.$$

Then $y = s(p)$ for some $p \in P$, and also $\pi(y) = 0$. Hence

$$0 = \pi(y) = \pi(s(p)) = (\pi \circ s)(p) = p.$$

Therefore

$$y = s(p) = s(0) = 0.$$

Thus

$$\mathrm{Im}(s) \cap \ker \pi = 0.$$

Hence

$$R^t = \mathrm{Im}(s) \oplus \ker \pi.$$

Since $s : P \to \mathrm{Im}(s)$ is an isomorphism, we obtain

$$R^t \cong P \oplus \ker \pi.$$

Taking

$$Q := \ker \pi,$$

we get

$$P \oplus Q \cong R^t.$$

$\square$

Remark 2.2.13. *This does not mean that P itself is free. It only says that P is a direct summand of a finite free module. Thus a finitely generated projective module is "part of" a* *finite free module:*

$$R^t \cong P \oplus Q.$$

Corollary 2.2.14. *Let R be a commutative ring with 1, let P be a finitely generated projective R-module, and let N be any R-module. Denote*

$$P^\vee := \operatorname{Hom}_R(P, R).$$

Then the canonical R-module homomorphism

$$\tau_{P,N} : P^\vee \otimes_R N \longrightarrow \operatorname{Hom}_R(P, N)$$

defined by

$$\tau_{P,N}(\varphi \otimes n)(p) = \varphi(p)n \qquad (\varphi \in P^\vee,\ n \in N,\ p \in P)$$

is an isomorphism.

Proof. Since P is finitely generated and projective, there exist elements

$$p_1, \ldots, p_t \in P$$

and homomorphisms

$$\varphi_1, \ldots, \varphi_t \in P^\vee$$

such that

$$p = \sum_{i=1}^{t} \varphi_i(p)p_i \qquad \text{for every } p \in P.$$

Indeed, since P is finitely generated projective, there exists an R-module Q and an integer $t \geq 0$ such that

$$P \oplus Q \cong R^t.$$

Let

$$\iota : P \longrightarrow R^t \qquad \text{and} \qquad \rho : R^t \longrightarrow P$$

be the corresponding inclusion and projection, so that

$$\rho \circ \iota = \mathrm{id}_P.$$

Let $e_1, \ldots, e_t$ be the standard basis of R^t, and let $e_1^\vee, \ldots, e_t^\vee$ be the dual basis of $(R^t)^\vee$. Define

$$p_i := \rho(e_i) \in P, \qquad \varphi_i := e_i^\vee \circ \iota \in P^\vee.$$

Then, for every $p \in P$,

$$p = \rho(\iota(p)) = \rho\left(\sum_{i=1}^{t} e_i^\vee(\iota(p))e_i\right) = \sum_{i=1}^{t} \varphi_i(p)p_i.$$

Now define a map

$$\sigma_{P,N} : \operatorname{Hom}_R(P,N) \longrightarrow P^\vee \otimes_R N$$

by

$$\sigma_{P,N}(f) = \sum_{i=1}^{t} \varphi_i \otimes f(p_i).$$

We show that $\sigma_{P,N}$ is the inverse of $\tau_{P,N}$.

First, let $f \in \operatorname{Hom}_R(P,N)$. For every $p \in P$, we have

$$\big(\tau_{P,N} \circ \sigma_{P,N}\big)(f)(p) = \tau_{P,N}\left(\sum_{i=1}^{t} \varphi_i \otimes f(p_i)\right)(p) = \sum_{i=1}^{t} \varphi_i(p) f(p_i).$$

Since f is R-linear,

$$\sum_{i=1}^{t} \varphi_i(p) f(p_i) = f\left(\sum_{i=1}^{t} \varphi_i(p) p_i\right) = f(p).$$

Hence

$$\tau_{P,N} \circ \sigma_{P,N} = \operatorname{id}_{\operatorname{Hom}_R(P,N)}.$$

Conversely, let $\psi \otimes n \in P^\vee \otimes_R N$ be an elementary tensor. Then

$$\big(\sigma_{P,N} \circ \tau_{P,N}\big)(\psi \otimes n) = \sigma_{P,N}\big(\tau_{P,N}(\psi \otimes n)\big) = \sum_{i=1}^{t} \varphi_i \otimes \tau_{P,N}(\psi \otimes n)(p_i).$$

By definition of $\tau_{P,N}$,

$$\tau_{P,N}(\psi \otimes n)(p_i) = \psi(p_i)n.$$

Therefore

$$\big(\sigma_{P,N} \circ \tau_{P,N}\big)(\psi \otimes n) = \sum_{i=1}^{t} \varphi_i \otimes \psi(p_i)n = \left(\sum_{i=1}^{t} \psi(p_i)\varphi_i\right) \otimes n.$$

But for every $p \in P$,

$$\left(\sum_{i=1}^{t} \psi(p_i)\varphi_i\right)(p) = \sum_{i=1}^{t} \psi(p_i)\varphi_i(p) = \sum_{i=1}^{t} \varphi_i(p)\psi(p_i) = \psi\left(\sum_{i=1}^{t} \varphi_i(p) p_i\right) = \psi(p).$$

Hence

$$\sum_{i=1}^{t} \psi(p_i)\varphi_i = \psi.$$

Thus

$$\big(\sigma_{P,N} \circ \tau_{P,N}\big)(\psi \otimes n) = \psi \otimes n.$$

Since elementary tensors generate $P^\vee \otimes_R N$, it follows that

$$\sigma_{P,N} \circ \tau_{P,N} = \operatorname{id}_{P^\vee \otimes_R N}.$$

Therefore $\tau_{P,N}$ is an isomorphism, with inverse $\sigma_{P,N}$. □

Proposition 2.2.15. *Assume that R is a subring of S, and M is an R-module. Then the bilinear map:*

$$S \times (S \otimes_R M) \to S \otimes_R M$$

defined by

$$(s, s' \otimes m) \mapsto ss' \otimes m$$

defines a structure of S-module on the R-module $S \otimes_R M$.

More generally, let $f : R \to T$ be a ring homomorphism. And view T as an R-module via $R \times T \to T, (\lambda, \mu) \mapsto f(\lambda)\mu$. Then the bilinear map:

$$T \times (T \otimes_R M) \to T \otimes_R M$$

defined by

$$(t, t' \otimes m) \mapsto tt' \otimes m$$

defines a structure of T-module on the R-module $T \otimes_R M$.

Proof. It suffices to prove the more general statement in (2), since (1) is the special case where $T = S$ and $f : R \hookrightarrow S$ is the inclusion map.

For each fixed $t \in T$, define a map

$$\beta_t : T \times M \longrightarrow T \otimes_R M, \qquad (t', m) \longmapsto tt' \otimes m.$$

This map is R-bilinear. Hence, by the universal property of the tensor product, there exists a unique R-linear map

$$\mu_t : T \otimes_R M \longrightarrow T \otimes_R M$$

such that

$$\mu_t(t' \otimes m) = tt' \otimes m \qquad \text{for all } t' \in T,\ m \in M.$$

Now define

$$t \cdot x := \mu_t(x), \qquad t \in T,\ x \in T \otimes_R M.$$

We claim that this gives a left T-module structure on $T \otimes_R M$.

Since the elementary tensors $t' \otimes m$ generate $T \otimes_R M$ as an abelian group, it is enough to verify the module axioms on elementary tensors.

Let $t_1, t_2, t' \in T$ and $m \in M$. Then

$$(t_1 + t_2) \cdot (t' \otimes m) = (t_1 + t_2)t' \otimes m = t_1t' \otimes m + t_2t' \otimes m = t_1 \cdot (t' \otimes m) + t_2 \cdot (t' \otimes m).$$

Also,

$$(t_1t_2) \cdot (t' \otimes m) = t_1t_2t' \otimes m = t_1 \cdot (t_2t' \otimes m) = t_1 \cdot \big(t_2 \cdot (t' \otimes m)\big).$$

Moreover,

$$1_T \cdot (t' \otimes m) = 1_Tt' \otimes m = t' \otimes m.$$

Finally, for any $x, y \in T \otimes_R M$,

$$t \cdot (x + y) = \mu_t(x + y) = \mu_t(x) + \mu_t(y) = t \cdot x + t \cdot y,$$

because μ_t is R-linear, hence additive.

Therefore the operation

$$T \times (T \otimes_R M) \longrightarrow T \otimes_R M, \qquad (t, t' \otimes m) \longmapsto tt' \otimes m$$

defines a left T-module structure on $T \otimes_R M$.

This proves (2), and hence also (1). □

Proposition 2.2.16. *Let I be an ideal of R. Let M be an R-module. Then the morphism of R-modules:*

$$M \to (R/I) \otimes_R M$$

defined by

$$m \mapsto 1_{R/I} \otimes m$$

induces an isomorphism of R-modules:

$$M/(IM) \cong (R/I) \otimes_R M$$

as R/I-modules.

Proof. Define

$$\varphi : M \longrightarrow (R/I) \otimes_R M, \qquad m \longmapsto 1_{R/I} \otimes m.$$

This is an R-module homomorphism. For any $a \in I$ and $m \in M$, we have

$$\varphi(am) = 1_{R/I} \otimes am = (1_{R/I} \cdot a) \otimes m = (a + I) \otimes m = 0,$$

since $a + I = 0$ in R/I. Hence $\varphi(IM) = 0$, so $IM \subseteq \ker(\varphi)$.

Therefore, by the universal property of the quotient, φ induces an R-module homomorphism

$$\overline{\varphi} : M/IM \longrightarrow (R/I) \otimes_R M, \qquad m + IM \longmapsto 1_{R/I} \otimes m.$$

Now define

$$\beta : (R/I) \times M \longrightarrow M/IM, \qquad (r + I, m) \longmapsto rm + IM.$$

We first check that β is well defined. If $r + I = r' + I$, then $r - r' \in I$, so

$$(r - r')m \in IM,$$

hence

$$rm + IM = r'm + IM.$$

Also, β is R-balanced, since for any $a \in R$,

$$\beta((r+I)a, m) = \beta(ra+I, m) = ram + IM = \beta(r+I, am).$$

Thus, by the universal property of the tensor product, there exists a unique homomorphism

$$\psi : (R/I) \otimes_R M \longrightarrow M/IM$$

such that

$$\psi((r+I) \otimes m) = rm + IM \qquad \text{for all } r \in R,\ m \in M.$$

We now show that $\overline{\varphi}$ and ψ are inverse to each other.

For any $m \in M$,

$$(\psi \circ \overline{\varphi})(m + IM) = \psi(1_{R/I} \otimes m) = 1m + IM = m + IM.$$

So

$$\psi \circ \overline{\varphi} = \mathrm{id}_{M/IM}.$$

For any elementary tensor $(r+I) \otimes m \in (R/I) \otimes_R M$,

$$(\overline{\varphi} \circ \psi)\big((r+I) \otimes m\big) = \overline{\varphi}(rm + IM) = 1_{R/I} \otimes rm = (r+I) \otimes m.$$

Since elementary tensors generate $(R/I) \otimes_R M$, it follows that

$$\overline{\varphi} \circ \psi = \mathrm{id}_{(R/I) \otimes_R M}.$$

Hence $\overline{\varphi}$ is an isomorphism:

$$M/IM \cong (R/I) \otimes_R M.$$

Finally, both modules carry natural R/I-module structures, and the maps $\overline{\varphi}$ and ψ are R/I-linear. Therefore this is an isomorphism of R/I-modules. □

2.3 Balanced product

Definition 2.3.1. *Let R be a ring (not necessarily commutative). Let M_R be a right R-module and ${}_RN$ be a left R-module. The balanced product of M and N is defined as a pair (P, f) where P is an abelian group and $f : M \times N \to P$ is a bilinear map satisfies:*

- $f(mr, n) = f(m, rn)$ *for all* $r \in R$, $m \in M$, $n \in N$.
- $f(m + m', n) = f(m, n) + f(m', n)$ *for all* $m, m' \in M$, $n \in N$.
- $f(m, n + n') = f(m, n) + f(m, n')$ *for all* $m \in M$, $n, n' \in N$.

Definition 2.3.2. *A tensor product of M_R and ${}_RN$ is a balanced product $(M \otimes_R N, t_{M,N})$ such that if (P, f) is any balanced product of M_R and ${}_RN$, then there exists a unique abelian group homomorphism $\tilde{f} : M \otimes_R N \to P$ such that the following diagram commutes:*

$$\begin{tikzcd} M \times N \arrow[r, "t_{M,N}"] \arrow[d, "f"'] & M \otimes_R N \arrow[dl, dashed, "\tilde{f}"'] \\ P & \end{tikzcd}$$

Chapter 3

Lecture 3: Categories, Bimodules

3.1 Categories

Definition 3.1.1. *A category $\mathcal{C}$ consists of:*

- *A class of objects* $\mathrm{ob}(\mathcal{C})$*.*
- *A set of morphisms* $\mathrm{Hom}_{\mathcal{C}}(A, B)$ *for each pair of objects* $A, B \in \mathcal{C}$*.*
- *A composition map* $\mathrm{Hom}_{\mathcal{C}}(B, C) \times \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{C}}(A, C)$*,* $(\phi, \psi) \mapsto \phi \circ \psi$ *for each triple of objects* $A, B, C \in \mathcal{C}$*.*

It is assumed that objects and morphisms satisfy the following conditions:

- If $(A, B) \neq (C, D)$, then
$$\mathrm{Hom}(A, B) \cap \mathrm{Hom}(C, D) = \varnothing.$$
- If
$$f \in \mathrm{Hom}(A, B), \qquad g \in \mathrm{Hom}(B, C), \qquad h \in \mathrm{Hom}(C, D),$$
then
$$(h \circ g) \circ f = h \circ (g \circ f).$$
- (Unit) For every object A we have an element
$$1_A \in \mathrm{Hom}(A, A)$$
such that
$$f \circ 1_A = f \qquad \text{for any } f \in \mathrm{Hom}(A, B),$$
and
$$1_A \circ g = g \qquad \text{for any } g \in \mathrm{Hom}(B, A).$$

Example 3.1.2. *The following are examples of categories.*

1. *The category of sets: the objects are sets, and the morphisms are maps. It is*

denoted by **Set***.*

2. *The category of groups: the objects are groups, and the morphisms are group homomorphisms. It is denoted by* **Grp***.*

3. *The category of abelian groups: the objects are abelian groups, and the morphisms are group homomorphisms. It is denoted by* **Ab***.*

4. *The category of R-modules: the objects are R-modules, and the morphisms are R-module homomorphisms. It is denoted by R-*$\mathbf{Mod}$*(left R-modules, for right R-modules, we use $\mathbf{Mod} - R$).*

5. *The category of finitely generated R-modules: the objects are finitely generated R-modules, and the morphisms are R-module homomorphisms. It is denoted by R-*$\mathbf{mod}$*.*

6. *The category of topological spaces: the objects are topological spaces, and the morphisms are continuous maps. It is denoted by* **Top***.*

Definition 3.1.3 (Subcategory)**.** *A category $\mathcal{D}$ is called a* subcategory *of a category $\mathcal{C}$ if*

1. *the objects of $\mathcal{D}$ form a subclass of the objects of $\mathcal{C}$;*

2. *for any objects A, B of $\mathcal{D}$, we have*

   - $\mathrm{Hom}_{\mathcal{D}}(A, B) \subseteq \mathrm{Hom}_{\mathcal{C}}(A, B)$*;*
   - *the identity morphisms and the composition of morphisms in $\mathcal{D}$ are the same as those in $\mathcal{C}$.*

The subcategory $\mathcal{D}$ of $\mathcal{C}$ is called full *if*

$$\mathrm{Hom}_{\mathcal{D}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(A, B)$$

for any objects A, B of $\mathcal{D}$.

Example 3.1.4. *The category* **Ab** *is a full subcategory of* **Grp***.*

The category R-$\mathbf{mod}$ *is a full subcategory of R-*$\mathbf{Mod}$*.*

The category **Grp** *is not a full subcategory of* **Set***.*

Definition 3.1.5. *Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A* covariant functor

$$F : \mathcal{C} \to \mathcal{D}$$

consists of the following data:

- *for every object A of $\mathcal{C}$, an object $F(A)$ of $\mathcal{D}$;*

- *for each pair of objects A, B of $\mathcal{C}$, a map*

$$F : \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F(A), F(B)), \qquad f \mapsto F(f).$$

These data are required to satisfy the following conditions:

1. *If $g \circ f$ is defined in $\mathcal{C}$, then*

$$F(g) \circ F(f) = F(g \circ f).$$

2. *For every object A of $\mathcal{C}$,*

$$F(1_A) = 1_{F(A)}.$$

Proposition 3.1.6. *Functors preserve diagram:*

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "g"'] & B \arrow[d, "h"] \\ & C \end{tikzcd} \qquad \xrightarrow{F} \qquad \begin{tikzcd} F(A) \arrow[r, "F(f)"] \arrow[dr, "F(g)"'] & F(B) \arrow[d, "F(h)"] \\ & F(AC \end{tikzcd}$$

Let M and N be R-modules. Then

$$\mathrm{Hom}_R(M, N) = \{ f : M \to N \mid f \text{ is an } R\text{-module homomorphism} \}.$$

This is an abelian group.

Moreover,

$$\mathrm{End}_R(M) = \mathrm{Hom}_R(M, M),$$

and $\mathrm{End}_R(M)$ is a ring.

Example 3.1.7. *Let M be an R-module. Then*

$$\mathrm{Hom}_R(M, -) : R\text{-Mod} \to \mathrm{Ab}$$

is a functor.

More precisely, it acts as follows:

$$N \longmapsto \mathrm{Hom}_R(M, N), \qquad f : N \to N' \longmapsto \mathrm{Hom}_R(M, f).$$

Given an R-module homomorphism

$$f : N \to N',$$

the induced map

$$\mathrm{Hom}_R(M, f) : \mathrm{Hom}_R(M, N) \to \mathrm{Hom}_R(M, N')$$

is defined by

$$\mathrm{Hom}_R(M, f)(h) = f \circ h \qquad (h \in \mathrm{Hom}_R(M, N)).$$

This corresponds to the commutative diagram

$$\begin{tikzcd} & M \arrow[dl, "h"'] \arrow[dr, "f\circ h"] & \\ N \arrow[rr, "f"'] & & N' \end{tikzcd}$$

Hence

$$\mathrm{Hom}_R(M, -)$$

is a covariant functor.

Definition 3.1.8. *Let $\mathcal{C}$ be a category. The* opposite category *(or* dual category*) $\mathcal{C}^{\mathrm{op}}$ is defined as follows:*

- $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C})$*;*
- *for any objects A, B,*

$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A).$$

The composition in $\mathcal{C}^{\mathrm{op}}$ is defined by reversing the order of composition in $\mathcal{C}$: if

$$f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B), \qquad g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(B, C),$$

then

$$g \circ f \text{ in } \mathcal{C}^{\mathrm{op}}$$

is defined to be

$$f \circ g \text{ in } \mathcal{C}.$$

We call $\mathcal{C}^{\mathrm{op}}$ the dual category of $\mathcal{C}$.

This is illustrated by the following diagrams:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "g\circ f"'] & B \arrow[d, "g"] \\ & C \end{tikzcd} \qquad \text{in } \mathcal{C}^{\mathrm{op}},$$

while the corresponding picture in $\mathcal{C}$ is

$$\begin{tikzcd} A & B \arrow[l, "f"'] \\ & C \arrow[u, "g"'] \arrow[ul, "f\circ g"] \end{tikzcd} \qquad \text{in } \mathcal{C}.$$

Definition 3.1.9. *A* contravariant functor *from a category $\mathcal{C}$ to a category $\mathcal{D}$ is a covariant functor*

$$\mathcal{C}^{\mathrm{op}} \to \mathcal{D}.$$

Example 3.1.10. *Let M be an R-module. Then*

$$\mathrm{Hom}_R(-, M) : R\text{-Mod} \to \mathrm{Ab}$$

is a contravariant functor.

More precisely, it is defined on objects by

$$N \longmapsto \mathrm{Hom}_R(N, M),$$

and on morphisms by sending an R-module homomorphism

$$f : N \to N'$$

to the induced group homomorphism

$$\mathrm{Hom}_R(f, M) : \mathrm{Hom}_R(N', M) \to \mathrm{Hom}_R(N, M),$$

given by

$$\mathrm{Hom}_R(f, M)(h) = h \circ f \qquad (h \in \mathrm{Hom}_R(N', M)).$$

This corresponds to the commutative diagram

$$\begin{tikzcd} N \arrow[d, "f"'] \arrow[dr, "h\circ f"] & \\ N' \arrow[r, "h"'] & M \end{tikzcd}$$

3.2 Exact Sequences

Definition 3.2.1. *Let $\{M_i\}_{i\in\mathbb{Z}}$ be a family of R-modules, and let*

$$\varphi_i : M_i \to M_{i+1}$$

be R-module homomorphisms for each $i \in \mathbb{Z}$. Then

$$\cdots \longrightarrow M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} M_{i+1} \longrightarrow \cdots$$

is called a sequence *of R-modules.*

Definition 3.2.2. *The sequence*

$$\cdots \longrightarrow M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} M_{i+1} \longrightarrow \cdots$$

is said to be exact at M_i *if*

$$\mathrm{Im}(\varphi_{i-1}) = \ker(\varphi_i).$$

It is called an exact sequence *if it is exact at every term.*

A short exact sequence is a sequence of the form

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0.$$

It is exact if and only if

$$\ker(\iota) = 0, \qquad \mathrm{Im}(\pi) = M'', \qquad \mathrm{Im}(\iota) = \ker(\pi).$$

Equivalently, ι is injective, π is surjective, and

$$\mathrm{Im}(\iota) = \ker(\pi).$$

Hence, for every short exact sequence

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0,$$

we have

$$M/M' \cong M''.$$

Definition 3.2.3. *Let R, S be rings, and let*

$$F : R\text{-Mod} \to S\text{-Mod}$$

be a covariant functor.

We say that F is left exact *if for any exact sequence*

$$0 \to N \xrightarrow{\varphi} M \xrightarrow{\psi} P,$$

the sequence

$$0 \to F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P)$$

is exact.

We say that F is right exact *if for any exact sequence*

$$N \xrightarrow{\varphi} M \xrightarrow{\psi} P \to 0,$$

the sequence

$$F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P) \to 0$$

is exact.

Definition 3.2.4. *Let*

$$G : R\text{-Mod} \to S\text{-Mod}$$

be a contravariant functor. We say that G is left exact *(respectively,* right exact*) if it is left exact (respectively, right exact) when regarded as a covariant functor*

$$G : (R\text{-Mod})^{\text{op}} \to S\text{-Mod}.$$

A functor is called exact *if it is both left exact and right exact.*

Let M be an R-module. Then:

$$\mathrm{Hom}_R(M, -) : R\text{-Mod} \to \mathrm{Ab} \quad \text{is left exact,}$$

$$\mathrm{Hom}_R(-, M) : R\text{-Mod} \to \mathrm{Ab} \quad \text{is left exact,}$$

and

$$M \otimes_R - : R\text{-Mod} \to \mathrm{Ab} \quad \text{is right exact.}$$

Definition 3.2.5. *A functor $F : \mathcal{C} \to \mathcal{D}$ is called* exact *if it is both left exact and right exact.*

Definition 3.2.6. *Let M be an R-module.*

1. *M is called* projective *if the functor*

$$\mathrm{Hom}_R(M, -)$$

   is exact.

2. *M is called* injective *if the functor*

$$\mathrm{Hom}_R(-, M)$$

   is exact.

3. *M is called* flat *if the functor*

$$- \otimes_R M$$

   is exact.

If R is noncommutative, then in order for

$$M \otimes_R -$$

to be well-defined, M must be a right R-module.

3.3 Bimodules

Definition 3.3.1. *Let R and S be rings. An abelian group M is called a* left-R-right-S-bimodule *if M is both a left R-module and a right S-module such that the two scalar multiplications satisfy*

$$r(xs) = (rx)s \qquad \text{for all } r \in R,\ s \in S,\ x \in M.$$

Notation.

$$_RM \qquad \text{left } R\text{-module},$$

$$M_S \qquad \text{right } S\text{-module},$$

$$_RM_S \qquad \text{left } R\text{-right } S\text{-bimodule}.$$

Let R, S be rings, and let $_RU_S$ be a bimodule.

For each left R-module $_RM$, there are two naturally induced S-module structures:

$$\mathrm{Hom}_R({_RU_S}, {_RM}), \qquad \text{a left } S\text{-module},$$

$$\mathrm{Hom}_R({_RM}, {_RU_S}), \qquad \text{a right } S\text{-module}.$$

More precisely, for each $s \in S$:

if

$$f \in \mathrm{Hom}_R({_RU_S}, {_RM}),$$

define

$$s \cdot f : U \to M, \qquad u \mapsto f(us).$$

Similarly, if

$$f \in \mathrm{Hom}_R({_RM}, {_RU_S}),$$

define

$$f \cdot s : M \to U, \qquad m \mapsto f(m)s.$$

Thus we obtain functors

$$\mathrm{Hom}_R({_RU_S}, -) : R\text{-}\mathbf{Mod} \longrightarrow S\text{-}\mathbf{Mod},$$

given on objects by

$$_RM \longmapsto \mathrm{Hom}_R({_RU_S}, {_RM}),$$

which is covariant, and

$$\mathrm{Hom}_R(-, {_RU_S}) : R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Mod}\text{-}S,$$

given on objects by

$$_RM \longmapsto \mathrm{Hom}_R({_RM}, {_RU_S}),$$

which is contravariant.

On morphisms, for

$$f : M \to M',$$

the induced map

$$\mathrm{Hom}_R({}_RU_S, {}_RM) \longrightarrow \mathrm{Hom}_R({}_RU_S, {}_RM')$$

is given by

$$h \longmapsto f \circ h.$$

the induced map

$$\mathrm{Hom}_R({}_RM', {}_RU_S) \longrightarrow \mathrm{Hom}_R({}_RM, {}_RU_S)$$

is given by

$$h \longmapsto h \circ f.$$

Theorem 3.3.2. *The functor*

$$\mathrm{Hom}_R({}_RU_S, -) : R\text{-}\mathbf{Mod} \longrightarrow S\text{-}\mathbf{Mod}$$

is left exact.

Proof. Let

$$0 \longrightarrow N' \xrightarrow{f} N \xrightarrow{g} N''$$

be an exact sequence in R-**Mod**. Consider the induced sequence

$$0 \longrightarrow \mathrm{Hom}_R({}_RU_S, N') \xrightarrow{\mathrm{Hom}_R(U,f)} \mathrm{Hom}_R({}_RU_S, N) \xrightarrow{\mathrm{Hom}_R(U,g)} \mathrm{Hom}_R({}_RU_S, N'').$$

We show that this sequence is exact.

First, $\mathrm{Hom}_R(U, f)$ is injective. Let

$$\varphi \in \mathrm{Hom}_R(U, N')$$

and suppose that

$$\mathrm{Hom}_R(U, f)(\varphi) = f \circ \varphi = 0.$$

Then for every $u \in U$,

$$f(\varphi(u)) = 0.$$

Since f is injective, it follows that $\varphi(u) = 0$ for all $u \in U$. Hence $\varphi = 0$, so $\mathrm{Hom}_R(U, f)$ is injective.

Next, we show that

$$\mathrm{Im}\big(\mathrm{Hom}_R(U, f)\big) \subseteq \ker\big(\mathrm{Hom}_R(U, g)\big).$$

Indeed, if

$$\varphi \in \mathrm{Hom}_R(U, N'),$$

then

$$\mathrm{Hom}_R(U,g)\big(\mathrm{Hom}_R(U,f)(\varphi)\big) = g\circ f\circ\varphi = 0,$$

because $g\circ f = 0$. Therefore

$$\mathrm{Im}\big(\mathrm{Hom}_R(U,f)\big) \subseteq \ker\big(\mathrm{Hom}_R(U,g)\big).$$

Finally, let

$$\psi \in \ker\big(\mathrm{Hom}_R(U,g)\big).$$

Then $\psi \in \mathrm{Hom}_R(U,N)$ and

$$g\circ\psi = 0.$$

Thus for every $u \in U$,

$$g(\psi(u)) = 0,$$

so $\psi(u) \in \ker g = \mathrm{Im}\, f$ by exactness. Hence for each $u \in U$ there exists some $y \in N'$ such that

$$f(y) = \psi(u).$$

Since f is injective, such y is unique. Define

$$\varphi : U \to N', \qquad u \mapsto y,$$

where y is the unique element satisfying $f(y) = \psi(u)$.

Then

$$f\circ\varphi = \psi.$$

It remains to check that φ is R-linear. For $u, v \in U$,

$$f(\varphi(u+v)) = \psi(u+v) = \psi(u)+\psi(v) = f(\varphi(u)) + f(\varphi(v)) = f(\varphi(u)+\varphi(v)).$$

Since f is injective,

$$\varphi(u+v) = \varphi(u)+\varphi(v).$$

Similarly, for $r \in R$ and $u \in U$,

$$f(\varphi(ru)) = \psi(ru) = r\psi(u) = rf(\varphi(u)) = f(r\varphi(u)),$$

so again by injectivity of f,

$$\varphi(ru) = r\varphi(u).$$

Thus $\varphi \in \mathrm{Hom}_R(U,N')$, and

$$\mathrm{Hom}_R(U,f)(\varphi) = f\circ\varphi = \psi.$$

Therefore

$$\ker\big(\mathrm{Hom}_R(U,g)\big) \subseteq \mathrm{Im}\big(\mathrm{Hom}_R(U,f)\big).$$

Combining the two inclusions, we get

$$\operatorname{Im}\big(\operatorname{Hom}_R(U, f)\big) = \ker\big(\operatorname{Hom}_R(U, g)\big).$$

Hence

$$0 \longrightarrow \operatorname{Hom}_R({}_RU_S, N') \xrightarrow{\operatorname{Hom}_R(U,f)} \operatorname{Hom}_R({}_RU_S, N) \xrightarrow{\operatorname{Hom}_R(U,g)} \operatorname{Hom}_R({}_RU_S, N'')$$

is exact, and so the functor $\operatorname{Hom}_R({}_RU_S, -)$ is left exact. □

Theorem 3.3.3. *Let ${}_RU_S$ be an R-S-bimodule. Then the contravariant functor*

$$\operatorname{Hom}_R(-, U) : R\text{-Mod} \longrightarrow \text{Mod-}S$$

is left exact.

Proof. Since $\operatorname{Hom}_R(-, U)$ is contravariant, to say that it is left exact means the following: for every short exact sequence of left R-modules

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0,$$

the induced sequence of right S-modules

$$0 \longrightarrow \operatorname{Hom}_R(C, U) \xrightarrow{\operatorname{Hom}_R(g,U)} \operatorname{Hom}_R(B, U) \xrightarrow{\operatorname{Hom}_R(f,U)} \operatorname{Hom}_R(A, U)$$

is exact.

We verify this.

First, $\operatorname{Hom}_R(g, U)$ is injective. Indeed, let $\varphi \in \operatorname{Hom}_R(C, U)$ and suppose

$$\operatorname{Hom}_R(g, U)(\varphi) = \varphi \circ g = 0.$$

Since g is surjective, it follows that $\varphi = 0$. Hence $\operatorname{Hom}_R(g, U)$ is injective.

Next, we show

$$\operatorname{Im}\big(\operatorname{Hom}_R(g, U)\big) \subseteq \ker\big(\operatorname{Hom}_R(f, U)\big).$$

Let $\varphi \in \operatorname{Hom}_R(C, U)$. Then

$$\operatorname{Hom}_R(f, U)\big(\operatorname{Hom}_R(g, U)(\varphi)\big) = (\varphi \circ g) \circ f = \varphi \circ (g \circ f) = 0,$$

because $g \circ f = 0$. Therefore

$$\operatorname{Im}\big(\operatorname{Hom}_R(g, U)\big) \subseteq \ker\big(\operatorname{Hom}_R(f, U)\big).$$

Finally, we prove the reverse inclusion. Let $\psi \in \operatorname{Hom}_R(B, U)$ satisfy

$$\operatorname{Hom}_R(f, U)(\psi) = \psi \circ f = 0.$$

Then ψ vanishes on $\mathrm{Im}(f) = \ker(g)$. Hence ψ factors through the quotient

$$B/\ker(g).$$

Since g is surjective, there exists a unique R-module homomorphism $\overline{\psi} : C \to U$ such that

$$\psi = \overline{\psi} \circ g.$$

Thus

$$\psi \in \mathrm{Im}\big(\mathrm{Hom}_R(g, U)\big),$$

so

$$\ker\big(\mathrm{Hom}_R(f, U)\big) \subseteq \mathrm{Im}\big(\mathrm{Hom}_R(g, U)\big).$$

Combining the two inclusions, we obtain

$$\mathrm{Im}\big(\mathrm{Hom}_R(g, U)\big) = \ker\big(\mathrm{Hom}_R(f, U)\big).$$

Hence the sequence

$$0 \longrightarrow \mathrm{Hom}_R(C, U) \xrightarrow{\mathrm{Hom}_R(g,U)} \mathrm{Hom}_R(B, U) \xrightarrow{\mathrm{Hom}_R(f,U)} \mathrm{Hom}_R(A, U)$$

is exact. This proves that $\mathrm{Hom}_R(-, U)$ is left exact. $\square$

Let ${}_SM_R$ be a left-S-right-R-bimodule, ${}_RN$ be a left R-module, consider the tensor product $M \otimes_R N$.

For each $s \in S$, the mapping

$$M \times N \longrightarrow M \otimes_R N, \qquad (m, n) \longmapsto (sm) \otimes n$$

is balanced.

So there is a unique abelian group homomorphism

$$u(s) : M \otimes_R N \longrightarrow M \otimes_R N$$

such that

$$u(s)(m \otimes n) = (sm) \otimes n.$$

Only need to check:

$$u : S \longrightarrow \mathrm{End}_{\mathbb{Z}}(M \otimes_R N)$$

is a ring homomorphism.

If M_R, ${}_RN_T$, then $M \otimes_R N$ is a right T-module:

$$(m \otimes n)t = m \otimes (nt).$$

Lemma 3.3.4. *Let ${}_SM_R$ and ${}_RN_T$ be bimodules. Then $M \otimes_R N$ carries a natural structure of a left S-right T-bimodule, defined on elementary tensors by*

$$s(m \otimes n) = (sm) \otimes n, \qquad (m \otimes n)t = m \otimes (nt),$$

for all $s \in S$, $t \in T$, $m \in M$, and $n \in N$.

Proof. Fix $s \in S$. Define

$$\beta_s : M \times N \to M \otimes_R N, \qquad \beta_s(m, n) = sm \otimes n.$$

We claim that β_s is R-balanced. Indeed, it is additive in each variable, and for $r \in R$,

$$\beta_s(mr, n) = s(mr) \otimes n = (sm)r \otimes n = sm \otimes rn = \beta_s(m, rn).$$

Hence, by the universal property of the tensor product, there exists a unique group homomorphism

$$\lambda_s : M \otimes_R N \to M \otimes_R N$$

such that

$$\lambda_s(m \otimes n) = sm \otimes n.$$

We write $\lambda_s(x) = sx$. Hence $\lambda : S \to \mathrm{End}_{\mathbb{Z}}(M \otimes_R N)$ is a ring homomorphism. Which means that $M \otimes_R N$ is a left S-module.

Similarly, for fixed $t \in T$, define

$$\gamma_t : M \times N \to M \otimes_R N, \qquad \gamma_t(m, n) = m \otimes nt.$$

Again γ_t is R-balanced, since for $r \in R$,

$$\gamma_t(mr, n) = mr \otimes nt = m \otimes r(nt) = m \otimes (rn)t = \gamma_t(m, rn).$$

Thus there exists a unique group homomorphism

$$\rho_t : M \otimes_R N \to M \otimes_R N$$

such that

$$\rho_t(m \otimes n) = m \otimes nt.$$

We write $\rho_t(x) = xt$. Hence $\rho : T \to \mathrm{End}_{\mathbb{Z}}(M \otimes_R N)$ is a ring homomorphism. Which means that $M \otimes_R N$ is a right T-module.

It remains to verify the bimodule axioms. Since elementary tensors generate $M \otimes_R N$, it suffices to check them on $m \otimes n$:

$$(s_1 + s_2)(m \otimes n) = ((s_1 + s_2)m) \otimes n = (s_1m) \otimes n + (s_2m) \otimes n = s_1(m \otimes n) + s_2(m \otimes n),$$

$$(s_1s_2)(m \otimes n) = ((s_1s_2)m) \otimes n = s_1((s_2m) \otimes n) = s_1\big(s_2(m \otimes n)\big),$$

$$1_S(m \otimes n) = (1_S m) \otimes n = m \otimes n,$$

and similarly

$$(m \otimes n)(t_1 + t_2) = m \otimes n(t_1 + t_2) = (m \otimes nt_1) + (m \otimes nt_2),$$

$$((m \otimes n)t_1)t_2 = (m \otimes nt_1)t_2 = m \otimes (nt_1)t_2 = m \otimes n(t_1 t_2) = (m \otimes n)(t_1 t_2),$$

$$(m \otimes n)1_T = m \otimes n.$$

Finally,

$$\big(s(m \otimes n)\big)t = ((sm) \otimes n)t = (sm) \otimes nt = s(m \otimes nt) = s\big((m \otimes n)t\big).$$

Therefore $M \otimes_R N$ is a left S-right T-bimodule. □

Lemma 3.3.5. *Let M_R, M'_R be right R-modules, and let ${}_R N$, ${}_R N'$ be left R-modules. Suppose*

$$f_1, f_2, f \in \mathrm{Hom}_R(M, M'), \qquad g_1, g_2, g \in \mathrm{Hom}_R(N, N').$$

Then for each pair (f, g) there exists a unique group homomorphism

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n) \qquad (m \in M,\ n \in N).$$

Moreover,

1. *$(f_1 + f_2) \otimes g = (f_1 \otimes g) + (f_2 \otimes g)$;*
2. *$f \otimes (g_1 + g_2) = (f \otimes g_1) + (f \otimes g_2)$;*
3. *$f \otimes 0 = 0$ and $0 \otimes g = 0$;*
4. *$1_M \otimes 1_N = 1_{M \otimes_R N}$.*

Proof. Define

$$\beta : M \times N \to M' \otimes_R N', \qquad \beta(m, n) = f(m) \otimes g(n).$$

We check that β is R-balanced. It is additive in each variable because f and g are module homomorphisms. Also, for $r \in R$,

$$\beta(mr, n) = f(mr) \otimes g(n) = f(m)r \otimes g(n) = f(m) \otimes rg(n) = f(m) \otimes g(rn) = \beta(m, rn).$$

Hence, by the universal property of $M \otimes_R N$, there exists a unique group homomorphism

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

Now each asserted identity is verified on elementary tensors, and therefore on all of $M \otimes_R N$.

For (1), for all $m \in M$ and $n \in N$,

$$\big(((f_1 + f_2) \otimes g)(m \otimes n)\big) = (f_1 + f_2)(m) \otimes g(n) = f_1(m) \otimes g(n) + f_2(m) \otimes g(n),$$

while

$$\big((f_1 \otimes g) + (f_2 \otimes g)\big)(m \otimes n) = (f_1 \otimes g)(m \otimes n) + (f_2 \otimes g)(m \otimes n) = f_1(m) \otimes g(n) + f_2(m) \otimes g(n).$$

Hence

$$(f_1 + f_2) \otimes g = (f_1 \otimes g) + (f_2 \otimes g).$$

The proof of (2) is analogous:

$$(f \otimes (g_1 + g_2))(m \otimes n) = f(m) \otimes (g_1 + g_2)(n) = f(m) \otimes g_1(n) + f(m) \otimes g_2(n),$$

which equals

$$((f \otimes g_1) + (f \otimes g_2))(m \otimes n).$$

For (3),

$$(f \otimes 0)(m \otimes n) = f(m) \otimes 0 = 0, \qquad (0 \otimes g)(m \otimes n) = 0 \otimes g(n) = 0,$$

so both maps are the zero homomorphism.

For (4),

$$(1_M \otimes 1_N)(m \otimes n) = 1_M(m) \otimes 1_N(n) = m \otimes n,$$

hence $1_M \otimes 1_N$ is the identity on $M \otimes_R N$. □

Lemma 3.3.6. *Let M_R, M'_R, M''_R be right R-modules, and let ${}_RN$, ${}_RN'$, ${}_RN''$ be left R-modules. Let*

$$f : M \to M', \qquad f' : M' \to M'', \qquad g : N \to N', \qquad g' : N' \to N''$$

be R-module homomorphisms. Then

$$(f' \otimes g') \circ (f \otimes g) = (f' \circ f) \otimes (g' \circ g).$$

Proof. Again it suffices to check the equality on elementary tensors. For $m \in M$ and $n \in N$,

$$\big((f' \otimes g') \circ (f \otimes g)\big)(m \otimes n) = (f' \otimes g')\big(f(m) \otimes g(n)\big) = f'(f(m)) \otimes g'(g(n)),$$

while

$$\big((f'\circ f)\otimes(g'\circ g)\big)(m\otimes n) = (f'\circ f)(m)\otimes(g'\circ g)(n) = f'(f(m))\otimes g'(g(n)).$$

Thus both homomorphisms agree on the generators $m\otimes n$ of $M\otimes_R N$, hence they are equal. $\square$

Definition 3.3.7. *Let ${}_SU_R$ be a bimodule. Define*

$$U\otimes_R - : R\text{-Mod} \longrightarrow S\text{-Mod}$$

by

$$M \longmapsto U\otimes_R M, \qquad f \longmapsto \mathrm{id}_U \otimes f.$$

Similarly, Let ${}_RU_S$ be a bimodule.

$$-\otimes_R U : \text{Mod-}R \longrightarrow \text{Mod-}S$$

by

$$N \longmapsto N\otimes_R U, \qquad g \longmapsto g\otimes \mathrm{id}_U\,.$$

Proposition 3.3.8. *Let ${}_SU_R$ be a bimodule. Then*

$$U\otimes_R - : R\text{-Mod} \longrightarrow S\text{-Mod}$$

is a covariant functor.

Similarly, if ${}_RU_S$ is a bimodule, then

$$-\otimes_R U : \text{Mod-}R \longrightarrow \text{Mod-}S$$

is a covariant functor.

Proof. We prove the first statement. For each left R-module M, the tensor product

$$U\otimes_R M$$

is naturally a left S-module via

$$s(u\otimes m) = (su)\otimes m \qquad (s\in S,\ u\in U,\ m\in M).$$

For each R-module homomorphism $f: M\to N$, define

$$U\otimes_R f : U\otimes_R M \to U\otimes_R N$$

by

$$(U\otimes_R f)(u\otimes m) = u\otimes f(m).$$

Equivalently,

$$U \otimes_R f = \mathrm{id}_U \otimes f.$$

This is well-defined and is an S-module homomorphism.

It remains to verify the functorial properties. For the identity map $\mathrm{id}_M : M \to M$, we have

$$U \otimes_R \mathrm{id}_M = \mathrm{id}_U \otimes \mathrm{id}_M = \mathrm{id}_{U \otimes_R M}.$$

If

$$M \xrightarrow{f} N \xrightarrow{g} P$$

are R-module homomorphisms, then for every $u \otimes m \in U \otimes_R M$,

$$(U \otimes_R (g \circ f))(u \otimes m) = u \otimes (g \circ f)(m) = g(u \otimes f(m)) = ((U \otimes_R g) \circ (U \otimes_R f))(u \otimes m).$$

Hence

$$U \otimes_R (g \circ f) = (U \otimes_R g) \circ (U \otimes_R f).$$

Therefore $U \otimes_R -$ is a covariant functor from R-Mod to S-Mod.

The proof of the second statement is entirely similar. If ${}_R U_S$ is a bimodule and N is a right R-module, then

$$N \otimes_R U$$

is naturally a right S-module via

$$(n \otimes u)s = n \otimes (us).$$

For a homomorphism $g : N \to N'$ in Mod-R, define

$$g \otimes_R U = g \otimes \mathrm{id}_U : N \otimes_R U \to N' \otimes_R U$$

by

$$(g \otimes_R U)(n \otimes u) = g(n) \otimes u.$$

One checks immediately that identities and compositions are preserved. Thus

$$- \otimes_R U : \text{Mod-}R \to \text{Mod-}S$$

is also a covariant functor. □

Theorem 3.3.9. *Let ${}_S U_R$ be a bimodule. Then the functor*

$$U \otimes_R - \ : \ R\text{-}Mod \longrightarrow S\text{-}Mod$$

is right exact. In other words, if

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \longrightarrow 0$$

is an exact sequence of left R-modules, then

$$U \otimes_R N' \xrightarrow{\mathrm{id}_U \otimes f} U \otimes_R N \xrightarrow{\mathrm{id}_U \otimes g} U \otimes_R N'' \longrightarrow 0$$

is exact.

Proof. Since

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \longrightarrow 0$$

is exact, we have

$$\ker g = \operatorname{Im} f \qquad \text{and} \qquad N'' \cong N/\operatorname{Im} f.$$

Let

$$\pi : N \longrightarrow N/\operatorname{Im} f$$

be the canonical projection. Then there exists an isomorphism

$$\alpha : N/\operatorname{Im} f \longrightarrow N''$$

such that

$$g = \alpha \circ \pi.$$

Now consider the map

$$\Phi : U \otimes_R N \longrightarrow U \otimes_R (N/\operatorname{Im} f), \qquad u \otimes n \longmapsto u \otimes \pi(n).$$

By the universal property of tensor products, Φ is a well-defined S-module homomorphism.

We claim that

$$\ker \Phi = \operatorname{Im}(\mathrm{id}_U \otimes f).$$

First, since $\pi \circ f = 0$, functoriality gives

$$\Phi \circ (\mathrm{id}_U \otimes f) = \mathrm{id}_U \otimes (\pi \circ f) = 0.$$

Hence

$$\operatorname{Im}(\mathrm{id}_U \otimes f) \subseteq \ker \Phi.$$

To prove the reverse inclusion, define

$$\overline{\Phi} : (U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f) \longrightarrow U \otimes_R (N/\operatorname{Im} f)$$

by

$$\overline{\Phi}\big((u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f)\big) = u \otimes \pi(n).$$

This is well defined because Φ vanishes on $\operatorname{Im}(\mathrm{id}_U \otimes f)$.

Next define

$$\Psi : U \times (N/\operatorname{Im} f) \longrightarrow (U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f)$$

by

$$\Psi(u, \overline{n}) = (u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f),$$

where $\overline{n} = n + \operatorname{Im} f$.

We check that Ψ is well defined. If $\overline{n} = \overline{n'}$, then

$$n - n' \in \operatorname{Im} f,$$

so there exists $x \in N'$ such that

$$n - n' = f(x).$$

Hence

$$(u \otimes n) - (u \otimes n') = u \otimes (n - n') = u \otimes f(x) = (\mathrm{id}_U \otimes f)(u \otimes x) \in \operatorname{Im}(\mathrm{id}_U \otimes f).$$

Thus

$$(u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f) = (u \otimes n') + \operatorname{Im}(\mathrm{id}_U \otimes f).$$

So Ψ is well defined. It is clearly R-balanced, hence induces an S-module homomorphism

$$\overline{\Psi} : U \otimes_R (N/\operatorname{Im} f) \longrightarrow (U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f)$$

such that

$$\overline{\Psi}(u \otimes \overline{n}) = (u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f).$$

It is immediate that $\overline{\Phi}$ and $\overline{\Psi}$ are inverse to each other. Therefore

$$(U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f) \cong U \otimes_R (N/\operatorname{Im} f).$$

It follows that

$$\ker \Phi = \operatorname{Im}(\mathrm{id}_U \otimes f).$$

Finally, since $\alpha : N/\operatorname{Im} f \to N''$ is an isomorphism, the induced map

$$\mathrm{id}_U \otimes \alpha : U \otimes_R (N/\operatorname{Im} f) \longrightarrow U \otimes_R N''$$

is an isomorphism. Moreover,

$$\mathrm{id}_U \otimes g = (\mathrm{id}_U \otimes \alpha) \circ \Phi.$$

Hence

$$\ker(\mathrm{id}_U \otimes g) = \ker \Phi = \operatorname{Im}(\mathrm{id}_U \otimes f),$$

and $\mathrm{id}_U \otimes g$ is surjective because Φ is surjective and $\mathrm{id}_U \otimes \alpha$ is an isomorphism.

Therefore

$$U \otimes_R N' \xrightarrow{\mathrm{id}_U \otimes f} U \otimes_R N \xrightarrow{\mathrm{id}_U \otimes g} U \otimes_R N'' \longrightarrow 0$$

is exact. $\square$

Theorem 3.3.10. *Let ${}_RU_S$ be a left-R-right-S-bimodule. Then the functor*

$$- \otimes_R U \ : \ Mod - R \longrightarrow Mod - S$$

is right exact.

Chapter 4

Lecture 4: Projectives, Injectives

4.1 Projective, Injective and Flat Modules

Definition 4.1.1. *Let R be a ring.*

- *A left R-module P is called* projective *if the functor*

$$\mathrm{Hom}_R(P,-) : R-\mathrm{Mod} \longrightarrow \mathbf{Ab}$$

is exact.

- *A left R-module I is called* injective *if the contravariant functor*

$$\mathrm{Hom}_R(-,I) : R-\mathrm{Mod} \longrightarrow \mathbf{Ab}$$

is exact.

- *A left R-module F is called* flat *if the functor*

$$-\otimes_R F : \text{Mod-}R \longrightarrow \mathbf{Ab}$$

is exact.

Theorem 4.1.2. *Let ${}_RP$ be a left R-module. The following statements are equivalent.*

1. *Whenever we are given a commutative diagram*

$$\begin{tikzcd} & P \arrow[dl, dashed, "\widetilde{\gamma}"'] \arrow[d, "\gamma"] & \\ M \arrow[r, two heads, "g"'] & N \arrow[r] & 0 \end{tikzcd}$$

with g an epimorphism, there exists an R-homomorphism

$$\widetilde{\gamma} : P \to M$$

such that $g \circ \widetilde{\gamma} = \gamma$.

2. *For every epimorphism $f : M \to N$ of left R-modules, the induced map*

$$\mathrm{Hom}_R(P, f) : \mathrm{Hom}_R(P, M) \longrightarrow \mathrm{Hom}_R(P, N)$$

is surjective.

3. *For every ring S and every right S-module structure on P making ${}_R P_S$ an R-S-bimodule, the functor*

$$\mathrm{Hom}_R(P, -) : R - \mathrm{Mod} \longrightarrow S - \mathrm{Mod}$$

is exact.

4. *For every short exact sequence*

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

of left R-modules, the sequence

$$0 \longrightarrow \mathrm{Hom}_R(P, M') \xrightarrow{\mathrm{Hom}_R(P,u)} \mathrm{Hom}_R(P, M) \xrightarrow{\mathrm{Hom}_R(P,v)} \mathrm{Hom}_R(P, M'') \longrightarrow 0$$

is exact.

Proof. We first recall two standard facts.

1. For every left R-module P, the functor

$$\mathrm{Hom}_R(P, -) : R\text{-Mod} \to \mathbf{Ab}$$

is left exact.

2. If ${}_R P_S$ is an R-S-bimodule and M is a left R-module, then $\mathrm{Hom}_R(P, M)$ is naturally a left S-module via

$$(s \cdot \varphi)(p) := \varphi(ps) \qquad (s \in S,\ \varphi \in \mathrm{Hom}_R(P, M),\ p \in P).$$

Moreover, for every R-homomorphism $h : M \to N$, the induced map

$$\mathrm{Hom}_R(P, h) : \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N)$$

is S-linear.

(1) $\Longleftrightarrow$(2). Let $f : M \to N$ be an epimorphism. For $\gamma \in \mathrm{Hom}_R(P, N)$, saying that there exists $\widetilde{\gamma} : P \to M$ such that
$$f \circ \widetilde{\gamma} = \gamma$$
is exactly saying that γ lies in the image of

$$\mathrm{Hom}_R(P, f) : \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N).$$

Hence (1) and (2) are equivalent.

(2) $\Longrightarrow$(4). Let

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

be a short exact sequence of left R-modules. Since $\mathrm{Hom}_R(P, -)$ is left exact, the sequence

$$0 \longrightarrow \mathrm{Hom}_R(P, M') \xrightarrow{\mathrm{Hom}_R(P,u)} \mathrm{Hom}_R(P, M) \xrightarrow{\mathrm{Hom}_R(P,v)} \mathrm{Hom}_R(P, M'')$$

is exact. Since v is an epimorphism, (2) implies that $\mathrm{Hom}_R(P, v)$ is surjective. Therefore

$$0 \longrightarrow \mathrm{Hom}_R(P, M') \xrightarrow{\mathrm{Hom}_R(P,u)} \mathrm{Hom}_R(P, M) \xrightarrow{\mathrm{Hom}_R(P,v)} \mathrm{Hom}_R(P, M'') \longrightarrow 0$$

is exact, proving (4).

(4) $\Longrightarrow$(2). Let $f : M \to N$ be an epimorphism. Apply (4) to the short exact sequence

$$0 \longrightarrow \ker(f) \longrightarrow M \xrightarrow{f} N \longrightarrow 0.$$

Then

$$\mathrm{Hom}_R(P, f) : \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N)$$

is surjective. Thus (2) holds.

(3) $\Longrightarrow$(4). Take $S = \mathbb{Z}$, with the canonical right $\mathbb{Z}$-module structure on P. Then ${}_RP_{\mathbb{Z}}$ is an R-$\mathbb{Z}$-bimodule, and $\mathbb{Z}$-Mod $=$ **Ab**. Hence (3) gives that

$$\mathrm{Hom}_R(P, -) : R\text{-Mod} \to \mathbf{Ab}$$

is exact, which is exactly statement (4).

(4) $\Longrightarrow$(3). Fix a ring S and a right S-module structure on P making ${}_RP_S$ an R-S-bimodule. Let

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

be a short exact sequence of left R-modules. By (4), the induced sequence

$$0 \longrightarrow \mathrm{Hom}_R(P, M') \xrightarrow{\mathrm{Hom}_R(P,u)} \mathrm{Hom}_R(P, M) \xrightarrow{\mathrm{Hom}_R(P,v)} \mathrm{Hom}_R(P, M'') \longrightarrow 0$$

is exact as a sequence of abelian groups. By the remark above, each term carries a natural left S-module structure and each map is S-linear. Since exactness in S-Mod is equivalent to exactness of the underlying abelian groups, the above sequence is exact in S-Mod. Therefore

$$\mathrm{Hom}_R(P, -) : R\text{-Mod} \to S\text{-Mod}$$

is exact. Hence (3) holds.

Therefore (1), (2), (3), and (4) are equivalent. □

Proposition 4.1.3. *Free modules are projective.*

Proof. Let $F = \bigoplus_{i\in I} Re_i$ be a free left R-module with basis $\{e_i\}_{i\in I}$. Let $g : M \to N$ be an epimorphism and let $\gamma : F \to N$ be an R-homomorphism. For each $i \in I$, choose $m_i \in M$ such that

$$g(m_i) = \gamma(e_i).$$

Since F is free, there exists a unique R-homomorphism

$$\widetilde{\gamma} : F \to M$$

such that $\widetilde{\gamma}(e_i) = m_i$ for all $i \in I$. Then

$$g\widetilde{\gamma}(e_i) = g(m_i) = \gamma(e_i)$$

for every basis element, so $g\widetilde{\gamma} = \gamma$. Hence F is projective. □

Theorem 4.1.4. *Let ${}_RP$ be a left R-module. The following statements are equivalent.*

1. *P is projective.*
2. *Every epimorphism*

$$f : M \longrightarrow P \longrightarrow 0$$

splits, that is, there exists an R-homomorphism $g : P \to M$ such that

$$f \circ g = \mathrm{id}_P\,.$$

3. *P is isomorphic to a direct summand of a free module.*

Proof. **(1) $\Longrightarrow$(2).** Apply the lifting property to the diagram

$$\begin{tikzcd} & P \arrow[dl, dashed, "g"'] \arrow[d, "\mathrm{id}_P"] & \\ M \arrow[r, two heads, "f"'] & P \arrow[r] & 0 \end{tikzcd}$$

to obtain $f \circ g = \mathrm{id}_P$.

(2) $\Longrightarrow$(3). Every module is a homomorphic image of a free module, so choose an epimorphism

$$\pi : F \to P$$

with F free. By (2), π splits, so there exists $s : P \to F$ such that

$$\pi \circ s = \mathrm{id}_P\,.$$

Therefore

$$F \cong \ker\pi \oplus s(P) \cong \ker\pi \oplus P,$$

and hence P is a direct summand of the free module F.

(3) $\Longrightarrow$(1). Write

$$F \cong P \oplus Q$$

with F free. Since free modules are projective, it suffices to show that a direct summand of a projective module is projective. Let $f : M \to N$ be an epimorphism and

$$\alpha : P \to N$$

an R-homomorphism. Define

$$\widehat{\alpha} : P \oplus Q \to N, \qquad \widehat{\alpha}(p, q) = \alpha(p).$$

Because F is projective, there exists

$$\widehat{\beta} : P \oplus Q \to M$$

such that $f \circ \widehat{\beta} = \widehat{\alpha}$. Let

$$\iota : P \to P \oplus Q, \qquad p \mapsto (p, 0).$$

Then

$$f \circ (\widehat{\beta} \circ \iota) = \widehat{\alpha} \circ \iota = \alpha.$$

Hence P is projective. ☐

Corollary 4.1.5. *A left R-module P is finitely generated and projective if and only if for some R-module P' and some integer $n \geq 0$ there is an R-isomorphism*

$$P \oplus P' \cong R^n.$$

Proof. If $P \oplus P' \cong R^n$, then P is a direct summand of a finite free module, hence finitely generated and projective.

Conversely, assume that P is finitely generated and projective. By the theorem above, there exists a free module

$$F = \bigoplus_{i \in I} Re_i$$

and an R-module Q such that

$$F \cong P \oplus Q.$$

Let $p : F \to P$ and $s : P \to F$ be the projection and inclusion corresponding to this direct sum decomposition. Choose generators $x_1, \ldots, x_m$ of P. Each $s(x_j)$ is a finite linear combination of the basis elements $\{e_i\}_{i \in I}$, so there exists a finite subset $J \subseteq I$ such that

$$s(x_1), \ldots, s(x_m) \in F_J := \bigoplus_{j \in J} Re_j.$$

Since the x_j generate P, we have $s(P) \subseteq F_J$. Hence the restriction

$$p|_{F_J} : F_J \to P$$

is surjective, and the map $s : P \to F_J$ is a section of $p|_{F_J}$. Therefore

$$F_J \cong P \oplus \ker(p|_{F_J}).$$

Since $F_J \cong R^{|J|}$, the result follows. $\square$

4.2 Schanuel Lemma

Theorem 4.2.1 (Schanuel Lemma). *Let*

$$0 \longrightarrow K \longrightarrow P \xrightarrow{\lambda} M \longrightarrow 0$$

and

$$0 \longrightarrow K' \longrightarrow P' \xrightarrow{\lambda'} M \longrightarrow 0$$

be short exact sequences of left R-modules, with P and P' projective. Then

$$K' \oplus P \cong K \oplus P'.$$

Proof. Define

$$Y := P \times_M P' = \{(p, p') \in P \oplus P' \mid \lambda(p) = \lambda'(p')\}.$$

Let

$$\mu : Y \to P, \qquad \mu(p, p') = p,$$

and

$$\mu' : Y \to P', \qquad \mu'(p, p') = p'.$$

Then Y fits into the pullback diagram

$$\begin{tikzcd} Y \arrow[r, "\mu'"] \arrow[d, "\mu"'] & P' \arrow[d, "\lambda'"] \\ P \arrow[r, "\lambda"'] & M \end{tikzcd}$$

that is, the square commutes and Y is the fiber product of λ and λ'.

We claim that there are short exact sequences

$$0 \longrightarrow K' \xrightarrow{i'} Y \xrightarrow{\mu} P \longrightarrow 0, \qquad i'(k') = (0, k'),$$

and

$$0 \longrightarrow K \xrightarrow{i} Y \xrightarrow{\mu'} P' \longrightarrow 0, \qquad i(k) = (k, 0).$$

First, μ is surjective. Indeed, let $p \in P$. Since λ' is surjective, there exists $p' \in P'$ such that

$$\lambda'(p') = \lambda(p).$$

Then $(p, p') \in Y$, so $\mu(p, p') = p$.

Next,
$$\ker(\mu) = \{(0,p') \in Y \mid \lambda'(p') = 0\} = \{(0,k') \mid k' \in K'\} = \mathrm{Im}(i').$$
Hence
$$0 \longrightarrow K' \xrightarrow{i'} Y \xrightarrow{\mu} P \longrightarrow 0$$
is exact.

Similarly, μ' is surjective, and
$$\ker(\mu') = \{(p,0) \in Y \mid \lambda(p) = 0\} = \{(k,0) \mid k \in K\} = \mathrm{Im}(i).$$
Thus
$$0 \longrightarrow K \xrightarrow{i} Y \xrightarrow{\mu'} P' \longrightarrow 0$$
is exact.

Since P is projective and $\mu : Y \to P$ is an epimorphism, there exists an R-homomorphism $s : P \to Y$ such that
$$\mu \circ s = \mathrm{id}_P.$$
This is expressed by the commutative diagram
$$\begin{tikzcd} & Y \arrow[d, two heads, "\mu"] \\ P \arrow[ur, dashed, "s"] \arrow[r, "\mathrm{id}_P"'] & P \end{tikzcd}$$
so the sequence
$$0 \longrightarrow K' \longrightarrow Y \xrightarrow{\mu} P \longrightarrow 0$$
splits. Therefore,
$$Y \cong K' \oplus P.$$

Likewise, since P' is projective and $\mu' : Y \to P'$ is an epimorphism, there exists $s' : P' \to Y$ such that
$$\mu' \circ s' = \mathrm{id}_{P'},$$
and hence
$$0 \longrightarrow K \longrightarrow Y \xrightarrow{\mu'} P' \longrightarrow 0$$
splits. Therefore,
$$Y \cong K \oplus P'.$$

Combining the two decompositions of Y, we obtain
$$K' \oplus P \cong Y \cong K \oplus P',$$
and hence
$$K' \oplus P \cong K \oplus P'.$$
□

Lemma 4.2.2. *Let M be a left R-module. Then*

$$\mathrm{Hom}_R({}_RR, M) \cong M$$

as left R-modules.

Proof. Define

$$\Phi : \mathrm{Hom}_R({}_RR, M) \to M, \qquad f \mapsto f(1).$$

This map is R-linear. Define also

$$\Psi : M \to \mathrm{Hom}_R({}_RR, M), \qquad m \mapsto (r \mapsto rm).$$

For $f \in \mathrm{Hom}_R({}_RR, M)$ and $r \in R$,

$$\Psi(\Phi(f))(r) = rf(1) = f(r \cdot 1) = f(r),$$

so $\Psi\Phi = \mathrm{id}$. Also

$$\Phi(\Psi(m)) = \Psi(m)(1) = 1m = m,$$

hence Φ and Ψ are inverse isomorphisms. □

Corollary 4.2.3. *For any ring R with 1,*

$$\mathrm{End}_R({}_RR) \cong R^{\mathrm{op}}$$

as rings.

Proof. By the lemma, every $f \in \mathrm{End}_R({}_RR)$ is determined by $f(1)$, and

$$f(r) = rf(1)$$

for all $r \in R$. Thus

$$\Theta : \mathrm{End}_R({}_RR) \to R^{\mathrm{op}}, \qquad f \mapsto f(1)$$

is bijective. Moreover, for $f, g \in \mathrm{End}_R({}_RR)$,

$$(f \circ g)(1) = f(g(1)) = g(1)f(1),$$

which is multiplication in the opposite ring. Therefore Θ is a ring isomorphism. □

4.3 Projective Morphisms

Lemma 4.3.1. *Let R be a commutative ring with 1, and let X, Y, M be R-modules. Then there is an R-module homomorphism*

$$\Theta_M^{X,Y} : \mathrm{Hom}_R(X, M) \otimes_R \mathrm{Hom}_R(M, Y) \longrightarrow \mathrm{Hom}_R(X, Y), \qquad \varphi \otimes \psi \longmapsto \psi \circ \varphi.$$

Its image is precisely the set of all morphisms $f : X \to Y$ which factor through some finite direct power M^n of M.

Proof. First define

$$b : \mathrm{Hom}_R(X, M) \times \mathrm{Hom}_R(M, Y) \longrightarrow \mathrm{Hom}_R(X, Y), \qquad b(\varphi, \psi) = \psi \circ \varphi.$$

Since R is commutative, composition is R-balanced in the sense that

$$b(r\varphi, \psi) = \psi \circ (r\varphi) = r(\psi \circ \varphi) = b(\varphi, r\psi)$$

for all $r \in R$, $\varphi \in \mathrm{Hom}_R(X, M)$, and $\psi \in \mathrm{Hom}_R(M, Y)$; moreover b is additive in each variable. Hence b induces a unique R-linear map

$$\Theta_M^{X,Y} : \mathrm{Hom}_R(X, M) \otimes_R \mathrm{Hom}_R(M, Y) \to \mathrm{Hom}_R(X, Y)$$

such that

$$\Theta_M^{X,Y}(\varphi \otimes \psi) = \psi \circ \varphi.$$

Now let

$$f = \Theta_M^{X,Y}\Big(\sum_{i=1}^n \varphi_i \otimes \psi_i\Big) = \sum_{i=1}^n \psi_i \circ \varphi_i.$$

Define morphisms

$$\alpha : X \to M^n, \qquad x \longmapsto (\varphi_1(x), \dots, \varphi_n(x)),$$

and

$$\beta : M^n \to Y, \qquad (m_1, \dots, m_n) \longmapsto \sum_{i=1}^n \psi_i(m_i).$$

Then for every $x \in X$,

$$(\beta \circ \alpha)(x) = \beta\big(\varphi_1(x), \dots, \varphi_n(x)\big) = \sum_{i=1}^n \psi_i(\varphi_i(x)) = f(x).$$

Thus $f = \beta \circ \alpha$, so every element in the image of $\Theta_M^{X,Y}$ factors through some finite direct power M^n.

Conversely, suppose that a morphism $f : X \to Y$ factors through M^n, say

$$f : X \xrightarrow{u} M^n \xrightarrow{v} Y.$$

Let $\pi_i : M^n \to M$ be the i th projection and $\iota_i : M \to M^n$ the i th inclusion. Since

$$\mathrm{id}_{M^n} = \sum_{i=1}^n \iota_i \circ \pi_i,$$

we obtain

$$f = v \circ u = v \circ \mathrm{id}_{M^n} \circ u = \sum_{i=1}^{n} v \circ \iota_i \circ \pi_i \circ u = \sum_{i=1}^{n} (v \circ \iota_i) \circ (\pi_i \circ u).$$

Therefore

$$f = \Theta_M^{X,Y}\Big(\sum_{i=1}^{n} (\pi_i \circ u) \otimes (v \circ \iota_i)\Big),$$

so f lies in the image of $\Theta_M^{X,Y}$.

Hence the image of $\Theta_M^{X,Y}$ is exactly the set of morphisms $X \to Y$ that factor through some finite direct power M^n. $\square$

Definition 4.3.2. *Let R be a commutative ring with 1. For an R-module X, write*

$$X^* := \mathrm{Hom}_R(X, R).$$

For R-modules X and Y, define

$$\tau_{X,Y} : X^* \otimes_R Y \longrightarrow \mathrm{Hom}_R(X, Y), \qquad \varphi \otimes y \longmapsto \big(x \mapsto \varphi(x)y\big).$$

Definition 4.3.3. *A morphism $f : X \to Y$ is called a* projective morphism *if it factors through a finitely generated projective module.*

Proposition 4.3.4. *Let R be a commutative ring with 1. A morphism $f : X \to Y$ is projective if and only if*

$$f \in \mathrm{Im}(\tau_{X,Y}).$$

Proof. By the previous lemma with $M = R$, the image of $\tau_{X,Y}$ consists exactly of those morphisms that factor through some finite free module R^n.

Since finite free modules are finitely generated projective, every morphism in $\mathrm{Im}(\tau_{X,Y})$ is projective.

Conversely, suppose $f : X \to Y$ factors through a finitely generated projective module P:

$$X \xrightarrow{u} P \xrightarrow{v} Y.$$

Choose Q and n such that

$$P \oplus Q \cong R^n.$$

Let $i : P \to R^n$ and $p : R^n \to P$ be the inclusion and projection corresponding to this decomposition, so that $p \circ i = \mathrm{id}_P$. Then

$$f = v \circ p \circ i \circ u,$$

hence f factors through R^n. Therefore $f \in \mathrm{Im}(\tau_{X,Y})$. $\square$

Definition 4.3.5. *For R-modules X and Y, we write*

$$\mathrm{Hom}_R^{\mathrm{pr}}(X, Y)$$

for the set of projective morphisms from X to Y.

Example 4.3.6. *Let k be a field, and consider the infinite-dimensional k-vector space*

$$V = \bigoplus_{n \geq 1} k e_n.$$

Then the identity morphism

$$\mathrm{id}_V : V \longrightarrow V$$

is not a projective morphism. In other words,

$$\mathrm{id}_V \notin \mathrm{Hom}_k^{\mathrm{pr}}(V, V).$$

Indeed, suppose for contradiction that id_V is a projective morphism. Then, by definition, there exists a finitely generated projective k-module P and k-linear maps

$$V \xrightarrow{\alpha} P \xrightarrow{\beta} V$$

such that

$$\mathrm{id}_V = \beta \circ \alpha.$$

Since k is a field, every finitely generated projective k-module is a finite-dimensional k-vector space. Hence P is finite-dimensional over k.

Therefore

$$\mathrm{Im}(\beta \circ \alpha) \subseteq \mathrm{Im}\, \beta$$

is finite-dimensional. But

$$\mathrm{Im}(\beta \circ \alpha) = \mathrm{Im}(\mathrm{id}_V) = V.$$

Thus V would be finite-dimensional, contradicting the definition

$$V = \bigoplus_{n \geq 1} k e_n.$$

Therefore id_V is not a projective morphism.

Equivalently, over a field k, projective morphisms are precisely finite-rank linear maps. Since id_V has infinite rank, it is not projective.

Lemma 4.3.7. *For any R-module M, the set*

$$\mathrm{Hom}_R^{\mathrm{pr}}(M, M)$$

is a two-sided ideal of the ring $\mathrm{End}_R(M)$*.*

Proof. We first show that $\mathrm{Hom}_R^{\mathrm{pr}}(M,M)$ is an additive subgroup of $\mathrm{End}_R(M)$.

Let $f,g \in \mathrm{Hom}_R^{\mathrm{pr}}(M,M)$. Then there exist finitely generated projective R-modules P,Q and morphisms

$$M \xrightarrow{u} P \xrightarrow{v} M, \qquad M \xrightarrow{u'} Q \xrightarrow{v'} M$$

such that

$$f = v \circ u, \qquad g = v' \circ u'.$$

Consider the morphisms

$$\alpha : M \to P \oplus Q, \qquad \alpha(x) = (u(x), u'(x)),$$

and

$$\beta : P \oplus Q \to M, \qquad \beta(p,q) = v(p) + v'(q).$$

Then

$$(f+g)(x) = v(u(x)) + v'(u'(x)) = \beta(\alpha(x))$$

for all $x \in M$, so

$$f + g = \beta \circ \alpha.$$

Since $P \oplus Q$ is again a finitely generated projective R-module, it follows that $f+g \in \mathrm{Hom}_R^{\mathrm{pr}}(M,M)$.

Also, the zero endomorphism factors through the zero module, and if $f = v \circ u$, then

$$-f = (-v) \circ u,$$

so $\mathrm{Hom}_R^{\mathrm{pr}}(M,M)$ is an additive subgroup of $\mathrm{End}_R(M)$.

Now let $f \in \mathrm{Hom}_R^{\mathrm{pr}}(M,M)$ and let $a,b \in \mathrm{End}_R(M)$. Choose a factorization

$$M \xrightarrow{u} P \xrightarrow{v} M$$

with P finitely generated projective and $f = v \circ u$. Then

$$a \circ f \circ b = a \circ v \circ u \circ b = (a \circ v) \circ (u \circ b),$$

so $a \circ f \circ b$ still factors through the same finitely generated projective module P. Hence

$$a \circ f \circ b \in \mathrm{Hom}_R^{\mathrm{pr}}(M,M).$$

Therefore $\mathrm{Hom}_R^{\mathrm{pr}}(M,M)$ is a two-sided ideal of $\mathrm{End}_R(M)$. □

Theorem 4.3.8. *Let R be a commutative ring with 1, and let P be an R-module. The following statements are equivalent.*

1. *P is a finitely generated projective module.*

2.
$$\mathrm{Hom}_R^{\mathrm{pr}}(P,P)=\mathrm{End}_R(P).$$

3. *For every R-module X, the morphism*
$$\tau_{X,P}:X^*\otimes_R P\longrightarrow \mathrm{Hom}_R(X,P)$$
is an isomorphism.

4. *For every R-module X, the morphism*
$$\tau_{P,X}:P^*\otimes_R X\longrightarrow \mathrm{Hom}_R(P,X)$$
is an isomorphism.

5. *The morphism*
$$\tau_{P,P}:P^*\otimes_R P\longrightarrow \mathrm{End}_R(P)$$
is an isomorphism.

Proof. **(1) $\Longrightarrow$(2).** Since P itself is finitely generated projective, the identity morphism
$$\mathrm{id}_P:P\to P$$
is projective. By the previous lemma, $\mathrm{Hom}_R^{\mathrm{pr}}(P,P)$ is a two-sided ideal of $\mathrm{End}_R(P)$ containing id_P, and therefore equals the whole endomorphism ring.

(3) $\Longrightarrow$(5) and **(4) $\Longrightarrow$(5)** are immediate by taking $X=P$.

(2) $\Longrightarrow$(3). By (2), id_P is projective, so there exist $\varphi_1,\dots,\varphi_n\in P^*$ and $p_1,\dots,p_n\in P$ such that
$$\mathrm{id}_P=\tau_{P,P}\Big(\sum_{i=1}^n\varphi_i\otimes p_i\Big).$$
Equivalently,
$$p=\sum_{i=1}^n\varphi_i(p)p_i \qquad \text{for all } p\in P.$$
For any R-module X, define
$$\sigma_X:\mathrm{Hom}_R(X,P)\to X^*\otimes_R P,\qquad \alpha\mapsto\sum_{i=1}^n(\varphi_i\circ\alpha)\otimes p_i.$$
Then for $\alpha\in\mathrm{Hom}_R(X,P)$ and $x\in X$,
$$\tau_{X,P}(\sigma_X(\alpha))(x)=\sum_{i=1}^n(\varphi_i\circ\alpha)(x)p_i=\sum_{i=1}^n\varphi_i(\alpha(x))p_i=\alpha(x),$$
so $\tau_{X,P}\circ\sigma_X=\mathrm{id}$.

Conversely, let

$$\xi = \sum_{j=1}^{m} \psi_j \otimes q_j \in X^* \otimes_R P.$$

Then

$$\begin{aligned}
\sigma_X(\tau_{X,P}(\xi)) &= \sum_{i=1}^{n} \Big(\varphi_i \circ \tau_{X,P}(\xi)\Big) \otimes p_i \\
&= \sum_{i=1}^{n} \left(x \mapsto \varphi_i\Big(\sum_{j=1}^{m} \psi_j(x) q_j\Big) \right) \otimes p_i \\
&= \sum_{i=1}^{n} \sum_{j=1}^{m} \big(\psi_j \varphi_i(q_j)\big) \otimes p_i \\
&= \sum_{j=1}^{m} \psi_j \otimes \Big(\sum_{i=1}^{n} \varphi_i(q_j) p_i\Big) \\
&= \sum_{j=1}^{m} \psi_j \otimes q_j \\
&= \xi.
\end{aligned}$$

Therefore σ_X is the inverse of $\tau_{X,P}$, and $\tau_{X,P}$ is an isomorphism.

(2) $\Longrightarrow$(4). Using the same decomposition of id_P, define for any R-module X

$$\rho_X : \mathrm{Hom}_R(P, X) \to P^* \otimes_R X, \qquad \beta \mapsto \sum_{i=1}^{n} \varphi_i \otimes \beta(p_i).$$

For $\beta \in \mathrm{Hom}_R(P, X)$ and $p \in P$,

$$\tau_{P,X}(\rho_X(\beta))(p) = \sum_{i=1}^{n} \varphi_i(p)\beta(p_i) = \beta\Big(\sum_{i=1}^{n} \varphi_i(p) p_i\Big) = \beta(p),$$

so $\tau_{P,X}\rho_X = \mathrm{id}$. Conversely, for

$$\eta = \sum_{j=1}^{m} \varphi'_j \otimes x_j \in P^* \otimes_R X,$$

we have

$$\begin{aligned}
\rho_X(\tau_{P,X}(\eta)) &= \sum_{i=1}^{n} \varphi_i \otimes \tau_{P,X}(\eta)(p_i) \\
&= \sum_{i=1}^{n} \varphi_i \otimes \Big(\sum_{j=1}^{m} \varphi'_j(p_i)x_j\Big) \\
&= \sum_{j=1}^{m} \Big(\sum_{i=1}^{n} \varphi'_j(p_i)\varphi_i\Big) \otimes x_j \\
&= \sum_{j=1}^{m} \varphi'_j \otimes x_j \\
&= \eta,
\end{aligned}$$

because

$$\varphi'_j(p) = \varphi'_j\Big(\sum_{i=1}^{n} \varphi_i(p)p_i\Big) = \sum_{i=1}^{n} \varphi_i(p)\varphi'_j(p_i)$$

for all $p \in P$. Hence $\tau_{P,X}$ is an isomorphism.

(5) $\implies$(1). Since $\tau_{P,P}$ is surjective, we can write

$$\mathrm{id}_P = \tau_{P,P}\Big(\sum_{i=1}^{n} \varphi_i \otimes p_i\Big)$$

for some $\varphi_i \in P^*$ and $p_i \in P$. Define

$$\iota : P \to R^n, \qquad \iota(p) = (\varphi_1(p), \dots, \varphi_n(p)),$$

and

$$\pi : R^n \to P, \qquad \pi(a_1, \dots, a_n) = \sum_{i=1}^{n} a_i p_i.$$

Then

$$(\pi \circ \iota)(p) = \sum_{i=1}^{n} \varphi_i(p)p_i = p$$

for all $p \in P$. Thus $\pi \circ \iota = \mathrm{id}_P$, so P is a direct summand of the finite free module R^n. Therefore P is finitely generated and projective. □

4.4 Projective Covers

Definition 4.4.1. *Let R be a ring with 1. An epimorphism*

$$f : U \to V$$

is called an essential epimorphism *if no proper submodule of U is mapped surjectively* *onto V by f. Equivalently, whenever*

$$g : W \to U$$

is a morphism such that $f \circ g$ is an epimorphism, then g is an epimorphism.

Proof. Suppose first that no proper submodule of U maps onto V. Let $g : W \to U$ satisfy $f \circ g$ epimorphic. Then

$$f(g(W)) = V.$$

By assumption, $g(W)$ cannot be a proper submodule of U, so $g(W) = U$ and g is surjective.

Conversely, assume that whenever $f \circ g$ is epi, then g is epi. Let $U_0 \subsetneq U$ be a proper submodule and let

$$\iota : U_0 \hookrightarrow U$$

be the inclusion. If $f(U_0) = V$, then $f \circ \iota$ is epi, hence ι is epi, which is impossible. Thus $f(U_0) \neq V$. □

Definition 4.4.2. *A* projective cover *of an R-module U is an essential epimorphism*

$$P \to U$$

with P projective.

Proposition 4.4.3. *Let*

$$U \xrightarrow{f} V \xrightarrow{g} W$$

be R-homomorphisms. If any two of f, g, and $g \circ f$ are essential epimorphisms, then so is the third.

Proof. **If f and g are essential, then $g \circ f$ is essential.** Let $U_0 \subsetneq U$ be a proper submodule. Since f is essential,

$$f(U_0) \neq V.$$

As g is essential, $g(f(U_0)) \neq W$. Hence $(g \circ f)(U_0) \neq W$, so $g \circ f$ is essential.

If g and $g \circ f$ are essential, then f is essential. Since $g \circ f$ is epic, f is epic. Let $U_0 \subsetneq U$ be proper. If $f(U_0) = V$, then

$$(g \circ f)(U_0) = W,$$

contradicting that $g \circ f$ is essential. Thus $f(U_0) \neq V$, so f is essential.

If f and $g \circ f$ are essential, then g is essential. Since $g \circ f$ is epic and f is epic, g is epic. Let $V_0 \subsetneq V$ be a proper submodule, and set

$$U_0 := f^{-1}(V_0).$$

Because f is surjective, U_0 is a proper submodule of U. Since $g \circ f$ is essential,

$$(g \circ f)(U_0) \neq W.$$

But

$$(g \circ f)(U_0) = g(V_0),$$

so $g(V_0) \neq W$. Hence g is essential. $\square$

4.5 Noetherian and Artinian Modules

Definition 4.5.1. *An R-module M is called* Noetherian *if for every ascending chain of submodules*

$$L_1 \subseteq L_2 \subseteq \cdots$$

there exists n_0 such that

$$L_n = L_{n_0} \qquad \text{for all } n \geq n_0.$$

Lemma 4.5.2. *For an R-module M, the following are equivalent.*

1. *M is Noetherian.*
2. *Every submodule of M is finitely generated.*
3. *Every non-empty set of submodules of M has a maximal element.*

Proof. **(1) $\Longrightarrow$(3).** Suppose $\mathcal{S}$ is a non-empty set of submodules of M having no maximal element. Choose $L_1 \in \mathcal{S}$. Since L_1 is not maximal, there exists $L_2 \in \mathcal{S}$ with

$$L_1 \subsetneq L_2.$$

Continuing inductively, we obtain a strictly ascending chain

$$L_1 \subsetneq L_2 \subsetneq L_3 \subsetneq \cdots,$$

contradicting that M is Noetherian.

(3) $\Longrightarrow$(2). Let $N \leq M$. Consider the set

$$\mathcal{T} := \{L \leq N \mid L \text{ is finitely generated}\}.$$

This set is non-empty because $0 \in \mathcal{T}$. By (3), $\mathcal{T}$ has a maximal element, say L. If $L \neq N$, choose $x \in N \setminus L$. Then

$$L + Rx$$

is a finitely generated submodule of N strictly containing L, contradicting the maximality of L. Hence $L = N$, so N is finitely generated.

(2) $\Longrightarrow$(1). Let
$$L_1 \subseteq L_2 \subseteq \cdots$$
be an ascending chain of submodules, and set
$$L := \bigcup_{n\geq 1} L_n.$$
Then L is a submodule of M, so by (2) it is finitely generated, say
$$L = Rx_1 + \cdots + Rx_m.$$
Each generator x_j lies in some L_{n_j}. If $N = \max\{n_1, \ldots, n_m\}$, then
$$x_1, \ldots, x_m \in L_N,$$
so $L = L_N$. Hence
$$L_n = L_N \qquad \text{for all } n \geq N,$$
and the chain stabilizes. $\square$

Proposition 4.5.3. *Let*
$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$
be a short exact sequence of R-modules. Then M is Noetherian if and only if both M' and M'' are Noetherian.

Proof. Let $\pi : M \to M''$ be the quotient map, and identify M' with its image in M.

First assume that M is Noetherian. Since M' is a submodule of M, it is Noetherian. Now let $N'' \leq M''$ be any submodule. Then $\pi^{-1}(N'')$ is a submodule of M, hence is finitely generated. Since $\pi(\pi^{-1}(N'')) = N''$, it follows that N'' is finitely generated. Therefore M'' is Noetherian.

Conversely, assume that both M' and M'' are Noetherian. Let $N \leq M$ be any submodule. Then $N \cap M'$ is a submodule of M', so it is finitely generated. Also $\pi(N)$ is a submodule of M'', so it is finitely generated. Choose elements $x_1, \ldots, x_r \in N$ such that
$$\pi(x_1), \ldots, \pi(x_r)$$
generate $\pi(N)$, and choose elements $y_1, \ldots, y_s \in N \cap M'$ which generate $N \cap M'$.

We claim that
$$N = Rx_1 + \cdots + Rx_r + Ry_1 + \cdots + Ry_s.$$
Indeed, let $x \in N$. Since $\pi(x) \in \pi(N)$, there exist $a_1, \ldots, a_r \in R$ such that
$$\pi(x) = \sum_{i=1}^{r} a_i \pi(x_i) = \pi\left(\sum_{i=1}^{r} a_i x_i\right).$$

Hence

$$x - \sum_{i=1}^{r} a_i x_i \in \ker(\pi) \cap N = M' \cap N.$$

Since $y_1, \ldots, y_s$ generate $N \cap M'$, there exist $b_1, \ldots, b_s \in R$ such that

$$x - \sum_{i=1}^{r} a_i x_i = \sum_{j=1}^{s} b_j y_j.$$

Therefore

$$x = \sum_{i=1}^{r} a_i x_i + \sum_{j=1}^{s} b_j y_j,$$

proving the claim. Thus N is finitely generated.

Since every submodule $N \leq M$ is finitely generated, M is Noetherian. □

Definition 4.5.4. *An R-module M is called* Artinian *if for every descending chain of submodules*

$$L_1 \supseteq L_2 \supseteq \cdots$$

there exists n_0 such that

$$L_n = L_{n_0} \qquad \text{for all } n \geq n_0.$$

Lemma 4.5.5. *An R-module M is Artinian if and only if every non-empty set of submodules of M contains a minimal element.*

Proof. Recall that M is Artinian if every descending chain of submodules

$$L_1 \supseteq L_2 \supseteq L_3 \supseteq \cdots$$

stabilizes.

First assume that M is Artinian. Let $\mathcal{S}$ be a non-empty set of submodules of M. We prove that $\mathcal{S}$ contains a minimal element.

Suppose, for contradiction, that $\mathcal{S}$ has no minimal element. Choose $L_1 \in \mathcal{S}$. Since L_1 is not minimal in $\mathcal{S}$, there exists $L_2 \in \mathcal{S}$ such that

$$L_2 \subsetneq L_1.$$

Similarly, since L_2 is not minimal in $\mathcal{S}$, there exists $L_3 \in \mathcal{S}$ such that

$$L_3 \subsetneq L_2.$$

Continuing inductively, we obtain a strictly descending chain

$$L_1 \supsetneq L_2 \supsetneq L_3 \supsetneq \cdots$$

of submodules of M. This contradicts the assumption that M is Artinian. Hence $\mathcal{S}$ must contain a minimal element.

Conversely, assume that every non-empty set of submodules of M contains a minimal element. Let

$$L_1 \supseteq L_2 \supseteq L_3 \supseteq \cdots$$

be a descending chain of submodules of M. Consider the non-empty set

$$\mathcal{S} := \{L_n \mid n \geq 1\}.$$

By assumption, $\mathcal{S}$ contains a minimal element, say L_N.

Since the chain is descending, for every $n \geq N$ we have

$$L_n \subseteq L_N.$$

But L_N is minimal among the elements of $\mathcal{S}$, and each L_n also belongs to $\mathcal{S}$. Therefore

$$L_n = L_N$$

for every $n \geq N$. Hence the descending chain stabilizes.

Therefore M is Artinian. □

Remark 4.5.6. *If an R-module M is Noetherian (respectively Artinian), then we also say that M satisfies the ascending chain condition (respectively descending chain condition) on submodules, abbreviated as ACC (respectively DCC).*

Proposition 4.5.7. *Let*

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0$$

be a short exact sequence of R-modules. Then M is Artinian if and only if both M' and M'' are Artinian.

Proof. Since the sequence is exact, we may identify M' with its image $\iota(M') \subseteq M$. Thus M' is regarded as a submodule of M, and $M'' \cong M/M'$ via the quotient map

$$\pi : M \longrightarrow M''.$$

First assume that M is Artinian.

We show that M' is Artinian. Let

$$L_1 \supseteq L_2 \supseteq L_3 \supseteq \cdots$$

be a descending chain of submodules of M'. Since each L_i is also a submodule of M, this is a descending chain of submodules of M. Because M is Artinian, the chain stabilizes. Hence M' is Artinian.

Next we show that M'' is Artinian. Let

$$N_1 \supseteq N_2 \supseteq N_3 \supseteq \cdots$$

be a descending chain of submodules of M''. Consider their inverse images under π:

$$\pi^{-1}(N_1) \supseteq \pi^{-1}(N_2) \supseteq \pi^{-1}(N_3) \supseteq \cdots .$$

This is a descending chain of submodules of M. Since M is Artinian, there exists n_0 such that

$$\pi^{-1}(N_n) = \pi^{-1}(N_{n_0}) \qquad \text{for all } n \geq n_0.$$

Applying π to both sides, and using the fact that each N_n is contained in $\mathrm{Im}(\pi) = M''$, we get

$$N_n = N_{n_0} \qquad \text{for all } n \geq n_0.$$

Therefore M'' is Artinian.

Conversely, assume that both M' and M'' are Artinian. We prove that M is Artinian.

Let

$$L_1 \supseteq L_2 \supseteq L_3 \supseteq \cdots$$

be a descending chain of submodules of M. Intersecting with M', we obtain a descending chain of submodules of M':

$$L_1 \cap M' \supseteq L_2 \cap M' \supseteq L_3 \cap M' \supseteq \cdots .$$

Since M' is Artinian, this chain stabilizes. Hence there exists a such that

$$L_n \cap M' = L_a \cap M' \qquad \text{for all } n \geq a.$$

Applying π to the original chain, we obtain a descending chain of submodules of M'':

$$\pi(L_1) \supseteq \pi(L_2) \supseteq \pi(L_3) \supseteq \cdots .$$

Since M'' is Artinian, this chain stabilizes. Hence there exists b such that

$$\pi(L_n) = \pi(L_b) \qquad \text{for all } n \geq b.$$

Let

$$N = \max\{a, b\}.$$

We claim that

$$L_n = L_N \qquad \text{for all } n \geq N.$$

Since the chain is descending, we automatically have

$$L_n \subseteq L_N \qquad \text{for all } n \geq N.$$

It remains to prove the reverse inclusion.

Fix $n \geq N$, and take any element $x \in L_N$. Since

$$\pi(L_n) = \pi(L_N),$$

there exists $y \in L_n$ such that

$$\pi(y) = \pi(x).$$

Therefore

$$\pi(x - y) = 0,$$

so

$$x - y \in \ker(\pi) = M'.$$

Also, since $x \in L_N$ and $y \in L_n \subseteq L_N$, we have

$$x - y \in L_N.$$

Hence

$$x - y \in L_N \cap M'.$$

But the chain of intersections has stabilized, so

$$L_N \cap M' = L_n \cap M'.$$

Thus

$$x - y \in L_n \cap M' \subseteq L_n.$$

Since $y \in L_n$, it follows that

$$x = y + (x - y) \in L_n.$$

Therefore $L_N \subseteq L_n$. Combining this with $L_n \subseteq L_N$, we get

$$L_n = L_N \qquad \text{for all } n \geq N.$$

Hence every descending chain of submodules of M stabilizes, so M is Artinian.

Therefore M is Artinian if and only if both M' and M'' are Artinian. □

Definition 4.5.8. *An R-module M is called* simple *(or* irreducible*) if it has exactly two submodules:* 0 *and* M*.*

Chapter 5

Lecture 5: Semisimplicity

5.1 Semisimple Modules and the Radical

Definition 5.1.1. *An R-module M is called* semisimple *if it is a direct sum of simple submodules.*

Theorem 5.1.2 (Schur's lemma)**.** *Let S_1 and S_2 be simple R-modules, and let*

$$\varphi : S_1 \to S_2$$

be an R-homomorphism. Then either $\varphi = 0$ or φ is an isomorphism. In particular, $\mathrm{End}_R(S)$ is a division ring for every simple R-module S.

Proof. Assume $\varphi \neq 0$. Since $\ker(\varphi)$ is a submodule of the simple module S_1, we have $\ker(\varphi) = 0$ or $\ker(\varphi) = S_1$. The latter is impossible because $\varphi \neq 0$, hence $\ker(\varphi) = 0$ and φ is injective.

Likewise, $\mathrm{Im}(\varphi)$ is a nonzero submodule of the simple module S_2, so $\mathrm{Im}(\varphi) = S_2$. Thus φ is surjective and therefore an isomorphism.

Taking $S_1 = S_2 = S$, every nonzero endomorphism of S is invertible, which means that $\mathrm{End}_R(S)$ is a division ring. □

Lemma 5.1.3. *Let S be a simple left R-module. Then S is a quotient of the regular left module ${}_RR$. More precisely, for every nonzero $x \in S$ there is an isomorphism*

$$S \cong R/\operatorname{Ann}_R(x).$$

In particular, $\operatorname{Ann}_R(x)$ is a maximal left ideal of R.

Proof. Fix $0 \neq x \in S$ and define

$$\pi_x : R \to S, \qquad r \mapsto rx.$$

Since S is simple and Rx is a nonzero submodule of S, we have $Rx = S$. Hence π_x is surjective. Its kernel is exactly

$$\ker(\pi_x) = \mathrm{Ann}_R(x) := \{r \in R \mid rx = 0\}.$$

By the first isomorphism theorem,

$$S \cong R/\operatorname{Ann}_R(x).$$

Because S is simple, the quotient $R/\operatorname{Ann}_R(x)$ is simple as a left R-module, so $\mathrm{Ann}_R(x)$ is a maximal left ideal. □

Definition 5.1.4. *Let U be an R-module. A proper submodule $M \subsetneq U$ is called a* maximal submodule *if whenever*

$$M \subseteq N \subseteq U,$$

then either $N = M$ or $N = U$.

Lemma 5.1.5. *Let W be a proper submodule of an R-module U. Assume that U/W is finitely generated. Then there exists a maximal submodule M of U such that*

$$W \subseteq M \subsetneq U.$$

Proof. Consider the set

$$\mathcal{S} := \{N \leq U \mid W \subseteq N \subsetneq U\},$$

ordered by inclusion. Since $W \in \mathcal{S}$, the set is non-empty. Let $\{N_\lambda\}_{\lambda\in\Lambda}$ be a totally ordered subset of $\mathcal{S}$, and set

$$N := \bigcup_{\lambda\in\Lambda} N_\lambda.$$

Then N is a submodule containing W. We claim that $N \neq U$. Indeed, write

$$U/W = R\overline{u}_1 + \cdots + R\overline{u}_n.$$

If $N = U$, then each u_i lies in some N_{λ_i}. Because the family is totally ordered, there exists μ such that

$$N_{\lambda_1}, \ldots, N_{\lambda_n} \subseteq N_\mu.$$

Hence $u_1, \ldots, u_n \in N_\mu$, so $U/W \subseteq N_\mu/W$, which implies $N_\mu = U$, contradicting $N_\mu \in \mathcal{S}$. Therefore N is an upper bound for the chain in $\mathcal{S}$.

By Zorn's lemma, $\mathcal{S}$ has a maximal element M, and this is precisely a maximal submodule of U containing W. □

Corollary 5.1.6. *Every proper left ideal of R is contained in a maximal left ideal.*

Proof. Apply the lemma to the left regular module ${}_RR$ and to the proper submodule $W = I$. □

Definition 5.1.7. *Let U be an R-module. The* radical *of U is the intersection of all maximal submodules of U, and is denoted by $\operatorname{Rad}(U)$. If U has no maximal submodules, we set $\operatorname{Rad}(U) = U$.*

Theorem 5.1.8 (Nakayama's lemma)**.** *Let U be an R-module and let $V \subseteq U$ be a submodule such that U/V is finitely generated. Then*

$$V + \operatorname{Rad}(U) = U \qquad \Longleftrightarrow \qquad V = U.$$

Proof. If $V = U$, then the equality is obvious.

Conversely, assume $V \neq U$. Since U/V is finitely generated, the previous lemma yields a maximal submodule M of U with

$$V \subseteq M \subsetneq U.$$

By definition,

$$\operatorname{Rad}(U) \subseteq M.$$

Therefore

$$V + \operatorname{Rad}(U) \subseteq M \subsetneq U,$$

so $V + \operatorname{Rad}(U) \neq U$. This proves the contrapositive. $\square$

Corollary 5.1.9. *If U is a finitely generated R-module, then the quotient map*

$$\pi : U \to U/\operatorname{Rad}(U)$$

is an essential epimorphism.

Proof. Let $W \leq U$ be a submodule such that $\pi(W) = U/\operatorname{Rad}(U)$. Then

$$W + \operatorname{Rad}(U) = U.$$

Since U/W is a quotient of the finitely generated module U, it is finitely generated. Nakayama's lemma therefore gives $W = U$, so π is essential. $\square$

Lemma 5.1.10. *Let M and N be finitely generated R-modules, and let*

$$f : M \to N$$

be an R-homomorphism.

1. *We have*

$$f(\operatorname{Rad}(M)) \subseteq \operatorname{Rad}(N).$$

2. *If f is surjective and* $\ker(f) \subseteq \mathrm{Rad}(M)$, *then*

$$\mathrm{Rad}(N) = f(\mathrm{Rad}(M)).$$

Proof. **(1).** Let $x \in \mathrm{Rad}(M)$. Suppose that $f(x) \notin \mathrm{Rad}(N)$. Then there exists a maximal submodule L of N such that

$$f(x) \notin L.$$

Let

$$\rho : N \to N/L$$

be the quotient map. Since N/L is simple and $\rho(f(x)) \neq 0$, the morphism $\rho \circ f : M \to N/L$ is nonzero, hence surjective. Therefore

$$M/\ker(\rho \circ f) \cong N/L$$

is simple, so $\ker(\rho \circ f)$ is a maximal submodule of M. But

$$x \notin \ker(\rho \circ f),$$

contradicting $x \in \mathrm{Rad}(M)$. Hence $f(x) \in \mathrm{Rad}(N)$.

(2). By (1),

$$f(\mathrm{Rad}(M)) \subseteq \mathrm{Rad}(N).$$

For the reverse inclusion, let $x \in M$ with $f(x) \in \mathrm{Rad}(N)$. If $x \notin \mathrm{Rad}(M)$, choose a maximal submodule K of M with

$$x \notin K.$$

Since $\ker(f) \subseteq \mathrm{Rad}(M) \subseteq K$, the map f induces a surjection

$$\overline{f} : M/K \to N/f(K).$$

Because M/K is simple and $x \notin K$, the quotient $N/f(K)$ is nonzero and hence simple. Thus $f(K)$ is a maximal submodule of N. But $f(x) \notin f(K)$, so $f(x) \notin \mathrm{Rad}(N)$, contradiction. Therefore

$$f^{-1}(\mathrm{Rad}(N)) = \mathrm{Rad}(M),$$

and applying f gives the desired equality

$$\mathrm{Rad}(N) = f(\mathrm{Rad}(M)).$$

□

Lemma 5.1.11. *Let M and N be finitely generated R-modules, and let*

$$f : M \to N$$

be an R-homomorphism. Then f is an essential epimorphism if and only if the induced morphism

$$\overline{f} : M/\operatorname{Rad}(M) \to N/\operatorname{Rad}(N)$$

is an essential epimorphism.

Proof. First assume that f is an essential epimorphism. In particular, f is surjective. By the previous lemma,

$$f(\operatorname{Rad}(M)) \subseteq \operatorname{Rad}(N),$$

so $\overline{f}$ is well defined and surjective. Let

$$X/\operatorname{Rad}(M) \subseteq M/\operatorname{Rad}(M)$$

be a submodule such that $\overline{f}(X/\operatorname{Rad}(M)) = N/\operatorname{Rad}(N)$. Then

$$f(X) + \operatorname{Rad}(N) = N.$$

Since $N/f(X)$ is a quotient of the finitely generated module N, Nakayama's lemma gives $f(X) = N$. As f is essential, we conclude that $X = M$. Hence $\overline{f}$ is essential.

Conversely, assume that $\overline{f}$ is an essential epimorphism. Since $\overline{f}$ is surjective, we have

$$f(M) + \operatorname{Rad}(N) = N.$$

By Nakayama's lemma applied to the submodule $f(M) \subseteq N$, it follows that $f(M) = N$, so f is surjective. Now let $X \leq M$ satisfy $f(X) = N$. Then

$$\overline{f}\big((X + \operatorname{Rad}(M))/\operatorname{Rad}(M)\big) = N/\operatorname{Rad}(N).$$

Since $\overline{f}$ is essential, we obtain

$$X + \operatorname{Rad}(M) = M.$$

Again Nakayama's lemma yields $X = M$. Hence f is an essential epimorphism. $\square$

5.2 Composition Series and Jordan–Hölder Theory

Definition 5.2.1. *A finite chain of submodules*

$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_n = 0$$

is called a composition series *of U if each factor*

$$U_i/U_{i+1}$$

is simple. The integer n is called the composition length *of U.*

Lemma 5.2.2. *A nonzero R-module U has a composition series if and only if U is both Noetherian and Artinian.*

Proof. Suppose first that
$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_n = 0$$
is a composition series. We argue by induction on n. If $n = 1$, then U is simple, hence both Noetherian and Artinian. If $n > 1$, then U_1 has a composition series of length $n-1$, so by induction U_1 is Noetherian and Artinian. The quotient U/U_1 is simple, hence also Noetherian and Artinian. By the permanence of both properties in short exact sequences, U is Noetherian and Artinian.

Conversely, assume that U is Noetherian and Artinian. Since $U \neq 0$, choose a maximal proper submodule U_1 of U. Then U/U_1 is simple. Because U_1 is a submodule of both a Noetherian and an Artinian module, it is again Noetherian and Artinian. Repeating the argument inductively and using the Artinian condition to rule out an infinite descending chain, we obtain a composition series. $\square$

Lemma 5.2.3. *Suppose that*
$$U = S_1 + \cdots + S_n$$
where each S_i is a simple submodule of U. If $V \leq U$, then there exists a subset $I \subseteq \{1, \dots, n\}$ such that
$$U = V \oplus \bigoplus_{i \in I} S_i.$$
In particular, every submodule of a finite direct sum of simple modules is a direct summand.

Proof. Choose a subset $I \subseteq \{1, \dots, n\}$ maximal with the property that
$$V \oplus \bigoplus_{i \in I} S_i$$
is a direct sum, and set
$$W := V \oplus \bigoplus_{i \in I} S_i.$$
We claim that $W = U$. If not, then there exists $j \notin I$ such that
$$S_j \nsubseteq W.$$
Now $W \cap S_j$ is a submodule of the simple module S_j, so either
$$W \cap S_j = 0 \qquad \text{or} \qquad W \cap S_j = S_j.$$
The second case is impossible because it would imply $S_j \subseteq W$. Hence $W \cap S_j = 0$, and

therefore

$$W \oplus S_j$$

is a direct sum, contradicting the maximality of I. Thus $W = U$. $\square$

Theorem 5.2.4. *Let U be an R-module with a composition series. The following are equivalent.*

1. *U is semisimple.*

2. *U is a sum of simple submodules.*

3. *Every submodule of U is a direct summand of U.*

Proof. **(1) $\Longrightarrow$(2).** This is immediate from the definition of a semisimple module.

(2) $\Longrightarrow$(3). Since U has finite composition length, a finite number of simple summands already generate U. Thus

$$U = S_1 + \cdots + S_n$$

for simple submodules S_i. The previous lemma shows that every submodule of U is a direct summand.

(3) $\Longrightarrow$(1). Let

$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_n = 0$$

be a composition series. Since U_1 is a direct summand of U, there exists a submodule S_1 such that

$$U = U_1 \oplus S_1.$$

Because U/U_1 is simple, the submodule S_1 is simple. Repeating the argument inside U_1, we obtain simple submodules $S_2, \ldots, S_n$ with

$$U_i = U_{i+1} \oplus S_{i+1}$$

for all i. Hence

$$U = S_1 \oplus \cdots \oplus S_n,$$

so U is semisimple. $\square$

Corollary 5.2.5. *Let U be an R-module of finite composition length.*

1. *The sum of all simple submodules of U is semisimple, hence is the unique largest semisimple submodule of U.*

2. *If S is a simple submodule of U, then the sum of all submodules of U isomorphic to S is a submodule of U isomorphic to a direct sum of copies of S. It is the unique largest submodule of U with this property.*

Proof. Since U has finite composition length, U is both Noetherian and Artinian. In particular, every submodule of U is finitely generated.

(1). Let

$$\Sigma := \sum_{\substack{T \leq U \\ T \text{ simple}}} T$$

be the sum of all simple submodules of U.

Since $\Sigma \leq U$ and U is Noetherian, the submodule Σ is finitely generated. Thus there exist elements

$$x_1, \ldots, x_m \in \Sigma$$

such that

$$\Sigma = Rx_1 + \cdots + Rx_m.$$

By the definition of a sum of submodules, each $x_j \in \Sigma$ lies in a finite sum of simple submodules of U. Hence for each j, there exist simple submodules

$$T_{j1}, \ldots, T_{jr_j} \leq U$$

such that

$$x_j \in T_{j1} + \cdots + T_{jr_j}.$$

Collecting all these finitely many simple submodules, we obtain simple submodules

$$T_1, \ldots, T_n \leq U$$

such that

$$x_1, \ldots, x_m \in T_1 + \cdots + T_n.$$

Therefore

$$\Sigma = Rx_1 + \cdots + Rx_m \subseteq T_1 + \cdots + T_n \subseteq \Sigma.$$

Hence

$$\Sigma = T_1 + \cdots + T_n.$$

Since Σ is a finite sum of simple submodules, by the previous theorem Σ is semisimple.

Now let $W \leq U$ be any semisimple submodule. Then W is a sum of simple submodules of W. But every simple submodule of W is also a simple submodule of U. Hence every simple summand appearing in W is contained in the defining sum of Σ. Therefore

$$W \subseteq \Sigma.$$

Thus Σ contains every semisimple submodule of U, and so Σ is the unique largest semisimple submodule of U.

(2). Let $S \leq U$ be a simple submodule, and define

$$\Sigma_S := \sum_{\substack{T \leq U \\ T \cong S}} T.$$

This is the sum of all submodules of U which are isomorphic to S.

Since $\Sigma_S \leq U$ and U is Noetherian, Σ_S is finitely generated. Thus there exist elements

$$y_1, \dots, y_\ell \in \Sigma_S$$

such that

$$\Sigma_S = Ry_1 + \cdots + Ry_\ell.$$

Again, by the definition of a sum of submodules, each y_j lies in a finite sum of submodules of U isomorphic to S. Hence, after collecting all the finitely many such submodules involved, there exist submodules

$$S_1, \dots, S_m \leq U$$

with

$$S_i \cong S \qquad \text{for all } i,$$

such that

$$\Sigma_S = S_1 + \cdots + S_m.$$

Each S_i is simple, since it is isomorphic to the simple module S. Hence Σ_S is a finite sum of simple submodules. By the previous theorem, Σ_S is semisimple. More precisely, applying the preceding lemma to the finite sum

$$\Sigma_S = S_1 + \cdots + S_m$$

with $V = 0$, we may choose a subset $I \subseteq \{1, \dots, m\}$ such that

$$\Sigma_S = \bigoplus_{i \in I} S_i.$$

Since each $S_i \cong S$, we obtain

$$\Sigma_S \cong S^{\oplus |I|}.$$

Thus Σ_S is isomorphic to a direct sum of copies of S.

Finally, let $W \leq U$ be any submodule which is isomorphic to a direct sum of copies of S. Then W is the sum of its simple submodules, each of which is isomorphic to S. Since these simple submodules are also submodules of U, they are all contained in the defining sum of Σ_S. Hence

$$W \subseteq \Sigma_S.$$

Therefore Σ_S contains every submodule of U which is isomorphic to a direct sum of copies of S. Hence Σ_S is the unique largest submodule of U with this property. □

Theorem 5.2.6 (Zassenhaus lemma). *Let $A' \subseteq A$ and $B' \subseteq B$ be submodules of an R-module M. Then*

$$\frac{A' + (A \cap B)}{A' + (A \cap B')} \cong \frac{B' + (A \cap B)}{B' + (A' \cap B)}.$$

Proof. Set

$$C := (A \cap B') + (A' \cap B).$$

We shall show that both quotient modules are naturally isomorphic to

$$\frac{A \cap B}{C}.$$

First consider the quotient

$$\frac{A' + (A \cap B)}{A' + (A \cap B')}.$$

Apply the second isomorphism theorem to the submodules

$$X := A \cap B, \qquad Y := A' + (A \cap B')$$

of M. Since

$$X + Y = (A \cap B) + A' + (A \cap B') = A' + (A \cap B),$$

we obtain

$$\frac{A' + (A \cap B)}{A' + (A \cap B')} = \frac{X + Y}{Y} \cong \frac{X}{X \cap Y}.$$

It remains to compute $X \cap Y$. We claim that

$$(A \cap B) \cap \big(A' + (A \cap B')\big) = (A' \cap B) + (A \cap B').$$

The inclusion "$\supseteq$" is immediate, because

$$A' \cap B \subseteq A \cap B, \qquad A \cap B' \subseteq A \cap B,$$

and both summands are contained in $A' + (A \cap B')$.

Conversely, let

$$x \in (A \cap B) \cap \big(A' + (A \cap B')\big).$$

Then $x \in A \cap B$, and we may write

$$x = a' + c$$

with $a' \in A'$ and $c \in A \cap B'$. Since $x \in B$ and $c \in B' \subseteq B$, we have

$$a' = x - c \in B.$$

Hence

$$a' \in A' \cap B.$$

Therefore

$$x = a' + c \in (A' \cap B) + (A \cap B').$$

This proves the claimed equality. Consequently,

$$\frac{A' + (A \cap B)}{A' + (A \cap B')} \cong \frac{A \cap B}{(A' \cap B) + (A \cap B')}.$$

Similarly, consider the quotient

$$\frac{B' + (A \cap B)}{B' + (A' \cap B)}.$$

Apply the second isomorphism theorem to

$$X := A \cap B, \qquad Y := B' + (A' \cap B).$$

Then

$$X + Y = (A \cap B) + B' + (A' \cap B) = B' + (A \cap B),$$

so

$$\frac{B' + (A \cap B)}{B' + (A' \cap B)} \cong \frac{A \cap B}{(A \cap B) \cap \big(B' + (A' \cap B)\big)}.$$

We now compute the intersection. We claim that

$$(A \cap B) \cap \big(B' + (A' \cap B)\big) = (A \cap B') + (A' \cap B).$$

Again the inclusion "$\supseteq$" is clear. Conversely, let

$$x \in (A \cap B) \cap \big(B' + (A' \cap B)\big).$$

Then $x \in A \cap B$, and we may write

$$x = b' + d$$

with $b' \in B'$ and $d \in A' \cap B$. Since $x \in A$ and $d \in A' \subseteq A$, we get

$$b' = x - d \in A.$$

Hence

$$b' \in A \cap B'.$$

Therefore

$$x = b' + d \in (A \cap B') + (A' \cap B).$$

This proves the second intersection equality. Thus

$$\frac{B' + (A \cap B)}{B' + (A' \cap B)} \cong \frac{A \cap B}{(A \cap B') + (A' \cap B)}.$$

We have shown that both quotient modules are naturally isomorphic to the same quotient

$$\frac{A \cap B}{(A \cap B') + (A' \cap B)}.$$

Hence

$$\frac{A' + (A \cap B)}{A' + (A \cap B')} \cong \frac{B' + (A \cap B)}{B' + (A' \cap B)}.$$

□

Definition 5.2.7. *Let*
$$U = U_0 \supseteq U_1 \supseteq \cdots \supseteq U_m = 0$$
be a finite submodule series of an R-module U. A refinement of this series is another finite submodule series obtained by inserting additional submodules between the terms U_{i-1} and U_i.

Two finite submodule series are said to be equivalent if their nonzero successive quotient modules are isomorphic up to a permutation.

Theorem 5.2.8 (Schreier refinement theorem)**.** *Any two finite submodule series of an R-module admit equivalent refinements.*

Proof. Let the two finite submodule series be
$$U = U_0 \supseteq U_1 \supseteq \cdots \supseteq U_m = 0$$
and
$$U = V_0 \supseteq V_1 \supseteq \cdots \supseteq V_n = 0.$$
We shall construct refinements of both series and then compare their successive quotient modules.

For each $1 \leq i \leq m$ and $0 \leq j \leq n$, define
$$U_{i,j} := U_i + (U_{i-1} \cap V_j).$$
Since
$$V_0 = U \qquad \text{and} \qquad V_n = 0,$$
we have
$$U_{i,0} = U_i + (U_{i-1} \cap V_0) = U_i + U_{i-1} = U_{i-1},$$
and
$$U_{i,n} = U_i + (U_{i-1} \cap V_n) = U_i.$$
Moreover, since
$$V_0 \supseteq V_1 \supseteq \cdots \supseteq V_n,$$
we get
$$U_{i,0} \supseteq U_{i,1} \supseteq \cdots \supseteq U_{i,n}.$$
Thus, for each fixed i, the chain
$$U_{i-1} = U_{i,0} \supseteq U_{i,1} \supseteq \cdots \supseteq U_{i,n} = U_i$$
refines the interval
$$U_{i-1} \supseteq U_i.$$

Splicing these chains together, we obtain a refinement of the first series:

$$U = U_{1,0} \supseteq U_{1,1} \supseteq \cdots \supseteq U_{1,n} = U_1 = U_{2,0} \supseteq \cdots \supseteq U_{m,n} = 0.$$

Similarly, for each $1 \le j \le n$ and $0 \le i \le m$, define

$$V_{j,i} := V_j + (V_{j-1} \cap U_i).$$

Then

$$V_{j,0} = V_j + (V_{j-1} \cap U_0) = V_j + V_{j-1} = V_{j-1},$$

and

$$V_{j,m} = V_j + (V_{j-1} \cap U_m) = V_j.$$

Hence, for each fixed j, we have a chain

$$V_{j-1} = V_{j,0} \supseteq V_{j,1} \supseteq \cdots \supseteq V_{j,m} = V_j.$$

Splicing these chains together gives a refinement of the second series:

$$U = V_{1,0} \supseteq V_{1,1} \supseteq \cdots \supseteq V_{1,m} = V_1 = V_{2,0} \supseteq \cdots \supseteq V_{n,m} = 0.$$

We now compare the successive quotients of these two refinements. Fix indices

$$1 \le i \le m, \qquad 1 \le j \le n.$$

Apply the Zassenhaus lemma to the four submodules

$$A' = U_i, \qquad A = U_{i-1}, \qquad B' = V_j, \qquad B = V_{j-1}.$$

Since $U_i \subseteq U_{i-1}$ and $V_j \subseteq V_{j-1}$, the Zassenhaus lemma gives

$$\frac{U_i + (U_{i-1} \cap V_{j-1})}{U_i + (U_{i-1} \cap V_j)} \cong \frac{V_j + (U_{i-1} \cap V_{j-1})}{V_j + (U_i \cap V_{j-1})}.$$

In terms of the notation introduced above, this is precisely

$$\frac{U_{i,j-1}}{U_{i,j}} \cong \frac{V_{j,i-1}}{V_{j,i}}.$$

Therefore every successive quotient in the refinement of the first series is isomorphic to a corresponding successive quotient in the refinement of the second series. The correspondence is given by the pairs of indices

$$(i, j) \longleftrightarrow (j, i).$$

Hence the two constructed refinements are equivalent.

If some consecutive terms in the constructed refinements coincide, then the corresponding quotient is zero. Deleting these repeated terms on both sides does not affect the list of nonzero quotient modules. Thus the two original finite submodule series admit equivalent refinements. ∎

Theorem 5.2.9 (Jordan–Hölder theorem). *Let U be an R-module with a composition series. Then any two composition series of U have the same length, and their composition factors are the same up to permutation and isomorphism.*

Proof. Let

$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_m = 0$$

and

$$U = V_0 \supsetneq V_1 \supsetneq \cdots \supsetneq V_n = 0$$

be two composition series of U. Thus each quotient

$$U_{i-1}/U_i$$

and each quotient

$$V_{j-1}/V_j$$

is simple.

By the Schreier refinement theorem, these two finite submodule series admit equivalent refinements. We now observe that a composition series has no proper refinement.

Indeed, suppose that one could insert a submodule W strictly between two consecutive terms of the first composition series:

$$U_{i-1} \supsetneq W \supsetneq U_i.$$

Then W/U_i would be a nonzero proper submodule of

$$U_{i-1}/U_i.$$

This contradicts the fact that U_{i-1}/U_i is simple. Hence no such intermediate submodule W exists. Therefore any refinement of the first composition series only inserts repeated terms. After deleting repetitions, the refinement is just the original composition series. The same argument applies to the second composition series.

Since the Schreier refinements of the two composition series are equivalent, and since those refinements reduce back to the original composition series after deleting repeated terms, the original composition series themselves are equivalent.

Therefore the number of nonzero successive quotient modules in the two series is the same. Hence

$$m = n.$$

Moreover, there exists a permutation $\sigma \in S_m$ such that

$$U_{i-1}/U_i \cong V_{\sigma(i)-1}/V_{\sigma(i)}$$

for every $1 \leq i \leq m$.

Thus any two composition series of U have the same length and the same composition factors up to permutation and isomorphism. $\square$

5.3 Socles, Tops and Projective Covers

Definition 5.3.1 (Soc of a module). *The unique largest semisimple submodule of an R-module U is called the* socle *of U and is denoted by*

$$\mathrm{Soc}(U).$$

Definition 5.3.2 (Top of a module). *Let U be an R-module. The top, or head, of U is defined by*

$$\mathrm{Top}(U) := U/\,\mathrm{Rad}(U).$$

Example 5.3.3. *If U is semisimple, then*

$$\mathrm{Rad}(U) = 0, \qquad \mathrm{Soc}(U) = U, \qquad \mathrm{Top}(U) = U.$$

In particular, if S is simple, then

$$\mathrm{Rad}(S) = 0, \qquad \mathrm{Soc}(S) = S, \qquad \mathrm{Top}(S) = S.$$

Example 5.3.4. *Let*

$$A = k[x]/(x^n),$$

and regard A as a left A-module. Then

$$J(A) = (x),$$

so

$$\mathrm{Rad}(A) = xA, \qquad \mathrm{Top}(A) = A/xA \cong k.$$

On the other hand,

$$\mathrm{Soc}(A) = \{a \in A \mid xa = 0\} = kx^{n-1}.$$

Thus

$$\mathrm{Top}(A) \cong k, \qquad \mathrm{Soc}(A) \cong k,$$

but they occur at opposite ends of the Loewy filtration.

Remark 5.3.5. *In general,* $\mathrm{Top}(U)$ *need not be semisimple for an arbitrary module U. However, if U is finitely generated and Artinian, then* $\mathrm{Top}(U)$ *is semisimple.*

Lemma 5.3.6. *Let*

$$U = S_1^{\oplus a_1} \oplus \cdots \oplus S_r^{\oplus a_r}$$

be a semisimple R-module, where the S_i are pairwise nonisomorphic simple modules.

Then the summand $S_i^{\oplus a_i}$ is the S_i-isotypic component of U, namely the sum of all submodules of U isomorphic to S_i. In particular, it is uniquely determined by U and the isomorphism class of S_i.

Proof. This is precisely Corollary 5.2.5(2) applied to $S = S_i$. $\square$

Lemma 5.3.7. *Let U be an R-module.*

1. *Suppose $M_1, \ldots, M_n$ are maximal submodules of U. Then there exists a subset $I \subseteq \{1, \ldots, n\}$ such that*

$$U/(M_1 \cap \cdots \cap M_n) \cong \bigoplus_{i \in I} U/M_i.$$

In particular, $U/(M_1 \cap \cdots \cap M_n)$ is semisimple.

2. *If U is finitely generated and Artinian, then $U/\operatorname{Rad}(U)$ is semisimple, and $\operatorname{Rad}(U)$ is the unique smallest submodule of U with semisimple quotient.*

Proof. **(1).** Choose a subset $I \subseteq \{1, \ldots, n\}$ maximal with the property that the diagonal map

$$\delta_I : U \longrightarrow \bigoplus_{i \in I} U/M_i, \qquad u \longmapsto (u + M_i)_{i \in I}$$

is surjective. Such a subset exists because $\{1, \ldots, n\}$ is finite; for instance $I = \varnothing$ is allowed.

We have

$$\ker(\delta_I) = \bigcap_{i \in I} M_i.$$

Put

$$K := \bigcap_{i \in I} M_i.$$

We claim that

$$K \subseteq M_j \qquad \text{for every } j \notin I.$$

Suppose not. Then for some $j \notin I$ we have

$$K \nsubseteq M_j.$$

Since M_j is maximal, this implies

$$K + M_j = U.$$

Hence the natural map

$$K \longrightarrow U/M_j$$

is surjective.

We now show that

$$U \longrightarrow \left(\bigoplus_{i \in I} U/M_i \right) \oplus U/M_j$$

is surjective. Indeed, given

$$\big((u_i + M_i)_{i\in I},\, u_j + M_j\big),$$

first use the surjectivity of δ_I to choose $u \in U$ such that

$$u + M_i = u_i + M_i \qquad \text{for all } i \in I.$$

Since $K \to U/M_j$ is surjective, choose $k \in K$ such that

$$k + M_j = (u_j - u) + M_j.$$

Then $u+k$ maps to the prescribed element in every component: for $i \in I$ we have $k \in K \subseteq M_i$, so

$$u + k + M_i = u + M_i = u_i + M_i,$$

and for the j-component,

$$u + k + M_j = u_j + M_j.$$

This contradicts the maximality of I.

Therefore

$$\bigcap_{i\in I} M_i \subseteq M_j \qquad \text{for every } j \notin I.$$

Hence

$$\bigcap_{i\in I} M_i = M_1 \cap \cdots \cap M_n.$$

By the first isomorphism theorem,

$$U/(M_1 \cap \cdots \cap M_n) = U/\ker(\delta_I) \cong \bigoplus_{i\in I} U/M_i.$$

Since each M_i is maximal, each quotient U/M_i is simple. Therefore the right hand side is semisimple.

(2). If $U = 0$, the statement is trivial. Assume $U \neq 0$.

Since U is finitely generated, every proper submodule of U is contained in a maximal submodule. Indeed, if $W \subsetneq U$, then U/W is finitely generated, so by the maximal-submodule lemma there exists a maximal submodule M of U such that

$$W \subseteq M \subsetneq U.$$

We now show that the intersection of all maximal submodules is in fact a finite intersection. Consider the set

$$\mathcal{F} := \{M_1 \cap \cdots \cap M_t \mid t \geq 1,\ M_1, \ldots, M_t \text{ maximal submodules of } U\}.$$

This is a nonempty set of submodules of U. Since U is Artinian, every nonempty set of submodules has a minimal element. Choose a minimal element

$$K = M_1 \cap \cdots \cap M_t$$

in $\mathcal{F}$.

Let M be any maximal submodule of U. Then

$$K \cap M = M_1 \cap \cdots \cap M_t \cap M$$

also belongs to $\mathcal{F}$, and

$$K \cap M \subseteq K.$$

By the minimality of K, we must have

$$K \cap M = K.$$

Hence

$$K \subseteq M.$$

Since this holds for every maximal submodule M of U, we get

$$K \subseteq \operatorname{Rad}(U).$$

The opposite inclusion is immediate because K is an intersection of maximal submodules. Therefore

$$\operatorname{Rad}(U) = K = M_1 \cap \cdots \cap M_t.$$

By part (1), it follows that

$$U/\operatorname{Rad}(U)$$

is semisimple.

It remains to prove the minimality statement. Let $V \leq U$ be a submodule such that U/V is semisimple. We show that

$$\operatorname{Rad}(U) \subseteq V.$$

Since U/V is semisimple, its radical is zero:

$$\operatorname{Rad}(U/V) = 0.$$

Maximal submodules of U/V correspond exactly to maximal submodules of U containing V. Therefore

$$\operatorname{Rad}(U/V) = \bigcap_{\substack{M \text{ maximal in } U \\ V \subseteq M}} M/V.$$

Thus

$$0 = \operatorname{Rad}(U/V) = \left(\bigcap_{\substack{M \text{ maximal in } U \\ V \subseteq M}} M \right) /V.$$

Hence
$$\bigcap_{\substack{M \text{ maximal in } U\\ V\subseteq M}} M = V.$$

Since $\operatorname{Rad}(U)$ is the intersection of all maximal submodules of U, it is contained in the intersection over the smaller family of maximal submodules containing V. Therefore

$$\operatorname{Rad}(U) \subseteq V.$$

Hence $\operatorname{Rad}(U)$ is contained in every submodule V such that U/V is semisimple. Since we already proved that $U/\operatorname{Rad}(U)$ is semisimple, $\operatorname{Rad}(U)$ is the unique smallest submodule of U with semisimple quotient. □

Corollary 5.3.8 (Universal property of the top)**.** *Let U be a finitely generated Artinian R-module. Then $\operatorname{Top}(U)$ is the largest semisimple quotient of U in the following sense: if S is semisimple and*
$$f : U \to S$$
is an R-module homomorphism, then f factors uniquely through the canonical quotient map
$$\pi : U \to \operatorname{Top}(U).$$
Equivalently, there exists a unique homomorphism

$$\overline{f} : \operatorname{Top}(U) \to S$$

such that
$$f = \overline{f} \circ \pi.$$

Proof. Recall that
$$\operatorname{Top}(U) = U/\operatorname{Rad}(U),$$

and let
$$\pi : U \longrightarrow \operatorname{Top}(U)$$

be the canonical quotient map.

Let S be semisimple and let
$$f : U \longrightarrow S$$

be an R-module homomorphism. Since S is semisimple, every submodule of S is semisimple. Hence
$$\operatorname{Im}(f) \leq S$$

is semisimple.

By the first isomorphism theorem, we have

$$U/\ker(f) \cong \operatorname{Im}(f).$$

Therefore $U/\ker(f)$ is semisimple. Thus $\ker(f)$ is a submodule of U such that the quotient $U/\ker(f)$ is semisimple.

Since U is finitely generated and Artinian, $\operatorname{Rad}(U)$ is the unique smallest submodule of U with semisimple quotient. Hence

$$\operatorname{Rad}(U) \subseteq \ker(f).$$

Therefore, by the universal property of the quotient module, the map f factors through $U/\operatorname{Rad}(U)$. That is, there exists an R-module homomorphism

$$\overline{f} : \operatorname{Top}(U) \longrightarrow S$$

such that

$$f = \overline{f} \circ \pi.$$

Finally, since π is surjective, such a homomorphism $\overline{f}$ is unique. Indeed, if

$$g : \operatorname{Top}(U) \longrightarrow S$$

also satisfies

$$f = g \circ \pi,$$

then

$$\overline{f} \circ \pi = g \circ \pi.$$

Since π is surjective, it follows that

$$\overline{f} = g.$$

Hence f factors uniquely through π, and therefore $\operatorname{Top}(U)$ is the largest semisimple quotient of U. $\square$

Example 5.3.9. *Suppose U has finite length. Since $\operatorname{Top}(U)$ is semisimple, we may write*

$$\operatorname{Top}(U) \cong S_1^{\oplus a_1} \oplus \cdots \oplus S_r^{\oplus a_r},$$

where the S_i are pairwise nonisomorphic simple modules. The integers a_i measure how many copies of S_i occur in the top layer of U.

Lemma 5.3.10. *For all R-modules U and V,*

$$\operatorname{Rad}(U \oplus V) = \operatorname{Rad}(U) \oplus \operatorname{Rad}(V),$$

$$\operatorname{Soc}(U \oplus V) = \operatorname{Soc}(U) \oplus \operatorname{Soc}(V),$$

$$\operatorname{Top}(U \oplus V) \cong \operatorname{Top}(U) \oplus \operatorname{Top}(V).$$

Proof. We first prove the statement for radicals.

Let

$$p_U : U \oplus V \longrightarrow U, \qquad p_V : U \oplus V \longrightarrow V$$

be the canonical projections. We claim that

$$p_U\big(\mathrm{Rad}(U \oplus V)\big) \subseteq \mathrm{Rad}(U), \qquad p_V\big(\mathrm{Rad}(U \oplus V)\big) \subseteq \mathrm{Rad}(V).$$

Indeed, let $x \in \mathrm{Rad}(U \oplus V)$. Suppose, for contradiction, that $p_U(x) \notin \mathrm{Rad}(U)$. Then there exists a maximal submodule $M \subsetneq U$ such that

$$p_U(x) \notin M.$$

Hence

$$M \oplus V$$

is a maximal submodule of $U \oplus V$, because

$$(U \oplus V)/(M \oplus V) \cong U/M$$

is simple. But $x \notin M \oplus V$, contradicting $x \in \mathrm{Rad}(U \oplus V)$. Therefore

$$p_U(x) \in \mathrm{Rad}(U).$$

The proof for p_V is identical. Thus

$$\mathrm{Rad}(U \oplus V) \subseteq \mathrm{Rad}(U) \oplus \mathrm{Rad}(V).$$

Conversely, let

$$u \in \mathrm{Rad}(U), \qquad v \in \mathrm{Rad}(V).$$

We show that $(u, v) \in \mathrm{Rad}(U \oplus V)$. Let

$$L \subsetneq U \oplus V$$

be a maximal submodule. Then

$$(U \oplus V)/L$$

is simple. Let

$$q : U \oplus V \longrightarrow (U \oplus V)/L$$

be the quotient map, and let

$$i_U : U \longrightarrow U \oplus V, \qquad i_V : V \longrightarrow U \oplus V$$

be the canonical inclusions.

Since $(U \oplus V)/L$ is simple, the homomorphism

$$q \circ i_U : U \longrightarrow (U \oplus V)/L$$

is either zero or surjective. In either case, its kernel contains $\mathrm{Rad}(U)$. Hence

$$q(i_U(u)) = 0.$$

Similarly,

$$q(i_V(v)) = 0.$$

Therefore

$$q(u,v) = q(i_U(u) + i_V(v)) = q(i_U(u)) + q(i_V(v)) = 0.$$

Hence

$$(u,v) \in L.$$

Since this holds for every maximal submodule $L \subsetneq U \oplus V$, we get

$$(u,v) \in \mathrm{Rad}(U \oplus V).$$

Thus

$$\mathrm{Rad}(U) \oplus \mathrm{Rad}(V) \subseteq \mathrm{Rad}(U \oplus V).$$

Combining the two inclusions gives

$$\mathrm{Rad}(U \oplus V) = \mathrm{Rad}(U) \oplus \mathrm{Rad}(V).$$

Next we prove the statement for socles.

Since $\mathrm{Soc}(U)$ and $\mathrm{Soc}(V)$ are semisimple submodules of U and V, respectively, their direct sum

$$\mathrm{Soc}(U) \oplus \mathrm{Soc}(V)$$

is a semisimple submodule of $U \oplus V$. Hence

$$\mathrm{Soc}(U) \oplus \mathrm{Soc}(V) \subseteq \mathrm{Soc}(U \oplus V).$$

Conversely, let S be a simple submodule of $U \oplus V$. Consider the restrictions of the two projections:

$$p_U|_S : S \longrightarrow U, \qquad p_V|_S : S \longrightarrow V.$$

Since S is simple, the images

$$p_U(S) \leq U, \qquad p_V(S) \leq V$$

are either zero or simple submodules. Hence

$$p_U(S) \subseteq \mathrm{Soc}(U), \qquad p_V(S) \subseteq \mathrm{Soc}(V).$$

For every $s \in S$, we have

$$s = (p_U(s), p_V(s)).$$

Therefore

$$s \in \mathrm{Soc}(U) \oplus \mathrm{Soc}(V),$$

and so

$$S \subseteq \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Since $\mathrm{Soc}(U \oplus V)$ is the sum of all simple submodules of $U \oplus V$, it follows that

$$\mathrm{Soc}(U \oplus V) \subseteq \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Hence

$$\mathrm{Soc}(U \oplus V) = \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Finally, by definition,

$$\mathrm{Top}(X) = X/\operatorname{Rad}(X)$$

for any R-module X. Therefore, using the radical formula proved above, we get

$$\begin{aligned}\mathrm{Top}(U \oplus V) &= (U \oplus V)/\operatorname{Rad}(U \oplus V)\\ &= (U \oplus V)/\big(\operatorname{Rad}(U) \oplus \operatorname{Rad}(V)\big).\end{aligned}$$

The canonical quotient isomorphism gives

$$(U \oplus V)/\big(\operatorname{Rad}(U) \oplus \operatorname{Rad}(V)\big) \cong U/\operatorname{Rad}(U) \oplus V/\operatorname{Rad}(V).$$

Hence

$$\mathrm{Top}(U \oplus V) \cong \mathrm{Top}(U) \oplus \mathrm{Top}(V).$$

□

Proposition 5.3.11. *Let*

$$f : P \to U$$

be a projective cover, and assume that P and U are finitely generated Artinian. Then the induced morphism

$$\overline{f} : \mathrm{Top}(P) \to \mathrm{Top}(U)$$

is an isomorphism.

Proof. Let

$$\pi_P : P \longrightarrow \mathrm{Top}(P) = P/\operatorname{Rad}(P)$$

and

$$\pi_U : U \longrightarrow \mathrm{Top}(U) = U/\operatorname{Rad}(U)$$

be the canonical quotient maps.

Since P and U are finitely generated, we have

$$f(\operatorname{Rad}(P)) \subseteq \operatorname{Rad}(U).$$

Hence f induces a well-defined homomorphism

$$\overline{f} : \mathrm{Top}(P) \longrightarrow \mathrm{Top}(U)$$

such that

$$\overline{f} \circ \pi_P = \pi_U \circ f.$$

Since $f : P \to U$ is a projective cover, it is in particular an epimorphism. Therefore the induced map

$$\overline{f} : \mathrm{Top}(P) \to \mathrm{Top}(U)$$

is also an epimorphism.

It remains to prove that $\overline{f}$ is injective. Suppose, for contradiction, that

$$K := \ker(\overline{f}) \neq 0.$$

Since P is finitely generated and Artinian, $\mathrm{Top}(P)$ is semisimple. Therefore K is a direct summand of $\mathrm{Top}(P)$. Hence there exists a submodule $C \leq \mathrm{Top}(P)$ such that

$$\mathrm{Top}(P) = K \oplus C.$$

Since $K \neq 0$, we have $C \subsetneq \mathrm{Top}(P)$.

Let

$$W := \pi_P^{-1}(C).$$

Then W is a proper submodule of P, because

$$\pi_P(W) = C \subsetneq \mathrm{Top}(P).$$

We claim that $f(W) = U$. Since $\overline{f}$ is surjective and

$$\mathrm{Top}(P) = K \oplus C,$$

while $K = \ker(\overline{f})$, the restriction

$$\overline{f}|_C : C \longrightarrow \mathrm{Top}(U)$$

is still surjective. Hence

$$\overline{f}(C) = \mathrm{Top}(U).$$

Using the commutative relation

$$\overline{f} \circ \pi_P = \pi_U \circ f,$$

we get

$$\pi_U(f(W)) = \overline{f}(\pi_P(W)) = \overline{f}(C) = \mathrm{Top}(U).$$

Therefore

$$f(W) + \mathrm{Rad}(U) = U.$$

Since U is finitely generated, Nakayama's lemma gives

$$f(W) = U.$$

But $f : P \to U$ is an essential epimorphism, and $W \subseteq P$ is a submodule satisfying $f(W) = U$. Hence

$$W = P,$$

contradicting the fact that $W \subsetneq P$.

Therefore

$$\ker(\overline{f}) = 0.$$

Thus $\overline{f}$ is both injective and surjective, hence an isomorphism:

$$\mathrm{Top}(P) \cong \mathrm{Top}(U).$$

□

Proposition 5.3.12. *Suppose that*

$$f : P \to U$$

is a projective cover of an R-module U, and that

$$g : Q \to U$$

is an epimorphism with Q projective.

1. *There exists a decomposition*

$$Q = Q_1 \oplus Q_2$$

such that, with respect to this decomposition,

$$g = (g_1, 0),$$

and there is an isomorphism $\gamma : Q_1 \to P$ satisfying

$$f \circ \gamma = g_1.$$

2. *If a projective cover of U exists, then it is unique up to isomorphism.*

Proof. Since Q is projective and $f : P \to U$ is surjective, there exists

$$\alpha : Q \to P$$

such that

$$f \circ \alpha = g.$$

Since P is projective and $g : Q \to U$ is surjective, there exists

$$\beta : P \to Q$$

such that

$$g \circ \beta = f.$$

Consequently,

$$f = f \circ (\alpha\beta).$$

Because f is an essential epimorphism, the endomorphism $\alpha\beta$ of P must be surjective. Since P is projective, $\alpha\beta$ splits. Let

$$P = \ker(\alpha\beta) \oplus P'.$$

If $\ker(\alpha\beta) \neq 0$, then P' is a proper submodule of P and

$$f(P') = f(\alpha\beta(P')) = f(P) = U,$$

contradicting the essentiality of f. Hence $\ker(\alpha\beta) = 0$, so $\alpha\beta$ is an automorphism.

Replacing β by $\beta \circ (\alpha\beta)^{-1}$, we may assume that

$$\alpha\beta = \mathrm{id}_P .$$

Thus β is a split monomorphism and

$$Q = \beta(P) \oplus \ker(\alpha).$$

Set

$$Q_1 := \beta(P), \qquad Q_2 := \ker(\alpha),$$

and let $\gamma : Q_1 \to P$ be the inverse of the isomorphism $\beta : P \to Q_1$. Then

$$g|_{Q_1} = f \circ \gamma.$$

Moreover,

$$g(Q_2) = f(\alpha(Q_2)) = 0,$$

so $g = (g_1, 0)$ with $g_1 = f \circ \gamma$. This proves (1).

For (2), if

$$f : P \to U \qquad \text{and} \qquad f' : P' \to U$$

are projective covers, apply (1) twice: once with $Q = P'$ and once with $Q = P$. We deduce that P is isomorphic to a direct summand of P' and P' is isomorphic to a direct summand of P. The decompositions obtained in (1) force the complementary summands to map trivially to U, hence to vanish by essentiality. Therefore

$$P \cong P',$$

so the projective cover is unique up to isomorphism. $\square$

Chapter 6

Lecture 6

6.1 Projective covers, radicals, and finiteness conditions

Corollary 6.1.1. *For finitely generated projective modules P and Q,*

$$P \cong Q \iff \operatorname{Top}(P) \cong \operatorname{Top}(Q).$$

Proof. Let

$$T := P/\operatorname{Rad}(P).$$

By Nakayama lemma, the quotient map $P \to T$ is an essential epimorphism, so P is the projective cover of T. Similarly $Q \to Q/\operatorname{Rad}(Q)$ is a projective cover. Then apply uniqueness of projective covers from the previous lecture. □

Definition 6.1.2. *A unital ring R is called left Noetherian if ${}_RR$ is Noetherian as a left R-module.*

Remark 6.1.3. *Similar definitions are given for right Noetherian and for left/right Artinian rings.*

Remark 6.1.4. *one-sided Noetherian / Artinian conditions generally do not imply the opposite-sided versions*

Example 6.1.5. *There exists a ring which is left Noetherian but not right Noetherian.*

Let

$$R = \left\{ \begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \,\middle|\, a, b \in \mathbb{Q},\ n \in \mathbb{Z} \right\}.$$

Then R is left Noetherian, but not right Noetherian.

Proof. Let

$$e_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad e_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then, as left R-modules,

$${}_RR = Re_{11} \oplus Re_{22}.$$

We first show that both Re_{11} and Re_{22} are Noetherian as left R-modules.

For Re_{11}, we have
$$Re_{11} = \left\{ \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix} \;\middle|\; q \in \mathbb{Q} \right\}.$$
This is a simple left R-module. Indeed, if
$$0 \neq \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix} \in Re_{11},$$
then for any $x \in \mathbb{Q}$,
$$\begin{pmatrix} xq^{-1} & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix}.$$
Hence every nonzero submodule of Re_{11} is all of Re_{11}, so Re_{11} is simple, in particular Noetherian.

Now consider
$$Re_{22} = \left\{ \begin{pmatrix} 0 & q \\ 0 & n \end{pmatrix} \;\middle|\; q \in \mathbb{Q},\ n \in \mathbb{Z} \right\}.$$
Define
$$\pi : Re_{22} \longrightarrow \mathbb{Z}, \qquad \pi\left(\begin{pmatrix} 0 & q \\ 0 & n \end{pmatrix} \right) = n.$$
We regard $\mathbb{Z}$ as a left R-module via
$$\begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \cdot m = nm.$$
Then π is a surjective R-homomorphism. Its kernel is
$$\ker \pi = \left\{ \begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \;\middle|\; q \in \mathbb{Q} \right\}.$$
This is again a simple left R-module, because
$$\begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & aq \\ 0 & 0 \end{pmatrix},$$
so the action is through multiplication by elements of $\mathbb{Q}$. Moreover, ${}_R\mathbb{Z}$ is Noetherian, since its submodules are exactly the ideals of $\mathbb{Z}$, and $\mathbb{Z}$ is Noetherian.

Therefore Re_{22} is an extension of two Noetherian left R-modules, hence is Noetherian. It follows that
$${}_RR = Re_{11} \oplus Re_{22}$$
is a Noetherian left R-module. Thus R is left Noetherian.

Next we show that R is not right Noetherian.

For each integer $k \geq 0$, define

$$I_k = \left\{ \begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \,\middle|\, q \in 2^{-k}\mathbb{Z} \right\}.$$

We claim that each I_k is a right ideal of R. Indeed, if

$$\begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \in I_k, \qquad \begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \in R,$$

then

$$\begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & n \end{pmatrix} = \begin{pmatrix} 0 & qn \\ 0 & 0 \end{pmatrix}.$$

Since $q \in 2^{-k}\mathbb{Z}$ and $n \in \mathbb{Z}$, we have $qn \in 2^{-k}\mathbb{Z}$, so the product again lies in I_k.

Moreover,

$$I_0 \subsetneq I_1 \subsetneq I_2 \subsetneq \cdots$$

is a strictly ascending chain of right ideals, because

$$2^{-(k+1)} \in 2^{-(k+1)}\mathbb{Z} \quad \text{but} \quad 2^{-(k+1)} \notin 2^{-k}\mathbb{Z}.$$

Hence R does not satisfy the ascending chain condition on right ideals.

Therefore R is not right Noetherian. □

Lemma 6.1.6. *Let R be a left Artinian ring. Then its Jacobson radical $J(R)$ is nilpotent.*

Proof. Set

$$J := J(R).$$

Since R is left Artinian, the descending chain of left ideals

$$J \supseteq J^2 \supseteq J^3 \supseteq \cdots$$

stabilizes. Hence there exists some $n \geq 1$ such that

$$J^n = J^{n+1}.$$

Let

$$N := J^n.$$

We claim that $N = 0$.

Suppose, for contradiction, that $N \neq 0$. Then

$$N^2 = J^{2n} = J^n = N,$$

since the chain has stabilized.

Now consider the set

$$\Sigma = \{ I \leq {}_R R \mid NI \neq 0 \}$$

of left ideals of R. This set is nonempty because

$$N^2 = N \neq 0,$$

so $N \in \Sigma$.

Since R is left Artinian, the set of left ideals satisfies the descending chain condition, so Σ contains a minimal element. Let $I_0 \in \Sigma$ be such a minimal left ideal.

Because $NI_0 \neq 0$, there exists some $a \in I_0$ such that

$$Na \neq 0.$$

Then Na is a left ideal of R contained in I_0, and moreover

$$N(Na) = N^2 a = Na \neq 0.$$

Thus $Na \in \Sigma$. By the minimality of I_0, we must have

$$Na = I_0.$$

Since $a \in I_0 = Na$, there exists $b \in N$ such that

$$a = ba.$$

Therefore

$$(1 - b)a = 0.$$

But $b \in N \subseteq J(R)$, and for every element of the Jacobson radical, $1 - b$ has a left inverse. Hence there exists $c \in R$ such that

$$c(1 - b) = 1.$$

Multiplying the equation $(1 - b)a = 0$ on the left by c, we get

$$a = c(1 - b)a = 0,$$

contradicting the choice of a with $Na \neq 0$.

This contradiction shows that $N = 0$. Hence

$$J^n = 0,$$

so $J(R)$ is nilpotent. □

Theorem 6.1.7 (Levitzki). *Every left Artinian ring is left Noetherian.*

Proof. Let R be a left Artinian ring, and let $J = J(R)$ be its Jacobson radical.

Since R is left Artinian, the quotient ring R/J is also left Artinian. Moreover, $J(R/J) = 0$, hence R/J is semisimple. Therefore every left R/J-module is semisimple.

Also, since R is left Artinian, J is nilpotent. Thus there exists $n \geq 1$ such that

$$J^n = 0.$$

Consider the finite chain of left ideals

$$0 = J^n \subseteq J^{n-1} \subseteq \cdots \subseteq J \subseteq R.$$

For each $i = 0, 1, \ldots, n-1$, the factor module J^i/J^{i+1} is naturally a left R/J-module, because

$$J \cdot J^i \subseteq J^{i+1}.$$

Hence J^i/J^{i+1} is semisimple.

On the other hand, J^i/J^{i+1} is Artinian, being a quotient of the Artinian left R-module ${}_RR$. A semisimple Artinian module must be a finite direct sum of simple modules, so it is Noetherian. Therefore each factor

$$J^i/J^{i+1}$$

is Noetherian.

Now we argue by descending induction on i. Since $J^n = 0$ is Noetherian, and for each $i = 0, 1, \ldots, n-1$ we have a short exact sequence

$$0 \longrightarrow J^{i+1} \longrightarrow J^i \longrightarrow J^i/J^{i+1} \longrightarrow 0,$$

it follows that if J^{i+1} is Noetherian, then so is J^i. Hence

$$J^{n-1},\ J^{n-2},\ \ldots,\ J,\ R$$

are all Noetherian as left R-modules.

In particular, ${}_RR$ is Noetherian. Therefore R is a left Noetherian ring. □

Theorem 6.1.8. *If R is left Noetherian/Artinian, then every finitely generated left R-module is Noetherian/Artinian.*

Proof. Prove for Noetherian case: If M is generated by m elements, then $M = Rm_1 + \ldots + Rm_n$, each Rm_i is a homomorphic image of ${}_RR$. Hence Rm_i is Noetherian. Finite sums of Noetherian modules are Noetherian, so M is Noetherian. □

6.2 Artin-Wedderburn

Theorem 6.2.1. *Let*

$$ {}_RV = V_1 \oplus \cdots \oplus V_n $$

be a direct sum of R-modules, and suppose all V_i are isomorphic to each other. Then

$$ \mathrm{End}_R({}_RV) \cong M_n\big(\mathrm{End}_R(V_i)\big). $$

Proof. For each i, let

$$ \iota_i : V_i \hookrightarrow V \qquad \text{and} \qquad \pi_i : V \to V_i $$

be the canonical inclusion and projection. Fix isomorphisms

$$ \theta_i : V_1 \xrightarrow{\sim} V_i \qquad (i = 1, \dots, n). $$

For $\sigma \in \mathrm{End}_R(V)$, define

$$ \varphi_{ij}(\sigma) := \theta_i^{-1}\pi_i\,\sigma\,\iota_j\theta_j \in \mathrm{End}_R(V_1). $$

Thus we obtain a map

$$ \Phi : \mathrm{End}_R(V) \longrightarrow M_n(E_1), \qquad \sigma \longmapsto \big(\varphi_{ij}(\sigma)\big), $$

where

$$ E_1 := \mathrm{End}_R(V_1). $$

We first check that Φ is a ring homomorphism.

Additivity is immediate. For multiplicativity, let $\sigma, \beta \in \mathrm{End}_R(V)$. Since

$$ 1_V = \sum_{j=1}^{n} \iota_j\pi_j, $$

we have

$$ \begin{aligned} \varphi_{ik}(\beta\sigma) &= \theta_i^{-1}\pi_i\,\beta\sigma\,\iota_k\theta_k \\ &= \theta_i^{-1}\pi_i\,\beta\left(\sum_{j=1}^{n}\iota_j\pi_j\right)\sigma\,\iota_k\theta_k \\ &= \sum_{j=1}^{n}\theta_i^{-1}\pi_i\,\beta\,\iota_j\pi_j\,\sigma\,\iota_k\theta_k \\ &= \sum_{j=1}^{n}\big(\theta_i^{-1}\pi_i\,\beta\,\iota_j\theta_j\big)\big(\theta_j^{-1}\pi_j\,\sigma\,\iota_k\theta_k\big) \\ &= \sum_{j=1}^{n}\varphi_{ij}(\beta)\varphi_{jk}(\sigma). \end{aligned} $$

Hence
$$\Phi(\beta\sigma) = \Phi(\beta)\Phi(\sigma).$$
So Φ is a ring homomorphism.

Next we show that Φ is injective. Suppose $\Phi(\sigma) = 0$. Then
$$\varphi_{ij}(\sigma) = 0 \qquad \text{for all } i, j,$$
so
$$\theta_i^{-1}\pi_i\,\sigma\,\iota_j\theta_j = 0 \qquad \text{for all } i, j.$$
Since each θ_i and θ_j is an isomorphism, this implies
$$\pi_i\,\sigma\,\iota_j = 0 \qquad \text{for all } i, j.$$
Therefore $\pi_i(\sigma(v)) = 0$ for every $v \in V_j$ and every i, j. Hence
$$\sigma(V_j) = 0 \qquad \text{for all } j.$$
Because
$$V = V_1 \oplus \cdots \oplus V_n,$$
it follows that $\sigma = 0$. Thus Φ is injective.

Finally we prove surjectivity. Let
$$(a_{ij}) \in M_n(E_1).$$
Define
$$\sigma := \sum_{i,j=1}^{n} \iota_i\theta_i\,a_{ij}\,\theta_j^{-1}\pi_j \in \mathrm{End}_R(V).$$
Then for each i, j,
$$\begin{aligned}\varphi_{ij}(\sigma) &= \theta_i^{-1}\pi_i\left(\sum_{r,s=1}^{n} \iota_r\theta_r\,a_{rs}\,\theta_s^{-1}\pi_s\right)\iota_j\theta_j \\ &= \sum_{r,s=1}^{n} \theta_i^{-1}(\pi_i\iota_r)\theta_r\,a_{rs}\,\theta_s^{-1}(\pi_s\iota_j)\theta_j.\end{aligned}$$
Now
$$\pi_i\iota_r = \begin{cases} 1_{V_i}, & i = r, \\ 0, & i \neq r, \end{cases} \qquad \pi_s\iota_j = \begin{cases} 1_{V_j}, & s = j, \\ 0, & s \neq j, \end{cases}$$
so all terms vanish except the one with $r = i$ and $s = j$. Hence
$$\varphi_{ij}(\sigma) = \theta_i^{-1}\theta_i\,a_{ij}\,\theta_j^{-1}\theta_j = a_{ij}.$$
Therefore
$$\Phi(\sigma) = (a_{ij}),$$

and Φ is surjective.

Thus Φ is a ring isomorphism, and we conclude that

$$\mathrm{End}_R(V) \cong M_n\big(\mathrm{End}_R(V_1)\big).$$

□

Definition 6.2.2. *A ring R is called semisimple if ${}_RR$ is semisimple as a left R-module.*

Lemma 6.2.3. *For a unital ring R, the following are equivalent: (1) R is semisimple. (2) Every left R-module is semisimple.*

Proof. If ${}_RR$ is semisimple, then every free module is semisimple, and every module is a quotient of a free module. Use closure properties of semisimple modules to deduce that every left R-module is semisimple. The converse is immediate by applying the statement to ${}_RR$ itself. □

Lemma 6.2.4. *For an R-module ${}_RM$, there is a standard identification*

$$\mathrm{Hom}_R({}_RR_R, {}_RM) \cong {}_RM, \qquad f \longmapsto f(1).$$

Proof. Define

$$\Phi : \mathrm{Hom}_R({}_RR_R, {}_RM) \longrightarrow {}_RM, \qquad \Phi(f) = f(1).$$

We first show that Φ is bijective.

For any $f \in \mathrm{Hom}_R({}_RR_R, {}_RM)$ and any $r \in R$, since f is left R-linear, we have

$$f(r) = f(r \cdot 1) = r\, f(1).$$

Thus f is completely determined by the element $f(1) \in M$.

Conversely, for each $m \in M$, define

$$\Psi(m) : {}_RR_R \longrightarrow {}_RM, \qquad \Psi(m)(r) = rm \quad (r \in R).$$

Then $\Psi(m)$ is left R-linear, because for all $a, r \in R$,

$$\Psi(m)(ar) = (ar)m = a(rm) = a\,\Psi(m)(r).$$

Hence $\Psi(m) \in \mathrm{Hom}_R({}_RR_R, {}_RM)$.

Now for $m \in M$,

$$(\Phi \circ \Psi)(m) = \Phi(\Psi(m)) = \Psi(m)(1) = 1m = m.$$

And for $f \in \mathrm{Hom}_R({}_RR_R, {}_RM)$ and $r \in R$,

$$(\Psi \circ \Phi)(f)(r) = \Psi(f(1))(r) = r\, f(1) = f(r).$$

So $\Psi \circ \Phi = \mathrm{id}$ and $\Phi \circ \Psi = \mathrm{id}$, hence Φ is a bijection.

Finally, we check that this identification is R-linear. Recall that $\mathrm{Hom}_R({}_R R_R, {}_R M)$ carries the left R-module structure

$$(a \cdot f)(r) = f(ra) \qquad (a, r \in R).$$

Then for $a \in R$ and $f \in \mathrm{Hom}_R({}_R R_R, {}_R M)$,

$$\Phi(a \cdot f) = (a \cdot f)(1) = f(1a) = f(a) = a\, f(1) = a\, \Phi(f).$$

Therefore Φ is an isomorphism of left R-modules:

$$\mathrm{Hom}_R({}_R R_R, {}_R M) \cong {}_R M.$$

$\square$

Corollary 6.2.5. *For any ring R, there is a ring isomorphism*

$$\mathrm{End}_R({}_R R) \cong R^{\mathrm{op}}, \qquad f \longmapsto f(1).$$

Proof. Consider the map

$$\Phi : \mathrm{End}_R({}_R R) \longrightarrow R, \qquad \Phi(f) = f(1).$$

By the preceding lemma with $M = R$, this map is a bijection of sets.

We now check the ring structure. Let $f_1, f_2 \in \mathrm{End}_R({}_R R)$. Then

$$\Phi(f_1 \circ f_2) = (f_1 \circ f_2)(1) = f_1\big(f_2(1)\big).$$

Since f_1 is left R-linear, for any $r \in R$ we have

$$f_1(r) = f_1(r \cdot 1) = r\, f_1(1).$$

Hence

$$f_1\big(f_2(1)\big) = f_2(1)\, f_1(1).$$

Therefore

$$\Phi(f_1 \circ f_2) = \Phi(f_2)\Phi(f_1),$$

where the product on the right is the multiplication in R. This means exactly that

$$\Phi(f_1 \circ f_2) = \Phi(f_1) \cdot_{\mathrm{op}} \Phi(f_2)$$

when the codomain is viewed as R^{op}.

Thus Φ is a ring isomorphism

$$\mathrm{End}_R({}_R R) \cong R^{\mathrm{op}}.$$

$\square$

Lemma 6.2.6. *For any ring R and any $n \geq 1$, there is a ring isomorphism*

$$M_n(R)^{\mathrm{op}} \cong M_n\big(R^{\mathrm{op}}\big), \qquad A \longmapsto A^\top.$$

Proof. Define

$$\Psi : M_n(R)^{\mathrm{op}} \longrightarrow M_n\big(R^{\mathrm{op}}\big), \qquad \Psi(A) = A^\top.$$

This map is clearly additive and bijective, since transpose is its own inverse.

It remains to check multiplicativity. Let

$$A = (a_{ij}), \qquad B = (b_{ij}) \in M_n(R).$$

Recall that multiplication in $M_n(R)^{\mathrm{op}}$ is reversed, so

$$A \cdot_{\mathrm{op}} B = BA$$

with the usual matrix product on the right-hand side.

Now fix i, j. Then

$$\big(\Psi(A \cdot_{\mathrm{op}} B)\big)_{ij} = \big((BA)^\top\big)_{ij} = (BA)_{ji} = \sum_{k=1}^{n} b_{jk} a_{ki}.$$

On the other hand, in the ring R^{op} the product is reversed, so

$$\big(\Psi(A)\Psi(B)\big)_{ij} = \sum_{k=1}^{n} (A^\top)_{ik} \cdot_{R^{\mathrm{op}}} (B^\top)_{kj} = \sum_{k=1}^{n} a_{ki} \cdot_{R^{\mathrm{op}}} b_{jk} = \sum_{k=1}^{n} b_{jk} a_{ki}.$$

Thus

$$\Psi(A \cdot_{\mathrm{op}} B) = \Psi(A)\Psi(B).$$

So Ψ is a ring homomorphism, hence a ring isomorphism.

Therefore

$$M_n(R)^{\mathrm{op}} \cong M_n\big(R^{\mathrm{op}}\big).$$

$\square$

Theorem 6.2.7. *(Artin-Wedderburn) Let A be a semisimple Artinian ring. Then*

$$A \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r),$$

where each D_i is a division ring. More concretely, if

$${}_A A \cong S_1^{\oplus n_1} \oplus \cdots \oplus S_r^{\oplus n_r}$$

with pairwise nonisomorphic simple A-modules S_i, then

$$D_i = \mathrm{End}_A(S_i)^{op}.$$

Proof. Decompose $\mathrm{End}_A(A)$ using the decomposition of the regular module. Use

$$\mathrm{Hom}_A(S_i, S_j) = 0 \quad (i \neq j)$$

and Schur's lemma to identify the diagonal blocks with division rings. Then use the previous matrix-ring result to rewrite

$$\mathrm{End}_A\big(S_i^{\oplus n_i}\big)$$

as

$$M_{n_i}\big(\mathrm{End}_A(S_i)\big) = M_{n_i}(D_i^{op}).$$

Therefore

$$A \cong \mathrm{End}_A(A)^{op} \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r).$$

□

Definition 6.2.8. *Let R be a commutative ring with 1, and let A be an associative ring with 1. An R-algebra is an R-module A whose multiplication is compatible with the R-action. Equivalently, there is a ring homomorphism*

$$R \to A$$

whose image lies in the center $Z(A)$.

Remark 6.2.9. *If $R = k$ is a field and A is finite-dimensional over k, then A is called a finite-dimensional k-algebra. Such an algebra is Noetherian and Artinian as a ring/module object.*

Example 6.2.10. *Let R be a commutative ring with 1. The following are R-algebras:*

1. *$R[x_1, \ldots, x_n]$.*
2. *$M_n(R)$.*
3. *Let G be a finite group. The group algebra*

$$RG = \left\{ \sum_{g \in G} a_g g \;\middle|\; a_g \in R \right\}$$

is an R-algebra, with multiplication defined by

$$\left(\sum_{g \in G} a_g g\right)\left(\sum_{h \in G} b_h h\right) = \sum_{r \in G}\left(\sum_{\substack{g,h \in G \\ gh = r}} a_g b_h\right) r.$$

Lemma 6.2.11. *Let A be a k-algebra, where k is a field, and let S be a simple A-module. Then $\operatorname{End}_A(S)$ is a division ring, and it is naturally a k-algebra.*

Proof. We first show that $\operatorname{End}_A(S)$ is a division ring.

Let
$$0 \neq f \in \operatorname{End}_A(S).$$

Since f is A-linear, both $\ker(f)$ and $\operatorname{im}(f)$ are A-submodules of S. Because S is simple and $f \neq 0$, we have $\operatorname{im}(f) \neq 0$, hence
$$\operatorname{im}(f) = S.$$

Also $\ker(f) \neq S$ since $f \neq 0$, so by simplicity,
$$\ker(f) = 0.$$

Thus f is bijective. Since the inverse of a bijective A-linear map is again A-linear, $f^{-1} \in \operatorname{End}_A(S)$. Therefore every nonzero element of $\operatorname{End}_A(S)$ is invertible, so $\operatorname{End}_A(S)$ is a division ring.

Next we show that $\operatorname{End}_A(S)$ is naturally a k-algebra.

Because A is a k-algebra, S becomes a k-vector space via
$$\lambda \cdot s := (\lambda 1_A)s \qquad (\lambda \in k,\ s \in S).$$

Now let $f \in \operatorname{End}_A(S)$. Then for $\lambda \in k$ and $s \in S$,
$$f(\lambda s) = f\big((\lambda 1_A)s\big) = (\lambda 1_A)f(s) = \lambda f(s).$$

Hence every A-linear endomorphism of S is automatically k-linear. So
$$\operatorname{End}_A(S) \subseteq \operatorname{End}_k(S)$$

as a k-subspace. Equivalently, there is a k-algebra homomorphism
$$k \longrightarrow \operatorname{End}_A(S), \qquad \lambda \longmapsto \lambda \operatorname{id}_S,$$

whose image lies in the center of $\operatorname{End}_A(S)$. Therefore $\operatorname{End}_A(S)$ is naturally a k-algebra. $\square$

Lemma 6.2.12. *If A is finite-dimensional over k, then $\operatorname{End}_A(S)$ is finite-dimensional over k.*

Proof. Choose $0 \neq s \in S$. Since S is simple, the cyclic submodule As is nonzero, hence
$$As = S.$$

Consider the map
$$\varphi : A \longrightarrow S, \qquad a \longmapsto as.$$

This is a surjective k-linear map. Therefore

$$\dim_k S \leq \dim_k A < \infty.$$

So S is finite-dimensional over k.

Since every A-endomorphism of S is k-linear, we have

$$\operatorname{End}_A(S) \subseteq \operatorname{End}_k(S)$$

as k-vector spaces. Hence

$$\dim_k \operatorname{End}_A(S) \leq \dim_k \operatorname{End}_k(S) = (\dim_k S)^2 < \infty.$$

Thus $\operatorname{End}_A(S)$ is finite-dimensional over k. $\square$

Lemma 6.2.13. *If A is finite-dimensional over k and k is algebraically closed, then*

$$\operatorname{End}_A(S) = k.$$

Proof. By the previous lemma, S is finite-dimensional over k, and $\operatorname{End}_A(S)$ is a finite-dimensional k-algebra.

Let

$$f \in \operatorname{End}_A(S).$$

Since S is a finite-dimensional k-vector space and k is algebraically closed, the k-linear operator $f : S \to S$ has an eigenvalue $\lambda \in k$. Hence

$$f - \lambda \operatorname{id}_S$$

has a nonzero kernel, so it is not injective.

But $f - \lambda \operatorname{id}_S$ is still an element of $\operatorname{End}_A(S)$. Since $\operatorname{End}_A(S)$ is a division ring by the first lemma, every nonzero element is invertible, hence injective. Therefore

$$f - \lambda \operatorname{id}_S = 0.$$

So

$$f = \lambda \operatorname{id}_S .$$

This shows that every A-endomorphism of S is scalar. Under the natural embedding

$$k \hookrightarrow \operatorname{End}_A(S), \qquad \lambda \mapsto \lambda \operatorname{id}_S,$$

we therefore obtain

$$\operatorname{End}_A(S) = k.$$

$\square$

Theorem 6.2.14. *If k is algebraically closed and A is a finite-dimensional semisimple k-algebra, then*

$$A \cong M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k).$$

Proof. Since A is finite-dimensional over k, it is Artinian. Hence A is a semisimple Artinian ring.

Decompose the left regular A-module into simple summands:

$$_AA \cong S_1^{\oplus n_1} \oplus \cdots \oplus S_r^{\oplus n_r},$$

Here $S_1, \ldots, S_r$ are pairwise non-isomorphic simple left A-modules. By the Artin–Wedderburn theorem,

$$A \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r),$$

where

$$D_i = \mathrm{End}_A(S_i)^{\mathrm{op}}.$$

It remains to identify the division rings D_i. Since A is a finite-dimensional k-algebra, each simple A-module S_i is finite-dimensional over k. Indeed, for any $0 \neq s \in S_i$, the map

$$A \longrightarrow S_i, \qquad a \longmapsto as$$

is a surjective k-linear map, because S_i is simple. Therefore

$$\dim_k S_i \leq \dim_k A < \infty.$$

Now $\mathrm{End}_A(S_i)$ is a division ring by Schur's lemma. Moreover, every A-endomorphism of S_i is k-linear. Since k is algebraically closed and S_i is finite-dimensional over k, every $f \in \mathrm{End}_A(S_i)$ has an eigenvalue $\lambda \in k$. Thus

$$f - \lambda\,\mathrm{id}_{S_i}$$

has nonzero kernel, so it is not injective. But $f - \lambda\,\mathrm{id}_{S_i} \in \mathrm{End}_A(S_i)$, and $\mathrm{End}_A(S_i)$ is a division ring. Hence a nonzero element of $\mathrm{End}_A(S_i)$ must be invertible, in particular injective. Therefore

$$f - \lambda\,\mathrm{id}_{S_i} = 0,$$

and so

$$f = \lambda\,\mathrm{id}_{S_i}.$$

This proves that

$$\mathrm{End}_A(S_i) = k.$$

Hence

$$D_i = \mathrm{End}_A(S_i)^{\mathrm{op}} \cong k^{\mathrm{op}} = k,$$

since k is commutative. Substituting this into the Artin–Wedderburn decomposition gives

$$A \cong M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k).$$

□

Corollary 6.2.15. *Let A be a finite-dimensional semisimple k-algebra, where k is a field.*

In any decomposition

$${}_A A \cong S_1^{n_1} \oplus \cdots \oplus S_r^{n_r},$$

where the S_i's are pairwise non-isomorphic simple A-modules, we have that $S_1, \ldots, S_r$ form a complete set of representatives of the isomorphism classes of simple A-modules.

If moreover $k = \overline{k}$, then $n_i = \dim_k S_i$ and

$$\dim_k A = n_1^2 + \cdots + n_r^2.$$

Proof. Let S be a simple module. Then

$$S \cong A/M$$

where M is a maximal submodule.

By semisimplicity,

$${}_A A = M \oplus S.$$

Hence all simple submodules appear in $\{S_1, \ldots, S_r\}$.

Since $k = \overline{k}$, we have

$$A \cong M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k).$$

Moreover,

$$M_{n_i}(k) = \underbrace{\left\{ \begin{pmatrix} * & \cdots & 0 \\ \vdots & \ddots & \vdots \\ * & \cdots & 0 \end{pmatrix} \right\} \oplus \cdots \oplus \left\{ \begin{pmatrix} 0 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & * \end{pmatrix} \right\}}_{\text{each of them is a simple } M_{n_i}(k)\text{-module.}}$$

□

6.3 k-algebras, indecomposables, and idempotents

Definition 6.3.1. *A nonzero R-module M is called indecomposable if every decomposition*

$$M = M_1 \oplus M_2$$

forces $M_1 = 0$ or $M_2 = 0$.

Theorem 6.3.2 (Krull-Schmidt)**.** *Let M be an R-module with a composition series. Suppose*

$$M \cong V_1 \oplus \cdots \oplus V_m \cong U_1 \oplus \cdots \oplus U_n$$

where all V_i and U_j are indecomposable. Then

$$m = n,$$

and after reordering the summands one has

$$V_i \cong U_i \quad \textit{for all } i.$$

Proof. Replacing the given isomorphisms by identifications, we may assume

$$M = V_1 \oplus \cdots \oplus V_m = U_1 \oplus \cdots \oplus U_n.$$

If $M = 0$, the statement is trivial. So assume $M \neq 0$.

Since M has a composition series, it has finite length. Hence each direct summand V_i and U_j also has finite length. Moreover, for every indecomposable module X of finite length, Fitting's lemma implies that every endomorphism of X is either invertible or nilpotent; therefore $\mathrm{End}_R(X)$ is a local ring.

We prove the theorem by induction on the length $\ell(M)$ of M. If $\ell(M) = 1$, then M is simple, hence indecomposable, so necessarily $m = n = 1$ and $V_1 \cong U_1$.

Now assume $\ell(M) > 1$ and that the statement is known for all modules of smaller length. Set

$$A := V_1, \qquad B := V_2 \oplus \cdots \oplus V_m,$$

so that

$$M = A \oplus B.$$

Let

$$\iota : A \to M, \qquad \pi : M \to A$$

be the canonical inclusion and projection. For the decomposition

$$M = U_1 \oplus \cdots \oplus U_n,$$

let

$$\kappa_j : U_j \to M, \qquad p_j : M \to U_j$$

be the canonical inclusion and projection for $1 \leq j \leq n$.

Since

$$1_M = \sum_{j=1}^{n} \kappa_j p_j,$$

we get

$$1_A = \pi\iota = \pi\Big(\sum_{j=1}^{n} \kappa_j p_j\Big)\iota = \sum_{j=1}^{n} (\pi\kappa_j)(p_j\iota).$$

For each j, put

$$u_j := (\pi\kappa_j)(p_j\iota) \in \mathrm{End}_R(A).$$

If all u_j were noninvertible, then, since $\mathrm{End}_R(A)$ is local, their sum would also be noninvertible. But their sum is 1_A, a contradiction. Hence u_j is a unit for some j. After reordering the U_j, we may assume that u_1 is a unit.

Set

$$a := \pi\kappa_1 : U_1 \to A, \qquad b := p_1\iota : A \to U_1.$$

Then

$$ab = u_1$$

is an automorphism of A. Consider

$$e := b(ab)^{-1}a \in \mathrm{End}_R(U_1).$$

Then

$$e^2 = b(ab)^{-1}a\, b(ab)^{-1}a = b(ab)^{-1}(ab)(ab)^{-1}a = b(ab)^{-1}a = e,$$

so e is an idempotent. Also,

$$eb = b(ab)^{-1}ab = b \neq 0,$$

hence $e \neq 0$. Since $\mathrm{End}_R(U_1)$ is local, its only idempotents are 0 and 1, therefore $e = 1_{U_1}$.

Thus

$$b(ab)^{-1}a = 1_{U_1},$$

so a is injective. On the other hand, ab is surjective, hence a is surjective as well. Therefore a is an isomorphism, and we obtain

$$A \cong U_1.$$

Now define

$$f := \kappa_1 a^{-1} : A \to M.$$

Then

$$\pi f = \pi\kappa_1 a^{-1} = aa^{-1} = 1_A.$$

Hence

$$M = f(A) \oplus B.$$

Indeed, for any $x \in M$,

$$x = f(\pi(x)) + \big(x - f(\pi(x))\big),$$

and

$$\pi\big(x - f(\pi(x))\big) = \pi(x) - \pi f(\pi(x)) = 0,$$

so $x - f(\pi(x)) \in \ker\pi = B$. Also, if $f(a) \in B = \ker\pi$, then

$$a = \pi f(a) = 0,$$

so the sum is direct. Since $f(A) = \kappa_1(U_1) = U_1$, we have

$$M = U_1 \oplus B.$$

Therefore

$$B \cong M/U_1 \cong U_2 \oplus \cdots \oplus U_n.$$

But also

$$B = V_2 \oplus \cdots \oplus V_m.$$

Now $\ell(B) < \ell(M)$, so by the induction hypothesis,

$$m - 1 = n - 1,$$

and after reordering $U_2, \ldots, U_n$, we have

$$V_i \cong U_i \qquad (i = 2, \ldots, m).$$

Together with $V_1 \cong U_1$, this gives

$$m = n, \qquad V_i \cong U_i \quad \text{for all } i$$

after a suitable reordering.

This completes the proof. $\square$

Definition 6.3.3. *An idempotent of a ring R is a nonzero element*

$$e \in R$$

such that

$$e^2 = e.$$

Remark 6.3.4. *If e is an idempotent, then the corner ring*

$$eRe$$

is a ring with identity element e.

Definition 6.3.5. *Two idempotents $e_1, e_2 \in R$ are called orthogonal if*

$$e_1 e_2 = e_2 e_1 = 0.$$

More generally, finitely many idempotents are orthogonal if they are pairwise orthogonal.

Lemma 6.3.6. *If $e_1, \ldots, e_n$ are orthogonal idempotents, then*

$$e_1 + \cdots + e_n$$

is again an idempotent.

Proof. Let

$$e := e_1 + \cdots + e_n.$$

Since each e_i is an idempotent, we have $e_i^2 = e_i$ for all i. Since the *idempotents* are orthogonal, for $i \neq j$ we have

$$e_i e_j = 0.$$

Therefore

$$e^2 = (e_1 + \cdots + e_n)^2 = \sum_{i=1}^{n} e_i^2 + \sum_{i \neq j} e_i e_j = \sum_{i=1}^{n} e_i + \sum_{i \neq j} 0 = e_1 + \cdots + e_n = e.$$

Hence e is an idempotent. ☐

Definition 6.3.7. *An idempotent decomposition of an idempotent e is an expression*

$$e = e_1 + \cdots + e_n$$

where the e_i are orthogonal idempotents.

Remark 6.3.8. *If $e = e_1 + e_2$ is an idempotent decomposition, then e_1 and e_2 are in eRe.*

Definition 6.3.9. *An idempotent is called primitive if it cannot be written as a sum of two nonzero orthogonal idempotents.*

Definition 6.3.10. *A primitive orthogonal decomposition is an idempotent decomposition in which all summands are primitive.*

Example 6.3.11. *If e is an idempotent in a unital ring, then*

$$1 = e + (1 - e)$$

is an idempotent decomposition of 1.

Example 6.3.12. *Let e be an idempotent, $Re = \{ae | a \in R\}$ is a left module of R.*

Theorem 6.3.13. *Let e be an idempotent of R.*

If

$$e = e_1 + \cdots + e_n$$

is an idempotent decomposition, then

$$Re = Re_1 \oplus \cdots \oplus Re_n.$$

Conversely, if Re is a direct sum of left ideals

$$Re = I_1 \oplus \cdots \oplus I_n,$$

then there exists an idempotent decomposition

$$e = e_1 + \cdots + e_n$$

with $e_i \in I_i$ and $I_i = Re_i$.

In this way, the idempotent decompositions of e are in one-to-one correspondence with the direct sum decompositions of the left R-module Re.

Proof. Assume first that

$$e = e_1 + \cdots + e_n$$

is an idempotent decomposition. Since

$$e_i e = e e_i = e_i,$$

we have

$$Re_i \subseteq Re.$$

Indeed, if $x \in Re_i$, then $xe = x$.

Now let $a \in R$. Then

$$ae = ae_1 + \cdots + ae_n \in Re_1 + \cdots + Re_n,$$

so

$$Re \subseteq Re_1 + \cdots + Re_n.$$

The reverse inclusion is clear, hence

$$Re = Re_1 + \cdots + Re_n.$$

To show that the sum is direct, suppose

$$a_1 e_1 + \cdots + a_n e_n = 0.$$

Multiplying on the right by e_i, we get

$$a_i e_i = 0$$

for each i, since $e_j e_i = 0$ for $j \neq i$. Therefore the expression is unique, and

$$Re = Re_1 \oplus \cdots \oplus Re_n.$$

Conversely, suppose

$$Re = I_1 \oplus \cdots \oplus I_n.$$

Since $e \in Re$, we may write

$$e = e_1 + \cdots + e_n, \qquad e_i \in I_i.$$

For any $a_i \in I_i \subseteq Re$, we have

$$a_i = a_i e = a_i e_1 + \cdots + a_i e_n.$$

Since $e_j \in I_j$, we have

$$Re_j \subseteq I_j,$$

hence

$$a_i e_j \in I_j.$$

But $a_i \in I_i$, and the decomposition

$$Re = I_1 \oplus \cdots \oplus I_n$$

is direct. Therefore

$$a_i = a_i e_i, \qquad a_i e_j = 0 \quad (j \neq i).$$

It follows that

$$I_i \subseteq Re_i.$$

On the other hand, since $e_i \in I_i$ and I_i is a left ideal, we have

$$Re_i \subseteq I_i.$$

Thus

$$I_i = Re_i.$$

Now take $a_i = e_i$. Then

$$e_i = e_i^2, \qquad e_i e_j = 0 \quad (j \neq i).$$

Hence

$$e = e_1 + \cdots + e_n$$

is an idempotent decomposition. □

Corollary 6.3.14. *Let e be an idempotent of R. Then the following are equivalent:*

(1) e is primitive.

(2) Re is indecomposable as a left R-module.

(3) The only idempotents in eRe are 0 and e.

Proof. **(1) $\iff$ (2).** This follows immediately from the previous theorem, which gives a bijection between idempotent decompositions of e and direct sum decompositions of the left ideal Re.

(1) $\implies$ (3). Suppose that $e' \in eRe$ is an idempotent with

$$e' \neq 0, \qquad e' \neq e.$$

Since $e' \in eRe$, we have

$$ee' = e' = e'e.$$

Hence

$$e = e' + (e - e').$$

We claim that this is an idempotent decomposition of e.

First,

$$(e - e')^2 = e^2 - ee' - e'e + (e')^2 = e - e' - e' + e' = e - e'.$$

So $e - e'$ is an idempotent.

Next,

$$e'(e - e') = e'e - (e')^2 = e' - e' = 0,$$

and similarly

$$(e - e')e' = ee' - (e')^2 = e' - e' = 0.$$

Thus e' and $e - e'$ are orthogonal idempotents. Since both are nonzero, this contradicts the primitiveness of e.

Therefore no such e' exists, and the only idempotents in eRe are 0 and e.

(3) $\implies$ (1). Suppose that e is not primitive. Then there exist nonzero orthogonal idempotents e_1, e_2 such that

$$e = e_1 + e_2.$$

Since

$$ee_1e = e_1, \qquad ee_2e = e_2,$$

we have

$$e_1, e_2 \in eRe.$$

But e_1 is a nonzero idempotent different from e, contradicting (3). Hence e must be primitive. $\square$

Chapter 7

Lecture 7

7.1 Idempotents and Module Homomorphisms

Theorem 7.1.1. *Let R be a ring, let $e, f \in R$ be idempotents, and let ${}_R V$ be a left R-module.*

1. *There is a natural isomorphism of abelian groups*
$$\mathrm{Hom}_R(Re, V) \cong eV.$$
In particular,
$$\mathrm{Hom}_R(Re, Rf) \cong eRf.$$

2. *There is an isomorphism of rings*
$$\mathrm{End}_R(Re) \cong (eRe)^{op}.$$

Proof. **(1)** Define
$$g : \mathrm{Hom}_R(Re, V) \longrightarrow eV, \qquad \varphi \longmapsto \varphi(e).$$
First we check that g is well-defined. If $\varphi \in \mathrm{Hom}_R(Re, V)$, then
$$\varphi(e) = \varphi(e^2) = e\varphi(e),$$
so indeed $\varphi(e) \in eV$.

Next, for any $a \in R$ and any $\varphi \in \mathrm{Hom}_R(Re, V)$, we have
$$\varphi(ae) = a\varphi(e).$$
Thus φ is uniquely determined by the single value $\varphi(e)$. Hence g is injective.

To show surjectivity, let $v \in eV$. Define
$$\psi_v : Re \longrightarrow V, \qquad ae \longmapsto av.$$

This is well-defined, since if $ae = be$, then

$$av = (ae)v = (be)v = bv.$$

It is also R-linear. Moreover,

$$g(\psi_v) = \psi_v(e) = ev = v,$$

because $v \in eV$. Hence g is surjective.

Finally, g is a homomorphism of abelian groups, since for $\varphi, \psi \in \mathrm{Hom}_R(Re, V)$,

$$g(\varphi + \psi) = (\varphi + \psi)(e) = \varphi(e) + \psi(e) = g(\varphi) + g(\psi).$$

Therefore g is an isomorphism of abelian groups:

$$\mathrm{Hom}_R(Re, V) \cong eV.$$

Taking $V = Rf$, we obtain

$$\mathrm{Hom}_R(Re, Rf) \cong e(Rf) = eRf.$$

(2) Apply part **(1)** with $V = Re$. Then we get an isomorphism of abelian groups

$$\Phi : \mathrm{End}_R(Re) \longrightarrow eRe, \qquad \sigma \longmapsto \sigma(e).$$

Now let $\sigma, \tau \in \mathrm{End}_R(Re)$. Then

$$\Phi(\sigma \circ \tau) = (\sigma \circ \tau)(e) = \sigma(\tau(e)).$$

Since $\tau(e) \in Re$, we have $\tau(e) = \tau(e)e$, and hence

$$\sigma(\tau(e)) = \sigma(\tau(e)e) = \tau(e)\sigma(e).$$

Therefore

$$\Phi(\sigma \circ \tau) = \Phi(\tau)\Phi(\sigma).$$

So composition in $\mathrm{End}_R(Re)$ corresponds to multiplication in eRe with the order reversed. Hence

$$\mathrm{End}_R(Re) \cong (eRe)^{op}$$

as rings. ☐

Theorem 7.1.2. *Let e, f be idempotents of a ring R. Then the followings are equivalent:*

1. *$Re \cong Rf$ as left R-modules.*
2. *$eR \cong fR$ as right R-modules.*

3. *There exist elements*
$$a \in eRf, \qquad b \in fRe$$
such that
$$ab = e, \qquad ba = f.$$

Proof. By passing to the opposite ring R^{op}, statement (2) is the right-handed analogue of (1). Thus it suffices to prove that (1) and (3) are equivalent.

(1) $\Rightarrow$ (3). Assume that there is an isomorphism of left R-modules
$$\varphi : Re \xrightarrow{\sim} Rf.$$
Set
$$a := \varphi(e), \qquad b := \varphi^{-1}(f).$$
Then
$$a = \varphi(e) = \varphi(ee) = e\varphi(e) = ea,$$
and since $a \in Rf$, we also have $af = a$. Hence
$$a \in eRf.$$
Similarly,
$$b = \varphi^{-1}(f) = \varphi^{-1}(ff) = f\varphi^{-1}(f) = fb,$$
and since $b \in Re$, we have $be = b$. Therefore
$$b \in fRe.$$

Now compute:
$$f = \varphi\big(\varphi^{-1}(f)\big) = \varphi(b) = \varphi(be) = b\varphi(e) = ba,$$
and
$$e = \varphi^{-1}(\varphi(e)) = \varphi^{-1}(a) = \varphi^{-1}(af) = a\varphi^{-1}(f) = ab.$$
Thus
$$ab = e, \qquad ba = f.$$

(3) $\Rightarrow$ (1). Assume there exist $a \in eRf$ and $b \in fRe$ such that
$$ab = e, \qquad ba = f.$$
Since $a \in eRf$ and $b \in fRe$, we have
$$ea = a, \qquad af = a, \qquad fb = b, \qquad be = b.$$

Define maps
$$\varphi_a : Re \longrightarrow Rf, \qquad xe \longmapsto xa,$$

and

$$\varphi_b : Rf \longrightarrow Re, \qquad xf \longmapsto xb.$$

These are well-defined left R-module homomorphisms.

Now for any $xe \in Re$,

$$(\varphi_b \circ \varphi_a)(xe) = \varphi_b(xa) = xab = xe,$$

since $ab = e$. Likewise, for any $xf \in Rf$,

$$(\varphi_a \circ \varphi_b)(xf) = \varphi_a(xb) = xba = xf,$$

since $ba = f$.

Hence φ_a and φ_b are inverse isomorphisms, so

$$Re \cong Rf$$

as left R-modules.

Therefore (1) and (3) are equivalent, and by symmetry so are all three statements. □

7.2 Jacobson Radical of a Ring

Definition 7.2.1 (Jacobson radical). *Let R be a ring with identity. The* Jacobson radical *of R is defined by*

$$J(R) := \bigcap \{\, M \mid M \text{ is a maximal left ideal of } R \,\}.$$

Remark 7.2.2 (Annihilators). *Let ${}_R M$ be a left R-module.*

1. *For $x \in M$, define*
$$\mathrm{Ann}_R(x) := \{\, r \in R \mid rx = 0 \,\}.$$
This is a left ideal of R.

2. *For a subset $X \subseteq M$, define*
$$\mathrm{Ann}_R(X) := \bigcap_{x \in X} \mathrm{Ann}_R(x).$$

3. *If $N \leq M$ is a submodule, then*
$$\mathrm{Ann}_R(N) := \bigcap_{x \in N} \mathrm{Ann}_R(x)$$
is a two-sided ideal of R.

Theorem 7.2.3 (Properties of the Jacobson radical). *Let R be a ring with identity. Then the following hold.*

1.
$$J(R) = \bigcap \{ \operatorname{Ann}_R(S) \mid S \textit{ is a simple left } R\textit{-module} \}.$$

In particular, $J(R)$ is a two-sided ideal of R.

2. *For $a \in R$,*
$$a \in J(R) \quad \Longleftrightarrow \quad 1 - xa \textit{ has a left inverse for every } x \in R.$$

Equivalently, for every $x \in R$ there exists $b \in R$ such that
$$b(1 - xa) = 1.$$

3. *$J(R)$ is the largest two-sided ideal I of R satisfying the condition*
$$a \in I \implies 1 - a \textit{ is a unit in } R.$$

4. *$J(R)$ coincides with the intersection of all maximal right ideals of R.*

Proof. **(1)** Let S be a simple left R-module, and choose $0 \neq v \in S$. Since S is simple,
$$S = Rv.$$

Consider the surjective R-homomorphism
$$R \longrightarrow S, \qquad r \longmapsto rv.$$

Its kernel is $\operatorname{Ann}_R(v)$, so
$$S \cong R/\operatorname{Ann}_R(v).$$

Because S is simple, $\operatorname{Ann}_R(v)$ is a maximal left ideal of R. Hence
$$J(R) \subseteq \operatorname{Ann}_R(v)$$

for every nonzero $v \in S$, and therefore
$$J(R) \subseteq \operatorname{Ann}_R(S)$$

for every simple left R-module S.

Conversely, suppose that
$$x \in \bigcap \{ \operatorname{Ann}_R(S) \mid S \text{ simple left } R\text{-module} \}.$$

Let M be any maximal left ideal of R. Then R/M is a simple left R-module, so
$$x \in \operatorname{Ann}_R(R/M).$$

In particular,

$$x(1+M)=0,$$

hence

$$x+M=0,$$

so $x \in M$. Since this holds for every maximal left ideal M, we obtain

$$x \in J(R).$$

Thus

$$J(R)=\bigcap\{\operatorname{Ann}_R(S) \mid S \text{ simple left } R\text{-module}\}.$$

Finally, each $\operatorname{Ann}_R(S)$ is a two-sided ideal, so their intersection $J(R)$ is also a two-sided ideal.

(2) Assume first that $a \in J(R)$. Suppose, for contradiction, that there exists $x \in R$ such that $1-xa$ has no left inverse. Then the principal left ideal

$$R(1-xa)$$

is a proper left ideal of R, so it is contained in some maximal left ideal M.

Since $a \in J(R)$, we have $a \in M$, and because M is a left ideal,

$$xa \in M.$$

Also,

$$1-xa \in M.$$

Therefore

$$1=(1-xa)+xa \in M,$$

a contradiction. Hence $1-xa$ has a left inverse for every $x \in R$.

Conversely, assume that for every $x \in R$, the element $1-xa$ has a left inverse. Suppose that $a \notin J(R)$. Then there exists a maximal left ideal M such that

$$a \notin M.$$

Hence

$$M+Ra=R,$$

so there exist $b \in M$ and $x \in R$ such that

$$b+xa=1.$$

Thus

$$1-xa=b \in M.$$

But an element of a proper left ideal cannot have a left inverse, for if $c(1-xa)=1$, then

$$1 \in M,$$

contradicting the propriety of M. This contradiction shows that $a \in J(R)$.

(3) First let $a \in J(R)$. By part **(2)** with $x=1$, there exists $b \in R$ such that

$$b(1-a)=1.$$

Then

$$b = 1 + ba = 1 - (-b)a.$$

Since $a \in J(R)$, part **(2)** again implies that b has a left inverse. So there exists $c \in R$ such that

$$cb = 1.$$

Now

$$c = cb(1-a) = 1-a.$$

Hence

$$(1-a)b = 1,$$

and together with $b(1-a)=1$, this shows that $1-a$ is a unit in R.

Thus $J(R)$ satisfies the stated property.

Now let I be a two-sided ideal of R such that

$$a \in I \implies 1-a \text{ is a unit in } R.$$

Suppose $I \not\subseteq J(R)$. Then there exists a maximal left ideal M such that

$$I \not\subseteq M.$$

Choose $a \in I \setminus M$. Then

$$M + Ra = R,$$

so there exist $m \in M$ and $r \in R$ such that

$$m + ra = 1.$$

Since I is a two-sided ideal and $a \in I$, we have

$$ra \in I.$$

Therefore

$$1 - ra$$

is a unit in R. But

$$1 - ra = m \in M,$$

and a proper left ideal cannot contain a unit. This is a contradiction. Hence

$$I \subseteq J(R).$$

So $J(R)$ is the largest such two-sided ideal.

(4) Let

$$J_r(R) := \bigcap \{\, N \mid N \text{ is a maximal right ideal of } R \,\}.$$

Applying the preceding arguments to the opposite ring R^{op}, we see that $J_r(R)$ is also the largest two-sided ideal I of R such that

$$a \in I \implies 1 - a \text{ is a unit in } R.$$

By part **(3)**, this property characterizes $J(R)$ uniquely. Therefore

$$J_r(R) = J(R).$$

Hence $J(R)$ coincides with the intersection of all maximal right ideals of R. $\square$

Definition 7.2.4. *Let R be a ring with 1.*

1. *An element $a \in R$ is called* nilpotent *if*

$$a^n = 0$$

for some integer $n \geq 1$.

2. *A left ideal $I \subseteq R$ is called* nilpotent *if*

$$I^n = 0$$

for some integer $n \geq 1$.

3. *A left ideal $I \subseteq R$ is called a* nil left ideal *if every element of I is nilpotent.*

Lemma 7.2.5. *If $a \in R$ is nilpotent, say $a^n = 0$, then $1 - a$ is a unit in R, with inverse*

$$(1 - a)^{-1} = 1 + a + a^2 + \cdots + a^{n-1}.$$

Proof. Since $a^n = 0$, we have

$$(1 - a)(1 + a + a^2 + \cdots + a^{n-1}) = 1 - a^n = 1,$$

and similarly

$$(1 + a + a^2 + \cdots + a^{n-1})(1 - a) = 1 - a^n = 1.$$

Hence $1 - a$ is a unit. $\square$

Lemma 7.2.6. *Every nilpotent left ideal is a nil left ideal.*

Proof. Suppose $I^n = 0$ and let $a \in I$. Then

$$a^n \in I^n = 0,$$

so a is nilpotent. Hence every element of I is nilpotent. $\square$

Remark 7.2.7. *We shall use the following standard characterization of the Jacobson radical:*

$$a \in J(R) \quad \Longleftrightarrow \quad \text{for every } x \in R,\ 1 - xa \text{ has a left inverse in } R.$$

In particular, if $a \in J(R)$, then $1 - a$ is a unit.

Theorem 7.2.8. *Let I be a nil left ideal of a ring R. Then*

$$I \subseteq J(R).$$

Proof. Let $a \in I$. For every $x \in R$, since I is a left ideal we have $xa \in I$. Because I is nil, the element xa is nilpotent. By the previous lemma, $1 - xa$ is a unit in R, hence in particular it has a left inverse.

Therefore, by the characterization of $J(R)$,

$$a \in J(R).$$

Since $a \in I$ was arbitrary, it follows that

$$I \subseteq J(R).$$

$\square$

7.3 Artinian Rings and the Jacobson Radical

Theorem 7.3.1. *Let A be an Artinian ring. Then $J(A)$ is nilpotent.*

Proof. Consider the descending chain of left ideals

$$J(A) \supseteq J(A)^2 \supseteq J(A)^3 \supseteq \cdots .$$

Since A is Artinian, this chain stabilizes. Hence there exists $n \geq 1$ such that

$$J(A)^n = J(A)^{n+1} = J(A)^{n+2} = \cdots .$$

Set

$$N := J(A)^n.$$

Then

$$N^2 = J(A)^{2n} = J(A)^n = N,$$

because all powers from the nth onward are equal.

We claim that $N = 0$. Suppose not. Consider the set

$$\Sigma := \{ L \leq_A A \mid NL \neq 0 \}$$

of left ideals L of A such that $NL \neq 0$. Since

$$NN = N \neq 0,$$

the set Σ is nonempty. As A is Artinian, Σ has a minimal element; call it L_0.

Since $NL_0 \neq 0$, there exists $a \in L_0$ such that

$$Na \neq 0.$$

Now Na is a left ideal of A contained in L_0, and

$$N(Na) = N^2 a = Na \neq 0.$$

Thus $Na \in \Sigma$. By minimality of L_0, we must have

$$L_0 = Na.$$

In particular, $a \in Na$, so there exists $b \in N$ such that

$$a = ba.$$

Hence

$$(1 - b)a = 0.$$

But $b \in N \subseteq J(A)$, so $1 - b$ is a unit. Multiplying by its inverse gives $a = 0$, a contradiction to $Na \neq 0$.

Therefore $N = 0$, that is,

$$J(A)^n = 0.$$

So $J(A)$ is nilpotent. $\square$

> **Corollary 7.3.2.** *Let A be an Artinian ring. Then $J(A)$ is the largest nilpotent left ideal of A.*

Proof. By the theorem above, $J(A)$ is itself nilpotent.

Now let I be any nilpotent left ideal of A. By the earlier lemma, I is a nil left ideal. Hence by the theorem on nil left ideals,

$$I \subseteq J(A).$$

Thus $J(A)$ contains every nilpotent left ideal, so it is the largest one. $\square$

Theorem 7.3.3. *Let R be a ring and let I be a two-sided ideal of R such that*

$$I \subseteq J(R).$$

Then

$$J(R/I) = J(R)/I.$$

Proof. We prove the two inclusions separately.

First inclusion: $J(R)/I \subseteq J(R/I)$.

Let $a \in J(R)$. For any $x \in R$, the element $1 - xa$ has a left inverse in R. So there exists $b \in R$ such that

$$b(1 - xa) = 1.$$

Passing to the quotient modulo I, we get

$$(b + I)\big((1 - xa) + I\big) = 1 + I.$$

Hence $(1 - xa) + I$ has a left inverse in R/I. Since this holds for every $x \in R$, the characterization of the Jacobson radical gives

$$a + I \in J(R/I).$$

Thus

$$J(R)/I \subseteq J(R/I).$$

Second inclusion: $J(R/I) \subseteq J(R)/I$.

Let $a + I \in J(R/I)$. We show that $a \in J(R)$. Let $x \in R$ be arbitrary. Since $a + I \in J(R/I)$, the element

$$(1 - xa) + I$$

has a left inverse in R/I. Therefore there exists $b \in R$ such that

$$(b + I)\big((1 - xa) + I\big) = 1 + I,$$

that is,

$$b(1 - xa) = 1 - c$$

for some $c \in I$.

Since $I \subseteq J(R)$, we have $c \in J(R)$, so $1 - c$ has a left inverse in R. Thus there exists $d \in R$ such that

$$d(1 - c) = 1.$$

Multiplying the equation $b(1 - xa) = 1 - c$ on the left by d, we obtain

$$(db)(1 - xa) = 1.$$

Hence $1 - xa$ has a left inverse in R. Since $x \in R$ was arbitrary, the characterization of $J(R)$ implies

$$a \in J(R).$$

Therefore

$$a + I \in J(R)/I.$$

Combining the two inclusions, we conclude that

$$J(R/I) = J(R)/I.$$

$\square$

Definition 7.3.4. *A ring A is called* semisimple *if the left regular module ${}_AA$ is semisimple.*

Theorem 7.3.5. *Let A be an Artinian ring. Then A is semisimple if and only if*

$$J(A) = 0.$$

Proof. *($\Rightarrow$)* Suppose A is semisimple. Then ${}_AA$ is a direct sum of simple left A-modules:

$${}_AA \cong S_1 \oplus \cdots \oplus S_r.$$

By the standard description of $J(A)$ as the intersection of the annihilators of all simple left A-modules, every element of $J(A)$ annihilates each S_i. Hence

$$J(A)A = 0.$$

But $1 \in A$, so for any $x \in J(A)$ we have

$$x = x \cdot 1 = 0.$$

Therefore

$$J(A) = 0.$$

($\Leftarrow$) Suppose now that $J(A) = 0$. Let

$$\mathcal{S} := \{ M_1 \cap \cdots \cap M_r \mid r \geq 1,\ M_1, \ldots, M_r \text{ maximal left ideals of } A \}.$$

Since A is Artinian, the set $\mathcal{S}$ has a minimal element with respect to inclusion. Choose such a minimal element and write it as

$$L = M_1 \cap \cdots \cap M_r.$$

We claim that $L = 0$. Suppose $L \neq 0$. Since

$$J(A) = \bigcap \{ M \mid M \text{ maximal left ideal of } A \} = 0,$$

there exists a maximal left ideal M such that

$$L \nsubseteq M.$$

Hence

$$L \cap M \subsetneq L.$$

But $L \cap M$ is again a finite intersection of maximal left ideals, contradicting the minimality of L. Therefore $L = 0$.

Now consider the natural A-module homomorphism

$$\varphi : A \longrightarrow A/M_1 \oplus \cdots \oplus A/M_r, \qquad a \longmapsto (a + M_1, \ldots, a + M_r).$$

Its kernel is

$$\ker \varphi = M_1 \cap \cdots \cap M_r = L = 0,$$

so φ is injective.

Each A/M_i is a simple left A-module, hence the direct sum

$$A/M_1 \oplus \cdots \oplus A/M_r$$

is semisimple. Since every submodule of a semisimple module is semisimple, it follows that ${}_A A$ is semisimple. Therefore A is semisimple. □

Corollary 7.3.6. *If A is an Artinian ring, then*

$$A/J(A)$$

is semisimple.

Proof. Since A is Artinian, so is the quotient ring $A/J(A)$. By the previous theorem, it suffices to show that

$$J\big(A/J(A)\big) = 0.$$

But by the quotient formula,

$$J\big(A/J(A)\big) = J(A)/J(A) = 0.$$

Hence $A/J(A)$ is semisimple. □

Corollary 7.3.7. *Let A be an Artinian ring. Then $J(A)$ is the unique two-sided ideal U of A such that*

1. *U is nilpotent;*

2. A/U *is semisimple.*

Proof. First, $J(A)$ has these two properties: it is nilpotent by the nilpotency theorem above, and $A/J(A)$ is semisimple by the previous corollary.

Now let U be a two-sided ideal satisfying (1) and (2). Since U is nilpotent, it is a nilpotent left ideal, hence

$$U \subseteq J(A).$$

Because A/U is semisimple, we have

$$J(A/U) = 0.$$

On the other hand, since $U \subseteq J(A)$, the quotient formula gives

$$J(A/U) = J(A)/U.$$

Therefore

$$J(A)/U = 0,$$

so

$$J(A) = U.$$

Thus $J(A)$ is unique. $\square$

7.4 Radical and Socle Series

Proposition 7.4.1. *Let A be an Artinian ring, and let U be a finitely generated left A-module.*

1. *For a submodule $V \leq U$, the following are equivalent:*
   - *(a)* $V = \mathrm{Rad}(U)$*;*
   - *(b)* V *is the smallest submodule of* U *such that* U/V *is semisimple;*
   - *(c)* $V = J(A)U$*.*
2. *For a submodule $V \leq U$, the following are equivalent:*
   - *(a)* $V = \mathrm{Soc}(U)$*;*
   - *(b)* V *is the largest semisimple submodule of* U*;*
   - *(c)*
$$V = \{\, u \in U \mid J(A)u = 0 \,\}.$$

Proof. We first record two standard facts.

Fact 1. Since A is Artinian, the quotient ring $A/J(A)$ is semisimple. Hence every $A/J(A)$-module is semisimple.

Fact 2. If S is a simple left A-module, then

$$J(A)S = 0.$$

Indeed, by the characterization

$$J(A) = \bigcap \{ \operatorname{Ann}_A(T) \mid T \text{ simple left } A\text{-module} \},$$

we have $J(A) \subseteq \operatorname{Ann}_A(S)$.

(1) We prove the equivalence of (a), (b), (c).

(c) $\Rightarrow$ (b). Set

$$W := J(A)U.$$

Then

$$J(A)(U/W) = 0,$$

so U/W is naturally an $A/J(A)$-module. By Fact 1, U/W is semisimple.

Now let $V' \leq U$ be any submodule such that U/V' is semisimple. Since every simple module is annihilated by $J(A)$, every semisimple module is annihilated by $J(A)$ as well. Hence

$$J(A)(U/V') = 0.$$

Equivalently,

$$J(A)U \subseteq V'.$$

Thus $W = J(A)U$ is the smallest submodule of U with semisimple quotient.

(b) $\Rightarrow$ (c). Assume that V is the smallest submodule of U such that U/V is semisimple.

Since $U/J(A)U$ is semisimple by the previous paragraph, the minimality of V gives

$$V \subseteq J(A)U.$$

On the other hand, U/V is semisimple, so it is annihilated by $J(A)$. Hence

$$J(A)(U/V) = 0,$$

which means

$$J(A)U \subseteq V.$$

Therefore

$$V = J(A)U.$$

(c) $\Leftrightarrow$ (a). Let $W = J(A)U$. We show that $W = \operatorname{Rad}(U)$.

First, let M be a maximal submodule of U. Then U/M is simple, hence semisimple. By the minimality proved above for $J(A)U$, we must have

$$J(A)U \subseteq M.$$

Since this is true for every maximal submodule M of U, it follows that

$$J(A)U \subseteq \mathrm{Rad}(U).$$

For the reverse inclusion, suppose there exists

$$u \in \mathrm{Rad}(U) \setminus J(A)U.$$

Let

$$\overline{U} := U/J(A)U, \qquad \overline{u} := u + J(A)U \neq 0.$$

We already know that $\overline{U}$ is semisimple, so we may write

$$\overline{U} = \bigoplus_{i \in I} S_i$$

with each S_i simple. Since $\overline{u} \neq 0$, some component of $\overline{u}$ in this direct sum is nonzero; choose $i_0 \in I$ such that the S_{i_0}-component of $\overline{u}$ is nonzero.

Set

$$\overline{M} := \bigoplus_{i \neq i_0} S_i.$$

Then $\overline{M}$ is a maximal submodule of $\overline{U}$ and

$$\overline{u} \notin \overline{M}.$$

Let M be the inverse image of $\overline{M}$ under the quotient map

$$\pi : U \twoheadrightarrow \overline{U}.$$

Then M is a maximal submodule of U, because

$$U/M \cong \overline{U}/\overline{M} \cong S_{i_0}$$

is simple. But $u \notin M$, contradicting $u \in \mathrm{Rad}(U)$, since $\mathrm{Rad}(U)$ is contained in every maximal submodule of U.

Hence

$$\mathrm{Rad}(U) \subseteq J(A)U.$$

Together with the opposite inclusion, we get

$$\mathrm{Rad}(U) = J(A)U.$$

Combining the implications above, we conclude that (a), (b), and (c) are equivalent.

(2) Recall that

$$\mathrm{Soc}(U) = \sum \{\, S \leq U \mid S \text{ is a simple submodule} \,\}.$$

Let

$$T := \{\, u \in U \mid J(A)u = 0 \,\}.$$

We first show that T is a submodule of U. If $u, v \in T$ and $a \in A$, then

$$J(A)(u+v) = J(A)u + J(A)v = 0,$$

and for every $j \in J(A)$,

$$j(au) = (ja)u = 0,$$

because $J(A)$ is a two-sided ideal and $ja \in J(A)$. Thus $au \in T$, so $T \leq U$.

(c) $\Rightarrow$ (b). Assume $V = T$.

Since $J(A)T = 0$, the module T is naturally an $A/J(A)$-module. By Fact 1, T is semisimple.

Now let $W \leq U$ be any semisimple submodule. Then W is a sum of simple submodules, and by Fact 2 each simple submodule is annihilated by $J(A)$. Hence

$$J(A)W = 0,$$

so

$$W \subseteq T.$$

Therefore T is the largest semisimple submodule of U.

(b) $\Rightarrow$ (c). Assume that V is the largest semisimple submodule of U.

The submodule

$$T = \{\, u \in U \mid J(A)u = 0 \,\}$$

is semisimple by the previous paragraph, so by maximality of V we get

$$T \subseteq V.$$

On the other hand, V is semisimple, hence $J(A)V = 0$, so

$$V \subseteq T.$$

Thus

$$V = T.$$

(a) $\Leftrightarrow$ (c). Let $T = \{u \in U \mid J(A)u = 0\}$ as above.

Every simple submodule $S \leq U$ satisfies $J(A)S = 0$ by Fact 2, so

$$S \subseteq T.$$

Taking the sum over all simple submodules of U, we obtain

$$\mathrm{Soc}(U) \subseteq T.$$

Conversely, T is semisimple, so it is a sum of simple submodules of U. Hence

$$T \subseteq \mathrm{Soc}(U).$$

Therefore

$$\mathrm{Soc}(U) = T.$$

So (a), (b), and (c) are equivalent. □

Let A be an Artinian ring, and let U be a finitely generated left A-module. Recall the basic identities

$$\mathrm{Rad}(U) = J(A)U, \qquad \mathrm{Soc}(U) = \{u \in U \mid J(A)u = 0\}.$$

Definition 7.4.2. *Define inductively*

$$\mathrm{Rad}^0(U) := U, \qquad \mathrm{Rad}^n(U) := \mathrm{Rad}(\mathrm{Rad}^{n-1}(U)) \quad (n \geq 1),$$

and

$$\mathrm{Soc}^0(U) := 0, \qquad \mathrm{Soc}^n(U)/\,\mathrm{Soc}^{n-1}(U) := \mathrm{Soc}\big(U/\,\mathrm{Soc}^{n-1}(U)\big) \quad (n \geq 1).$$

The descending chain

$$U = \mathrm{Rad}^0(U) \supseteq \mathrm{Rad}(U) \supseteq \mathrm{Rad}^2(U) \supseteq \cdots$$

is called the radical series *of U, and the ascending chain*

$$0 = \mathrm{Soc}^0(U) \subseteq \mathrm{Soc}(U) \subseteq \mathrm{Soc}^2(U) \subseteq \cdots$$

is called the socle series *of U.*

Proposition 7.4.3. *For every integer $n \geq 1$,*

$$\mathrm{Rad}^n(U) = J(A)^n U,$$

and

$$\mathrm{Soc}^n(U) = \{u \in U \mid J(A)^n u = 0\}.$$

Proof. We prove both formulas by induction on n.

Radical formula. For $n = 1$, this is exactly the identity

$$\mathrm{Rad}(U) = J(A)U.$$

Assume now that $\mathrm{Rad}^{n-1}(U) = J(A)^{n-1}U$. Since $\mathrm{Rad}^{n-1}(U)$ is a finitely generated A-

module, we may apply the same identity to it:

$$\operatorname{Rad}^n(U) = \operatorname{Rad}(\operatorname{Rad}^{n-1}(U)) = J(A)\operatorname{Rad}^{n-1}(U) = J(A)\big(J(A)^{n-1}U\big) = J(A)^nU.$$

Socle formula. For $n = 1$, this is exactly the identity

$$\operatorname{Soc}(U) = \{u \in U \mid J(A)u = 0\}.$$

Assume now that

$$\operatorname{Soc}^{n-1}(U) = \{u \in U \mid J(A)^{n-1}u = 0\}.$$

By definition,

$$\operatorname{Soc}^n(U)/\operatorname{Soc}^{n-1}(U) = \operatorname{Soc}\big(U/\operatorname{Soc}^{n-1}(U)\big).$$

Applying the case $n = 1$ to the module $U/\operatorname{Soc}^{n-1}(U)$, we get

$$\operatorname{Soc}\big(U/\operatorname{Soc}^{n-1}(U)\big) = \left\{u + \operatorname{Soc}^{n-1}(U) \in U/\operatorname{Soc}^{n-1}(U) \;\middle|\; J(A)u \subseteq \operatorname{Soc}^{n-1}(U)\right\}.$$

By the induction hypothesis, the condition

$$J(A)u \subseteq \operatorname{Soc}^{n-1}(U)$$

is equivalent to

$$J(A)^{n-1}(J(A)u) = 0,$$

that is,

$$J(A)^n u = 0.$$

Hence the preimage of $\operatorname{Soc}(U/\operatorname{Soc}^{n-1}(U))$ in U is precisely

$$\{u \in U \mid J(A)^n u = 0\}.$$

Therefore

$$\operatorname{Soc}^n(U) = \{u \in U \mid J(A)^n u = 0\}.$$

This completes the induction. □

Remark 7.4.4. *For $n \geq 1$, the quotient*

$$\operatorname{Rad}^{n-1}(U)/\operatorname{Rad}^n(U)$$

is called the n-th radical layer *of U, and*

$$\operatorname{Soc}^n(U)/\operatorname{Soc}^{n-1}(U)$$

is called the n-th socle layer *of U.*

Remark 7.4.5. *The n-th radical layer is the top of the previous radical term:*

$$\operatorname{Rad}^{n-1}(U)/\operatorname{Rad}^{n}(U) = \operatorname{Top}\big(\operatorname{Rad}^{n-1}(U)\big).$$

Lemma 7.4.6. *For every $n \geq 1$ and any finitely generated A-modules U, V,*

$$\operatorname{Rad}^{n}(U \oplus V) = \operatorname{Rad}^{n}(U) \oplus \operatorname{Rad}^{n}(V),$$

and

$$\operatorname{Soc}^{n}(U \oplus V) = \operatorname{Soc}^{n}(U) \oplus \operatorname{Soc}^{n}(V).$$

Proof. Using Proposition 7.4.3,

$$\operatorname{Rad}^{n}(U \oplus V) = J(A)^{n}(U \oplus V) = J(A)^{n}U \oplus J(A)^{n}V = \operatorname{Rad}^{n}(U) \oplus \operatorname{Rad}^{n}(V).$$

Similarly,

$$\operatorname{Soc}^{n}(U \oplus V) = \{(u, v) \in U \oplus V \mid J(A)^{n}(u, v) = 0\}.$$

But

$$J(A)^{n}(u, v) = 0 \iff J(A)^{n}u = 0 \text{ and } J(A)^{n}v = 0.$$

Hence

$$\operatorname{Soc}^{n}(U \oplus V) = \{u \in U \mid J(A)^{n}u = 0\} \oplus \{v \in V \mid J(A)^{n}v = 0\} = \operatorname{Soc}^{n}(U) \oplus \operatorname{Soc}^{n}(V).$$

□

Theorem 7.4.7. *Let A be an Artinian ring, and let U be a finitely generated left A-module.*

1. *The radical series of U is the fastest descending chain*

$$U = U_0 \supseteq U_1 \supseteq U_2 \supseteq \cdots$$

such that every factor U_{r-1}/U_r is semisimple. More precisely, if

$$U = U_0 \supseteq U_1 \supseteq U_2 \supseteq \cdots$$

is any descending chain with each U_{r-1}/U_r semisimple, then for every $r \geq 0$,

$$\operatorname{Rad}^{r}(U) \subseteq U_r.$$

2. *The socle series of U is the fastest ascending chain*

$$0 = V_0 \subseteq V_1 \subseteq V_2 \subseteq \cdots$$

such that every factor V_r/V_{r-1} is semisimple. More precisely, if

$$0 = V_0 \subseteq V_1 \subseteq V_2 \subseteq \cdots$$

is any ascending chain with each V_r/V_{r-1} semisimple, then for every $r \geq 0$,

$$V_r \subseteq \mathrm{Soc}^r(U).$$

3. *Both series terminate. If m and n are the minimal integers such that*

$$\mathrm{Rad}^m(U) = 0 \qquad \textit{and} \qquad \mathrm{Soc}^n(U) = U,$$

then

$$m = n.$$

Proof. **(1)** Let

$$U = U_0 \supseteq U_1 \supseteq U_2 \supseteq \cdots$$

be a descending chain such that each quotient U_{r-1}/U_r is semisimple. We prove by induction on r that

$$\mathrm{Rad}^r(U) \subseteq U_r.$$

For $r = 0$, this is clear, since $\mathrm{Rad}^0(U) = U = U_0$.

Assume $r \geq 1$ and that

$$\mathrm{Rad}^{r-1}(U) \subseteq U_{r-1}.$$

Consider the quotient

$$\mathrm{Rad}^{r-1}(U)/\big(\mathrm{Rad}^{r-1}(U) \cap U_r\big).$$

By the second isomorphism theorem,

$$\mathrm{Rad}^{r-1}(U)/\big(\mathrm{Rad}^{r-1}(U) \cap U_r\big) \cong \big(\mathrm{Rad}^{r-1}(U) + U_r\big)/U_r.$$

Since $\mathrm{Rad}^{r-1}(U) \subseteq U_{r-1}$, we have

$$\big(\mathrm{Rad}^{r-1}(U) + U_r\big)/U_r \subseteq U_{r-1}/U_r.$$

But U_{r-1}/U_r is semisimple, and any submodule of a semisimple module is semisimple. Hence

$$\mathrm{Rad}^{r-1}(U)/\big(\mathrm{Rad}^{r-1}(U) \cap U_r\big)$$

is semisimple.

By definition of the radical as the smallest submodule with semisimple quotient, it follows that

$$\mathrm{Rad}\big(\mathrm{Rad}^{r-1}(U)\big) \subseteq \mathrm{Rad}^{r-1}(U) \cap U_r.$$

That is,

$$\mathrm{Rad}^r(U) \subseteq U_r.$$

This proves (1).

(2) Let
$$0 = V_0 \subseteq V_1 \subseteq V_2 \subseteq \cdots$$
be an ascending chain such that each quotient V_r/V_{r-1} is semisimple. We prove by induction on r that
$$V_r \subseteq \operatorname{Soc}^r(U).$$

For $r = 0$, this is clear.

Assume $r \geq 1$ and that
$$V_{r-1} \subseteq \operatorname{Soc}^{r-1}(U).$$
Consider the image of V_r in the quotient $U/\operatorname{Soc}^{r-1}(U)$:
$$\big(V_r + \operatorname{Soc}^{r-1}(U)\big)/\operatorname{Soc}^{r-1}(U).$$
By the second isomorphism theorem,
$$\big(V_r + \operatorname{Soc}^{r-1}(U)\big)/\operatorname{Soc}^{r-1}(U) \cong V_r/\big(V_r \cap \operatorname{Soc}^{r-1}(U)\big).$$
Since $V_{r-1} \subseteq \operatorname{Soc}^{r-1}(U)$, we have
$$V_{r-1} \subseteq V_r \cap \operatorname{Soc}^{r-1}(U),$$
so
$$V_r/\big(V_r \cap \operatorname{Soc}^{r-1}(U)\big)$$
is a quotient of V_r/V_{r-1}. Because V_r/V_{r-1} is semisimple, so is every quotient of it. Hence
$$\big(V_r + \operatorname{Soc}^{r-1}(U)\big)/\operatorname{Soc}^{r-1}(U)$$
is a semisimple submodule of
$$U/\operatorname{Soc}^{r-1}(U).$$
By the maximality of the socle,
$$\big(V_r + \operatorname{Soc}^{r-1}(U)\big)/\operatorname{Soc}^{r-1}(U) \subseteq \operatorname{Soc}\big(U/\operatorname{Soc}^{r-1}(U)\big) = \operatorname{Soc}^r(U)/\operatorname{Soc}^{r-1}(U).$$
Therefore
$$V_r \subseteq \operatorname{Soc}^r(U).$$
This proves (2).

(3) Since A is Artinian, $J(A)$ is nilpotent. Choose $N \geq 1$ such that
$$J(A)^N = 0.$$

Then by Proposition 7.4.3,

$$\operatorname{Rad}^N(U) = J(A)^N U = 0,$$

and

$$\operatorname{Soc}^N(U) = \{u \in U \mid J(A)^N u = 0\} = U.$$

So both series terminate.

Now let m be minimal such that $\operatorname{Rad}^m(U) = 0$. By Proposition 7.4.3,

$$J(A)^m U = 0.$$

Hence every $u \in U$ satisfies $J(A)^m u = 0$, so again by Proposition 7.4.3,

$$\operatorname{Soc}^m(U) = U.$$

Thus $n \leq m$.

Conversely, let n be minimal such that $\operatorname{Soc}^n(U) = U$. Then for every $u \in U$,

$$J(A)^n u = 0.$$

So

$$J(A)^n U = 0,$$

hence

$$\operatorname{Rad}^n(U) = J(A)^n U = 0.$$

Therefore $m \leq n$.

Combining the two inequalities, we obtain

$$m = n.$$

$\square$

Definition 7.4.8. *The common integer in Theorem 7.4.7(3) is called the* Loewy length *of U.*

7.5 Lifting Idempotents

Definition 7.5.1. *Let R be a ring and let I be a two-sided ideal of R.*

If $f \in R$ is an idempotent and $f \notin I$, then $e = f + I$ is an idempotent of R/I. We say that an idempotent $f \in R$ lifts *e.*

Remark 7.5.2. *For a general ideal I, an idempotent of R/I need not lift to an idempotent of R. The next theorem shows that lifting is always possible when I is nilpotent.*

Theorem 7.5.3. *Let R be a ring, let I be a nilpotent two-sided ideal of R, and let*

$$e \in R/I$$

be an idempotent. Then there exists an idempotent $f \in R$ such that

$$f + I = e.$$

Moreover, if e is primitive, then every lift f of e is primitive.

Proof. Since I is nilpotent, there exists $r \geq 1$ such that

$$I^r = 0.$$

We shall construct idempotents

$$e_i \in R/I^i \qquad (1 \leq i \leq r)$$

such that

$$e_1 = e$$

and for each $i \geq 2$, the image of e_i in R/I^{i-1} is e_{i-1}.

Assume inductively that $e_{i-1} \in R/I^{i-1}$ is an idempotent, with $2 \leq i \leq r$. Choose any element

$$a \in R/I^i$$

whose image in R/I^{i-1} is e_{i-1}. Since e_{i-1} is idempotent, we have

$$a^2 - a \in I^{i-1}/I^i.$$

Now

$$\left(I^{i-1}/I^i\right)^2 = 0,$$

because

$$I^{2i-2} \subseteq I^i \qquad (i \geq 2).$$

Hence

$$(a^2 - a)^2 = 0 \qquad \text{in } R/I^i.$$

Define

$$e_i := 3a^2 - 2a^3 \in R/I^i.$$

First, e_i maps to e_{i-1} in R/I^{i-1}, because

$$e_i - a = 3a^2 - 2a^3 - a = -(2a-1)(a^2 - a) \in I^{i-1}/I^i.$$

Next, e_i is idempotent. Indeed, the polynomial identity

$$(3t^2 - 2t^3)^2 - (3t^2 - 2t^3) = -(3 - 2t)(1 + 2t)(t^2 - t)^2$$

gives

$$e_i^2 - e_i = -(3 - 2a)(1 + 2a)(a^2 - a)^2 = 0$$

in R/I^i.

Thus, by induction, we obtain an idempotent

$$e_r \in R/I^r = R.$$

Let

$$f := e_r.$$

Then f is an idempotent of R whose image in R/I is e.

Now suppose that e is primitive and that f is a lift of e. Assume that

$$f = f_1 + f_2$$

is a decomposition of f into orthogonal idempotents:

$$f_1^2 = f_1, \qquad f_2^2 = f_2, \qquad f_1 f_2 = f_2 f_1 = 0.$$

Reducing modulo I, we get

$$e = (f_1 + I) + (f_2 + I),$$

and this is again a decomposition into orthogonal idempotents in R/I. Since e is primitive, one of $f_1 + I$ or $f_2 + I$ must be zero. Without loss of generality, assume

$$f_1 + I = 0.$$

Then $f_1 \in I$.

But f_1 is idempotent, so for every $m \geq 1$,

$$f_1 = f_1^m \in I^m.$$

Taking $m = r$, we get

$$f_1 \in I^r = 0,$$

hence $f_1 = 0$. Therefore f cannot be written as a sum of two nonzero orthogonal idempotents, so f is primitive. □

Chapter 8

Lecture 8

8.1 Lifting Idempotent Decompositions

Corollary 8.1.1. *Let I be a nilpotent two-sided ideal of R, and let*

$$1 = e_1 + \cdots + e_n$$

be an idempotent decomposition in R/I. Then there exists an idempotent decomposition

$$1 = f_1 + \cdots + f_n$$

in R such that

$$f_i + I = e_i \qquad (1 \leq i \leq n).$$

If moreover the e_i are primitive, then so are the f_i.

Proof. We argue by induction on n.

If $n = 1$, this is exactly Theorem 7.5.3.

Assume the result holds for decompositions with fewer than n summands. Write

$$1 = e_1 + E, \qquad \text{where} \qquad E = e_2 + \cdots + e_n$$

in R/I. Then E is an idempotent orthogonal to e_1.

By Theorem 7.5.3, e_1 lifts to an idempotent $f \in R$ such that

$$e_1 = f + I.$$

Set

$$F := 1 - f.$$

Then F is an idempotent and

$$F + I = 1 - (f + I) = 1 - e_1 = E,$$

so F lifts E.

Now F is the identity element of the corner ring FRF, and FIF is a nilpotent two-sided ideal of FRF. Consider the composite homomorphism

$$FRF \hookrightarrow R \twoheadrightarrow R/I.$$

Its kernel is $FRF \cap I$. We claim that

$$FRF \cap I = FIF.$$

Indeed, if $x \in FIF$, then clearly $x \in FRF \cap I$. Conversely, if $x \in FRF \cap I$, then since $x \in FRF$ we have

$$x = FxF,$$

and because $x \in I$ and I is a two-sided ideal, it follows that

$$x = FxF \in FIF.$$

Thus the claim holds.

Hence we obtain an injective homomorphism

$$FRF/FIF \hookrightarrow R/I.$$

Its image is

$$E(R/I)E.$$

Indeed, the image of $FrF \in FRF$ is

$$(F + I)(r + I)(F + I) = E(r + I)E,$$

and every element of $E(R/I)E$ has this form. Therefore

$$FRF/FIF \cong E(R/I)E.$$

In the ring $E(R/I)E$, whose identity element is E, we have the idempotent decomposition

$$E = e_2 + \cdots + e_n.$$

So, by the induction hypothesis applied to the ring FRF and its nilpotent ideal FIF, there exists an idempotent decomposition

$$F = f_2 + \cdots + f_n$$

in FRF such that

$$f_i + I = e_i \qquad (2 \le i \le n).$$

Finally, letting

$$f_1 := f,$$

we get

$$1 = f_1 + F = f_1 + f_2 + \cdots + f_n,$$

which is an idempotent decomposition in R, and each f_i lifts e_i.

Moreover, since each $f_i \in FRF$ for $i \geq 2$, we have

$$Ff_i = f_i = f_iF, \qquad f_1f_i = f_if_1 = 0 \qquad (i \geq 2),$$

so the sum is orthogonal.

If the e_i are primitive, then f_1 is primitive by Theorem 7.5.3, and $f_2, \dots, f_n$ are primitive by the induction hypothesis. Hence all f_i are primitive. ☐

Corollary 8.1.2. *Let I be a nilpotent two-sided ideal of R, and let $f \in R$ be an idempotent. Then f is primitive in R if and only if $f + I$ is primitive in R/I.*

Proof. First suppose that $f + I$ is primitive in R/I. Since f is an idempotent lift of $f + I$, Theorem 7.5.3 implies that f is primitive in R.

Conversely, suppose that f is primitive in R. We show that $f + I$ is primitive in R/I. Assume that

$$f + I = \bar{e}_1 + \bar{e}_2$$

is an idempotent decomposition in R/I, where

$$\bar{e}_1^2 = \bar{e}_1, \qquad \bar{e}_2^2 = \bar{e}_2, \qquad \bar{e}_1\bar{e}_2 = \bar{e}_2\bar{e}_1 = 0.$$

Since $\bar{e}_1 + \bar{e}_2 = f + I$, we have

$$\bar{e}_j(f + I) = \bar{e}_j = (f + I)\bar{e}_j \qquad (j = 1, 2).$$

Hence $\bar{e}_1, \bar{e}_2$ lie in the corner ring

$$(f + I)(R/I)(f + I).$$

There is a natural isomorphism

$$fRf/fIf \cong (f + I)(R/I)(f + I).$$

Moreover, fIf is a nilpotent two-sided ideal of the ring fRf.

Thus, by the lifting theorem for idempotent decompositions, the decomposition

$$f + I = \bar{e}_1 + \bar{e}_2$$

lifts to an idempotent decomposition

$$f = f_1 + f_2$$

in $fRf \subseteq R$, with

$$f_j + I = \overline{e}_j \qquad (j = 1, 2).$$

Since f is primitive in R, one of f_1, f_2 must be zero. Therefore one of $\overline{e}_1, \overline{e}_2$ is zero.

Hence $f + I$ admits no nontrivial idempotent decomposition in R/I, so $f + I$ is primitive.

$\square$

8.2 Projective Covers over Artinian Rings

Lemma 8.2.1. *Let A be a semisimple Artinian ring. Then the following statements hold.*

1. *There exists a primitive idempotent decomposition*

   $$1 = e_1 + \cdots + e_n.$$

   This decomposition corresponds to a decomposition of the left regular module

   $${}_A A = Ae_1 \oplus \cdots \oplus Ae_n,$$

   where each Ae_i is a simple left A-module.

2. *For every simple left A-module S, there exists a primitive idempotent $e \in A$ such that*

   $$S \cong Ae.$$

3. *For every simple left A-module S, there exists some $1 \leq i \leq n$ such that*

   $$e_i S \neq 0.$$

   Moreover, if T is a simple left A-module with $T \ncong S$, then

   $$e_i T = 0.$$

Proof. **(1)** Since A is semisimple Artinian, the left regular module ${}_A A$ is semisimple and Artinian. Hence it is a finite direct sum of simple left ideals:

$${}_A A = I_1 \oplus \cdots \oplus I_n.$$

By the correspondence between direct sum decompositions of ${}_A A$ and idempotent decompositions of 1, there exist pairwise orthogonal idempotents

$$e_1, \ldots, e_n \in A$$

such that

$$1 = e_1 + \cdots + e_n$$

and

$$I_i = Ae_i \qquad (1 \leq i \leq n).$$

Since each I_i is simple, each Ae_i is simple and hence indecomposable. Therefore each e_i is primitive.

(2) Let S be a simple left A-module. Choose $0 \neq s \in S$. Then

$$S = As,$$

so the map

$$A \longrightarrow S, \qquad a \longmapsto as$$

is a surjective A-module homomorphism.

Using the decomposition from part (1), we have

$$A = Ae_1 \oplus \cdots \oplus Ae_n.$$

Since the map $A \to S$ is nonzero and surjective, its restriction to at least one summand Ae_i is nonzero. Thus there exists i such that we have a nonzero homomorphism

$$Ae_i \longrightarrow S.$$

Because both Ae_i and S are simple, this nonzero homomorphism is an isomorphism. Therefore

$$S \cong Ae_i.$$

Taking $e = e_i$, we obtain the desired primitive idempotent.

(3) Let S be a simple left A-module. Since

$$1 = e_1 + \cdots + e_n,$$

we have

$$S = 1S = (e_1 + \cdots + e_n)S = e_1S + \cdots + e_nS.$$

Since $S \neq 0$, there exists some i such that

$$e_iS \neq 0.$$

For this i, we claim that if T is a simple left A-module with $T \ncong S$, then

$$e_iT = 0.$$

Indeed, by the standard isomorphism

$$\mathrm{Hom}_A(Ae_i, T) \cong e_iT, \qquad \varphi \longmapsto \varphi(e_i),$$

it is enough to show that
$$\mathrm{Hom}_A(Ae_i, T) = 0.$$
Since $e_i S \neq 0$, the same isomorphism gives
$$\mathrm{Hom}_A(Ae_i, S) \cong e_i S \neq 0.$$
Thus there is a nonzero homomorphism
$$Ae_i \longrightarrow S.$$
Because Ae_i and S are simple, this homomorphism is an isomorphism. Hence
$$Ae_i \cong S.$$
If $T \ncong S$, then $T \ncong Ae_i$. Therefore, by Schur's lemma,
$$\mathrm{Hom}_A(Ae_i, T) = 0.$$
Consequently,
$$e_i T = 0.$$
□

Theorem 8.2.2. *Let A be an Artinian ring, and let S be a simple left A-module. Then the followings hold.*

1. *There exists an indecomposable projective left A-module P_S such that*
$$P_S / \mathrm{Rad}(P_S) \cong S.$$
Moreover, P_S is of the form
$$P_S = Af,$$
where f is a primitive idempotent of A.
2. *The idempotent f has the following property:*
$$fS \neq 0,$$
and if T is any simple left A-module with $T \ncong S$, then
$$fT = 0.$$
3. *The module P_S is the projective cover of S.*

Proof. Let
$$\overline{A} := A/J(A).$$

Since A is Artinian, the ideal $J(A)$ is nilpotent, and $\overline{A}$ is a semisimple Artinian ring.

By the structure theory of semisimple Artinian rings, there exists a primitive idempotent

$$e \in \overline{A}$$

such that

$$eS \neq 0.$$

Equivalently, the simple $\overline{A}$-module $\overline{A}e$ is isomorphic to S.

Since $J(A)$ is nilpotent, idempotents lift modulo $J(A)$. Hence there exists an idempotent $f \in A$ such that

$$f + J(A) = e.$$

Moreover, because e is primitive, the lift f is also primitive.

Now define

$$P_S := Af.$$

Since f is an idempotent, Af is a direct summand of the left regular module ${}_AA$. Indeed,

$$A = Af \oplus A(1-f).$$

Hence Af is projective. Since f is primitive, Af is indecomposable. Therefore $P_S = Af$ is an indecomposable projective left A-module.

Consider the natural map

$$Af \longrightarrow \overline{A}e$$

defined by

$$af \longmapsto (a + J(A))(f + J(A)).$$

Since $f + J(A) = e$, this map is explicitly

$$af \longmapsto (a + J(A))e.$$

It is a surjective A-module homomorphism. Its kernel is

$$J(A)f.$$

Indeed, af maps to zero if and only if

$$(a + J(A))(f + J(A)) = 0,$$

which is equivalent to

$$af \in J(A).$$

Since $af \in Af$, the kernel is

$$Af \cap J(A) = J(A)f.$$

Thus

$$Af/J(A)f \cong \overline{A}e.$$

Because $\overline{A}e \cong S$, we get

$$P_S/J(A)P_S \cong S.$$

Since A is Artinian and P_S is finitely generated, we have

$$\operatorname{Rad}(P_S) = J(A)P_S.$$

Therefore

$$P_S/\operatorname{Rad}(P_S) \cong S.$$

Next we verify the stated property of f. Since every simple A-module is annihilated by $J(A)$, the action of f on any simple A-module is the same as the action of

$$e = f + J(A)$$

through the quotient algebra $\overline{A} = A/J(A)$. Hence

$$fS = eS \neq 0.$$

If T is a simple A-module with $T \not\cong S$, then, by the choice of e in the semisimple algebra $\overline{A}$, we have

$$eT = 0.$$

Therefore

$$fT = 0.$$

Finally, the natural epimorphism

$$P_S = Af \longrightarrow P_S/\operatorname{Rad}(P_S) \cong S$$

is essential by Nakayama's lemma. Since P_S is projective, this epimorphism is a projective cover of S.

Thus P_S is the projective cover of S. □

Theorem 8.2.3. *Let A be an Artinian ring. Up to isomorphism, the indecomposable projective left A-modules are exactly the modules*

$$P_S,$$

where S runs through the isomorphism classes of simple left A-modules and P_S denotes the projective cover of S.

Moreover,

$$P_S \cong P_T \qquad \Longleftrightarrow \qquad S \cong T.$$

Each P_S occurs as a direct summand of the left regular module ${}_AA$, with multi-

plicity equal to the multiplicity of S as a direct summand of

$$A/J(A).$$

Equivalently, one has a decomposition

$$_AA \cong \bigoplus_S P_S^{n_S},$$

where S runs through the isomorphism classes of simple left A-modules and

$$A/J(A) \cong \bigoplus_S S^{n_S}.$$

Furthermore,

$$n_S = \dim_{D_S} S, \qquad D_S := \operatorname{End}_A(S),$$

where S is regarded as a right D_S-vector space by

$$s \cdot \varphi := \varphi(s).$$

In particular, if A is a finite-dimensional k-algebra over an algebraically closed field k, then

$$n_S = \dim_k S.$$

Proof. We first classify the indecomposable projective modules.

Let P be a finitely generated indecomposable projective left A-module. Since A is Artinian, P has finite length, and therefore

$$P/\operatorname{Rad}(P)$$

is semisimple. Hence we may write

$$P/\operatorname{Rad}(P) \cong S_1 \oplus \cdots \oplus S_t,$$

where each S_i is simple.

The canonical epimorphism

$$P \longrightarrow P/\operatorname{Rad}(P)$$

is essential by Nakayama's lemma. Since P is projective, this epimorphism is a projective cover of

$$S_1 \oplus \cdots \oplus S_t.$$

For each simple left A-module T, let

$$\pi_T : P_T \longrightarrow T$$

denote the projective cover of T, whose existence was proved in Theorem 8.2.2. Thus P_T is an indecomposable projective left A-module and

$$P_T/\operatorname{Rad}(P_T) \cong T.$$

In particular, for each $i = 1, \ldots, t$, we have a projective cover

$$\pi_{S_i} : P_{S_i} \longrightarrow S_i.$$

Therefore the direct sum map

$$\pi_{S_1} \oplus \cdots \oplus \pi_{S_t} : P_{S_1} \oplus \cdots \oplus P_{S_t} \longrightarrow S_1 \oplus \cdots \oplus S_t$$

is also a projective cover. By uniqueness of projective covers, we obtain

$$P \cong P_{S_1} \oplus \cdots \oplus P_{S_t}.$$

Since P is indecomposable, we must have $t = 1$. Hence

$$P \cong P_{S_1}.$$

Thus every indecomposable projective module is isomorphic to P_S for some simple module S.

Conversely, for every simple module S, the projective cover P_S is indecomposable. Indeed, by the previous theorem, P_S has the form

$$P_S \cong Af$$

for some primitive idempotent $f \in A$. Since f is primitive, the module Af is indecomposable.

Next we show that

$$P_S \cong P_T \qquad \Longleftrightarrow \qquad S \cong T.$$

If $P_S \cong P_T$, then passing to radical quotients gives

$$P_S/\operatorname{Rad}(P_S) \cong P_T/\operatorname{Rad}(P_T).$$

But

$$P_S/\operatorname{Rad}(P_S) \cong S, \qquad P_T/\operatorname{Rad}(P_T) \cong T,$$

so $S \cong T$. The converse is immediate from the uniqueness of projective covers.

Now consider the left regular module ${}_AA$. Since A is Artinian, the natural epimorphism

$$A \longrightarrow A/J(A)$$

is essential by Nakayama's lemma. Since ${}_AA$ is projective, this epimorphism is a projective cover of $A/J(A)$.

Write the semisimple A-module $A/J(A)$ as

$$A/J(A) \cong \bigoplus_S S^{n_S},$$

where S runs through the isomorphism classes of simple left A-modules. Then

$$\bigoplus_S P_S^{n_S} \longrightarrow \bigoplus_S S^{n_S}$$

is a projective cover. By uniqueness of projective covers, we get

$${}_A A \cong \bigoplus_S P_S^{n_S}.$$

Thus the multiplicity of P_S in ${}_A A$ is exactly the multiplicity of S in $A/J(A)$.

It remains to identify the integer n_S. Put

$$\overline{A} := A/J(A).$$

Since A is Artinian, the quotient $\overline{A}$ is semisimple Artinian. Hence the Wedderburn–Artin theorem applies to $\overline{A}$, not necessarily to A itself.

Every simple left A-module S is annihilated by $J(A)$, and therefore may be regarded naturally as a simple left $\overline{A}$-module. Conversely, every simple left $\overline{A}$-module is naturally a simple left A-module via the quotient map $A \to \overline{A}$. Moreover,

$$\mathrm{End}_{\overline{A}}(S) = \mathrm{End}_A(S).$$

By the Wedderburn–Artin theorem applied to $\overline{A}$, the multiplicity of S in the left regular $\overline{A}$-module ${}_{\overline{A}}\overline{A}$ is

$$n_S = \dim_{D_S} S,$$

where

$$D_S := \mathrm{End}_{\overline{A}}(S)^{\mathrm{op}} = \mathrm{End}_A(S)^{\mathrm{op}}.$$

Here S is regarded as a right D_S-vector space by

$$s \cdot \varphi = \varphi(s), \qquad s \in S,\ \varphi \in \mathrm{End}_A(S).$$

Equivalently, this is the usual right action of $\mathrm{End}_A(S)^{\mathrm{op}}$ on S.

Finally, if A is a finite-dimensional k-algebra over an algebraically closed field k, then Schur's lemma gives

$$\mathrm{End}_A(S) \cong k.$$

Thus also

$$D_S \cong k,$$

and hence

$$n_S = \dim_{D_S} S = \dim_k S.$$

$\square$

Corollary 8.2.4. *Let A be an Artinian ring, and let U be a finitely generated left A-module. Then U has a projective cover.*

Proof. Since A is Artinian and U is finitely generated, the quotient

$$U/\operatorname{Rad}(U)$$

is semisimple. Hence we may write

$$U/\operatorname{Rad}(U) \cong S_1 \oplus \cdots \oplus S_n,$$

where each S_i is a simple left A-module.

For each i, let

$$\pi_i : P_{S_i} \longrightarrow S_i$$

be the projective cover of S_i. Then the direct sum

$$\pi := \pi_1 \oplus \cdots \oplus \pi_n : P_{S_1} \oplus \cdots \oplus P_{S_n} \longrightarrow S_1 \oplus \cdots \oplus S_n$$

is a projective cover.

Let

$$q : U \longrightarrow U/\operatorname{Rad}(U)$$

be the canonical epimorphism. After identifying

$$U/\operatorname{Rad}(U) \cong S_1 \oplus \cdots \oplus S_n,$$

we may regard π as an epimorphism

$$\pi : P_{S_1} \oplus \cdots \oplus P_{S_n} \longrightarrow U/\operatorname{Rad}(U).$$

Since

$$P := P_{S_1} \oplus \cdots \oplus P_{S_n}$$

is projective and q is surjective, there exists an A-module homomorphism

$$\widetilde{\pi} : P \longrightarrow U$$

such that the following diagram commutes:

$$\begin{tikzcd} & P \arrow[dl, dashed, "\widetilde{\pi}"'] \arrow[d, "\pi"] & \\ U \arrow[r, "q"'] & U/\operatorname{Rad}(U) \arrow[r] & 0. \end{tikzcd}$$

That is,

$$q \circ \widetilde{\pi} = \pi.$$

We claim that $\widetilde{\pi}$ is an essential epimorphism. Since

$$q \circ \widetilde{\pi} = \pi,$$

and since both q and π are essential epimorphisms, it follows from the two-out-of-three property for essential epimorphisms that $\widetilde{\pi}$ is also an essential epimorphism.

Therefore

$$\widetilde{\pi} : P_{S_1} \oplus \cdots \oplus P_{S_n} \longrightarrow U$$

is an essential epimorphism from a projective module onto U. Hence it is a projective cover of U. $\square$

Lemma 8.2.5. *Let A be a ring with identity.*

1. *If A is semisimple, then A is left Artinian if and only if A is left Noetherian.*
2. *If A is left Artinian, then A is left Noetherian.*

Proof. **(1)** Assume that A is semisimple. Then the left regular module ${}_AA$ is semisimple, so it is a direct sum of simple left A-modules:

$${}_AA = \bigoplus_{\lambda \in \Lambda} S_\lambda.$$

For a semisimple module, being Noetherian is equivalent to being a finite direct sum of simple modules, and the same is true for being Artinian. Indeed, if the index set Λ is infinite, then we can form an infinite strictly increasing chain

$$S_{\lambda_1} \subset S_{\lambda_1} \oplus S_{\lambda_2} \subset S_{\lambda_1} \oplus S_{\lambda_2} \oplus S_{\lambda_3} \subset \cdots,$$

so the module is not Noetherian. Similarly, we can form an infinite strictly decreasing chain by removing summands one by one, so the module is not Artinian.

Thus a semisimple module is Noetherian if and only if it is Artinian, namely if and only if it is a finite direct sum of simple modules. Applying this to the left regular module ${}_AA$, we obtain that A is left Artinian if and only if A is left Noetherian.

(2) Assume that A is left Artinian. Let

$$J = J(A)$$

be the Jacobson radical of A. Since A is left Artinian, the radical J is nilpotent. Hence there exists $n \geq 1$ such that

$$J^n = 0.$$

Therefore we have a finite filtration of left A-modules

$$A \supseteq J \supseteq J^2 \supseteq \cdots \supseteq J^n = 0.$$

It is enough to show that each quotient

$$J^i/J^{i+1}$$

is Noetherian as a left A-module. Since J annihilates J^i/J^{i+1}, each quotient is naturally a module over

$$A/J.$$

Moreover, A/J is semisimple Artinian. Hence every A/J-module is semisimple.

Because J^i/J^{i+1} is a quotient of the left Artinian module J^i, it is Artinian. Thus J^i/J^{i+1} is an Artinian semisimple module. By part (1), it is Noetherian.

Now using the finite filtration

$$A \supseteq J \supseteq J^2 \supseteq \cdots \supseteq J^n = 0$$

and the fact that a module is Noetherian whenever it has a finite filtration whose successive quotients are Noetherian, we conclude that ${}_AA$ is Noetherian.

Therefore A is left Noetherian. □

8.3 Multiplicities and the Cartan Matrix

Definition 8.3.1 (Multiplicity of a simple module)**.** *Let A be an Artinian ring, and let V be a finitely generated left A-module. Then V has a composition series*

$$0 = V_n \subset V_{n-1} \subset \cdots \subset V_1 \subset V_0 = V,$$

where each quotient

$$V_{i-1}/V_i$$

is a simple A-module.

For a simple A-module S, the number of composition factors

$$V_{i-1}/V_i$$

which are isomorphic to S is called the multiplicity of S in V*. We denote it by*

$$[V : S].$$

Theorem 8.3.2. *Let A be an Artinian ring, and let $f \in A$ be a primitive idempotent. Put*

$$J = J(A), \qquad \overline{A} = A/J, \qquad e = f + J \in \overline{A}.$$

Assume that e is a primitive idempotent in $\overline{A}$. Then, for every finitely generated left A-module V, the multiplicity of the simple A-module $\overline{A}e$ in V is equal to the *composition length of fV as a left fAf-module. In other words,*

$$[V : \overline{A}e] = \ell_{fAf}(fV).$$

Proof. Let

$$V = V_0 \supset V_1 \supset \cdots \supset V_n = 0$$

be a composition series of V. Applying f gives a descending chain of left fAf-modules

$$fV = fV_0 \supseteq fV_1 \supseteq \cdots \supseteq fV_n = 0.$$

For each i, the quotient V_i/V_{i+1} is a simple A-module, hence it is also a simple $\overline{A}$-module because $J(A)$ annihilates every simple A-module. We claim that

$$V_i/V_{i+1} \cong \overline{A}e \quad \Longleftrightarrow \quad f(V_i/V_{i+1}) \neq 0.$$

Indeed, the action of f on a simple A-module is the same as the action of $e = f + J$. Hence, for a simple A-module S,

$$fS \neq 0 \quad \Longleftrightarrow \quad eS \neq 0.$$

Moreover,

$$eS \cong \mathrm{Hom}_{\overline{A}}(\overline{A}e, S),$$

and since $\overline{A}e$ and S are simple $\overline{A}$-modules, this space is nonzero if and only if

$$S \cong \overline{A}e.$$

Thus the claim follows.

Now for each i, we have

$$f(V_i/V_{i+1}) \cong \frac{fV_i + V_{i+1}}{V_{i+1}} \cong \frac{fV_i}{fV_i \cap V_{i+1}}.$$

Since f is idempotent, one has

$$fV_i \cap V_{i+1} = fV_{i+1}.$$

Indeed, if $x \in fV_i \cap V_{i+1}$, then $x = fv$ for some $v \in V_i$, and since $x \in V_{i+1}$, we get

$$x = fx \in fV_{i+1}.$$

The reverse inclusion is clear. Therefore

$$f(V_i/V_{i+1}) \cong \frac{fV_i}{fV_{i+1}}.$$

Hence every quotient

$$fV_i/fV_{i+1}$$

is either 0 or a simple fAf-module. Moreover, it is nonzero precisely when

$$V_i/V_{i+1} \cong \overline{A}e.$$

Therefore, after deleting the zero quotients from the chain

$$fV = fV_0 \supseteq fV_1 \supseteq \cdots \supseteq fV_n = 0,$$

we obtain a composition series of fV as an fAf-module.

Consequently, the composition length of fV as an fAf-module is exactly the number of composition factors of V isomorphic to $\overline{A}e$. Hence

$$\ell_{fAf}(fV) = [V : \overline{A}e].$$

□

Definition 8.3.3 (Cartan invariant and Cartan matrix)**.** *Let A be an Artinian ring, and let*

$$J = J(A), \qquad \overline{A} = A/J.$$

Let

$$1 = e_1 + \cdots + e_n$$

be a primitive idempotent decomposition of 1 *in* $\overline{A}$. *Choose a lifting*

$$1 = f_1 + \cdots + f_n$$

to a primitive idempotent decomposition of 1 *in* A, *so that*

$$e_i = f_i + J(A) \qquad \text{for all } 1 \le i \le n.$$

For $1 \le i, j \le n$, define

$$c_{ij} := \left[Af_j : \overline{A}e_i\right],$$

that is, c_{ij} is the multiplicity of the simple A-module $\overline{A}e_i$ as a composition factor of the A-module Af_j.

The integer c_{ij} is called the Cartan invariant *of A. The matrix*

$$C_A = (c_{ij})_{1 \le i,j \le n}$$

is called the Cartan matrix *of A.*

Theorem 8.3.4. *Let A be an Artinian ring, and let*

$$J = J(A), \qquad \overline{A} = A/J.$$

Let

$$1 = e_1 + \cdots + e_n$$

be a primitive idempotent decomposition in $\overline{A}$, and let

$$1 = f_1 + \cdots + f_n$$

be a primitive idempotent decomposition in A such that

$$e_i = f_i + J(A) \qquad \text{for all } 1 \leq i \leq n.$$

Let

$$c_{ij} = [Af_j : \overline{A}e_i]$$

be the Cartan invariant of A. Then

$$c_{ij} \neq 0 \quad \Longleftrightarrow \quad f_iAf_j \neq 0.$$

Proof. By the multiplicity theorem for primitive idempotents, for any finitely generated A-module V, the multiplicity of $\overline{A}e_i$ in V is equal to the composition length of f_iV as an f_iAf_i-module. That is,

$$[V : \overline{A}e_i] = \ell_{f_iAf_i}(f_iV).$$

Now take

$$V = Af_j.$$

Then

$$f_iV = f_i(Af_j) = f_iAf_j.$$

Therefore

$$c_{ij} = [Af_j : \overline{A}e_i] = \ell_{f_iAf_i}(f_iAf_j).$$

Since f_iAf_j is a finite length f_iAf_i-module, its composition length is nonzero if and only if the module itself is nonzero. Hence

$$c_{ij} \neq 0 \quad \Longleftrightarrow \quad \ell_{f_iAf_i}(f_iAf_j) \neq 0 \quad \Longleftrightarrow \quad f_iAf_j \neq 0.$$

This proves the theorem. □

Proposition 8.3.5. *Let A be a finite-dimensional k-algebra over a field k. Let S be a simple left A-module with projective cover*

$$P_S \twoheadrightarrow S,$$

and let U be a finite-dimensional left A-module.

1. *If T is a simple left A-module, then*

$$\dim_k \operatorname{Hom}_A(P_S, T) = \begin{cases} \dim_k \operatorname{End}_A(S), & S \cong T, \\ 0, & S \not\cong T. \end{cases}$$

2. *The multiplicity of S in U is given by*

$$[U : S] = \frac{\dim_k \operatorname{Hom}_A(P_S, U)}{\dim_k \operatorname{End}_A(S)}.$$

3. *If $e \in A$ is an idempotent, then*

$$\dim_k \operatorname{Hom}_A(Ae, U) = \dim_k eU.$$

4. *Let $e_S, e_T \in A$ be idempotents such that*

$$P_S = Ae_S, \qquad P_T = Ae_T$$

are projective covers of S and T, respectively. If

$$c_{ST} := [P_T : S]$$

denotes the Cartan invariant, then

$$c_{ST} = \frac{\dim_k \operatorname{Hom}_A(P_S, P_T)}{\dim_k \operatorname{End}_A(S)} = \frac{\dim_k e_S A e_T}{\dim_k \operatorname{End}_A(S)}.$$

In particular, if k is algebraically closed, then

$$c_{ST} = \dim_k \operatorname{Hom}_A(P_S, P_T) = \dim_k e_S A e_T.$$

Proof. Since $P_S \twoheadrightarrow S$ is a projective cover, we have

$$P_S / \operatorname{Rad}(P_S) \cong S.$$

(1) Let T be a simple left A-module. Every homomorphism

$$\varphi : P_S \to T$$

vanishes on $\operatorname{Rad}(P_S)$. Indeed, if $\varphi = 0$, this is clear. If $\varphi \neq 0$, then φ is surjective because T is simple, hence $\ker \varphi$ is a maximal submodule of P_S. Therefore

$$\operatorname{Rad}(P_S) \subseteq \ker \varphi.$$

Thus every map $P_S \to T$ factors uniquely through $P_S / \operatorname{Rad}(P_S)$. Hence

$$\operatorname{Hom}_A(P_S, T) \cong \operatorname{Hom}_A(P_S / \operatorname{Rad}(P_S), T) \cong \operatorname{Hom}_A(S, T).$$

By Schur's lemma,

$$\operatorname{Hom}_A(S,T)=0$$

if $S \ncong T$, while

$$\operatorname{Hom}_A(S,S)=\operatorname{End}_A(S)$$

if $S \cong T$. Therefore

$$\dim_k \operatorname{Hom}_A(P_S,T)=\begin{cases}\dim_k \operatorname{End}_A(S), & S\cong T,\\ 0, & S\ncong T.\end{cases}$$

(2) Let

$$U=U_0\supset U_1\supset\cdots\supset U_m=0$$

be a composition series of U. Since P_S is projective, the functor

$$\operatorname{Hom}_A(P_S,-)$$

is exact. Therefore, for each i, applying $\operatorname{Hom}_A(P_S,-)$ to

$$0\longrightarrow U_{i+1}\longrightarrow U_i\longrightarrow U_i/U_{i+1}\longrightarrow 0$$

gives an exact sequence

$$0\longrightarrow \operatorname{Hom}_A(P_S,U_{i+1})\longrightarrow \operatorname{Hom}_A(P_S,U_i)\longrightarrow \operatorname{Hom}_A(P_S,U_i/U_{i+1})\longrightarrow 0.$$

Hence

$$\dim_k \operatorname{Hom}_A(P_S,U)=\sum_{i=0}^{m-1}\dim_k \operatorname{Hom}_A(P_S,U_i/U_{i+1}).$$

By part (1), each composition factor U_i/U_{i+1} contributes $\dim_k \operatorname{End}_A(S)$ precisely when

$$U_i/U_{i+1}\cong S,$$

and contributes 0 otherwise. Therefore

$$\dim_k \operatorname{Hom}_A(P_S,U)=[U:S]\cdot\dim_k \operatorname{End}_A(S).$$

Thus

$$[U:S]=\frac{\dim_k \operatorname{Hom}_A(P_S,U)}{\dim_k \operatorname{End}_A(S)}.$$

(3) Define a map

$$\Phi:\operatorname{Hom}_A(Ae,U)\longrightarrow eU$$

by

$$\Phi(\varphi)=\varphi(e).$$

Since

$$\varphi(e) = \varphi(e^2) = e\varphi(e),$$

we indeed have $\varphi(e) \in eU$.

Conversely, for $u \in eU$, define

$$\psi_u : Ae \longrightarrow U$$

by

$$\psi_u(ae) = au.$$

This is well-defined and A-linear. Moreover,

$$\Phi(\psi_u) = \psi_u(e) = u.$$

Also, for $\varphi \in \mathrm{Hom}_A(Ae, U)$,

$$\psi_{\varphi(e)}(ae) = a\varphi(e) = \varphi(ae).$$

Hence Φ is an isomorphism of k-vector spaces:

$$\mathrm{Hom}_A(Ae, U) \cong eU.$$

Therefore

$$\dim_k \mathrm{Hom}_A(Ae, U) = \dim_k eU.$$

(4) By definition,

$$c_{ST} = [P_T : S].$$

Applying part (2) to $U = P_T$, we obtain

$$c_{ST} = \frac{\dim_k \mathrm{Hom}_A(P_S, P_T)}{\dim_k \mathrm{End}_A(S)}.$$

Since $P_S = Ae_S$ and $P_T = Ae_T$, part (3) gives

$$\mathrm{Hom}_A(P_S, P_T) = \mathrm{Hom}_A(Ae_S, Ae_T) \cong e_S A e_T.$$

Thus

$$\dim_k \mathrm{Hom}_A(P_S, P_T) = \dim_k e_S A e_T.$$

Therefore

$$c_{ST} = \frac{\dim_k \mathrm{Hom}_A(P_S, P_T)}{\dim_k \mathrm{End}_A(S)} = \frac{\dim_k e_S A e_T}{\dim_k \mathrm{End}_A(S)}.$$

If k is algebraically closed, then by Schur's lemma

$$\mathrm{End}_A(S) \cong k,$$

so

$$\dim_k \mathrm{End}_A(S) = 1.$$

Hence

$$c_{ST} = \dim_k \operatorname{Hom}_A(P_S, P_T) = \dim_k e_S A e_T.$$

□

8.4 Integral Extensions

Definition 8.4.1 (Integral element). *Let S be an integral domain, and let $R \subseteq S$ be a subring containing 1_S. Let $\alpha \in S$. We say that α is* integral over R *if there exists a monic polynomial*

$$f(X) \in R[X]$$

such that

$$f(\alpha) = 0.$$

Equivalently, there exist $a_0, a_1, \ldots, a_{n-1} \in R$ such that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0.$$

Proposition 8.4.2 (Equivalent characterizations of integrality). *Let S be an integral domain, and let $R \subseteq S$ be a subring containing 1_S. For $\alpha \in S$, the following conditions are equivalent:*

1. *α is integral over R.*
2. *$R[\alpha]$ is a finitely generated R-module.*
3. *There exists a subring T of S such that*

$$R \subseteq T, \qquad \alpha \in T,$$

and T is finitely generated as an R-module.

Proof. We prove the implications separately.

(2) ⇒ (3). Take

$$T = R[\alpha].$$

Then T is a subring of S, contains R, contains α, and is finitely generated as an R-module by assumption.

(1) ⇒ (2). Assume that α is integral over R. Then there exists a monic polynomial

$$f(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0 \in R[X]$$

such that $f(\alpha) = 0$. Hence

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0,$$

so

$$\alpha^n = -a_{n-1}\alpha^{n-1} - \cdots - a_1\alpha - a_0.$$

Thus α^n is an R-linear combination of

$$1, \alpha, \alpha^2, \ldots, \alpha^{n-1}.$$

Multiplying this relation by powers of α, we see inductively that every power α^m with $m \geq n$ is also an R-linear combination of

$$1, \alpha, \alpha^2, \ldots, \alpha^{n-1}.$$

Therefore every element of $R[\alpha]$ is an R-linear combination of

$$1, \alpha, \alpha^2, \ldots, \alpha^{n-1}.$$

Hence

$$R[\alpha] = R \cdot 1 + R\alpha + \cdots + R\alpha^{n-1},$$

so $R[\alpha]$ is a finitely generated R-module.

(3) $\Rightarrow$ (1). Assume that there exists a subring T of S such that

$$R \subseteq T, \qquad \alpha \in T,$$

and T is finitely generated as an R-module.

Let

$$\varepsilon_1, \ldots, \varepsilon_n$$

be a set of generators of T as an R-module. Since $\alpha \in T$ and T is a subring, for each i we have

$$\alpha\varepsilon_i \in T.$$

Hence there exist elements $a_{ji} \in R$ such that

$$\alpha\varepsilon_i = \sum_{j=1}^{n} a_{ji}\varepsilon_j, \qquad 1 \leq i \leq n.$$

Let

$$A = (a_{ji}) \in M_n(R).$$

Writing

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix},$$

the above relations can be written as

$$(\alpha I_n - A)\varepsilon = 0.$$

Multiplying both sides by the adjugate matrix

$$\operatorname{adj}(\alpha I_n - A),$$

we obtain

$$\det(\alpha I_n - A)\varepsilon = 0.$$

Thus

$$\det(\alpha I_n - A)\varepsilon_i = 0 \qquad \text{for all } i = 1, \dots, n.$$

Since $\varepsilon_1, \dots, \varepsilon_n$ generate T as an R-module, it follows that

$$\det(\alpha I_n - A)\,T = 0.$$

But $1 \in T$, so

$$\det(\alpha I_n - A) = 0.$$

Now define

$$p(X) = \det(XI_n - A) \in R[X].$$

This is a monic polynomial of degree n. Since

$$p(\alpha) = \det(\alpha I_n - A) = 0,$$

we conclude that α is integral over R. □

Lemma 8.4.3 (Finite generation of a generated subring)**.** *Let S be an integral domain, and let $R \subseteq S$ be a subring containing 1_S. Let M_1, M_2 be subrings of S such that*

$$R \subseteq M_i \subseteq S, \qquad i = 1, 2.$$

If M_1 and M_2 are finitely generated as R-modules, then the subring

$$R[M_1, M_2] \subseteq S$$

generated by R, M_1, M_2 is finitely generated as an R-module.

Proof. Suppose

$$M_1 = R\alpha_1 + \cdots + R\alpha_m$$

as an R-module, and

$$M_2 = R\beta_1 + \cdots + R\beta_n$$

as an R-module.

We claim that $R[M_1, M_2]$ is generated as an R-module by the finite set

$$\{\alpha_i\beta_j \mid 1 \leq i \leq m,\ 1 \leq j \leq n\}.$$

Set
$$N = \sum_{i=1}^{m}\sum_{j=1}^{n} R\alpha_i\beta_j.$$
Clearly $N \subseteq R[M_1, M_2]$. We show that N is a subring of S containing M_1 and M_2.

Since $1 \in M_2$, every $\alpha_i \in M_1$ can be written as
$$\alpha_i = \alpha_i \cdot 1 \in N.$$
Similarly, since $1 \in M_1$, every $\beta_j \in M_2$ belongs to N. Hence
$$M_1 \subseteq N, \qquad M_2 \subseteq N.$$

It remains to show that N is closed under multiplication. Take two generators $\alpha_i\beta_j$ and $\alpha_{i'}\beta_{j'}$. Then
$$(\alpha_i\beta_j)(\alpha_{i'}\beta_{j'}) = (\alpha_i\alpha_{i'})(\beta_j\beta_{j'}).$$
Since M_1 and M_2 are subrings, we have
$$\alpha_i\alpha_{i'} \in M_1, \qquad \beta_j\beta_{j'} \in M_2.$$
Therefore there exist $r_k, s_\ell \in R$ such that
$$\alpha_i\alpha_{i'} = \sum_{k=1}^{m} r_k\alpha_k, \qquad \beta_j\beta_{j'} = \sum_{\ell=1}^{n} s_\ell\beta_\ell.$$
Thus
$$(\alpha_i\alpha_{i'})(\beta_j\beta_{j'}) = \sum_{k=1}^{m}\sum_{\ell=1}^{n} r_k s_\ell\, \alpha_k\beta_\ell \in N.$$
Hence N is closed under multiplication.

Therefore N is a subring of S containing M_1 and M_2. By minimality of $R[M_1, M_2]$, we get
$$R[M_1, M_2] \subseteq N.$$
Since the reverse inclusion is clear, we have
$$R[M_1, M_2] = N.$$
Hence $R[M_1, M_2]$ is finitely generated as an R-module. □

Theorem 8.4.4 (Integral elements form a subring)**.** *Let S be an integral domain, and let $R \subseteq S$ be a subring containing 1_S. Then the set of integral elements of S over R forms a subring of S containing R.*

Proof. Let $\alpha, \beta \in S$ be integral over R. By the characterization of integral elements, $R[\alpha]$ and $R[\beta]$ are finitely generated R-modules. Hence, by the finite-generation lemma for generated subrings,

$$R[\alpha, \beta]$$

is also a finitely generated R-module.

Since

$$\alpha + \beta, \quad \alpha - \beta, \quad \alpha\beta \in R[\alpha, \beta],$$

again by the characterization of integrality, the elements

$$\alpha + \beta, \qquad \alpha - \beta, \qquad \alpha\beta$$

are integral over R.

Moreover, every element $r \in R$ is integral over R, since it satisfies the monic polynomial

$$X - r \in R[X].$$

Therefore the integral elements of S over R form a subring of S containing R. $\square$

Definition 8.4.5 (Algebraic integer)**.** *Let $\mathbb{Z} \subseteq \mathbb{C}$. If $z \in \mathbb{C}$ is integral over $\mathbb{Z}$, then z is called an* algebraic integer*.*

Definition 8.4.6 (Integrally closed subring)**.** *Let S be an integral domain, and let $R \subseteq S$ be a subring containing 1_S. We say that R is* integrally closed in S *if every element of S which is integral over R already lies in R.*

Definition 8.4.7 (Integrally closed domain)**.** *Let R be an integral domain with quotient field $K = \mathrm{Frac}(R)$. We say that R is* integrally closed *if R is integrally closed in its quotient field K. Equivalently, every element of K integral over R belongs to R.*

Theorem 8.4.8. *Every unique factorization domain is integrally closed.*

Proof. Let R be a unique factorization domain, and let

$$K = \mathrm{Frac}(R)$$

be its quotient field. We show that R is integrally closed in K.

Let $\alpha \in K$ be integral over R. Write

$$\alpha = \frac{b}{a},$$

where $a, b \in R$, $a \neq 0$, and $\gcd(a, b) = 1$.

Since α is integral over R, there exist $n \geq 1$ and $c_0, c_1, \ldots, c_{n-1} \in R$ such that

$$\alpha^n + c_{n-1}\alpha^{n-1} + \cdots + c_1\alpha + c_0 = 0.$$

Substituting $\alpha = b/a$, we get

$$\left(\frac{b}{a}\right)^n + c_{n-1}\left(\frac{b}{a}\right)^{n-1} + \cdots + c_1\left(\frac{b}{a}\right) + c_0 = 0.$$

Multiplying by a^n, we obtain

$$b^n + c_{n-1}b^{n-1}a + \cdots + c_1ba^{n-1} + c_0a^n = 0.$$

Thus

$$b^n = -a\left(c_{n-1}b^{n-1} + c_{n-2}b^{n-2}a + \cdots + c_1ba^{n-2} + c_0a^{n-1}\right).$$

Hence

$$a \mid b^n.$$

If a is not a unit, then a has a prime divisor p. Since $a \mid b^n$, we have

$$p \mid b^n.$$

Because p is prime in the UFD R, it follows that

$$p \mid b.$$

But $p \mid a$ as well, contradicting the assumption that $\gcd(a, b) = 1$.

Therefore a must be a unit in R. Hence

$$\alpha = \frac{b}{a} \in R.$$

So every element of K integral over R belongs to R. Therefore R is integrally closed. □

8.5 Valuations

Definition 8.5.1 (Valuation)**.** *Let K be a field. A* valuation *on K is a function*

$$v : K^\times \longrightarrow \mathbb{R},$$

where $K^\times = K \setminus \{0\}$, satisfying the following conditions:

1. *For all $a, b \in K^\times$,*
$$v(ab) = v(a) + v(b).$$

2. *For all $a, b \in K^\times$ with $a + b \neq 0$,*
$$v(a + b) \geq \min\{v(a), v(b)\}.$$

We extend v to K by setting

$$v(0) = +\infty.$$

With this convention, the second condition can be written as

$$v(a+b) \geq \min\{v(a), v(b)\}$$

for all $a, b \in K$.

Proposition 8.5.2. *Let v be a valuation on a field K. Then*

$$v : (K^\times, \cdot) \longrightarrow (\mathbb{R}, +)$$

is a group homomorphism.

Proof. By definition, for all $a, b \in K^\times$,

$$v(ab) = v(a) + v(b).$$

Also,

$$0 = v(1) = v(aa^{-1}) = v(a) + v(a^{-1}),$$

so

$$v(a^{-1}) = -v(a).$$

Therefore v is a group homomorphism from $K^\times$ to the additive group $(\mathbb{R}, +)$. $\square$

Definition 8.5.3 (Value group)**.** *Let (K, v) be a valued field. The subgroup*

$$v(K^\times) \subseteq (\mathbb{R}, +)$$

is called the value group *of v.*

A field K together with a valuation v is called a valued field, *and is denoted by*

$$(K, v).$$

Proposition 8.5.4 (Basic properties of valuations)**.** *Let v be a valuation on a field K. Then for all $a, b \in K^\times$, the following hold:*

1. $$v(1) = 0, \qquad v(-1) = 0.$$

2. $$v(-a) = v(a).$$

3. $$v(a^{-1}) = -v(a).$$

4. *If*
$$v(b) < v(a),$$

then
$$v(a+b) = v(b).$$

Proof. First,
$$v(1) = v(1 \cdot 1) = v(1) + v(1),$$

so
$$v(1) = 0.$$

Since $(-1)^2 = 1$, we have
$$2v(-1) = v((-1)^2) = v(1) = 0,$$

and hence
$$v(-1) = 0.$$

For $a \in K^\times$,
$$v(-a) = v((-1)a) = v(-1) + v(a) = v(a).$$

Also,
$$0 = v(1) = v(aa^{-1}) = v(a) + v(a^{-1}),$$

so
$$v(a^{-1}) = -v(a).$$

Finally, suppose that $v(b) < v(a)$. By the valuation inequality,

$$v(a+b) \geq \min\{v(a), v(b)\} = v(b).$$

We claim that equality holds. If instead

$$v(a+b) > v(b),$$

then using
$$b = (a+b) - a,$$

we get
$$v(b) \geq \min\{v(a+b), v(a)\}.$$

But both $v(a+b)$ and $v(a)$ are strictly larger than $v(b)$, so the right-hand side is strictly larger than $v(b)$, a contradiction. Therefore

$$v(a+b) = v(b).$$

□

Chapter 9

Lecture 9

9.1 Local rings

Definition 9.1.1 (Local ring)**.** *A ring R is called* local *if the set of non-units of R forms an ideal of R.*

Theorem 9.1.2. *Let R be a ring with identity. The following conditions are equivalent:*

1. *R is local.*
2. *The set of non-units of R coincides with $J(R)$. Equivalently, $J(R)$ is the unique maximal left ideal of R and also the unique maximal right ideal of R.*
3. *$R/J(R)$ is a division ring.*

Proof. Let M denote the set of non-units of R.

(1) ⇒ (2). Assume that R is local. By definition, M is an ideal of R. Since $1 \notin M$, the ideal M is proper.

We claim that M contains every proper left ideal of R. Indeed, let L be a proper left ideal of R. If L contained a unit u, then $1 = u^{-1}u \in L$, and hence $L = R$, a contradiction. Therefore every element of L is a non-unit, so

$$L \subseteq M.$$

Thus M is the unique maximal left ideal of R.

Similarly, M contains every proper right ideal of R, and hence M is also the unique maximal right ideal of R.

Since the Jacobson radical is the intersection of all maximal left ideals, we get

$$J(R) = M.$$

Hence the set of non-units of R coincides with $J(R)$.

(2) ⇒ (3). Assume that the set of non-units of R is exactly $J(R)$. Let

$$\overline{a} = a + J(R) \in R/J(R)$$

be nonzero. Then $a \notin J(R)$, so by assumption a is a unit in R. Hence there exists $a^{-1} \in R$ such that

$$aa^{-1} = a^{-1}a = 1.$$

Passing to the quotient gives

$$\overline{a}\,\overline{a^{-1}} = \overline{a^{-1}}\,\overline{a} = \overline{1}.$$

Thus every nonzero element of $R/J(R)$ is invertible, so $R/J(R)$ is a division ring.

(3) ⇒ (1). Assume that $R/J(R)$ is a division ring. We show that the set of non-units of R is precisely $J(R)$.

First, every element of $J(R)$ is a non-unit, because $J(R)$ is contained in every maximal left ideal, hence is a proper ideal.

Conversely, suppose $a \notin J(R)$. Then

$$\overline{a} = a + J(R) \neq 0$$

in $R/J(R)$. Since $R/J(R)$ is a division ring, $\overline{a}$ is invertible. Hence there exist $b, d \in R$ such that

$$\overline{b}\,\overline{a} = \overline{1}, \qquad \overline{a}\,\overline{d} = \overline{1}.$$

Equivalently, there exist $c, c' \in J(R)$ such that

$$ba = 1 - c, \qquad ad = 1 - c'.$$

Since $c, c' \in J(R)$, the elements $1 - c$ and $1 - c'$ are units in R. Therefore ba is a unit and ad is a unit.

Since ba is a unit, a has a left inverse. Indeed,

$$(ba)^{-1}ba = 1,$$

so

$$\big((ba)^{-1}b\big)a = 1.$$

Similarly, since ad is a unit, a has a right inverse:

$$a\big(d(ad)^{-1}\big) = 1.$$

Thus a has both a left inverse and a right inverse, hence a is a unit.

Therefore every element outside $J(R)$ is a unit, so the set of non-units is exactly $J(R)$. Since $J(R)$ is an ideal, the set of non-units forms an ideal. Hence R is local. $\square$

Theorem 9.1.3 (Idempotents in a local ring)**.** *Let R be a local ring. Then R has no non-trivial idempotents. In other words, if $e \in R$ satisfies*

$$e^2 = e,$$

then

$$e = 0 \qquad \text{or} \qquad e = 1.$$

Proof. Let $e \in R$ be an idempotent, so

$$e^2 = e.$$

Then

$$e(1-e) = e - e^2 = 0.$$

We first observe that if e is a unit, then $e = 1$. Indeed, multiplying $e^2 = e$ by e^{-1} gives

$$e = 1.$$

Similarly, if $1 - e$ is a unit, then from

$$e(1-e) = 0$$

we get

$$e = 0.$$

Therefore, if $e \neq 0$ and $e \neq 1$, then neither e nor $1 - e$ is a unit. Since R is local, the set of non-units of R is precisely the Jacobson radical $J(R)$. Hence

$$e \in J(R), \qquad 1 - e \in J(R).$$

Because $J(R)$ is an ideal, it follows that

$$1 = e + (1-e) \in J(R).$$

This is impossible, since $J(R)$ is a proper ideal of R.

Hence our assumption $e \neq 0, 1$ is false. Therefore every idempotent of R is either 0 or 1. □

9.2 Additive valuations, discrete valuations, and DVRs

Definition 9.2.1 (Valuation ring and valuation ideal)**.** *Let (K, v) be a valued field. Define*

$$R_v := \{a \in K \mid v(a) \geq 0\}$$

and

$$P_v := \{a \in K \mid v(a) > 0\}.$$

Then R_v *is called the* valuation ring *of* v*, and* P_v *is called the* valuation ideal *of* v*.*

Proposition 9.2.2. *Let* (K, v) *be a valued field. Then:*

1. R_v *is a subring of* K*.*

2. P_v *is an ideal of* R_v*.*

3. *The group of units of* R_v *is*

$$R_v^\times = \{a \in K^\times \mid v(a) = 0\}.$$

4. *One has*

$$K = R_v \cup R_v^{-1},$$

where

$$R_v^{-1} := \{a^{-1} \mid a \in R_v,\ a \neq 0\}.$$

In particular, K *is the field of fractions of* R_v*.*

5. R_v *is a local ring and*

$$J(R_v) = P_v.$$

Equivalently, P_v *is precisely the set of non-units of* R_v*.*

6. *The quotient field*

$$\kappa(v) := R_v/P_v$$

is called the residue field *of* v*.*

Proof. First, $0, 1 \in R_v$. If $a, b \in R_v$, then

$$v(ab) = v(a) + v(b) \geq 0,$$

so $ab \in R_v$. Also,

$$v(a + b) \geq \min\{v(a), v(b)\} \geq 0,$$

so $a + b \in R_v$. Finally, since $v(-a) = v(a)$, we have $-a \in R_v$. Therefore R_v is a subring of K.

Now let $a, b \in P_v$. Then

$$v(a + b) \geq \min\{v(a), v(b)\} > 0,$$

so $a + b \in P_v$. If $r \in R_v$ and $a \in P_v$, then

$$v(ra) = v(r) + v(a) > 0.$$

Thus $ra \in P_v$, and hence P_v is an ideal of R_v.

Next, let $a \in R_v$. If a is a unit of R_v, then $a^{-1} \in R_v$, so

$$0 = v(1) = v(aa^{-1}) = v(a) + v(a^{-1}).$$

Since $v(a) \geq 0$ and $v(a^{-1}) \geq 0$, we get $v(a) = 0$. Conversely, if $v(a) = 0$, then

$$v(a^{-1}) = -v(a) = 0,$$

so $a^{-1} \in R_v$. Therefore

$$R_v^{\times} = \{a \in K^{\times} \mid v(a) = 0\}.$$

For every $a \in K^{\times}$, either $v(a) \geq 0$, in which case $a \in R_v$, or $v(a) < 0$, in which case

$$v(a^{-1}) = -v(a) > 0,$$

so $a^{-1} \in R_v$. Hence

$$K = R_v \cup R_v^{-1}.$$

In particular, every element of K is a fraction of elements of R_v, so

$$K = \operatorname{Frac}(R_v).$$

Finally, the non-units of R_v are exactly the elements $a \in R_v$ satisfying $v(a) > 0$, namely the elements of P_v. Hence R_v has a unique maximal ideal P_v, so R_v is local and

$$J(R_v) = P_v.$$

Consequently,

$$\kappa(v) := R_v/P_v$$

is a field. $\square$

Definition 9.2.3 (Equivalent valuations)**.** *Let v be a valuation on K, and let $\lambda \in \mathbb{R}_{>0}$. Define*

$$v_\lambda(a) := \lambda v(a), \qquad a \in K^{\times}.$$

Then v_λ is again a valuation on K. We say that v is equivalent *to v_λ, and write*

$$v \sim v_\lambda.$$

Proposition 9.2.4. *Equivalent valuations have the same valuation ring, valuation ideal, group of units, and residue field.*

Proof. Let $v_\lambda = \lambda v$, where $\lambda > 0$. Then for $a \in K^{\times}$,

$$v_\lambda(a) \geq 0 \iff \lambda v(a) \geq 0 \iff v(a) \geq 0.$$

Hence

$$R_{v_\lambda} = R_v.$$

Similarly,

$$v_\lambda(a) > 0 \iff v(a) > 0,$$

so

$$P_{v_\lambda} = P_v.$$

Also,

$$v_\lambda(a) = 0 \iff v(a) = 0,$$

and therefore

$$R_{v_\lambda}^\times = R_v^\times.$$

Since the valuation rings and valuation ideals are the same, the residue fields are also the same:

$$R_{v_\lambda}/P_{v_\lambda} = R_v/P_v.$$

□

Definition 9.2.5 (Discrete valuation)**.** *A valuation v of K is called* discrete *if its value group is isomorphic to $\mathbb{Z}$, that is,*

$$v(K^\times) \cong \mathbb{Z}.$$

If moreover

$$v(K^\times) = \mathbb{Z},$$

then v is called a normalized discrete valuation*.*

Remark 9.2.6. *If v is a discrete valuation, then after replacing v by an equivalent valuation λv for a suitable $\lambda > 0$, one may assume that*

$$v(K^\times) = \mathbb{Z}.$$

Thus every discrete valuation is equivalent to a normalized discrete valuation.

Definition 9.2.7. *An element $\pi \in R$ satisfying*

$$v(\pi) = 1$$

is called a uniformizer.

Theorem 9.2.8. *Let (K, v) be a field with a normalized discrete valuation. Let*

$$R = \{x \in K \mid v(x) \geq 0\}$$

be its valuation ring, and let

$$P = \{x \in R \mid v(x) > 0\}$$

be its maximal ideal. Suppose $\pi \in R$ satisfies $v(\pi) = 1$. Then:

1. *For every $\alpha \in K^\times$, one can write*

$$\alpha = \pi^{v(\alpha)} u$$

for some unit $u \in R^\times$.

2. *Every nonzero ideal I of R is of the form*

$$I = (\pi^i)$$

for some integer $i \geq 0$. In particular,

$$P = (\pi), \qquad I = P^i,$$

In particular, R is a principal ideal domain.

3. *The ring R is integrally closed.*

Proof. **(1).** Let $\alpha \in K^\times$, and set

$$i = v(\alpha).$$

Since $v(\pi) = 1$, we have

$$v(\pi^{-i}\alpha) = v(\alpha) - iv(\pi) = i - i = 0.$$

Therefore

$$u := \pi^{-i}\alpha \in R^\times.$$

Hence

$$\alpha = \pi^i u = \pi^{v(\alpha)} u,$$

as desired.

(2). Let $I \neq 0$ be an ideal of R. Since $I \subseteq R$, for every nonzero $a \in I$ we have

$$v(a) \geq 0.$$

Thus the set

$$\{v(a) \mid 0 \neq a \in I\} \subseteq \mathbb{Z}_{\geq 0}$$

is nonempty, and therefore has a minimal element. Let

$$i = \min\{v(a) \mid 0 \neq a \in I\}.$$

Choose $a_0 \in I$ such that
$$v(a_0) = i.$$
By part (1), we may write
$$a_0 = \pi^i u$$
for some $u \in R^\times$. Hence
$$\pi^i = a_0 u^{-1} \in I.$$
Therefore
$$(\pi^i) \subseteq I.$$

Conversely, let $a \in I$ be nonzero. Write
$$v(a) = j.$$
By the minimality of i, we have $j \geq i$. Again by part (1), there exists $u \in R^\times$ such that
$$a = \pi^j u.$$
Then
$$a = \pi^i \big(\pi^{j-i} u\big).$$
Since $j - i \geq 0$, we have $\pi^{j-i} u \in R$. Therefore
$$a \in (\pi^i).$$
Hence
$$I \subseteq (\pi^i).$$
Combining the two inclusions, we obtain
$$I = (\pi^i).$$

In particular, applying this to the maximal ideal P, we get
$$P = (\pi).$$
Therefore every nonzero ideal of R is of the form
$$I = (\pi^i) = P^i.$$
Since the zero ideal is also principal, R is a principal ideal domain.

(3). By part (2), the ring R is a principal ideal domain. Hence R is a unique factorization domain. Since every unique factorization domain is integrally closed, it follows that R is integrally closed. $\square$

Corollary 9.2.9. *Let v and v' be two discrete valuations on a field K. Then v and v' are equivalent if and only if they have the same valuation ideal, that is,*

$$P_v = P_{v'},$$

where

$$P_v = \{x \in K \mid v(x) > 0\}, \qquad P_{v'} = \{x \in K \mid v'(x) > 0\}.$$

Proof. If v and v' are equivalent, then there exists $\lambda \in \mathbb{R}_{>0}$ such that

$$v' = \lambda v.$$

Hence, for every $x \in K$,

$$v'(x) > 0 \iff \lambda v(x) > 0 \iff v(x) > 0.$$

Therefore

$$P_v = P_{v'}.$$

Conversely, suppose that

$$P_v = P_{v'} =: P.$$

We first show that the valuation ring can be recovered from the valuation ideal. Namely,

$$R_v = K \setminus P^{-1},$$

where

$$P^{-1} := \{a^{-1} \mid a \in P,\ a \neq 0\}.$$

Indeed, if $x \in R_v$, then $v(x) \geq 0$. Hence x cannot be of the form a^{-1} with $a \in P$, because then

$$v(x) = v(a^{-1}) = -v(a) < 0.$$

Thus $x \notin P^{-1}$. Conversely, if $x \notin R_v$, then $x \neq 0$ and $v(x) < 0$. Hence

$$v(x^{-1}) = -v(x) > 0,$$

so $x^{-1} \in P$, and therefore

$$x = (x^{-1})^{-1} \in P^{-1}.$$

This proves

$$R_v = K \setminus P^{-1}.$$

Since $P_v = P_{v'}$, we get

$$R_v = R_{v'} =: R.$$

Now replace v and v' by equivalent normalized discrete valuations. This does not change the valuation ring or the valuation ideal. Hence we may assume that

$$v(K^\times) = \mathbb{Z}, \qquad v'(K^\times) = \mathbb{Z}.$$

Choose $\pi \in K^\times$ such that

$$v(\pi) = 1.$$

Then π is a uniformizer for v, so by Theorem 9.0.11,

$$P = (\pi)$$

as an ideal of R.

Let π' be a uniformizer for v'. Then similarly

$$P = (\pi').$$

Since

$$(\pi) = (\pi'),$$

there exist $r, s \in R$ such that

$$\pi = r\pi', \qquad \pi' = s\pi.$$

Applying v', we obtain

$$v'(\pi) = v'(r) + v'(\pi') \geq 1,$$

and

$$1 = v'(\pi') = v'(s) + v'(\pi) \geq v'(\pi).$$

Therefore

$$v'(\pi) = 1.$$

Now let $a \in K^\times$. By Theorem 9.0.11, there exists $u \in R^\times$ such that

$$a = \pi^{v(a)}u.$$

Since $R = R_{v'}$, the units of R are exactly the elements of valuation 0 with respect to v'. Hence

$$v'(u) = 0.$$

Therefore

$$v'(a) = v\,(a)\,v'(\pi) + v'(u) = v(a).$$

Thus $v' = v$ after normalization.

Hence the original discrete valuations differ by multiplication by a positive real constant. Therefore v and v' are equivalent. □

Theorem 9.2.10 (Discrete valuation rings and local PIDs)**.** *Let R be an integral domain which is not a field, and let $K = \mathrm{Frac}(R)$. Then R is a discrete valuation* *ring if and only if R is a local principal ideal domain.*

Proof. Suppose first that R is a discrete valuation ring. Then there exists a normalized discrete valuation

$$v : K^\times \longrightarrow \mathbb{Z}$$

such that

$$R = \{x \in K \mid v(x) \geq 0\}.$$

By the structure theorem for valuation rings of normalized discrete valuations, R is a local principal ideal domain. More precisely, if $\pi \in R$ satisfies $v(\pi) = 1$, then the maximal ideal of R is

$$P = \{x \in R \mid v(x) > 0\} = (\pi),$$

and every nonzero ideal of R is of the form

$$P^n = (\pi^n)$$

for some integer $n \geq 0$. Hence R is a local PID.

Conversely, suppose that R is a local principal ideal domain. Since R is not a field, its unique maximal ideal is nonzero. Denote it by

$$P = J(R).$$

Since R is a PID, there exists a nonzero nonunit $\pi \in R$ such that

$$P = (\pi).$$

We first claim that

$$\bigcap_{n \geq 1} P^n = 0.$$

Suppose, for contradiction, that

$$0 \neq a \in \bigcap_{n \geq 1} P^n.$$

Since $P^n = (\pi^n)$, for each $n \geq 1$ there exists $a_n \in R$ such that

$$a = \pi^n a_n.$$

Then

$$\pi^n a_n = \pi^{n+1} a_{n+1}.$$

Since R is an integral domain and $\pi \neq 0$, we may cancel π^n, obtaining

$$a_n = \pi a_{n+1}.$$

Hence

$$(a_n) \subseteq (a_{n+1}).$$

This inclusion is strict. Indeed, if $(a_n) = (a_{n+1})$, then $a_{n+1} = ra_n$ for some $r \in R$. Hence

$$a_{n+1} = r\pi a_{n+1}.$$

Since $a_{n+1} \neq 0$, we get $1 = r\pi$, which means that π is a unit, a contradiction. Thus we obtain a strictly ascending chain of ideals

$$(a_1) \subsetneq (a_2) \subsetneq (a_3) \subsetneq \cdots .$$

This contradicts the fact that R is Noetherian, because every PID is Noetherian. Therefore

$$\bigcap_{n \geq 1} P^n = 0.$$

Now let $0 \neq a \in R$. Since $P^0 = R$, the set

$$\{n \geq 0 \mid a \in P^n\}$$

is nonempty. By the claim above, a cannot belong to all powers P^n. Hence there exists a unique integer $n \geq 0$ such that

$$a \in P^n \qquad \text{and} \qquad a \notin P^{n+1}.$$

Since $P^n = (\pi^n)$, we may write

$$a = \pi^n u$$

for some $u \in R$. If $u \in P$, then

$$a = \pi^n u \in \pi^n P = P^{n+1},$$

contradicting the choice of n. Thus $u \notin P$. Since R is local and P is the set of nonunits, it follows that

$$u \in R^\times.$$

Therefore every nonzero element $a \in R$ can be written uniquely in the form

$$a = \pi^n u, \qquad n \geq 0, \quad u \in R^\times.$$

We now extend this to $K^\times$. Every $x \in K^\times$ can be written as

$$x = \frac{a}{b}, \qquad a, b \in R \setminus \{0\}.$$

Write

$$a = \pi^m u, \qquad b = \pi^n w,$$

where $m, n \geq 0$ and $u, w \in R^\times$. Then

$$x = \pi^{m-n} u w^{-1}.$$

Hence every $x \in K^\times$ has a representation

$$x = \pi^r u, \qquad r \in \mathbb{Z}, \quad u \in R^\times.$$

This representation is unique. Indeed, if

$$\pi^r u = \pi^s w, \qquad u, w \in R^\times,$$

and say $r < s$, then

$$u = \pi^{s-r} w \in P,$$

contradicting that u is a unit. Hence $r = s$.

Define

$$v : K^\times \longrightarrow \mathbb{Z}$$

by

$$v(\pi^r u) = r, \qquad r \in \mathbb{Z}, \quad u \in R^\times.$$

This is well-defined by the uniqueness just proved.

We verify that v is a valuation. If

$$x = \pi^r u, \qquad y = \pi^s w,$$

with $u, w \in R^\times$, then

$$xy = \pi^{r+s} uw,$$

and $uw \in R^\times$. Therefore

$$v(xy) = r + s = v(x) + v(y).$$

Next, suppose $x + y \neq 0$. Without loss of generality, assume $r \leq s$. Then

$$x + y = \pi^r \left(u + \pi^{s-r} w \right).$$

Since

$$u + \pi^{s-r} w \in R,$$

its valuation is nonnegative whenever it is nonzero. Thus

$$v(x + y) \geq r = \min\{r, s\} = \min\{v(x), v(y)\}.$$

Hence v is a valuation.

Moreover,

$$v(\pi) = 1,$$

so

$$v(K^\times) = \mathbb{Z}.$$

Therefore v is a normalized discrete valuation.

Finally, we show that R is exactly the valuation ring of v. Let

$$R_v := \{x \in K \mid v(x) \geq 0\}.$$

If $x \in R \setminus \{0\}$, then by the decomposition above,

$$x = \pi^n u$$

with $n \geq 0$, so $v(x) = n \geq 0$. Hence $R \subseteq R_v$.

Conversely, if $x \in R_v \setminus \{0\}$, write

$$x = \pi^r u, \qquad r \in \mathbb{Z}, \quad u \in R^\times.$$

Since $v(x) = r \geq 0$, we have

$$x = \pi^r u \in R.$$

Hence $R_v \subseteq R$.

Therefore

$$R = R_v.$$

Thus R is the valuation ring of a normalized discrete valuation, and hence R is a discrete valuation ring. □

9.3 Multiplicative valuations

Definition 9.3.1 (Multiplicative valuation associated to a discrete valuation)**.** *Let K be a field, and let*

$$v : K^\times \longrightarrow \mathbb{Z}$$

be a normalized discrete valuation. We extend v to K by setting

$$v(0) = +\infty.$$

Fix a real number

$$0 < \gamma < 1.$$

Define a function

$$\varphi : K \longrightarrow \mathbb{R}_{\geq 0}$$

by

$$\varphi(a) = \begin{cases} \gamma^{v(a)}, & a \neq 0, \\ 0, & a = 0. \end{cases}$$

Then φ is called the multiplicative valuation associated to v with parameter γ.

Proposition 9.3.2. *The function*

$$\varphi : K \longrightarrow \mathbb{R}_{\geq 0}$$

defined above satisfies the following properties:

1. $\varphi(a) = 0$ *if and only if* $a = 0$.
2. *For all* $a, b \in K$,
$$\varphi(ab) = \varphi(a)\varphi(b).$$
3. *For all* $a, b \in K$,
$$\varphi(a + b) \leq \max\{\varphi(a), \varphi(b)\}.$$

Proof. The first assertion follows immediately from the definition, because

$$\gamma^{v(a)} > 0$$

whenever $a \neq 0$, while $\varphi(0) = 0$.

For the multiplicativity, if $a, b \neq 0$, then

$$v(ab) = v(a) + v(b).$$

Hence

$$\varphi(ab) = \gamma^{v(ab)} = \gamma^{v(a)+v(b)} = \gamma^{v(a)}\gamma^{v(b)} = \varphi(a)\varphi(b).$$

If $a = 0$ or $b = 0$, both sides are zero, so the formula still holds.

Finally, assume first that $a + b \neq 0$. Since v is a valuation,

$$v(a + b) \geq \min\{v(a), v(b)\}.$$

Because $0 < \gamma < 1$, the function $x \mapsto \gamma^x$ is decreasing. Therefore

$$\gamma^{v(a+b)} \leq \gamma^{\min\{v(a),v(b)\}} = \max\{\gamma^{v(a)}, \gamma^{v(b)}\}.$$

Thus

$$\varphi(a + b) \leq \max\{\varphi(a), \varphi(b)\}.$$

If $a + b = 0$, then $\varphi(a + b) = 0$, and the same inequality is obvious. □

Definition 9.3.3 (Multiplicative valuation)**.** *Let* K *be a field. A function*

$$\varphi : K \longrightarrow \mathbb{R}_{\geq 0}$$

is called a multiplicative valuation, or a non-archimedean absolute value, if it satisfies:

1. $\varphi(a) = 0$ *if and only if* $a = 0$;

2. $$\varphi(ab) = \varphi(a)\varphi(b)$$

   for all $a, b \in K$*;*

3. $$\varphi(a + b) \le \max\{\varphi(a), \varphi(b)\}$$

   for all $a, b \in K$*.*

Proposition 9.3.4 (Basic properties of multiplicative valuations)**.** *Let*

$$\varphi : K \longrightarrow \mathbb{R}_{\ge 0}$$

be a multiplicative valuation. Then for all $a, b \in K^\times$*, the following hold:*

1. $$\varphi(1) = 1, \qquad \varphi(-1) = 1.$$

2. $$\varphi(-a) = \varphi(a).$$

3. $$\varphi(a^{-1}) = \varphi(a)^{-1}.$$

4. *If*

   $$\varphi(a) < \varphi(b),$$

   then

   $$\varphi(a + b) = \varphi(b).$$

Proof. Since $1 \cdot 1 = 1$, we have

$$\varphi(1) = \varphi(1)^2.$$

Since $\varphi(1) \neq 0$, it follows that

$$\varphi(1) = 1.$$

Also, since $(-1)^2 = 1$, we get

$$\varphi(-1)^2 = \varphi(1) = 1.$$

Since $\varphi(-1) > 0$, we have

$$\varphi(-1) = 1.$$

Hence, for $a \in K^\times$,

$$\varphi(-a) = \varphi((-1)a) = \varphi(-1)\varphi(a) = \varphi(a).$$

Also,

$$1 = \varphi(1) = \varphi(aa^{-1}) = \varphi(a)\varphi(a^{-1}),$$

so

$$\varphi(a^{-1}) = \varphi(a)^{-1}.$$

Finally, suppose that

$$\varphi(a) < \varphi(b).$$

By the non-archimedean triangle inequality,

$$\varphi(a+b) \leq \max\{\varphi(a), \varphi(b)\} = \varphi(b).$$

On the other hand,

$$b = (a+b) - a.$$

Hence

$$\varphi(b) \leq \max\{\varphi(a+b), \varphi(a)\}.$$

If $\varphi(a+b) < \varphi(b)$, then both $\varphi(a+b)$ and $\varphi(a)$ are strictly smaller than $\varphi(b)$, contradicting the above inequality. Therefore

$$\varphi(a+b) = \varphi(b).$$

□

Definition 9.3.5 (Valuation ring and valuation ideal)**.** *Let φ be a multiplicative valuation on K. Define*

$$R_\varphi := \{a \in K \mid \varphi(a) \leq 1\},$$

and

$$P_\varphi := \{a \in K \mid \varphi(a) < 1\}.$$

Then R_φ is called the valuation ring of φ, and P_φ is called the valuation ideal of φ.

Proposition 9.3.6. *Let φ be a multiplicative valuation on K. Then R_φ is a subring of K, and P_φ is an ideal of R_φ.*

Proof. We first show that R_φ is a subring. Clearly,

$$\varphi(0) = 0 \leq 1, \qquad \varphi(1) = 1,$$

so $0, 1 \in R_\varphi$. If $a, b \in R_\varphi$, then

$$\varphi(ab) = \varphi(a)\varphi(b) \leq 1,$$

hence $ab \in R_\varphi$. Also,

$$\varphi(-a) = \varphi(a) \leq 1,$$

so $-a \in R_\varphi$. Finally,

$$\varphi(a+b) \leq \max\{\varphi(a), \varphi(b)\} \leq 1,$$

so $a + b \in R_\varphi$. Therefore R_φ is a subring of K.

Next we show that P_φ is an ideal of R_φ. If $a, b \in P_\varphi$, then

$$\varphi(a+b) \leq \max\{\varphi(a), \varphi(b)\} < 1,$$

so $a+b \in P_\varphi$. Also, if $r \in R_\varphi$ and $a \in P_\varphi$, then

$$\varphi(ra) = \varphi(r)\varphi(a) < 1.$$

Hence $ra \in P_\varphi$. Thus P_φ is an ideal of R_φ. $\square$

Remark 9.3.7. *If $\varphi(a) = \gamma^{v(a)}$ for a normalized discrete valuation v, where $0 < \gamma < 1$, then*

$$R_\varphi = \{a \in K \mid \varphi(a) \leq 1\} = \{a \in K \mid v(a) \geq 0\} = R_v,$$

and

$$P_\varphi = \{a \in K \mid \varphi(a) < 1\} = \{a \in K \mid v(a) > 0\} = P_v.$$

Proposition 9.3.8 (From multiplicative valuations to additive valuations)**.** *Let φ be a multiplicative valuation on K, and fix*

$$0 < \gamma < 1.$$

Define

$$v_\varphi : K^\times \longrightarrow \mathbb{R}$$

by

$$v_\varphi(a) := \log_\gamma \varphi(a).$$

Then v_φ is an additive valuation on K, that is,

$$v_\varphi(ab) = v_\varphi(a) + v_\varphi(b),$$

and

$$v_\varphi(a+b) \geq \min\{v_\varphi(a), v_\varphi(b)\}$$

whenever $a + b \neq 0$.

Proof. For $a, b \in K^\times$, we have

$$\varphi(ab) = \varphi(a)\varphi(b).$$

Taking logarithms with base γ, we get

$$v_\varphi(ab) = \log_\gamma \varphi(ab) = \log_\gamma \varphi(a) + \log_\gamma \varphi(b) = v_\varphi(a) + v_\varphi(b).$$

Now suppose $a + b \neq 0$. Since φ is non-archimedean,

$$\varphi(a+b) \leq \max\{\varphi(a), \varphi(b)\}.$$

Because $0 < \gamma < 1$, the function $\log_\gamma(x)$ is decreasing. Therefore

$$\log_\gamma \varphi(a+b) \geq \log_\gamma \max\{\varphi(a), \varphi(b)\}.$$

But

$$\log_\gamma \max\{\varphi(a), \varphi(b)\} = \min\{\log_\gamma \varphi(a), \log_\gamma \varphi(b)\}.$$

Hence

$$v_\varphi(a+b) \geq \min\{v_\varphi(a), v_\varphi(b)\}.$$

Thus v_φ is an additive valuation. $\square$

Definition 9.3.9 (Value group)**.** *Let φ be a multiplicative valuation on K. Then*

$$\varphi(K^\times) \subseteq \mathbb{R}_{>0}$$

is a multiplicative subgroup of $\mathbb{R}_{>0}$. It is called the value group of φ.

Definition 9.3.10 (Equivalent multiplicative valuations)**.** *Let K be a field, and let*

$$\varphi, \psi : K \longrightarrow \mathbb{R}_{\geq 0}$$

be two multiplicative valuations on K. We say that φ and ψ are equivalent if there exists a positive real number

$$\gamma \in \mathbb{R}_{>0}$$

such that

$$\psi(a) = \varphi(a)^\gamma \qquad \text{for all } a \in K.$$

Definition 9.3.11 (Metric induced by a multiplicative valuation)**.** *Let φ be a multiplicative valuation on a field K. Define*

$$d : K \times K \longrightarrow \mathbb{R}_{\geq 0}$$

by

$$d(a, b) = \varphi(a - b).$$

Then d is a metric on K.

Proposition 9.3.12. *Let K be a field equipped with a multiplicative valuation φ. Then the function*

$$d(a, b) = \varphi(a - b)$$

defines a metric on K. In particular, K is a Hausdorff topological space with respect to the topology induced by d.

Proof. We verify the axioms of a metric.

First, for any $a, b \in K$, we have

$$d(a,b) = \varphi(a-b) \geq 0.$$

Moreover,

$$d(a,b) = 0 \Longleftrightarrow \varphi(a-b) = 0 \Longleftrightarrow a-b = 0 \Longleftrightarrow a = b.$$

Next, since $\varphi(-1) = 1$, we have

$$d(a,b) = \varphi(a-b) = \varphi\big(-(b-a)\big) = \varphi(b-a) = d(b,a).$$

Finally, for any $a, b, c \in K$, the valuation inequality gives

$$d(a,c) = \varphi(a-c) = \varphi\big((a-b)+(b-c)\big) \leq \varphi(a-b) + \varphi(b-c).$$

Hence

$$d(a,c) \leq d(a,b) + d(b,c).$$

Therefore d is a metric on K. Since every metric space is Hausdorff, the field K is Hausdorff with respect to the topology induced by d. $\square$

9.4 Completion of valued fields

Definition 9.4.1 (Cauchy sequence)**.** *Let K be a field equipped with a metric*

$$d : K \times K \longrightarrow \mathbb{R}_{\geq 0}.$$

A sequence

$$(a_n)_{n\geq 1} = (a_1, a_2, \ldots, a_n, \ldots)$$

of elements of K is called a Cauchy sequence if for every

$$\varepsilon > 0,$$

there exists an integer $N \geq 1$ such that

$$d(a_m, a_n) < \varepsilon \qquad \text{for all } m, n \geq N.$$

Definition 9.4.2 (Convergent sequence)**.** *Let K be a field equipped with a metric d. A sequence*

$$(a_n)_{n\geq 1}$$

in K is said to converge to an element $a \in K$ if

$$\lim_{n\to\infty} d(a_n, a) = 0.$$

In this case, we write

$$\lim_{n\to\infty} a_n = a.$$

If the metric d is induced by a multiplicative valuation φ, namely

$$d(x, y) = \varphi(x - y),$$

then the above condition is equivalently

$$\lim_{n\to\infty} \varphi(a_n - a) = 0.$$

Definition 9.4.3 (Complete valued field)**.** *Let K be a field equipped with a metric d, or equivalently with a valuation inducing d. We say that K is complete if every Cauchy sequence in K converges to an element of K.*

If K is complete with respect to the metric induced by a valuation φ, then K is also called a complete valued field.

Definition 9.4.4 (Complete valuation ring)**.** *Let (K, φ) be a valued field, and let*

$$R_\varphi = \{x \in K \mid \varphi(x) \leq 1\}$$

be its valuation ring in the multiplicative convention. If K is complete with respect to the metric induced by φ, then R_φ is called a complete valuation ring.

Definition 9.4.5 (Ring of Cauchy sequences)**.** *Let (K, φ) be a valued field, and let*

$$d(a, b) = \varphi(a - b)$$

be the metric induced by φ. Denote by $\mathcal{C}$ the set of all Cauchy sequences in K:

$$\mathcal{C} := \{(a_n)_{n\geq 1} \mid (a_n)_{n\geq 1} \text{ is a Cauchy sequence in } K\}.$$

For $(a_n), (b_n) \in \mathcal{C}$, define

$$(a_n) + (b_n) := (a_n + b_n),$$

and

$$(a_n)(b_n) := (a_n b_n).$$

With these operations, $\mathcal{C}$ is a commutative ring.

Definition 9.4.6 (Null sequences)**.** *Let*

$$\mathcal{I} := \left\{(a_n) \in \mathcal{C} \;\middle|\; \lim_{n\to\infty} a_n = 0\right\}.$$

Equivalently,

$$(a_n) \in \mathcal{I} \iff \lim_{n\to\infty} \varphi(a_n) = 0.$$

The elements of $\mathcal{I}$ are called null sequences.

Proposition 9.4.7. *The subset $\mathcal{I}$ is a maximal ideal of $\mathcal{C}$. Consequently, the quotient*

$$\widehat{K} := \mathcal{C}/\mathcal{I}$$

is a field.

Proof. It is straightforward to check that $\mathcal{I}$ is an ideal of $\mathcal{C}$.

To show that $\mathcal{I}$ is maximal, let

$$(a_n) \in \mathcal{C} \setminus \mathcal{I}.$$

Then (a_n) does not converge to 0. Since (a_n) is Cauchy, there exist $N \geq 1$ and $\varepsilon > 0$ such that

$$\varphi(a_n) \geq \varepsilon \qquad \text{for all } n \geq N.$$

Hence $a_n \neq 0$ for all sufficiently large n. Define a sequence

$$b_n = \begin{cases} 0, & n < N, \\ a_n^{-1}, & n \geq N. \end{cases}$$

Then (b_n) is again a Cauchy sequence in K, and

$$(a_n)(b_n) - 1 \in \mathcal{I}.$$

Therefore the class of (a_n) in $\mathcal{C}/\mathcal{I}$ is invertible. Thus every nonzero element of $\mathcal{C}/\mathcal{I}$ is invertible, so $\mathcal{C}/\mathcal{I}$ is a field. Hence $\mathcal{I}$ is maximal. $\square$

Definition 9.4.8 (Completion of a valued field)**.** *The field*

$$\widehat{K} := \mathcal{C}/\mathcal{I}$$

is called the completion of K with respect to the valuation φ.

Proposition 9.4.9 (Extension of the valuation)**.** *Let $(a_n) \in \mathcal{C}$. Then the sequence*

$$\big(\varphi(a_n)\big)_{n\geq 1}$$

converges in $\mathbb{R}_{\geq 0}$.

Moreover, if

$$(a_n) - (b_n) \in \mathcal{I},$$

then

$$\lim_{n\to\infty}\varphi(a_n)=\lim_{n\to\infty}\varphi(b_n).$$

Hence the rule

$$\widehat{\varphi}\big((a_n)+\mathcal{I}\big):=\lim_{n\to\infty}\varphi(a_n)$$

defines a well-defined multiplicative valuation on $\widehat{K}$.

Proof. Since (a_n) is Cauchy in K, for every $\varepsilon>0$ there exists $N\geq 1$ such that

$$\varphi(a_n-a_m)<\varepsilon \qquad \text{for all } m,n\geq N.$$

By the reverse triangle inequality,

$$|\varphi(a_n)-\varphi(a_m)|\leq\varphi(a_n-a_m).$$

Therefore $\big(\varphi(a_n)\big)_{n\geq 1}$ is a Cauchy sequence in $\mathbb{R}_{\geq 0}$. Since $\mathbb{R}$ is complete, the limit

$$\lim_{n\to\infty}\varphi(a_n)$$

exists.

Now assume that

$$(a_n)-(b_n)\in\mathcal{I}.$$

Then

$$\lim_{n\to\infty}\varphi(a_n-b_n)=0.$$

Again by the reverse triangle inequality,

$$|\varphi(a_n)-\varphi(b_n)|\leq\varphi(a_n-b_n).$$

Taking limits gives

$$\lim_{n\to\infty}\varphi(a_n)=\lim_{n\to\infty}\varphi(b_n).$$

Thus $\widehat{\varphi}$ is well-defined.

Finally, for $(a_n),(b_n)\in\mathcal{C}$, we have

$$\widehat{\varphi}\left(\big((a_n)+\mathcal{I}\big)\big((b_n)+\mathcal{I}\big)\right)=\lim_{n\to\infty}\varphi(a_nb_n)=\lim_{n\to\infty}\varphi(a_n)\varphi(b_n),$$

hence

$$\widehat{\varphi}\left(\big((a_n)+\mathcal{I}\big)\big((b_n)+\mathcal{I}\big)\right)=\widehat{\varphi}\big((a_n)+\mathcal{I}\big)\widehat{\varphi}\big((b_n)+\mathcal{I}\big).$$

Similarly, the triangle inequality for φ passes to the limit. Hence $\widehat{\varphi}$ is a multiplicative valuation on $\widehat{K}$. □

Proposition 9.4.10. *There is a natural embedding of valued fields*

$$K\hookrightarrow\widehat{K}.$$

Proof. Define

$$\iota : K \longrightarrow \widehat{K}$$

by sending $a \in K$ to the class of the constant sequence:

$$\iota(a) := (a, a, a, \dots) + \mathcal{I}.$$

This is a field homomorphism. If $\iota(a) = 0$, then the constant sequence $(a, a, a, \dots)$ belongs to $\mathcal{I}$, so

$$\lim_{n\to\infty} \varphi(a) = 0.$$

Hence $\varphi(a) = 0$, and therefore $a = 0$. Thus ι is injective.

Moreover,

$$\widehat{\varphi}(\iota(a)) = \widehat{\varphi}\big((a, a, a, \dots) + \mathcal{I}\big) = \varphi(a),$$

so the embedding preserves the valuation. ☐

Theorem 9.4.11 (Completion of a valued field)**.** *Let (K, φ) be a valued field. Then there exists a complete valued field*

$$(\widehat{K}, \widehat{\varphi})$$

together with a valuation-preserving embedding

$$K \hookrightarrow \widehat{K}$$

such that the image of K is dense in $\widehat{K}$.

Moreover, $\widehat{K}$ is unique up to a unique K-isomorphism of valued fields.

Proof. The field $\widehat{K} = \mathcal{C}/\mathcal{I}$, together with the valuation $\widehat{\varphi}$, is complete. Indeed, every Cauchy sequence in $\widehat{K}$ can be represented by a suitable diagonal sequence of Cauchy sequences in K, and this diagonal sequence gives its limit in $\widehat{K}$.

The image of K is dense in $\widehat{K}$. If

$$x = (a_n) + \mathcal{I} \in \widehat{K},$$

then the sequence of constant classes

$$\iota(a_n) \in \widehat{K}$$

converges to x.

Finally, if L is another complete valued field containing K densely and preserving the valuation, then every Cauchy sequence in K has a unique limit in L. Thus the rule

$$(a_n) + \mathcal{I} \longmapsto \lim_{n\to\infty} a_n$$

defines a unique K-isomorphism

$$\widehat{K} \cong L$$

of valued fields. Hence the completion is unique up to K-isomorphism of valued fields. $\square$

9.5 Residue fields and the discrete valuation topology

Theorem 9.5.1 (Residue field and completion)**.** *Let (K, φ) be a valued field with valuation ring*

$$R = \{x \in K \mid \varphi(x) \le 1\}$$

and valuation ideal

$$P = \{x \in K \mid \varphi(x) < 1\}.$$

Let $(\widehat{K}, \widehat{\varphi})$ be the completion of K, and let

$$\widehat{R} = \{x \in \widehat{K} \mid \widehat{\varphi}(x) \le 1\}$$

be its valuation ring, with valuation ideal

$$\widehat{P} = \{x \in \widehat{K} \mid \widehat{\varphi}(x) < 1\}.$$

Then there is a canonical isomorphism of residue fields

$$R/P \cong \widehat{R}/\widehat{P}.$$

Proof. We identify K with its image in $\widehat{K}$. Since the embedding

$$K \hookrightarrow \widehat{K}$$

preserves the valuation, we have

$$\widehat{\varphi}|_K = \varphi.$$

Hence

$$R \subseteq \widehat{R}, \qquad P = R \cap \widehat{P}.$$

It remains to show that

$$\widehat{R} = R + \widehat{P}.$$

The inclusion

$$R + \widehat{P} \subseteq \widehat{R}$$

is clear, since $R \subseteq \widehat{R}$ and $\widehat{P} \subseteq \widehat{R}$.

Conversely, let $a \in \widehat{R}$. If $a = 0$, then $a \in R + \widehat{P}$. Suppose $a \neq 0$. Since K is dense in $\widehat{K}$, there exists a sequence

$$(a_n)_{n \ge 1}$$

in K such that

$$\lim_{n\to\infty} a_n = a.$$

Thus

$$\lim_{n\to\infty} \widehat{\varphi}(a_n - a) = 0.$$

Since $a \neq 0$, we have

$$\widehat{\varphi}(a) > 0.$$

Therefore, for some sufficiently large n,

$$\widehat{\varphi}(a_n - a) < \widehat{\varphi}(a).$$

Since

$$a_n = a + (a_n - a)$$

and

$$\widehat{\varphi}(a_n - a) < \widehat{\varphi}(a),$$

the strong triangle inequality for a non-archimedean multiplicative valuation implies

$$\widehat{\varphi}(a_n) = \widehat{\varphi}\big(a + (a_n - a)\big) = \widehat{\varphi}(a).$$

Indeed, for a multiplicative non-archimedean valuation, if

$$\widehat{\varphi}(x) < \widehat{\varphi}(y),$$

then

$$\widehat{\varphi}(x + y) = \widehat{\varphi}(y).$$

Since $a \in \widehat{R}$, we have

$$\widehat{\varphi}(a) \leq 1.$$

Hence

$$\varphi(a_n) = \widehat{\varphi}(a_n) \leq 1,$$

so

$$a_n \in R.$$

Moreover,

$$\widehat{\varphi}(a - a_n) < \widehat{\varphi}(a) \leq 1,$$

hence

$$a - a_n \in \widehat{P}.$$

Therefore

$$a = a_n + (a - a_n) \in R + \widehat{P}.$$

Thus

$$\widehat{R} \subseteq R + \widehat{P}.$$

Hence
$$\widehat{R} = R + \widehat{P}.$$

Now by the second isomorphism theorem,

$$\widehat{R}/\widehat{P} = (R + \widehat{P})/\widehat{P} \cong R/(R \cap \widehat{P}).$$

Since
$$R \cap \widehat{P} = P,$$

we obtain
$$\widehat{R}/\widehat{P} \cong R/P.$$

□

Proposition 9.5.2 (Fundamental system of neighbourhoods)**.** *Let* (K, v) *be a field with a normalized discrete valuation. Let*

$$R = \{x \in K \mid v(x) \geq 0\}$$

be its valuation ring, and let

$$P = \{x \in K \mid v(x) > 0\}$$

be its valuation ideal. Then
$$P = J(R),$$

and the decreasing sequence of ideals

$$R = P^0 \supset P \supset P^2 \supset \cdots \supset P^\ell \supset \cdots$$

forms a fundamental system of neighbourhoods of 0 *in* K*.*

Proof. Since v is a normalized discrete valuation, we have

$$v(K^\times) = \mathbb{Z}.$$

Choose a uniformizer $\pi \in R$, so that

$$v(\pi) = 1.$$

Then
$$P = (\pi),$$

and more generally

$$P^\ell = (\pi^\ell) = \{x \in R \mid v(x) \geq \ell\} \qquad \text{for every } \ell \geq 0.$$

Thus saying that an element is sufficiently close to 0 is equivalent to saying that it lies in

P^ℓ for large ℓ. Hence the ideals

$$P^\ell, \qquad \ell \geq 0,$$

form a fundamental system of neighbourhoods of 0. $\square$

Corollary 9.5.3 (Cauchy criterion). *Let $(a_n)_{n\geq 1}$ be a sequence in K. Then (a_n) is a Cauchy sequence if and only if for every natural number $\ell \geq 1$, there exists a natural number $N \geq 1$ such that*

$$a_m - a_n \in P^\ell \qquad \textit{whenever } m, n \geq N.$$

Proof. By definition, (a_n) is Cauchy if and only if for every neighbourhood U of 0, there exists $N \geq 1$ such that

$$a_m - a_n \in U \qquad \text{for all } m, n \geq N.$$

Since the ideals P^ℓ form a fundamental system of neighbourhoods of 0, it is enough to test this condition on $U = P^\ell$. Hence (a_n) is Cauchy if and only if for every $\ell \geq 1$, there exists $N \geq 1$ such that

$$a_m - a_n \in P^\ell \qquad \text{for all } m, n \geq N.$$

$\square$

Corollary 9.5.4 (Convergence criterion). *Let $(a_n)_{n\geq 1}$ be a sequence in K, and let $a \in K$. Then*

$$\lim_{n\to\infty} a_n = a$$

if and only if for every natural number $\ell \geq 1$, there exists a natural number $N \geq 1$ such that

$$a_n - a \in P^\ell \qquad \textit{whenever } n \geq N.$$

Proof. The sequence (a_n) converges to a if and only if

$$a_n - a \longrightarrow 0.$$

Since the ideals P^ℓ form a fundamental system of neighbourhoods of 0, this is equivalent to saying that for every $\ell \geq 1$, there exists $N \geq 1$ such that

$$a_n - a \in P^\ell \qquad \text{for all } n \geq N.$$

$\square$

Lemma 9.5.5 (Equivalent discrete multiplicative valuations). *Let φ_1 and φ_2 be discrete multiplicative valuations of a field K. The following conditions are equivalent:*

1. *$\varphi_1 \sim \varphi_2$, that is, there exists $\lambda > 0$ such that*

$$\varphi_2(x) = \varphi_1(x)^\lambda \qquad \textit{for all } x \in K.$$

2. *The valuation ideals are the same:*

$$\{ x \in K \mid \varphi_1(x) < 1 \} = \{ x \in K \mid \varphi_2(x) < 1 \}.$$

3. *The topologies on K induced by φ_1 and φ_2 are the same.*

Proof. For $i = 1, 2$, let

$$R_i := \{ x \in K \mid \varphi_i(x) \leq 1 \}$$

be the valuation ring of φ_i, and let

$$P_i := \{ x \in K \mid \varphi_i(x) < 1 \}$$

be its valuation ideal.

We first note that condition (2) is precisely the assertion

$$P_1 = P_2.$$

(1) $\Rightarrow$ (2). Suppose that $\varphi_2 = \varphi_1^\lambda$ for some $\lambda > 0$. Then, for every $x \in K$,

$$\varphi_2(x) < 1 \iff \varphi_1(x)^\lambda < 1 \iff \varphi_1(x) < 1.$$

Hence $P_1 = P_2$.

(2) $\Rightarrow$ (1). Assume that

$$P_1 = P_2 =: P.$$

Since φ_1 and φ_2 are discrete valuations, their valuation rings are discrete valuation rings with maximal ideal P. Hence the common ideal P is principal. Choose a uniformizer

$$\pi \in P.$$

Then every element $x \in K^\times$ admits a unique expression

$$x = \pi^n u, \qquad n \in \mathbb{Z}, \quad u \in R_i^\times.$$

Therefore, for $i = 1, 2$,

$$\varphi_i(x) = \varphi_i(\pi)^n.$$

Put

$$\rho_i := \varphi_i(\pi).$$

Since $\pi \in P$, we have $0 < \rho_i < 1$. Choose

$$\lambda := \frac{\log \rho_2}{\log \rho_1} > 0.$$

Then

$$\rho_2 = \rho_1^\lambda.$$

Hence for every $x = \pi^n u \in K^\times$,

$$\varphi_2(x) = \rho_2^n = (\rho_1^\lambda)^n = (\rho_1^n)^\lambda = \varphi_1(x)^\lambda.$$

Thus

$$\varphi_2 = \varphi_1^\lambda,$$

so $\varphi_1 \sim \varphi_2$.

(1) $\Rightarrow$ (3). If $\varphi_2 = \varphi_1^\lambda$ with $\lambda > 0$, then for any sequence (x_n) in K,

$$\varphi_1(x_n) \to 0 \iff \varphi_1(x_n)^\lambda \to 0 \iff \varphi_2(x_n) \to 0.$$

Therefore the two valuations define the same convergent sequences to 0, and hence induce the same topology on K.

(3) $\Rightarrow$ (2). Assume that the topologies induced by φ_1 and φ_2 are the same. Let $a \in P_1$. Then

$$\varphi_1(a) < 1.$$

Hence

$$\lim_{n\to\infty} \varphi_1(a^n) = \lim_{n\to\infty} \varphi_1(a)^n = 0.$$

Thus

$$a^n \longrightarrow 0$$

with respect to the topology induced by φ_1. Since the two topologies are the same, we also have

$$a^n \longrightarrow 0$$

with respect to the topology induced by φ_2. Therefore

$$\lim_{n\to\infty} \varphi_2(a^n) = \lim_{n\to\infty} \varphi_2(a)^n = 0.$$

This implies

$$\varphi_2(a) < 1,$$

so $a \in P_2$. Hence

$$P_1 \subseteq P_2.$$

By symmetry, we also get

$$P_2 \subseteq P_1.$$

Therefore

$$P_1 = P_2.$$

We have proved

$$(1) \Rightarrow (2) \Rightarrow (1), \qquad (1) \Rightarrow (3), \qquad (3) \Rightarrow (2).$$

Hence the three conditions are equivalent. $\square$

9.6 Norms on finite-dimensional vector spaces

Definition 9.6.1 (Norm over a multiplicatively valued field)**.** *Let (K, φ) be a field equipped with a multiplicative valuation, and let V be a vector space over K. A function*

$$\|\cdot\| : V \longrightarrow \mathbb{R}_{\geq 0}$$

is called a non-archimedean norm on V compatible with φ if the following conditions hold:

1. *For every $v \in V$,*

$$\|v\| \geq 0,$$

   and

$$\|v\| = 0 \iff v = 0.$$

2. *For every $\alpha \in K$ and every $v \in V$,*

$$\|\alpha v\| = \varphi(\alpha)\|v\|.$$

3. *For every $u, v \in V$,*

$$\|u + v\| \leq \max\{\|u\|, \|v\|\}.$$

Proposition 9.6.2. *Let V be a vector space over a multiplicatively valued field (K, φ). Suppose that V is equipped with a non-archimedean norm $\|\cdot\|$. Then V becomes a metric space with metric*

$$d(u, v) := \|u - v\|.$$

Moreover, the maps

$$V \times V \longrightarrow V, \qquad (u, v) \longmapsto u + v,$$

and

$$K \times V \longrightarrow V, \qquad (\alpha, v) \longmapsto \alpha v,$$

are both continuous.

Proof. The function $d(u, v) = \|u - v\|$ is a metric. Indeed, for $u, v \in V$,

$$d(u, v) = 0 \iff \|u - v\| = 0 \iff u = v.$$

Also,

$$d(u, v) = \|u - v\| = \| - (v - u)\| = \|v - u\| = d(v, u),$$

because $\varphi(-1) = 1$. Finally, for $u, v, w \in V$,

$$d(u, w) = \|u - w\| = \|(u - v) + (v - w)\| \leq \max\{\|u - v\|, \|v - w\|\} = \max\{d(u, v), d(v, w)\}.$$

Hence d is a non-archimedean metric on V.

The addition map is continuous because

$$\|(u+v)-(u_0+v_0)\| = \|(u-u_0)+(v-v_0)\| \leq \max\{\|u-u_0\|, \|v-v_0\|\}.$$

The scalar multiplication map is continuous because

$$\|\alpha v - \alpha_0 v_0\| = \|\alpha(v-v_0) + (\alpha-\alpha_0)v_0\| \leq \max\{\varphi(\alpha)\|v-v_0\|, \varphi(\alpha-\alpha_0)\|v_0\|\}.$$

Since $\varphi(\alpha)$ is locally bounded near α_0, the right-hand side tends to 0 whenever $\alpha \to \alpha_0$ in K and $v \to v_0$ in V. Thus scalar multiplication is continuous. □

Lemma 9.6.3. *Let (K, φ) be a multiplicatively valued field, and let V be a finite-dimensional K-vector space with K-basis*

$$u_1, \ldots, u_n.$$

For

$$v = a_1 u_1 + \cdots + a_n u_n \in V,$$

define

$$\|v\|_0 := \max\{\, \varphi(a_i) \mid 1 \leq i \leq n \,\}.$$

Then $\|\cdot\|_0$ is a non-archimedean norm on V. Moreover, the following hold:

1. *Let*

$$v_r = a_{r,1} u_1 + \cdots + a_{r,n} u_n \in V \qquad (r \geq 1).$$

   Then the sequence (v_r) is a Cauchy sequence in V if and only if each coordinate sequence $(a_{r,i})_{r \geq 1}$ is a Cauchy sequence in K.

   Also, for

$$v = a_1 u_1 + \cdots + a_n u_n \in V,$$

   one has

$$\lim_{r\to\infty} v_r = v \iff \lim_{r\to\infty} a_{r,i} = a_i \quad \textit{for all } 1 \leq i \leq n.$$

2. *If K is complete, then V is complete with respect to the norm $\|\cdot\|_0$.*

Proof. First we verify that $\|\cdot\|_0$ is a norm. Clearly

$$\|v\|_0 \geq 0.$$

Moreover,

$$\|v\|_0 = 0 \iff \varphi(a_i) = 0 \quad \text{for all } i \iff a_i = 0 \quad \text{for all } i \iff v = 0.$$

For $\lambda \in K$, we have

$$\|\lambda v\|_0 = \max_i \varphi(\lambda a_i) = \varphi(\lambda) \max_i \varphi(a_i) = \varphi(\lambda)\|v\|_0.$$

Finally, if

$$v = a_1 u_1 + \cdots + a_n u_n, \qquad w = b_1 u_1 + \cdots + b_n u_n,$$

then

$$v + w = (a_1 + b_1)u_1 + \cdots + (a_n + b_n)u_n.$$

Hence

$$\|v + w\|_0 = \max_i \varphi(a_i + b_i) \leq \max_i \max\{\varphi(a_i), \varphi(b_i)\} = \max\{\|v\|_0, \|w\|_0\}.$$

Thus $\|\cdot\|_0$ is a non-archimedean norm.

Now let

$$v_r = a_{r,1}u_1 + \cdots + a_{r,n}u_n.$$

For $r, s \geq 1$, we have

$$v_r - v_s = \sum_{i=1}^{n} (a_{r,i} - a_{s,i})u_i.$$

Therefore

$$\|v_r - v_s\|_0 = \max_{1 \leq i \leq n} \varphi(a_{r,i} - a_{s,i}).$$

It follows immediately that (v_r) is Cauchy in V if and only if each $(a_{r,i})_{r \geq 1}$ is Cauchy in K.

Similarly,

$$v_r - v = \sum_{i=1}^{n} (a_{r,i} - a_i)u_i,$$

so

$$\|v_r - v\|_0 = \max_{1 \leq i \leq n} \varphi(a_{r,i} - a_i).$$

Hence

$$\lim_{r \to \infty} v_r = v \iff \lim_{r \to \infty} a_{r,i} = a_i \quad \text{for all } i.$$

Finally, assume that K is complete. Let (v_r) be a Cauchy sequence in V. By the first part, each coordinate sequence $(a_{r,i})$ is Cauchy in K. Since K is complete, there exists $a_i \in K$ such that

$$\lim_{r \to \infty} a_{r,i} = a_i.$$

Put

$$v = a_1 u_1 + \cdots + a_n u_n.$$

By the convergence criterion above,

$$\lim_{r \to \infty} v_r = v.$$

Therefore V is complete. □

Theorem 9.6.4 (Equivalence of norms over a complete non-archimedean field)**.** *Let (K, φ) be a complete non-archimedean normed field, and let V be a finite-dimensional*

K-vector space with basis

$$u_1, \ldots, u_n.$$

Define the associated supremum norm by

$$\left\| \sum_{i=1}^{n} a_i u_i \right\|_\infty := \max_{1 \leq i \leq n} \varphi(a_i).$$

Then, for any non-archimedean norm $\| \cdot \|$ on V, there exist constants $\lambda, \mu > 0$ such that

$$\|v\| \leq \lambda \|v\|_\infty, \qquad \|v\|_\infty \leq \mu \|v\|$$

for all $v \in V$. Consequently, $\| \cdot \|$ and $\| \cdot \|_\infty$ induce the same topology on V. In particular, any two norms on V induce the same topology.

Proof. Let

$$\lambda = \max_{1 \leq i \leq n} \|u_i\|.$$

For

$$v = \sum_{i=1}^{n} a_i u_i,$$

the non-archimedean triangle inequality gives

$$\|v\| \leq \max_{1 \leq i \leq n} \|a_i u_i\| = \max_{1 \leq i \leq n} \varphi(a_i) \|u_i\| \leq \lambda \|v\|_\infty.$$

Hence the first inequality holds.

It remains to prove that there exists $\mu > 0$ such that

$$\|v\|_\infty \leq \mu \|v\|$$

for all $v \in V$. We prove this by induction on $n = \dim_K V$.

If $n = 1$, then every $v \in V$ can be written as $v = a_1 u_1$. Hence

$$\|v\| = \varphi(a_1) \|u_1\|, \qquad \|v\|_\infty = \varphi(a_1).$$

Therefore

$$\|v\|_\infty = \frac{1}{\|u_1\|} \|v\|,$$

so the claim holds.

Now assume $n > 1$, and suppose the result is known for all vector spaces of dimension $n - 1$. For

$$v = \sum_{i=1}^{n} a_i u_i,$$

define

$$\psi_i(v) := \varphi(a_i).$$

We claim that, for each i, there exists a constant $\mu_i > 0$ such that

$$\psi_i(v) \le \mu_i \|v\|$$

for all $v \in V$. Once this is proved, taking

$$\mu = \max_{1 \le i \le n} \mu_i$$

gives

$$\|v\|_\infty = \max_{1 \le i \le n} \psi_i(v) \le \mu \|v\|.$$

It suffices to prove the claim for $i = n$, since the other cases follow by reordering the basis. Put

$$V' = K u_1 \oplus \cdots \oplus K u_{n-1}.$$

By the induction hypothesis, the restrictions of $\|\cdot\|$ and $\|\cdot\|_\infty$ to V' induce the same topology.

Since K is complete, V' is complete with respect to the norm $\|\cdot\|_\infty$. Hence V' is also complete with respect to $\|\cdot\|$. Therefore V' is closed in V with respect to the topology induced by $\|\cdot\|$.

Suppose, for contradiction, that there is no constant $\mu_n > 0$ such that

$$\psi_n(v) \le \mu_n \|v\|$$

for all $v \in V$. Then, for every integer $j \ge 1$, there exists $v_j \in V$ such that

$$\psi_n(v_j) > j \|v_j\|.$$

Since the inequality is homogeneous, we may multiply v_j by a nonzero scalar and assume that the coefficient of u_n in v_j is 1. Thus we may write

$$v_j = a_{1j} u_1 + \cdots + a_{n-1,j} u_{n-1} + u_n.$$

Then

$$\psi_n(v_j) = 1,$$

and the inequality above gives

$$\|v_j\| < \frac{1}{j}.$$

Hence

$$v_j \longrightarrow 0$$

with respect to $\|\cdot\|$.

On the other hand,

$$v_j - u_n \in V'$$

for every j, and

$$v_j - u_n \longrightarrow -u_n$$

with respect to $\|\cdot\|$. Since V' is closed in V, it follows that

$$-u_n \in V'.$$

This is impossible, because $u_1, \dots, u_n$ are linearly independent.

Therefore there exists $\mu_n > 0$ such that

$$\psi_n(v) \leq \mu_n \|v\|$$

for all $v \in V$. As explained above, the same argument applies to each coordinate ψ_i. Hence there exists $\mu > 0$ such that

$$\|v\|_\infty \leq \mu \|v\|$$

for all $v \in V$.

Combining the two inequalities, we conclude that $\|\cdot\|$ and $\|\cdot\|_\infty$ are equivalent norms. Therefore they induce the same topology on V. □

9.7 Extensions of complete discrete valuations

Definition 9.7.1 (Extension of a discretely valued field)**.** *Let $K \subseteq L$ be fields equipped with discrete multiplicative valuations*

$$\varphi : K \longrightarrow \mathbb{R}_{\geq 0}, \qquad \psi : L \longrightarrow \mathbb{R}_{\geq 0},$$

respectively.

We say that ψ is an extension of φ to L if

$$\psi|_K = \varphi.$$

In this case, we also say that the valued field (L, ψ) is an extension of the valued field (K, φ).

Theorem 9.7.2 (Extension of a complete discrete valuation)**.** *Let φ be a complete discrete multiplicative valuation on a field K, and let L/K be a finite field extension.*

Suppose that there exists an extension (L, ψ) of (K, φ). Then ψ is unique up to equivalence. Furthermore, the valued field

$$(L, \psi)$$

is complete.

Proof. Let ψ_1 and ψ_2 be two extensions of φ to L. We first prove uniqueness up to equivalence.

Since L/K is finite, L is a finite-dimensional K-vector space. Moreover, each ψ_i is a non-archimedean norm on L as a K-vector space, because for every $a \in K$ and $x \in L$,

$$\psi_i(ax) = \psi_i(a)\psi_i(x) = \varphi(a)\psi_i(x).$$

Thus ψ_i is compatible with the scalar norm φ on K.

Since (K, φ) is complete and L is finite-dimensional over K, the equivalence theorem for norms on finite-dimensional vector spaces over a complete non-archimedean field implies that the two norms ψ_1 and ψ_2 induce the same topology on L.

Now use the standard fact that two discrete valuations on a field which induce the same topology are equivalent. Indeed, if w_1, w_2 are two discrete valuations inducing the same topology, then their valuation ideals are determined topologically by

$$P_{w_i} = \{x \in L \mid x^n \to 0 \text{ as } n \to \infty\}.$$

Hence

$$P_{w_1} = P_{w_2}.$$

The valuation ring is recovered from the valuation ideal by

$$R_{w_i} = \{x \in L \mid xP_{w_i} \subseteq P_{w_i}\},$$

so

$$R_{w_1} = R_{w_2}.$$

Since the valuations are discrete, choosing a uniformizer π for w_1, every $x \in L^\times$ can be written uniquely in the form

$$x = \pi^m u, \qquad m \in \mathbb{Z}, \quad u \in R_{w_1}^\times.$$

Because the valuation rings agree, the unit groups also agree. Therefore $w_2(u) = 0$, and hence

$$w_2(x) = m\, w_2(\pi) = w_2(\pi)\, w_1(x).$$

Thus

$$w_2 = \lambda w_1$$

for some $\lambda > 0$, so w_1 and w_2 are equivalent.

Applying this to ψ_1 and ψ_2, we conclude that

$$\psi_1 \sim \psi_2.$$

Hence the extension of φ to L, if it exists, is unique up to equivalence.

It remains to prove that (L, ψ) is complete. Choose a K-basis

$$u_1, \ldots, u_n$$

of L, and define the associated supremum norm

$$\left\| \sum_{i=1}^{n} a_i u_i \right\|_{\infty} = \max_{1 \le i \le n} \varphi(a_i).$$

Since K is complete with respect to φ, the finite-dimensional normed K-vector space $(L, \|\cdot\|_{\infty})$ is complete.

On the other hand, ψ is another non-archimedean norm on the same finite-dimensional K-vector space L. Hence, by the equivalence theorem for finite-dimensional norms, ψ and $\|\cdot\|_{\infty}$ induce the same topology on L. In particular, a sequence in L is ψ-Cauchy if and only if it is $\|\cdot\|_{\infty}$-Cauchy.

Let (x_m) be a ψ-Cauchy sequence in L. Then it is $\|\cdot\|_{\infty}$-Cauchy, so by completeness of $(L, \|\cdot\|_{\infty})$, there exists $x \in L$ such that

$$x_m \longrightarrow x$$

with respect to $\|\cdot\|_{\infty}$. Since ψ and $\|\cdot\|_{\infty}$ induce the same topology, the same convergence holds with respect to ψ. Therefore every ψ-Cauchy sequence converges in L.

Hence (L, ψ) is complete. $\square$

Chapter 10

Lecture 10

10.1 The I-adic metric and completeness

Let R be a ring, let V be an R-module, and let I be an ideal of R. Assume that

$$\bigcap_{i=1}^{\infty} I^i V = 0.$$

We set $I^0 V := V$.

Fix a real number γ such that

$$0 < \gamma < 1.$$

For $v \in V$, define

$$\|0\|_I = 0.$$

If $v \neq 0$, then there exists a unique integer $i \geq 0$ such that

$$v \in I^i V \setminus I^{i+1} V.$$

We define

$$\|v\|_I = \gamma^i.$$

Equivalently, if

$$\operatorname{ord}_I(v) := \max\{\, i \geq 0 \mid v \in I^i V \,\},$$

then

$$\|v\|_I = \gamma^{\operatorname{ord}_I(v)} \qquad (v \neq 0).$$

Definition 10.1.1 (I-adic metric)**.** *Define*

$$d_I(v, w) = \|v - w\|_I, \qquad v, w \in V.$$

We call d_I the I-adic metric on V.

Proposition 10.1.2. *The function d_I is a metric on V. More precisely, for all $u, v, w \in V$, one has*

$$d_I(v,u) \leq \max\{\, d_I(v,w), d_I(w,u)\,\}.$$

In particular, d_I is a non-archimedean metric.

Proof. It is clear that

$$d_I(v,w) \geq 0$$

and that

$$d_I(v,w) = d_I(w,v).$$

Moreover,

$$d_I(v,w) = 0$$

if and only if $v - w \in \bigcap_{i\geq 1} I^i V$. By assumption,

$$\bigcap_{i\geq 1} I^i V = 0,$$

so $d_I(v,w) = 0$ if and only if $v = w$.

It remains to prove the triangle inequality. Let

$$a = v - w, \qquad b = w - u.$$

Then

$$v - u = a + b.$$

Suppose

$$a \in I^r V, \qquad b \in I^s V.$$

If $m = \min\{r, s\}$, then

$$a, b \in I^m V,$$

hence

$$a + b \in I^m V.$$

Therefore

$$\mathrm{ord}_I(a+b) \geq \min\{\mathrm{ord}_I(a), \mathrm{ord}_I(b)\}.$$

Since $0 < \gamma < 1$, this gives

$$\|a+b\|_I \leq \max\{\|a\|_I, \|b\|_I\}.$$

Thus

$$d_I(v,u) = \|v-u\|_I = \|a+b\|_I \leq \max\{\|a\|_I, \|b\|_I\} = \max\{d_I(v,w), d_I(w,u)\}.$$

Hence d_I is a non-archimedean metric on V. $\square$

Proposition 10.1.3 (Cauchy sequences and convergence)**.** *Let* $(v_i)_{i\geq 1}$ *be a sequence of elements of* V*. Then:*

1. *The sequence* (v_i) *is Cauchy with respect to* d_I *if and only if for every integer* $m > 0$*, there exists an integer* $n > 0$ *such that*

$$v_i - v_j \in I^m V$$

for all $i, j > n$*.*

2. *The sequence* (v_i) *converges to* $v \in V$ *with respect to* d_I *if and only if for every integer* $m > 0$*, there exists an integer* $n > 0$ *such that*

$$v_i - v \in I^m V$$

for all $i > n$*.*

Proof. We prove the first statement. The second is analogous.

Suppose first that (v_i) is Cauchy. Let $m > 0$. Since

$$\gamma^m > 0,$$

there exists $n > 0$ such that

$$d_I(v_i, v_j) < \gamma^m$$

for all $i, j > n$. This means

$$\|v_i - v_j\|_I < \gamma^m.$$

By the definition of the I-adic norm, this implies

$$v_i - v_j \in I^m V.$$

Conversely, suppose that for every $m > 0$, there exists $n > 0$ such that

$$v_i - v_j \in I^m V$$

for all $i, j > n$. Let $\varepsilon > 0$. Since $0 < \gamma < 1$, we may choose $m > 0$ such that

$$\gamma^m < \varepsilon.$$

By assumption, there exists $n > 0$ such that

$$v_i - v_j \in I^m V$$

for all $i, j > n$. Hence

$$d_I(v_i, v_j) = \|v_i - v_j\|_I \leq \gamma^m < \varepsilon.$$

Therefore (v_i) is a Cauchy sequence. □

Lemma 10.1.4. *With respect to the I-adic metrics, the following maps are continuous:*

$$V \times V \longrightarrow V, \qquad (a,b) \longmapsto a+b,$$

$$R \times V \longrightarrow V, \qquad (r,v) \longmapsto rv,$$

$$R \times R \longrightarrow R, \qquad (r,s) \longmapsto r+s,$$

$$R \times R \longrightarrow R, \qquad (r,s) \longmapsto rs.$$

Moreover, if W is another R-module endowed with its I-adic metric, then every R-module homomorphism

$$f \in \mathrm{Hom}_R(V,W)$$

is continuous.

Proof. We indicate the main estimates.

For addition in V, one has

$$\|(a+b)-(a'+b')\|_I = \|(a-a')+(b-b')\|_I \leq \max\{\|a-a'\|_I, \|b-b'\|_I\}.$$

Hence addition is continuous.

For scalar multiplication, suppose $r-r' \in I^m$ and $v-v' \in I^mV$. Then

$$rv - r'v' = r(v-v') + (r-r')v'.$$

Since

$$r(v-v') \in I^mV, \qquad (r-r')v' \in I^mV,$$

we get

$$rv - r'v' \in I^mV.$$

Thus scalar multiplication is continuous.

The proofs for addition and multiplication in R are the special cases obtained by taking $V = R$.

Finally, let $f : V \to W$ be an R-module homomorphism. If

$$v - v' \in I^mV,$$

then

$$f(v) - f(v') = f(v-v') \in I^mW.$$

Hence f is continuous. □

Before discussing completeness, one should first check that all the maps considered are well-defined. Once this is done, the continuity statements from the previous show that the I-adic topology is compatible with the algebraic operations.

Definition 10.1.5 (I-adic completeness)**.** *Let R be a ring and let I be an ideal of R. Suppose that the I-adic metric d_I is defined on R, namely*

$$\bigcap_{n\geq 1} I^n = 0.$$

We say that R is complete with respect to I if the metric space

$$(R, d_I)$$

is complete.

Similarly, if V is an R-module and the I-adic metric d_I is defined on V, namely

$$\bigcap_{n\geq 1} I^n V = 0,$$

then we say that V is complete with respect to I if the metric space

$$(V, d_I)$$

is complete.

Theorem 10.1.6. *Suppose that the I-adic metric d_I is defined on every finitely generated R-module. Let V be a finitely generated R-module. Then every submodule of V is closed with respect to the I-adic metric on V.*

Proof. Let $W \leq V$ be a submodule. Consider the quotient map

$$f : V \longrightarrow V/W.$$

Since V is finitely generated, the quotient V/W is also finitely generated. By assumption, the I-adic metric is defined on V/W.

The quotient map f is an R-module homomorphism, and hence is continuous with respect to the I-adic metrics. Moreover,

$$W = f^{-1}(0).$$

Since $\{0\}$ is closed in the metric space V/W, its inverse image under the continuous map f is closed in V. Therefore W is closed in V. ☐

Theorem 10.1.7. *Let V be a finitely generated R-module. Assume that the I-adic metric is defined on V. If R is complete with respect to the I-adic metric, then V is complete with respect to the I-adic metric.*

Proof. Let

$$v_1, \ldots, v_s$$

be a finite set of generators of V, and let

$$(w_n)_{n\geq 1}$$

be a Cauchy sequence in V.

Since (w_n) is Cauchy, there exists a nondecreasing sequence of nonnegative integers

$$m(n) \qquad (n \geq 1)$$

such that

$$m(n) \longrightarrow \infty$$

and

$$w_i - w_j \in I^{m(n)}V$$

for all $i, j > n$. In particular, if

$$\Delta_n := w_{n+1} - w_n,$$

then

$$\Delta_n \in I^{m(n)}V.$$

Write

$$w_1 = \sum_{t=1}^{s} b_{1t}v_t, \qquad b_{1t} \in R.$$

Since $\Delta_n \in I^{m(n)}V$, and since $v_1, \dots, v_s$ generate V, we may choose elements

$$a_{nt} \in I^{m(n)} \qquad (1 \leq t \leq s)$$

such that

$$\Delta_n = w_{n+1} - w_n = \sum_{t=1}^{s} a_{nt}v_t.$$

For $n \geq 1$, define

$$b_{nt} := b_{1t} + \sum_{j=1}^{n-1} a_{jt}.$$

We claim that

$$w_n = \sum_{t=1}^{s} b_{nt}v_t$$

for every $n \geq 1$. Indeed, this is clear for $n = 1$, and if it holds for n, then

$$w_{n+1} = w_n + \Delta_n = \sum_{t=1}^{s} b_{nt}v_t + \sum_{t=1}^{s} a_{nt}v_t = \sum_{t=1}^{s}(b_{nt} + a_{nt})v_t = \sum_{t=1}^{s} b_{n+1,t}v_t.$$

Hence the claim follows by induction.

We now show that each sequence $(b_{nt})_{n\geq 1}$ is Cauchy in R. Fix t and let $\ell \geq 1$. Since

$m(n) \to \infty$, there exists N such that

$$m(N) \geq \ell.$$

If $q > p \geq N$, then

$$b_{qt} - b_{pt} = \sum_{j=p}^{q-1} a_{jt}.$$

Since $m(j) \geq m(p) \geq \ell$ for $j \geq p$, we have

$$a_{jt} \in I^{m(j)} \subseteq I^{\ell}.$$

Therefore

$$b_{qt} - b_{pt} \in I^{\ell}.$$

Thus $(b_{nt})_{n \geq 1}$ is Cauchy in R.

Since R is complete, for each $t = 1, \ldots, s$, there exists $b_t \in R$ such that

$$b_{nt} \longrightarrow b_t$$

in the I-adic metric on R. Define

$$w := \sum_{t=1}^{s} b_t v_t \in V.$$

It remains to prove that $w_n \to w$. Let $\ell \geq 1$. For each t, since $b_{nt} \to b_t$, there exists N_t such that

$$b_{nt} - b_t \in I^{\ell}$$

for all $n \geq N_t$. Put

$$N = \max\{N_1, \ldots, N_s\}.$$

Then for $n \geq N$,

$$w_n - w = \sum_{t=1}^{s} (b_{nt} - b_t) v_t \in I^{\ell} V.$$

Since $\ell \geq 1$ was arbitrary, this shows that

$$w_n \longrightarrow w$$

in the I-adic metric on V.

Therefore every Cauchy sequence in V converges in V. Hence V is complete. $\square$

10.2 Complete discrete valuation rings and lifting idempotents

Let R be a complete discrete valuation ring with normalized valuation

$$v : K^\times \longrightarrow \mathbb{Z},$$

where $K = \mathrm{Frac}(R)$. Let $\pi \in R$ be a uniformizer, so that

$$v(\pi) = 1.$$

Denote the maximal ideal of R by

$$P = J(R) = (\pi),$$

and denote the residue field by

$$F = R/P.$$

Lemma 10.2.1. *We have*

$$\bigcap_{n \geq 1} P^n = 0.$$

Proof. Since R is a discrete valuation ring, every nonzero element $x \in R$ can be written uniquely in the form

$$x = \pi^m u,$$

where $m \geq 0$ and $u \in R^\times$. Hence $x \in P^m$, but

$$x \notin P^{m+1}.$$

Therefore no nonzero element can lie in every P^n. Thus

$$\bigcap_{n \geq 1} P^n = 0.$$

□

Proposition 10.2.2. *Let V be a finitely generated R-module. Then V is a finite direct sum of cyclic R-modules. Thus we may write*

$$V = Ru_1 \oplus \cdots \oplus Ru_r.$$

For every $n \geq 0$, one has

$$P^n V = \pi^n V = (\pi^n R)u_1 \oplus \cdots \oplus (\pi^n R)u_r.$$

Moreover,

$$\bigcap_{n\geq 1} P^n V = 0.$$

Proof. Since R is a discrete valuation ring, it is a principal ideal domain. Hence, by the structure theorem for finitely generated modules over a principal ideal domain, V is a finite direct sum of cyclic R-modules:

$$V = Ru_1 \oplus \cdots \oplus Ru_r.$$

Therefore, for every $n \geq 0$,

$$P^n V = (\pi^n)V = (\pi^n R)u_1 \oplus \cdots \oplus (\pi^n R)u_r.$$

Since

$$\bigcap_{n\geq 1} (\pi^n) = 0,$$

it follows componentwise that

$$\bigcap_{n\geq 1} P^n V = 0.$$

□

Definition 10.2.3 (P-adic metric)**.** *Let V be a finitely generated R-module. For $0 \neq x \in V$, define*

$$\nu_V(x) := \max\{\, n \geq 0 \mid x \in P^n V \,\}.$$

This is well-defined because

$$\bigcap_{n\geq 1} P^n V = 0.$$

We also set

$$\nu_V(0) = +\infty.$$

Fix a real number $0 < c < 1$. The P-adic metric on V is defined by

$$d_P(x, y) = c^{\nu_V(x-y)}.$$

Theorem 10.2.4. *Let V be a finitely generated R-module. If R is complete with respect to the P-adic metric, then V is complete with respect to the P-adic metric.*

Proof. Since V is finitely generated over the discrete valuation ring R, it is a finite direct sum of cyclic R-modules:

$$V = Ru_1 \oplus \cdots \oplus Ru_r.$$

Each cyclic summand Ru_i is isomorphic either to R or to $R/(\pi^{m_i})$ for some $m_i \geq 1$. Since R is complete, and quotients of complete P-adic modules by closed submodules are complete, each cyclic summand Ru_i is complete.

A finite direct sum of complete metric R-modules is complete. Hence V is complete with respect to the P-adic metric. $\square$

Now let A be an R-algebra which is finitely generated as an R-module. For $n \geq 0$, define

$$A_n := P^n A = \pi^n A.$$

Since A is a finitely generated R-module and R is complete, the R-module A is complete with respect to the P-adic metric.

Proposition 10.2.5. *The multiplication map*

$$A \times A \longrightarrow A, \qquad (a, b) \longmapsto ab$$

is continuous with respect to the P-adic topology.

Proof. It is enough to check continuity at an arbitrary point $(a, b) \in A \times A$. Let $\ell \geq 0$. Suppose

$$a' - a \in P^\ell A, \qquad b' - b \in P^\ell A.$$

Then

$$a'b' - ab = (a' - a)b + a(b' - b) + (a' - a)(b' - b).$$

Since $P^\ell A$ is a two-sided ideal of A, we have

$$(a' - a)b \in P^\ell A, \qquad a(b' - b) \in P^\ell A,$$

and also

$$(a' - a)(b' - b) \in P^{2\ell} A \subseteq P^\ell A.$$

Hence

$$a'b' - ab \in P^\ell A.$$

Therefore multiplication is continuous. $\square$

Proposition 10.2.6. *The quotient*

$$\overline{A} := A/A_1 = A/PA$$

is a finite-dimensional F-algebra, where

$$F = R/P.$$

Proof. Since A is finitely generated as an R-module, choose generators

$$a_1, \ldots, a_m \in A.$$

Then their images

$$\overline{a}_1, \ldots, \overline{a}_m \in A/PA$$

generate A/PA as an $F = R/P$-vector space. Hence A/PA is finite-dimensional over F.

The multiplication on A induces a multiplication on A/PA, because PA is a two-sided ideal of A. Therefore

$$A/PA$$

is a finite-dimensional F-algebra. □

Theorem 10.2.7. *Let A be a finitely generated R-algebra. Then:*

1. $$A\pi \subseteq J(A).$$

2. *There exists an integer $n > 0$ such that*

$$J(A)^n \subseteq A\pi.$$

Proof. Let V be a simple left A-module. Since $A\pi$ is a two-sided ideal of A, the subspace

$$\pi V$$

is an A-submodule of V. Hence, by simplicity,

$$\pi V = V \qquad \text{or} \qquad \pi V = 0.$$

Choose $0 \neq v \in V$. Since V is simple, we have

$$V = Av.$$

Since A is finitely generated over R, it follows that V is finitely generated as an R-module.

By Nakayama's lemma, because

$$\pi \in J(R),$$

the equality $\pi V = V$ is impossible. Therefore

$$\pi V = 0.$$

Thus $A\pi$ annihilates every simple A-module. Hence

$$A\pi \subseteq J(A).$$

This proves (1).

For (2), set

$$A^* := A/A\pi.$$

Since A is finitely generated as an R-module, A^* is a finite-dimensional algebra over the residue field

$$F = R/(\pi).$$

Hence A^* is an Artinian ring. Therefore its Jacobson radical is nilpotent.

Since $A\pi \subseteq J(A)$, we have

$$J(A^*) = J(A/A\pi) = J(A)/A\pi.$$

Hence there exists some $n > 0$ such that

$$\big(J(A)/A\pi\big)^n = 0.$$

Equivalently,

$$J(A)^n \subseteq A\pi.$$

This proves (2). □

Theorem 10.2.8. *Let R be a complete discrete valuation ring with uniformizer π, and let A be an R-algebra which is finitely generated as an R-module. Let I be a two-sided ideal of A such that*

$$I \subseteq J(A).$$

Write

$$\overline{A} := A/I.$$

Let $\overline{c} \in \overline{A}$ be an idempotent, and suppose

$$\overline{c} = \overline{c}_1 + \cdots + \overline{c}_n$$

is an idempotent decomposition in $\overline{A}$. Then:

1. *The idempotent $\overline{c}$ lifts to an idempotent $e \in A$, and there exist orthogonal idempotents*
$$e_1, \dots, e_n \in A$$
such that
$$\overline{e_i} = \overline{c_i}, \qquad 1 \leq i \leq n,$$
and
$$e = e_1 + \cdots + e_n.$$

2. *An idempotent $e \in A$ is primitive if and only if its image*
$$\overline{e} \in \overline{A}$$
is primitive.

3. *If $e, f \in A$ are primitive idempotents, then*
$$Ae \cong Af \qquad \Longleftrightarrow \qquad \overline{A}\,\overline{e} \cong \overline{A}\,\overline{f}.$$

Proof. Since $I \subseteq J(A)$, by the previous theorem there exists an integer $m > 0$ such that

$$I^m \subseteq A\pi.$$

Hence, for every $r \geq 1$,

$$I^{mr} \subseteq A\pi^r.$$

For each $r \geq 1$, set

$$A^{(r)} := A/I^{mr}.$$

We first prove that every idempotent of $\overline{A} = A/I$ lifts to an idempotent of A. Since I/I^m is a nilpotent ideal of A/I^m, the idempotent $\bar{c} \in A/I$ lifts to an idempotent in A/I^m. Thus there exists $c^{(1)} \in A$ such that

$$\left(c^{(1)}\right)^2 \equiv c^{(1)} \pmod{I^m}, \qquad c^{(1)} \equiv c \pmod{I}.$$

Next, since I^m/I^{2m} is a nilpotent ideal of A/I^{2m}, the idempotent $c^{(1)} + I^m \in A/I^m$ lifts to an idempotent in A/I^{2m}. Hence there exists $c^{(2)} \in A$ such that

$$\left(c^{(2)}\right)^2 \equiv c^{(2)} \pmod{I^{2m}}, \qquad c^{(2)} \equiv c^{(1)} \pmod{I^m}.$$

Repeating this argument, we obtain a sequence $\{c^{(r)}\}_{r\geq 1}$ in A such that

$$\left(c^{(r)}\right)^2 \equiv c^{(r)} \pmod{I^{rm}},$$

and

$$c^{(r)} \equiv c^{(r-1)} \pmod{I^{(r-1)m}} \qquad (r \geq 2).$$

Since

$$I^{(r-1)m} \subseteq A\pi^{r-1},$$

the sequence $\{c^{(r)}\}_{r\geq 1}$ is Cauchy with respect to the π-adic topology on A.

As A is finitely generated over the complete ring R, the R-module A is complete. Therefore the limit

$$e := \lim_{r\to\infty} c^{(r)}$$

exists in A. Since multiplication

$$A \times A \longrightarrow A$$

is continuous, passing to the limit in

$$\left(c^{(r)}\right)^2 - c^{(r)} \longrightarrow 0$$

gives

$$e^2 = e.$$

Moreover, $e \equiv c \pmod{I}$, so e lifts $\bar{c}$.

We now lift the idempotent decomposition. Let $e \in A$ be any idempotent lifting $\bar{c}$. We construct, for each $r \geq 1$, elements

$$c_1^{(r)}, \ldots, c_n^{(r)} \in A$$

such that their images in $A^{(r)} = A/I^{rm}$ are orthogonal idempotents satisfying

$$e \equiv c_1^{(r)} + \cdots + c_n^{(r)} \pmod{I^{rm}},$$

and

$$c_i^{(r)} \equiv c_i^{(r-1)} \pmod{I^{(r-1)m}} \qquad (r \geq 2).$$

This is obtained inductively by applying the lifting theorem for idempotent decompositions modulo nilpotent ideals. Indeed, the kernel of

$$A/I^{rm} \longrightarrow A/I^{(r-1)m}$$

is

$$I^{(r-1)m}/I^{rm},$$

which is nilpotent.

For each fixed i, the sequence

$$\{c_i^{(r)}\}_{r\geq 1}$$

is Cauchy, because

$$c_i^{(r)} - c_i^{(r-1)} \in I^{(r-1)m} \subseteq A\pi^{r-1}.$$

Hence the limit

$$e_i := \lim_{r\to\infty} c_i^{(r)}$$

exists in A.

Passing to the limit, and using the continuity of multiplication, we get

$$e_i^2 = e_i, \qquad e_i e_j = 0 \quad (i \neq j),$$

and

$$e = e_1 + \cdots + e_n.$$

Moreover,

$$\overline{e_i} = \overline{c_i}.$$

This proves (1).

We prove (2). Recall that $J(A)$ contains no nonzero idempotents. Indeed, if $x \in J(A)$ and $x^2 = x$, then $1 - x$ is a unit, and

$$x(1-x) = 0$$

implies $x = 0$.

Suppose first that $\overline{e}$ is primitive. If

$$e = e_1 + e_2$$

is an idempotent decomposition in A, then reducing modulo I gives

$$\overline{e} = \overline{e_1} + \overline{e_2}.$$

Since $\overline{e}$ is primitive, one of $\overline{e_1}$, $\overline{e_2}$ is zero. Suppose $\overline{e_1} = 0$. Then

$$e_1 \in I \subseteq J(A).$$

Since e_1 is an idempotent and $J(A)$ contains no nonzero idempotents, we get

$$e_1 = 0.$$

Hence e is primitive.

Conversely, suppose that e is primitive. If $\overline{e}$ were not primitive, then there would be a nontrivial idempotent decomposition

$$\overline{e} = \overline{d}_1 + \overline{d}_2$$

in $\overline{A}$. By part (1), this decomposition lifts to an idempotent decomposition

$$e = d_1 + d_2$$

in A. Since $\overline{d}_1 \neq 0$ and $\overline{d}_2 \neq 0$, both d_1 and d_2 are nonzero. This contradicts the primitiveness of e. Therefore $\overline{e}$ is primitive.

This proves (2).

Finally, we prove (3). Since $I \subseteq J(A)$, for every idempotent $e \in A$ we have

$$Ie \subseteq J(A)e = \mathrm{Rad}(Ae).$$

Therefore the natural epimorphism

$$Ae \longrightarrow Ae/Ie$$

is essential. Moreover,

$$Ae/Ie \cong \overline{A}\,\overline{e}.$$

Since Ae is projective, the map

$$Ae \longrightarrow \overline{A}\,\overline{e}$$

is a projective cover.

Similarly,

$$Af \longrightarrow \overline{A}\,\overline{f}$$

is a projective cover.

If

$$Ae \cong Af,$$

then passing to the quotient modulo I gives

$$\overline{A}\,\overline{e} \cong \overline{A}\,\overline{f}.$$

Conversely, if

$$\overline{A}\,\overline{e} \cong \overline{A}\,\overline{f},$$

then Ae and Af are projective covers of isomorphic modules. By uniqueness of projective covers, we obtain

$$Ae \cong Af.$$

Hence

$$Ae \cong Af \qquad \Longleftrightarrow \qquad \overline{A}\,\overline{e} \cong \overline{A}\,\overline{f}.$$

□

10.3 Extension of scalars and splitting fields

Let R be a commutative ring, let A be an R-algebra, and let U be an A-module. Suppose that

$$\varphi : R \longrightarrow R'$$

is a homomorphism of commutative rings. Then R' is naturally an R-algebra via φ.

Definition 10.3.1 (Extension of scalars)**.** *With the notation above, the tensor product*

$$R' \otimes_R A$$

is naturally an R'-algebra. Moreover,

$$R' \otimes_R U$$

is naturally a module over the R'-algebra $R' \otimes_R A$.

Explicitly, the action is given on elementary tensors by

$$(r' \otimes a) \cdot (s' \otimes u) := r's' \otimes au,$$

where $r', s' \in R'$, $a \in A$, and $u \in U$.

Example 10.3.2. *Let*

$$R = \mathbb{Q}, \qquad R' = \mathbb{C}, \qquad A = M_n(\mathbb{Q}).$$

Then

$$\mathbb{C} \otimes_{\mathbb{Q}} M_n(\mathbb{Q}) \cong M_n(\mathbb{C}).$$

Thus extending scalars from $\mathbb{Q}$ *to* $\mathbb{C}$ *turns the* $\mathbb{Q}$*-algebra* $M_n(\mathbb{Q})$ *into the* $\mathbb{C}$*-algebra* $M_n(\mathbb{C})$*.*

Remark 10.3.3. *If* $R \subseteq R'$ *is a subring and* U *is an* A*-module, then we say that the module*

$$V := R' \otimes_R U$$

is obtained from U *by extending scalars from* R *to* R'*.*

Definition 10.3.4. *Let* V *be an* $R' \otimes_R A$*-module. If there exists an* A*-module* U *such that*

$$V \cong R' \otimes_R U$$

as $R' \otimes_R A$*-modules, then we say that* V *can be written over* R*, or that* V *is obtained from* U *by extension of scalars.*

Example 10.3.5. *Let* R *be a discrete valuation ring, and let* $K = \mathrm{Frac}(R)$ *be its fraction field. If* A *is an* R*-algebra which is finitely generated as an* R*-module, then*

$$K \otimes_R A$$

is a finite-dimensional K*-algebra.*

Proof. Since A is finitely generated as an R-module, there exist elements

$$a_1, \ldots, a_m \in A$$

such that

$$A = Ra_1 + \cdots + Ra_m.$$

We first show that $K \otimes_R A$ is finite-dimensional as a K-vector space.

The K-vector space structure on $K \otimes_R A$ is given by

$$\lambda \cdot (\mu \otimes a) := (\lambda\mu) \otimes a, \qquad \lambda, \mu \in K,\ a \in A.$$

Every element of $K \otimes_R A$ is a finite sum of elementary tensors

$$\lambda \otimes a, \qquad \lambda \in K,\ a \in A.$$

Since $a_1, \ldots, a_m$ generate A as an R-module, for each $a \in A$ there exist $r_1, \ldots, r_m \in R$ such that

$$a = r_1 a_1 + \cdots + r_m a_m.$$

Hence

$$\lambda \otimes a = \lambda \otimes \left(\sum_{i=1}^{m} r_i a_i \right) = \sum_{i=1}^{m} \lambda r_i \otimes a_i.$$

Since

$$\lambda r_i \otimes a_i = (\lambda r_i) \cdot (1 \otimes a_i),$$

it follows that

$$1 \otimes a_1, \ldots, 1 \otimes a_m$$

span $K \otimes_R A$ as a K-vector space. Therefore

$$\dim_K(K \otimes_R A) \leq m < \infty.$$

It remains to describe the algebra structure. Since A is an R-algebra and K is an R-algebra via the inclusion

$$R \hookrightarrow K,$$

the tensor product $K \otimes_R A$ carries a natural multiplication defined on elementary tensors by

$$(\lambda \otimes a)(\mu \otimes b) := \lambda\mu \otimes ab, \qquad \lambda, \mu \in K,\ a, b \in A.$$

This multiplication is K-bilinear and associative, and its identity element is

$$1_K \otimes 1_A.$$

Hence $K \otimes_R A$ is a K-algebra.

Since it is finite-dimensional as a K-vector space, we conclude that

$$K \otimes_R A$$

is a finite-dimensional K-algebra. $\square$

Remark 10.3.6. *Suppose that $V = R' \otimes_R U$ can be written over R. If U is free as an R-module, then we identify U with its image*

$$1_{R'} \otimes_R U \subseteq R' \otimes_R U.$$

In this case, an R-basis of U becomes an R'-basis of $R' \otimes_R U$.

Example 10.3.7 (Reduction modulo an ideal)**.** *Let I be an ideal of R, and take*

$$R' = R/I.$$

Then applying the functor

$$R' \otimes_R -$$

to an R-module U is the same as reducing U modulo I. More precisely,

$$(R/I) \otimes_R U \cong U/IU.$$

Definition 10.3.8 (Lifting of modules)**.** *Let V be an $R' \otimes_R A$-module. If there exists an A-module U such that*

$$V \cong R' \otimes_R U,$$

then we say that V can be lifted to U, and that U is a lift of V.

Example 10.3.9. *Let R be a discrete valuation ring with maximal ideal*

$$(\pi) = J(R),$$

and let A be an R-algebra. Put

$$F = R/(\pi).$$

Then

$$F \otimes_R A \cong A/\pi A.$$

If V is an A-module, then

$$F \otimes_R V \cong V/\pi V.$$

Thus tensoring with the residue field F corresponds to reducing modulo π.

Proposition 10.3.10. *Let*

$$F \subseteq E$$

be an algebraic extension of fields. Let A be a finite-dimensional F-algebra, and let V be an $E \otimes_F A$-module which is finite-dimensional as an E-vector space.

Then there exists a field K such that

$$F \subseteq K \subseteq E, \qquad [K : F] < \infty,$$

and V can be written over K. More precisely, there exists a $K \otimes_F A$-module V_K such that

$$V \cong E \otimes_K V_K$$

as $E \otimes_F A$-modules.

Proof. If $V = 0$, the result is trivial. Assume $V \neq 0$, and write

$$d = \dim_E V.$$

Choose an E-basis

$$v_1, \ldots, v_d$$

of V.

Let

$$a^1, \dots, a^n$$

be an F-basis of A. Since V is an $E \otimes_F A$-module, each element $a^t \in A$ acts E-linearly on V. With respect to the chosen E-basis of V, write the action of a^t as

$$a^t v_j = \sum_{i=1}^{d} a_{ij}^t v_i, \qquad a_{ij}^t \in E.$$

Define

$$K := F\big(a_{ij}^t \mid 1 \le t \le n,\ 1 \le i, j \le d\big) \subseteq E.$$

Since E/F is algebraic, each a_{ij}^t is algebraic over F. Hence the field obtained by adjoining finitely many such elements is a finite extension of F. Therefore

$$[K : F] < \infty.$$

Now define

$$V_K := Kv_1 + \cdots + Kv_d.$$

This is a K-vector space with basis $v_1, \dots, v_d$. By construction, for every t and every j,

$$a^t v_j = \sum_{i=1}^{d} a_{ij}^t v_i \in V_K,$$

because all coefficients a_{ij}^t lie in K. Thus V_K is stable under the action of A.

Hence V_K becomes a $K \otimes_F A$-module by

$$(\lambda \otimes a) \cdot v := \lambda(av), \qquad \lambda \in K,\ a \in A,\ v \in V_K.$$

Finally, consider the natural E-linear map

$$E \otimes_K V_K \longrightarrow V, \qquad \lambda \otimes v \longmapsto \lambda v.$$

Since $v_1, \dots, v_d$ is both a K-basis of V_K and an E-basis of V, this map is an isomorphism of E-vector spaces.

Moreover, it is compatible with the $E \otimes_F A$-module structures. Indeed, for $\mu, \lambda \in E$, $a \in A$, and $v \in V_K$,

$$(\mu \otimes a) \cdot (\lambda \otimes v) = \mu\lambda \otimes av \longmapsto \mu\lambda av,$$

which is exactly the action of $\mu \otimes a$ on $\lambda v \in V$.

Therefore

$$V \cong E \otimes_K V_K$$

as $E \otimes_F A$-modules. Hence V can be written over the finite extension K/F. $\square$

Definition 10.3.11 (Absolutely simple module)**.** *Let F be a field, and let A be a finite-dimensional F-algebra. A simple A-module U is called absolutely simple if, for every field extension E/F, the $E \otimes_F A$-module*

$$E \otimes_F U$$

is simple.

Definition 10.3.12 (Splitting field)**.** *Let F be a field, and A be a finite-dimensional F-algebra. A field extension E/F is called a splitting field for A if every simple $E \otimes_F A$-module is absolutely simple.*

Proposition 10.3.13. *Let A be a finite-dimensional algebra over a field F, and let U be a simple A-module. Then the following conditions are equivalent:*

1. *U is absolutely simple.*
2. $$\operatorname{End}_A(U) = F.$$
3. *The matrix algebra summand of $A/J(A)$ corresponding to U has the form*
$$M_n(F),$$
where
$$n = \dim_F U.$$

Proof. We prove the equivalences.

(1) $\Rightarrow$ (2). Let $\overline{F}$ be an algebraic closure of F. Since U is absolutely simple,

$$\overline{F} \otimes_F U$$

is a simple $\overline{F} \otimes_F A$-module.

By Schur's lemma,

$$\operatorname{End}_{\overline{F} \otimes_F A}\big(\overline{F} \otimes_F U\big)$$

is a division algebra over $\overline{F}$. Since $\overline{F}$ is algebraically closed and $\overline{F} \otimes_F U$ is finite-dimensional over $\overline{F}$, we get

$$\operatorname{End}_{\overline{F} \otimes_F A}\big(\overline{F} \otimes_F U\big) = \overline{F}.$$

On the other hand, by base change for endomorphism rings,

$$\operatorname{End}_{\overline{F} \otimes_F A}\big(\overline{F} \otimes_F U\big) \cong \overline{F} \otimes_F \operatorname{End}_A(U).$$

Therefore

$$\overline{F} \otimes_F \operatorname{End}_A(U) \cong \overline{F}.$$

Taking dimensions over $\overline{F}$, we obtain

$$\dim_F \operatorname{End}_A(U) = 1.$$

Since F embeds into $\operatorname{End}_A(U)$ by scalar multiplication, it follows that

$$\operatorname{End}_A(U) = F.$$

(2) $\Rightarrow$ (3). By the Artin–Wedderburn theorem, the simple module U corresponds to a matrix algebra summand of $A/J(A)$ of the form

$$M_n(D),$$

where

$$D = \operatorname{End}_A(U)^{\mathrm{op}}.$$

If

$$\operatorname{End}_A(U) = F,$$

then

$$D = F.$$

Hence the corresponding summand is

$$M_n(F).$$

In this case the natural simple module for $M_n(F)$ is F^n, and therefore

$$n = \dim_F U.$$

(3) $\Rightarrow$ (1). Suppose that the matrix algebra summand of $A/J(A)$ corresponding to U is

$$M_n(F),$$

with

$$n = \dim_F U.$$

Then A acts on U through a surjective algebra homomorphism

$$A \longrightarrow M_n(F),$$

and under this action we may identify

$$U \cong F^n.$$

Let E/F be any field extension. After extending scalars, $E \otimes_F A$ acts on $E \otimes_F U$ through the induced homomorphism

$$E \otimes_F A \longrightarrow E \otimes_F M_n(F) \cong M_n(E).$$

Moreover,

$$E \otimes_F U \cong E \otimes_F F^n \cong E^n.$$

The natural $M_n(E)$-module E^n is simple. Therefore

$$E \otimes_F U$$

is a simple $E \otimes_F A$-module for every field extension E/F.

Hence U is absolutely simple. □

Example 10.3.14. *Let*

$$f(x) = x^4 - 2 \in \mathbb{Q}[x],$$

and let

$$\alpha = \sqrt[4]{2}, \qquad i = \sqrt{-1}.$$

The roots of $f(x)$ in $\mathbb{C}$ are

$$\alpha, \quad i\alpha, \quad -\alpha, \quad -i\alpha.$$

Thus the splitting field of $x^4 - 2$ over $\mathbb{Q}$ is

$$K = \mathbb{Q}(\alpha, i).$$

Since $x^4 - 2$ is irreducible over $\mathbb{Q}$ and $i \notin \mathbb{Q}(\alpha)$, we have

$$[K : \mathbb{Q}] = 8.$$

Therefore

$$|\mathrm{Gal}(K/\mathbb{Q})| = 8.$$

The Galois group acts on the set of roots

$$\Omega = \{\alpha, i\alpha, -\alpha, -i\alpha\}.$$

Consider the automorphisms $\sigma, \tau \in \mathrm{Gal}(K/\mathbb{Q})$ defined by

$$\sigma(\alpha) = -i\alpha, \qquad \sigma(i) = -i,$$

and

$$\tau(\alpha) = \alpha, \qquad \tau(i) = -i.$$

Their actions on the roots are

$$\sigma: \quad \alpha \leftrightarrow -i\alpha, \qquad -\alpha \leftrightarrow i\alpha,$$

and

$$\tau : \quad i\alpha \leftrightarrow -i\alpha, \qquad \alpha \mapsto \alpha, \qquad -\alpha \mapsto -\alpha.$$

Hence

$$\sigma^2 = \tau^2 = (\sigma\tau)^4 = 1.$$

Moreover,

$$\langle \sigma, \tau \rangle = \mathrm{Gal}(K/\mathbb{Q}).$$

Therefore

$$\mathrm{Gal}(K/\mathbb{Q}) \cong D_8,$$

where

$$D_8 = \big\langle \sigma, \tau \mid \sigma^2 = \tau^2 = (\sigma\tau)^4 = 1 \big\rangle$$

is the dihedral group of order 8.

The group D_8 *can be represented as a subgroup of*

$$\mathrm{GL}_2(\mathbb{R}).$$

Indeed, consider the real vector space

$$E = \mathbb{R}\alpha \oplus \mathbb{R}i\alpha$$

with ordered basis

$$(\alpha, i\alpha).$$

With respect to this basis, the above actions are represented by the matrices

$$\sigma \longmapsto \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \qquad \tau \longmapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These matrices satisfy

$$\sigma^2 = \tau^2 = I_2, \qquad (\sigma\tau)^4 = I_2.$$

Thus they generate a subgroup of $\mathrm{GL}_2(\mathbb{R})$ *isomorphic to* D_8.

Geometrically, this is the usual symmetry group of a square whose vertices are

$$\alpha, \quad i\alpha, \quad -\alpha, \quad -i\alpha.$$

10.4 Representations in categories

A representation of a group is a way to view the group as acting on some mathematical object while preserving the structure of that object.

For example, a group may act on:

- a set;
- the set of roots of a polynomial;

- a vector space;
- a topological space;
- an object of a category.

The key idea is that a group action should preserve the underlying structure of the object.

Let $\mathcal{C}$ be a category. Examples include:

- **Set**, the category of sets;
- **FinSet**, the category of finite sets;
- **Top**, the category of topological spaces;
- $\mathbf{Vect}_k$, the category of vector spaces over a field k;
- $\mathbf{vect}_k$, the category of finite-dimensional vector spaces over k.

If X is an object of $\mathcal{C}$, then the automorphism group of X is

$$\mathrm{Aut}_{\mathcal{C}}(X) = \{\, f : X \to X \mid f \text{ is an isomorphism in } \mathcal{C}\,\}.$$

Equivalently, $\mathrm{Aut}_{\mathcal{C}}(X)$ consists of all morphisms

$$f : X \to X$$

which admit an inverse morphism.

Definition 10.4.1 (Representation in a category)**.** *Let G be a group and let X be an object of a category $\mathcal{C}$. A representation of G on X is a group homomorphism*

$$\rho : G \longrightarrow \mathrm{Aut}_{\mathcal{C}}(X).$$

Example 10.4.2 (Permutation representations)**.** *If $\mathcal{C} = \mathbf{Set}$, then a representation of G on a set X is a group homomorphism*

$$\rho : G \longrightarrow \mathrm{Aut}_{\mathbf{Set}}(X).$$

Since

$$\mathrm{Aut}_{\mathbf{Set}}(X) = \mathrm{Sym}(X),$$

this is precisely a permutation representation

$$\rho : G \longrightarrow \mathrm{Sym}(X).$$

Example 10.4.3 (Linear representations)**.** *If $\mathcal{C} = \mathbf{Vect}_k$, then for a vector space V*

over k,

$$\mathrm{Aut}_{\mathbf{Vect}_k}(V) = \mathrm{GL}(V).$$

Therefore a linear representation of G on V is a group homomorphism

$$\rho : G \longrightarrow \mathrm{GL}(V).$$

Example 10.4.4 (Topological representations)**.** *If $\mathcal{C} = \mathbf{Top}$, then for a topological space X,*

$$\mathrm{Aut}_{\mathbf{Top}}(X) = \mathrm{Homeo}(X),$$

the group of homeomorphisms of X. Thus a topological action of G on X is a homomorphism

$$\rho : G \longrightarrow \mathrm{Homeo}(X).$$

Definition 10.4.5 ($\mathcal{C}$-representation of a group)**.** *Let G be a group, and let $\mathcal{C}$ be a category. A representation of G on $\mathcal{C}$, or a $\mathcal{C}$-representation of G, is a pair*

$$(X, \rho),$$

where

$$X \in \mathrm{Ob}(\mathcal{C})$$

is an object of $\mathcal{C}$, and

$$\rho : G \longrightarrow \mathrm{Aut}_{\mathcal{C}}(X)$$

is a group homomorphism.

Remark 10.4.6. *If (X, ρ) is a representation of G, we also say that G acts on X. For each $g \in G$, the morphism*

$$\rho(g) : X \longrightarrow X$$

is an automorphism of X in the category $\mathcal{C}$.

Definition 10.4.7 (Faithful representation)**.** *A representation (X, ρ) of G is called faithful if*

$$\rho : G \longrightarrow \mathrm{Aut}_{\mathcal{C}}(X)$$

is injective.

Definition 10.4.8 (Morphism of representations)**.** *Let (X, ρ) and (X', ρ') be two $\mathcal{C}$-representations of G. A morphism*

$$f : (X, \rho) \longrightarrow (X', \rho')$$

is a morphism

$$f : X \longrightarrow X'$$

in $\mathcal{C}$ such that for every $g \in G$, the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\rho(g)"'] & X' \arrow[d, "\rho'(g)"] \\ X \arrow[r, "f"'] & X' \end{tikzcd}$$

Equivalently,

$$f \circ \rho(g) = \rho'(g) \circ f \qquad \text{for all } g \in G.$$

Definition 10.4.9 (The category of representations)**.** *Let G be a group and let $\mathcal{C}$ be a category. We define*

$$\mathrm{Rep}(G, \mathcal{C})$$

to be the category whose objects are $\mathcal{C}$-representations of G, and whose morphisms are morphisms of representations.

More explicitly:

- *an object of $\mathrm{Rep}(G, \mathcal{C})$ is a pair*

$$(X, \rho),$$

  where $X \in \mathrm{Ob}(\mathcal{C})$ and

$$\rho : G \to \mathrm{Aut}_{\mathcal{C}}(X)$$

  is a group homomorphism;

- *a morphism*

$$(X, \rho) \longrightarrow (X', \rho')$$

  is a morphism $f : X \to X'$ in $\mathcal{C}$ satisfying

$$f \circ \rho(g) = \rho'(g) \circ f \qquad \text{for all } g \in G.$$

Remark 10.4.10. *The identity morphism of a representation (X, ρ) is*

$$\mathrm{id}_X : X \to X,$$

and the composition of morphisms is inherited from the category $\mathcal{C}$. Hence $\mathrm{Rep}(G, \mathcal{C})$ is itself a category.

Chapter 11

Lecture 11

11.1 Categorical representations and outer automorphisms

Recall the category

$$\mathrm{Rep}(G, \mathcal{C}),$$

whose objects are pairs (X, ρ), where $X \in \mathcal{C}$ and

$$\rho : G \longrightarrow \mathrm{Aut}_{\mathcal{C}}(X)$$

is a group homomorphism. A morphism

$$f : (X, \rho) \longrightarrow (X', \rho')$$

is a morphism $f : X \to X'$ in $\mathcal{C}$ such that, for every $g \in G$,

$$\rho'(g) \circ f = f \circ \rho(g).$$

Lemma 11.1.1. *A morphism*

$$f : (X, \rho) \longrightarrow (X', \rho')$$

in $\mathrm{Rep}(G, \mathcal{C})$ *is an isomorphism if and only if the underlying morphism*

$$f : X \longrightarrow X'$$

is an isomorphism in $\mathcal{C}$*.*

Proof. If f is an isomorphism in $\mathrm{Rep}(G, \mathcal{C})$, then its underlying morphism in $\mathcal{C}$ is clearly an isomorphism.

Conversely, suppose that $f : X \to X'$ is an isomorphism in $\mathcal{C}$. Since f is a morphism of representations, for every $g \in G$ we have

$$\rho'(g) \circ f = f \circ \rho(g).$$

Multiplying by f^{-1}, we obtain

$$f^{-1} \circ \rho'(g) = \rho(g) \circ f^{-1}.$$

Hence $f^{-1} : X' \to X$ is also a morphism of representations. Therefore f is an isomorphism in $\mathrm{Rep}(G, \mathcal{C})$. $\square$

Definition 11.1.2. *Let (X, ρ) and (X, ρ') be two representations of G on the same object X. We say that (X, ρ) and (X, ρ') are isomorphic if there exists an automorphism*

$$f : X \longrightarrow X$$

in $\mathcal{C}$ such that, for every $g \in G$,

$$\rho'(g) = f \circ \rho(g) \circ f^{-1}.$$

In this case, we also say that ρ and ρ' are uniformly conjugate.

If two representations (X, ρ) and (X', ρ') are isomorphic, we write

$$(X, \rho) \cong (X', \rho').$$

Let (X, ρ) and (Y, σ) be two $\mathcal{C}$-representations of G. Then the group

$$\mathrm{Aut}_{\mathcal{C}}(X) \times \mathrm{Aut}_{\mathcal{C}}(Y)$$

acts on the set

$$\mathrm{Hom}_{\mathcal{C}}(X, Y)$$

by

$$(\alpha, \beta) \cdot f := \beta \circ f \circ \alpha^{-1},$$

where

$$\alpha \in \mathrm{Aut}_{\mathcal{C}}(X), \qquad \beta \in \mathrm{Aut}_{\mathcal{C}}(Y), \qquad f \in \mathrm{Hom}_{\mathcal{C}}(X, Y).$$

The diagonal group homomorphism

$$G \longrightarrow \mathrm{Aut}_{\mathcal{C}}(X) \times \mathrm{Aut}_{\mathcal{C}}(Y)$$

is given by

$$g \longmapsto \big(\rho(g), \sigma(g)\big).$$

Therefore it induces an action of G on $\mathrm{Hom}_{\mathcal{C}}(X, Y)$:

$$g \cdot f := \sigma(g) \circ f \circ \rho(g)^{-1}, \qquad g \in G, \quad f \in \mathrm{Hom}_{\mathcal{C}}(X, Y).$$

Equivalently, the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\rho(g)"'] & Y \arrow[d, "\sigma(g)"] \\ X \arrow[r, "g\cdot f"'] & Y \end{tikzcd}$$

where

$$g \cdot f = \sigma(g) \circ f \circ \rho(g)^{-1}.$$

Remark 11.1.3. *A morphism $f : X \to Y$ is a morphism of representations*

$$f : (X, \rho) \longrightarrow (Y, \sigma)$$

if and only if it is fixed by the above G-action, that is,

$$g \cdot f = f \qquad \text{for all } g \in G.$$

Indeed, this condition is equivalent to

$$\sigma(g) \circ f = f \circ \rho(g),$$

which is exactly the equivariance condition.

Lemma 11.1.4. *Let (X, ρ) and (Y, σ) be two objects of $\mathrm{Rep}(G, \mathcal{C})$. Then the set of morphisms*

$$(X, \rho) \longrightarrow (Y, \sigma)$$

in $\mathrm{Rep}(G, \mathcal{C})$ is precisely the set of fixed points of G in its action on

$$\mathrm{Hom}_{\mathcal{C}}(X, Y).$$

Proof. Recall that G acts on $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ by

$$g \cdot f := \sigma(g) \circ f \circ \rho(g)^{-1}.$$

Thus f is fixed by every $g \in G$ if and only if

$$\sigma(g) \circ f \circ \rho(g)^{-1} = f$$

for every $g \in G$. This is equivalent to

$$\sigma(g) \circ f = f \circ \rho(g),$$

which is exactly the condition that f is a morphism of representations

$$f : (X, \rho) \longrightarrow (Y, \sigma).$$

Therefore the morphisms in $\mathrm{Rep}(G, \mathcal{C})$ are precisely the G-fixed points of $\mathrm{Hom}_{\mathcal{C}}(X, Y)$. $\square$

Assume further that $\mathcal{C}$ is a category whose objects are sets and whose morphisms are maps between sets.

Let (X, ρ) be a $\mathcal{C}$-representation of G. We write

$$g \cdot x := \rho(g)(x), \qquad x \in X, \quad g \in G.$$

Thus the representation $\rho : G \to \mathrm{Aut}_{\mathcal{C}}(X)$ gives an action of G on the set X.

A morphism

$$f : (X, \rho) \longrightarrow (Y, \sigma)$$

is a morphism $f : X \to Y$ in $\mathcal{C}$ satisfying

$$f(g \cdot x) = g \cdot f(x)$$

for all $x \in X$ and $g \in G$. Such a map is also called a G-equivariant map.

Definition 11.1.5. *Let G act on a set X. For $x \in X$, the stabilizer of x is the subgroup*

$$G_x := \{\, g \in G \mid g \cdot x = x \,\}.$$

Lemma 11.1.6. *Let (X, ρ) be a $\mathcal{C}$-representation of G. Then:*

1. *For every $g \in G$ and every $x \in X$,*

$$G_{g \cdot x} = g G_x g^{-1}.$$

2. *If*

$$f : (X, \rho) \longrightarrow (Y, \sigma)$$

is an isomorphism of G-representations, then

$$G_{f(x)} = G_x$$

for every $x \in X$.

Proof. (1) Let $h \in G$. Then

$$h \in G_{g \cdot x}$$

if and only if

$$h \cdot (g \cdot x) = g \cdot x.$$

This is equivalent to

$$(g^{-1} h g) \cdot x = x,$$

that is,

$$g^{-1} h g \in G_x.$$

Therefore

$$h \in gG_xg^{-1}.$$

Hence

$$G_{g\cdot x} = gG_xg^{-1}.$$

(2) Since f is an isomorphism of G-representations, it is G-equivariant and bijective. For $g \in G$, we have

$$g \in G_x \iff g \cdot x = x.$$

Applying f and using equivariance gives

$$f(g \cdot x) = f(x) \iff g \cdot f(x) = f(x).$$

Hence

$$g \in G_x \iff g \in G_{f(x)}.$$

Therefore

$$G_x = G_{f(x)}.$$

$\square$

Let G be a group. Denote by

$$\mathrm{Aut}(G)$$

the automorphism group of G.

For each $g \in G$, define a map

$$\mathrm{Inn}(g) : G \longrightarrow G$$

by

$$\mathrm{Inn}(g)(x) = gxg^{-1}, \qquad x \in G.$$

This is called the inner automorphism of G induced by g.

Definition 11.1.7. *The group of inner automorphisms of G is*

$$\mathrm{Inn}(G) := \{\, \mathrm{Inn}(g) \mid g \in G \,\}.$$

Remark 11.1.8. *For every $g \in G$, the map $\mathrm{Inn}(g)$ is an automorphism of G. Indeed,*

$$\mathrm{Inn}(g)(xy) = gxyg^{-1} = (gxg^{-1})(gyg^{-1}) = \mathrm{Inn}(g)(x)\,\mathrm{Inn}(g)(y),$$

and its inverse is

$$\mathrm{Inn}(g^{-1}).$$

Proposition 11.1.9. *There is a group isomorphism*

$$\mathrm{Inn}(G) \cong G/Z(G),$$

where

$$Z(G) := \{ z \in G \mid zg = gz \text{ for all } g \in G \}$$

is the center of G.

Proof. Define

$$\Phi : G \longrightarrow \operatorname{Aut}(G)$$

by

$$\Phi(g) = \operatorname{Inn}(g).$$

We first check that Φ is a group homomorphism. For $g, h \in G$ and $x \in G$, we have

$$\big(\operatorname{Inn}(g) \circ \operatorname{Inn}(h)\big)(x) = \operatorname{Inn}(g)(hxh^{-1}) = ghxh^{-1}g^{-1} = (gh)x(gh)^{-1}.$$

Hence

$$\operatorname{Inn}(g) \circ \operatorname{Inn}(h) = \operatorname{Inn}(gh),$$

so

$$\Phi(gh) = \Phi(g)\Phi(h).$$

Thus Φ is a group homomorphism.

By definition,

$$\operatorname{Im}(\Phi) = \operatorname{Inn}(G).$$

Now we compute the kernel. We have

$$g \in \operatorname{Ker}(\Phi)$$

if and only if

$$\operatorname{Inn}(g) = \operatorname{id}_G .$$

This means that for every $x \in G$,

$$gxg^{-1} = x.$$

Equivalently,

$$gx = xg$$

for every $x \in G$. Hence

$$\operatorname{Ker}(\Phi) = Z(G).$$

Therefore, by the first isomorphism theorem,

$$G/Z(G) \cong \operatorname{Im}(\Phi) = \operatorname{Inn}(G).$$

□

Proposition 11.1.10. *The group of inner automorphisms is a normal subgroup of* *the automorphism group:*

$$\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G).$$

More precisely, for every $g \in G$ and every $a \in \mathrm{Aut}(G)$, one has

$$a \circ \mathrm{Inn}(g) \circ a^{-1} = \mathrm{Inn}(a(g)).$$

Proof. Let $a \in \mathrm{Aut}(G)$, $g \in G$, and $x \in G$. Then

$$\big(a \circ \mathrm{Inn}(g) \circ a^{-1}\big)(x) = a\Big(g\, a^{-1}(x)\, g^{-1}\Big).$$

Since a is a group homomorphism, this equals

$$a(g)\, a(a^{-1}(x))\, a(g^{-1}).$$

Hence

$$\big(a \circ \mathrm{Inn}(g) \circ a^{-1}\big)(x) = a(g)\, x\, a(g)^{-1}.$$

Therefore

$$a \circ \mathrm{Inn}(g) \circ a^{-1} = \mathrm{Inn}(a(g)).$$

Since $a(g) \in G$, we have

$$\mathrm{Inn}(a(g)) \in \mathrm{Inn}(G).$$

Thus every conjugate of an inner automorphism by an automorphism is again an inner automorphism. Hence

$$\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G).$$

□

Let G be a group. Recall that

$$\mathrm{Aut}(G)$$

denotes the automorphism group of G.

Let (X, ρ) be a $\mathcal{C}$-representation of G, where

$$\rho : G \longrightarrow \mathrm{Aut}_{\mathcal{C}}(X)$$

is a group homomorphism. For $a \in \mathrm{Aut}(G)$, define a new representation

$${}^{a}(X, \rho) := (X, {}^{a}\rho),$$

where

$${}^{a}\rho : G \longrightarrow \mathrm{Aut}_{\mathcal{C}}(X)$$

is given by

$${}^{a}\rho = \rho \circ a^{-1}.$$

Equivalently, for $g \in G$,

$$ {}^{a}\rho(g) = \rho(a^{-1}(g)). $$

Lemma 11.1.11. *For every $a \in \mathrm{Aut}(G)$, the pair*

$$ {}^{a}(X, \rho) = (X, {}^{a}\rho) $$

is again a $\mathcal{C}$-representation of G.

Proof. Since $a^{-1} : G \to G$ is a group automorphism and $\rho : G \to \mathrm{Aut}_{\mathcal{C}}(X)$ is a group homomorphism, their composite

$$ {}^{a}\rho = \rho \circ a^{-1} $$

is again a group homomorphism. Therefore

$$ {}^{a}\rho : G \longrightarrow \mathrm{Aut}_{\mathcal{C}}(X) $$

defines a $\mathcal{C}$-representation of G. □

Lemma 11.1.12. *Let*

$$ f : (X, \rho) \longrightarrow (X', \rho') $$

be a morphism of $\mathcal{C}$-representations of G. Then, for every $a \in \mathrm{Aut}(G)$, the same underlying morphism

$$ f : X \longrightarrow X' $$

induces a morphism of representations

$$ f : {}^{a}(X, \rho) \longrightarrow {}^{a}(X', \rho'). $$

Proof. Since $f : (X, \rho) \to (X', \rho')$ is a morphism of representations, for every $h \in G$ one has

$$ \rho'(h) \circ f = f \circ \rho(h). $$

Now let $g \in G$. Applying the above identity to $h = a^{-1}(g)$, we get

$$ \rho'(a^{-1}(g)) \circ f = f \circ \rho(a^{-1}(g)). $$

By definition of the twisted representations, this is precisely

$$ {}^{a}\rho'(g) \circ f = f \circ {}^{a}\rho(g). $$

Hence f is a morphism

$$ {}^{a}(X, \rho) \longrightarrow {}^{a}(X', \rho'). $$

□

Therefore $\mathrm{Aut}(G)$ acts on the set of isomorphism classes of $\mathcal{C}$-representations of G by

$$a \cdot [(X,\rho)] := [\,{}^{a}(X,\rho)\,].$$

Remark 11.1.13. *This is indeed a left action. For $a, b \in \mathrm{Aut}(G)$, we have*

$${}^{a}({}^{b}\rho) = (\rho \circ b^{-1}) \circ a^{-1} = \rho \circ b^{-1}a^{-1} = \rho \circ (ab)^{-1} = {}^{ab}\rho.$$

Hence

$${}^{a}\big({}^{b}(X,\rho)\big) = {}^{ab}(X,\rho).$$

For $g \in G$, recall that the inner automorphism induced by g is

$$\mathrm{Inn}(g) : G \longrightarrow G, \qquad x \longmapsto gxg^{-1}.$$

Thus

$$\mathrm{Inn}(g)^{-1} = \mathrm{Inn}(g^{-1}).$$

Let (X,ρ) be a $\mathcal{C}$-representation of G. Then

$${}^{\mathrm{Inn}(g)}\rho(h) = \rho\big(\mathrm{Inn}(g)^{-1}(h)\big) = \rho(g^{-1}hg)$$

for every $h \in G$.

Proposition 11.1.14. *For every $g \in G$, the morphism*

$$\rho(g) : X \longrightarrow X$$

is an isomorphism of representations

$$\rho(g) : {}^{\mathrm{Inn}(g)}(X,\rho) \xrightarrow{\cong} (X,\rho).$$

Proof. Since $\rho(g) \in \mathrm{Aut}_{\mathcal{C}}(X)$, the underlying morphism $\rho(g) : X \to X$ is an isomorphism in $\mathcal{C}$.

It remains to check equivariance. For $h \in G$, we need to prove

$$\rho(h) \circ \rho(g) = \rho(g) \circ {}^{\mathrm{Inn}(g)}\rho(h).$$

The left-hand side is

$$\rho(h) \circ \rho(g) = \rho(hg).$$

On the other hand,

$$\rho(g) \circ {}^{\mathrm{Inn}(g)}\rho(h) = \rho(g) \circ \rho(g^{-1}hg) = \rho(hg).$$

Hence

$$\rho(h) \circ \rho(g) = \rho(g) \circ {}^{\mathrm{Inn}(g)}\rho(h).$$

Therefore $\rho(g)$ is an isomorphism of representations

$$^{\mathrm{Inn}(g)}(X, \rho) \cong (X, \rho).$$

$\square$

Corollary 11.1.15. *Every inner automorphism of G acts trivially on the set of isomorphism classes of $\mathcal{C}$-representations of G.*

Proof. By the previous proposition, for every $g \in G$ and every representation (X, ρ), we have an isomorphism

$$^{\mathrm{Inn}(g)}(X, \rho) \cong (X, \rho).$$

Hence

$$\mathrm{Inn}(g) \cdot [(X, \rho)] = [(X, \rho)].$$

Therefore inner automorphisms act trivially on isomorphism classes. $\square$

Since

$$\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G),$$

the action of $\mathrm{Aut}(G)$ on isomorphism classes descends to an action of the outer automorphism group

$$\mathrm{Out}(G) := \mathrm{Aut}(G)/\mathrm{Inn}(G).$$

Thus

$$\mathrm{Out}(G)$$

acts naturally on the set of isomorphism classes of $\mathcal{C}$-representations of G.

11.2 Permutation representations and transitive actions

Let

$$\mathcal{C} = \mathrm{Set}\,.$$

An object of $\mathrm{Rep}(G, \mathrm{Set})$ is a pair

$$(X, \rho),$$

where X is a set and

$$\rho : G \longrightarrow \mathrm{Sym}(X)$$

is a group homomorphism. Equivalently, G acts on X by

$$g \cdot x := \rho(g)(x), \qquad g \in G,\ x \in X.$$

A morphism

$$f : (X, \rho) \longrightarrow (Y, \sigma)$$

is a map of sets

$$f : X \longrightarrow Y$$

such that, for every $g \in G$, the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\rho(g)"'] & Y \arrow[d, "\sigma(g)"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

Equivalently,

$$f(\rho(g)(x)) = \sigma(g)(f(x)), \qquad g \in G, \ x \in X.$$

In action notation, this condition is written as

$$f(g \cdot x) = g \cdot f(x).$$

Lemma 11.2.1. *Let*

$$f : (X, \rho) \longrightarrow (Y, \sigma)$$

be a morphism in $\mathrm{Rep}(G, \mathrm{Set})$*. If the underlying map* $f : X \to Y$ *is bijective, then* f *is an isomorphism in* $\mathrm{Rep}(G, \mathrm{Set})$*.*

Proof. Since $f : X \to Y$ is bijective, it has an inverse map

$$f^{-1} : Y \longrightarrow X.$$

We need to show that f^{-1} is G-equivariant.

Let $y \in Y$. Write $y = f(x)$ for a unique $x \in X$. For every $g \in G$, since f is equivariant, we have

$$f(g \cdot x) = g \cdot f(x) = g \cdot y.$$

Applying f^{-1}, we get

$$f^{-1}(g \cdot y) = g \cdot x = g \cdot f^{-1}(y).$$

Hence f^{-1} is G-equivariant. Therefore f is an isomorphism in $\mathrm{Rep}(G, \mathrm{Set})$. $\square$

For sets X and Y, recall the product

$$X \times Y$$

and the disjoint union

$$X \sqcup Y.$$

If Z is another set, then there is a natural bijection

$$Z \times (X \sqcup Y) \cong (Z \times X) \sqcup (Z \times Y).$$

Let (X, ρ) and (Y, σ) be two G-sets.

Definition 11.2.2. *The disjoint union of* (X, ρ) *and* (Y, σ) *is the* G*-set*

$$(X, \rho) \sqcup (Y, \sigma) := (X \sqcup Y, \rho \sqcup \sigma),$$

where the action is defined by

$$(\rho \sqcup \sigma)(g)(z) = \begin{cases} \rho(g)(z), & z \in X, \\ \sigma(g)(z), & z \in Y. \end{cases}$$

Definition 11.2.3. *The product of* (X, ρ) *and* (Y, σ) *is the* G*-set*

$$(X, \rho) \times (Y, \sigma) := (X \times Y, \rho \times \sigma),$$

where

$$(\rho \times \sigma)(g)(x, y) = \big(\rho(g)(x), \sigma(g)(y)\big).$$

In action notation,

$$g \cdot (x, y) = (g \cdot x, g \cdot y).$$

Definition 11.2.4. *Let* (X, ρ) *be a set-representation of* G*. A subset*

$$X_1 \subseteq X$$

is called G*-stable if*

$$\rho(g)(X_1) \subseteq X_1$$

for every $g \in G$*.*

If $X_1 \subseteq X$ *is* G*-stable, then the restricted action*

$$\rho_1(g) := \rho(g)|_{X_1}$$

defines a representation

$$(X_1, \rho_1),$$

called a subrepresentation of (X, ρ)*.*

Remark 11.2.5. *Since every* $\rho(g)$ *is a bijection and* $\rho(g^{-1}) = \rho(g)^{-1}$*, the condition*

$$\rho(g)(X_1) \subseteq X_1$$

for every $g \in G$ *implies*

$$\rho(g)(X_1) = X_1$$

for every $g \in G$*.*

Example 11.2.6. *Let* (X, ρ) *be a* G*-set and let* $x \in X$*. The orbit of* x *is*

$$X_1 := Gx := \{ g \cdot x \mid g \in G \}.$$

Then X_1 *is* G*-stable. Indeed, for* $h \in G$*,*

$$h \cdot (g \cdot x) = (hg) \cdot x \in Gx.$$

Therefore

$$(X_1, \rho|_{X_1})$$

is a subrepresentation of (X, ρ)*.*

Proposition 11.2.7. *Let* (X, ρ) *be a* G*-set and let*

$$X_1 = Gx$$

be the orbit of some $x \in X$*. Put*

$$X_2 := X \setminus X_1.$$

Then X_2 *is also* G*-stable. Hence*

$$(X, \rho) = (X_1, \rho_1) \sqcup (X_2, \rho_2),$$

where

$$\rho_1 = \rho|_{X_1}, \qquad \rho_2 = \rho|_{X_2}.$$

Proof. We already know that X_1 is G-stable. We prove that X_2 is G-stable.

Let $y \in X_2$ and $g \in G$. Suppose, for contradiction, that

$$g \cdot y \in X_1.$$

Since X_1 is G-stable and $g^{-1} \in G$, applying g^{-1} gives

$$y = g^{-1} \cdot (g \cdot y) \in X_1,$$

contradicting $y \in X_2 = X \setminus X_1$. Hence

$$g \cdot y \in X_2.$$

Therefore X_2 is G-stable.

Since

$$X = X_1 \sqcup X_2$$

as sets and both X_1 and X_2 are G-stable, the action ρ restricts to actions ρ_1 and ρ_2, and we obtain

$$(X, \rho) = (X_1, \rho_1) \sqcup (X_2, \rho_2).$$

$\square$

Let (X, ρ) be a set-representation of G, that is,

$$\rho : G \longrightarrow \mathrm{Sym}(X).$$

We write

$$g \cdot x := \rho(g)(x), \qquad g \in G, \ x \in X.$$

Proposition 11.2.8. *Let (X, ρ) be a set-representation of G, and assume that*

$$X \neq \varnothing.$$

The following conditions are equivalent:

1. *(X, ρ) is simple, or irreducible. That is, if (X_1, ρ_1) is a subrepresentation of (X, ρ), then*

$$X_1 = X \qquad \text{or} \qquad X_1 = \varnothing.$$

2. *(X, ρ) is indecomposable. That is, if*

$$(X, \rho) = (X_1, \rho_1) \sqcup (X_2, \rho_2),$$

then

$$X_1 = \varnothing \qquad \text{or} \qquad X_2 = \varnothing.$$

3. *(X, ρ) is transitive. That is, for every $x, y \in X$, there exists $g \in G$ such that*

$$y = g \cdot x.$$

Proof. We prove the equivalence of the three conditions.

(1) $\Rightarrow$ (2). Suppose that (X, ρ) is simple and that

$$(X, \rho) = (X_1, \rho_1) \sqcup (X_2, \rho_2).$$

Then $X_1 \subseteq X$ is a G-stable subset, hence (X_1, ρ_1) is a subrepresentation of (X, ρ). By simplicity,

$$X_1 = X \qquad \text{or} \qquad X_1 = \varnothing.$$

If $X_1 = X$, then $X_2 = \varnothing$. Therefore (X, ρ) is indecomposable.

(2) $\Rightarrow$ (1). Suppose that (X, ρ) is indecomposable. Let $X_1 \subseteq X$ be a G-stable subset. Then its complement

$$X_2 := X \setminus X_1$$

is also G-stable. Hence

$$(X, \rho) = (X_1, \rho|_{X_1}) \sqcup (X_2, \rho|_{X_2}).$$

By indecomposability,

$$X_1 = \varnothing \qquad \text{or} \qquad X_2 = \varnothing.$$

In the second case $X_1 = X$. Thus (X, ρ) is simple.

(1) $\Leftrightarrow$ (3). For $x \in X$, the orbit

$$Gx := \{ g \cdot x \mid g \in G \}$$

is a G-stable subset of X. If (X, ρ) is simple, then since $Gx \neq \varnothing$, we must have

$$Gx = X.$$

Thus every element of X lies in the orbit of x, so the action is transitive.

Conversely, if the action is transitive and $X_1 \subseteq X$ is a nonempty G-stable subset, choose $x \in X_1$. For any $y \in X$, transitivity gives some $g \in G$ such that

$$y = g \cdot x.$$

Since X_1 is G-stable, $g \cdot x \in X_1$. Hence $y \in X_1$, so $X_1 = X$. Therefore there are no nontrivial subrepresentations, and (X, ρ) is simple. $\square$

Example 11.2.9 (The transitive G-set G/H)**.** *Let $H \leq G$. Consider the set of left cosets*

$$G/H := \{ xH \mid x \in G \}.$$

Define an action of G on G/H by

$$\rho_H(g)(xH) = gxH.$$

This action is transitive. Indeed, for any two cosets $xH, yH \in G/H$, we have

$$yH = (yx^{-1})xH.$$

Hence

$$(G/H, \rho_H)$$

is a simple set-representation of G.

Moreover, the stabilizer of the coset yH is

$$G_{yH} = yHy^{-1}.$$

Indeed,

$$g \in G_{yH} \iff gyH = yH \iff y^{-1}gy \in H \iff g \in yHy^{-1}.$$

Let (X, ρ) be a G-set. Define a relation $\sim_G$ on X by

$$x \sim_G y \iff \exists g \in G \text{ such that } y = g \cdot x.$$

This is an equivalence relation. Its equivalence classes are precisely the orbits of the G-action.

We denote by

$$G \backslash X$$

the set of equivalence classes, or equivalently, the set of G-orbits in X.

Proposition 11.2.10. *Let (X, ρ) be a G-set.*

1. *Any equivalence class $\Omega \in G \backslash X$ defines a simple subrepresentation*

$$(\Omega, \rho|_\Omega).$$

2. *There is a decomposition*

$$(X, \rho) = \bigsqcup_{\Omega \in G \backslash X} (\Omega, \rho|_\Omega).$$

Proof. Each equivalence class $\Omega \in G \backslash X$ is an orbit. Hence it is G-stable, and the restriction

$$\rho|_\Omega : G \longrightarrow \mathrm{Sym}(\Omega)$$

defines a subrepresentation $(\Omega, \rho|_\Omega)$.

The action of G on Ω is transitive by definition of an orbit. Therefore, by the previous proposition, $(\Omega, \rho|_\Omega)$ is simple.

Finally, the set X is the disjoint union of its equivalence classes:

$$X = \bigsqcup_{\Omega \in G \backslash X} \Omega.$$

Since each Ω is G-stable, this gives a decomposition of G-sets:

$$(X, \rho) = \bigsqcup_{\Omega \in G \backslash X} (\Omega, \rho|_\Omega).$$

□

Theorem 11.2.11. *Let (X, ρ) be a transitive set-representation of G. For $x \in X$, let*

$$G_x := \{ g \in G \mid g \cdot x = x \}$$

be the stabilizer of x. Then the map

$$\Phi : G/G_x \longrightarrow X, \qquad gG_x \longmapsto g \cdot x$$

defines an isomorphism of G-sets

$$(G/G_x, \rho_{G_x}) \cong (X, \rho).$$

Proof. First we show that Φ is well-defined. Suppose

$$gG_x = g'G_x.$$

Then $g' = gh$ for some $h \in G_x$. Hence

$$g' \cdot x = (gh) \cdot x = g \cdot (h \cdot x) = g \cdot x,$$

since $h \in G_x$. Thus $\Phi(gG_x) = \Phi(g'G_x)$.

The map Φ is surjective because the action of G on X is transitive.

It is injective as well. If

$$\Phi(gG_x) = \Phi(g'G_x),$$

then

$$g \cdot x = g' \cdot x.$$

Hence

$$(g'^{-1}g) \cdot x = x,$$

so $g'^{-1}g \in G_x$. Therefore

$$gG_x = g'G_x.$$

Finally, Φ is G-equivariant. For $a \in G$,

$$\Phi\big(\rho_{G_x}(a)(gG_x)\big) = \Phi(agG_x) = ag \cdot x,$$

while

$$\rho(a)\big(\Phi(gG_x)\big) = a \cdot (g \cdot x) = ag \cdot x.$$

Hence

$$\Phi \circ \rho_{G_x}(a) = \rho(a) \circ \Phi.$$

Thus Φ is an isomorphism of G-sets. $\square$

Theorem 11.2.12. *Let $H, H' \leq G$. Then the G-sets*

$$(G/H, \rho_H) \qquad \textit{and} \qquad (G/H', \rho_{H'})$$

are isomorphic if and only if H and H' are conjugate in G.

Proof. Suppose first that

$$F : G/H \longrightarrow G/H'$$

is an isomorphism of G-sets. Write

$$F(H) = aH'$$

for some $a \in G$. Since F is G-equivariant, the stabilizer of H must equal the stabilizer of $F(H)$. Now

$$G_H = H,$$

and

$$G_{aH'} = aH'a^{-1}.$$

Hence

$$H = aH'a^{-1}.$$

Thus H and H' are conjugate in G.

Conversely, suppose that H and H' are conjugate. Write

$$H = aH'a^{-1}$$

for some $a \in G$. Define

$$F : G/H \longrightarrow G/H'$$

by

$$F(gH) = gaH'.$$

This is well-defined. Indeed, if $gH = g_1H$, then $g_1 = gh$ for some $h \in H$. Since $H = aH'a^{-1}$, we have

$$a^{-1}ha \in H'.$$

Therefore

$$g_1aH' = ghaH' = ga(a^{-1}ha)H' = gaH'.$$

Thus $F(gH) = F(g_1H)$.

The map F is bijective, with inverse

$$F^{-1}(gH') = ga^{-1}H.$$

It is also G-equivariant, since for $t \in G$,

$$F(tgH) = tgaH' = tF(gH).$$

Hence F is an isomorphism of G-sets:

$$(G/H, \rho_H) \cong (G/H', \rho_{H'}).$$

$\square$

Definition 11.2.13 (Degree)**.** *Let (X, ρ) be a set-representation of G, that is,*

$$\rho : G \longrightarrow \mathrm{Sym}(X).$$

The degree of (X, ρ) is defined to be the cardinality of X, denoted by

$$\deg(X, \rho) := |X|.$$

Corollary 11.2.14. *Let G be a finite group. Then the degree of any irreducible set-representation of G divides $|G|$.*

Proof. Let (X, ρ) be an irreducible set-representation of G. Irreducible set-representations are transitive G-sets, the action of G on X is transitive.

Choose $x \in X$, and let

$$G_x := \{ g \in G \mid g \cdot x = x \}$$

be the stabilizer of x. By the classification of transitive G-sets, we have an isomorphism of G-sets

$$X \cong G/G_x.$$

Hence

$$|X| = |G/G_x| = [G : G_x].$$

Since G is finite, Lagrange's theorem gives

$$|G| = [G : G_x] \cdot |G_x|.$$

Therefore

$$|X| = [G : G_x]$$

divides $|G|$. Thus the degree of (X, ρ) divides $|G|$. □

11.3 Finite-dimensional linear representations

Let k be a field and let G be a finite group.

Definition 11.3.1 (Finite-dimensional k-linear representation)**.** *A finite dimensional k-linear representation of G is a pair*

$$(M, \rho),$$

where M is a finite-dimensional k-vector space and

$$\rho : G \longrightarrow \mathrm{GL}(M)$$

is a group homomorphism. Here

$$\mathrm{GL}(M) = \{\, f : M \to M \mid f \text{ is an invertible } k\text{-linear map} \,\}.$$

For $g \in G$ and $m \in M$, we often write

$$g \cdot m := \rho(g)(m).$$

Thus the representation condition says

$$(gh) \cdot m = g \cdot (h \cdot m), \qquad 1_G \cdot m = m.$$

Definition 11.3.2 (The category of k-linear representations)**.** *The category of finite-dimensional k-linear representations of G is denoted by*

$$\mathrm{Rep}(G, \mathrm{vect}_k).$$

Its objects are finite-dimensional k-linear representations

$$(M, \rho).$$

A morphism

$$f : (M, \rho) \longrightarrow (N, \sigma)$$

is a k-linear map

$$f : M \longrightarrow N$$

such that

$$f(g \cdot m) = g \cdot f(m)$$

for every $m \in M$ and $g \in G$. Equivalently,

$$\sigma(g) \circ f = f \circ \rho(g), \qquad g \in G.$$

Definition 11.3.3 (Degree)**.** *Let (M, ρ) be a finite-dimensional k-linear representation of G. The dimension of M is called the degree of the representation:*

$$\deg(M, \rho) := \dim_k M.$$

Example 11.3.4 (The trivial representation)**.** *The trivial representation, also called the principal representation, is the one-dimensional representation*

$$\rho_{\mathrm{triv}} : G \longrightarrow \mathrm{GL}_1(k) \cong k^{\times}$$

given by

$$\rho_{\mathrm{triv}}(g) = 1$$

for all $g \in G$. Equivalently, G acts on the one-dimensional vector space k by

$$g \cdot x = x, \qquad g \in G,\ x \in k.$$

Example 11.3.5 (One-dimensional representations)**.** *If (M, ρ) is a k-linear representation with*

$$\dim_k M = 1,$$

then after choosing a basis of M, we may identify

$$\mathrm{GL}(M) \cong k^\times.$$

Hence a one-dimensional representation is the same as a group homomorphism

$$\rho : G \longrightarrow k^\times.$$

Let (M, ρ) be a finite-dimensional k-linear representation of G, and assume

$$r := \dim_k M.$$

After fixing a basis of M, we obtain an isomorphism of groups

$$\mathrm{GL}(M) \cong \mathrm{GL}_r(k),$$

where

$$\mathrm{GL}_r(k)$$

is the group of invertible $r \times r$ matrices over k.

Therefore a k-linear representation of G may be represented by a pair

$$(r, \rho),$$

where

$$r \in \mathbb{Z}_{\geq 1}$$

and

$$\rho : G \longrightarrow \mathrm{GL}_r(k)$$

is a group homomorphism. Such a pair is called a matrix representation of G over k.

Remark 11.3.6. *Choosing a basis of M transforms each linear automorphism*

$$\rho(g) : M \longrightarrow M$$

into an invertible matrix. Thus a representation may be viewed concretely as assigning to every $g \in G$ an invertible matrix $\rho(g) \in \mathrm{GL}_r(k)$, satisfying

$$\rho(gh) = \rho(g)\rho(h), \qquad \rho(1_G) = I_r.$$

Proposition 11.3.7. *Let*

$$(r, \rho) \qquad \text{and} \qquad (s, \sigma)$$

be two matrix representations of G. Then they are isomorphic if and only if

$$r = s$$

and there exists a matrix

$$A \in \mathrm{GL}_r(k)$$

such that

$$\sigma(g) = A\rho(g)A^{-1}$$

for every $g \in G$.

Proof. Suppose first that the two representations are isomorphic. Then there exists a k-linear isomorphism

$$f : k^r \longrightarrow k^s$$

such that

$$\sigma(g) \circ f = f \circ \rho(g)$$

for every $g \in G$. Since f is an isomorphism, we must have

$$r = s.$$

Let $A \in \mathrm{GL}_r(k)$ be the matrix of f with respect to the chosen bases. Then the equivariance condition becomes

$$\sigma(g)A = A\rho(g).$$

Multiplying on the right by A^{-1}, we obtain

$$\sigma(g) = A\rho(g)A^{-1}.$$

Conversely, suppose that $r = s$ and that there exists $A \in \mathrm{GL}_r(k)$ such that

$$\sigma(g) = A\rho(g)A^{-1}$$

for all $g \in G$. Let

$$f : k^r \longrightarrow k^r$$

be the linear isomorphism whose matrix is A. Then

$$\sigma(g)A = A\rho(g),$$

which means precisely that

$$\sigma(g) \circ f = f \circ \rho(g)$$

for every $g \in G$. Hence f is an isomorphism of representations. □

Remark 11.3.8. *Thus two matrix representations* (r, ρ) *and* (r, σ) *are isomorphic precisely when the matrices*

$$\rho(g) \qquad \text{and} \qquad \sigma(g)$$

are uniformly similar, meaning that the same change-of-basis matrix $A \in \mathrm{GL}_r(k)$ *satisfies*

$$\sigma(g) = A\rho(g)A^{-1}$$

for every $g \in G$.

Let $k \subseteq k'$ be a field extension, and let G be a group.

Definition 11.3.9 (Extension of scalars)**.** *Let* (M, ρ) *be a finite-dimensional* k*-linear representation of* G*, where*

$$\rho : G \longrightarrow \mathrm{GL}(M).$$

The extension of scalars of (M, ρ) *from* k *to* k' *is the* k'*-linear representation*

$$\big(k' \otimes_k M,\ k' \otimes_k \rho\big),$$

where

$$k' \otimes_k \rho : G \longrightarrow \mathrm{GL}_{k'}(k' \otimes_k M)$$

is defined by

$$g \longmapsto 1_{k'} \otimes \rho(g).$$

Explicitly,

$$(1_{k'} \otimes \rho(g))(\lambda \otimes m) = \lambda \otimes \rho(g)(m), \qquad \lambda \in k',\ m \in M.$$

Remark 11.3.10. *If*

$$e_1, \dots, e_n$$

is a k*-basis of* M*, then*

$$1 \otimes e_1, \dots, 1 \otimes e_n$$

is a k'*-basis of* $k' \otimes_k M$*. In particular,*

$$\dim_{k'}(k' \otimes_k M) = \dim_k M.$$

Remark 11.3.11 (Matrix form)**.** *Suppose that* (M, ρ) *is given by a matrix representation*

$$\rho : G \longrightarrow \mathrm{GL}_n(k).$$

Extending scalars to k' *amounts to viewing each matrix*

$$\rho(g) \in \mathrm{GL}_n(k)$$

as a matrix with entries in k'*:*

$$\rho(g) \in \mathrm{GL}_n(k')$$

via the inclusion $k \subseteq k'$.

Definition 11.3.12 (Representations rational over a subfield)**.** *Let (M', ρ') be a k'-linear representation of G. We say that (M', ρ') is rational over k if there exists a k-linear representation (M, ρ) of G such that*

$$(M', \rho') \cong \big(k' \otimes_k M,\ k' \otimes_k \rho\big)$$

as k'-linear representations of G.

Proposition 11.3.13 (Matrix criterion for rationality)**.** *Let*

$$\rho : G \longrightarrow \mathrm{GL}_n(k')$$

be a matrix representation of G over k'. Then ρ is rational over k if and only if there exists a matrix

$$U \in \mathrm{GL}_n(k')$$

such that, for every $g \in G$, all entries of

$$U\rho(g)U^{-1}$$

belong to k.

Proof. The representation ρ is rational over k precisely when it is isomorphic over k' to the scalar extension of some representation

$$\rho_0 : G \longrightarrow \mathrm{GL}_n(k).$$

An isomorphism of matrix representations over k' is given by a change of basis matrix $U \in \mathrm{GL}_n(k')$, and the isomorphism condition is exactly

$$U\rho(g)U^{-1} = \rho_0(g)$$

for every $g \in G$. Since $\rho_0(g)$ has entries in k, this proves one direction.

Conversely, if such a matrix U exists, then the matrices

$$U\rho(g)U^{-1}$$

define a representation of G over k, and ρ is obtained from it by extension of scalars to k'. Hence ρ is rational over k. $\square$

Let

$$(M, \rho), \qquad (N, \sigma)$$

be two k-linear representations of G.

Definition 11.3.14 (Direct sum)**.** *The direct sum of* (M, ρ) *and* (N, σ) *is the representation*

$$(M, \rho) \oplus (N, \sigma) := (M \oplus N, \rho \oplus \sigma),$$

where

$$(\rho \oplus \sigma)(g)(m, n) = (\rho(g)m, \sigma(g)n).$$

Definition 11.3.15 (Tensor product)**.** *The tensor product of* (M, ρ) *and* (N, σ) *is the representation*

$$(M, \rho) \otimes (N, \sigma) := (M \otimes_k N, \rho \otimes \sigma),$$

where

$$(\rho \otimes \sigma)(g)(m \otimes n) = \rho(g)m \otimes \sigma(g)n.$$

Definition 11.3.16 (Subrepresentation)**.** *Let* (M, ρ) *be a* k*-linear representation of* G*. A* k*-subspace*

$$M_1 \subseteq M$$

is called G*-stable if*

$$\rho(g)(M_1) \subseteq M_1$$

for every $g \in G$*.*

If $M_1 \subseteq M$ *is* G*-stable, then the restricted maps*

$$\rho_1(g) := \rho(g)|_{M_1}$$

define a representation

$$(M_1, \rho_1),$$

called a subrepresentation of (M, ρ)*.*

Remark 11.3.17. *Since each* $\rho(g)$ *is invertible and*

$$\rho(g)^{-1} = \rho(g^{-1}),$$

the condition

$$\rho(g)(M_1) \subseteq M_1$$

for all $g \in G$ *implies*

$$\rho(g)(M_1) = M_1$$

for all $g \in G$*.*

Definition 11.3.18 (Quotient representation)**.** *Let* (M, ρ) *be a* k*-linear representation of* G*, and let* $M_1 \subseteq M$ *be a* G*-stable subspace. Then the quotient vector space*

$$M/M_1$$

carries a natural representation

$$\overline{\rho} : G \longrightarrow \mathrm{GL}(M/M_1)$$

defined by

$$\overline{\rho}(g)(m + M_1) := \rho(g)(m) + M_1.$$

This is called the quotient representation of (M, ρ) by (M_1, ρ_1).

Definition 11.3.19 (Simple representation)**.** *A k-linear representation (M, ρ) of G is called simple, or irreducible, if its only G-stable subspaces are*

$$0 \qquad \textit{and} \qquad M.$$

Definition 11.3.20 (Indecomposable representation)**.** *A k-linear representation of G is called indecomposable if whenever*

$$(M, \rho) \cong (M_1, \rho_1) \oplus (M_2, \rho_2),$$

one has

$$M_1 = 0 \qquad \textit{or} \qquad M_2 = 0.$$

Remark 11.3.21. *Every simple representation is indecomposable. The converse need not hold in general.*

Theorem 11.3.22 (Krull–Schmidt theorem for finite-dimensional representations)**.** *Let G be a finite group, let k be a field, and let*

$$(M, \rho)$$

be a finite-dimensional k-linear representation of G.

1. *The representation (M, ρ) is a finite direct sum of indecomposable representations.*

2. *Suppose that*

$$(M, \rho) \cong \bigoplus_{i \in I} (M_i, \rho_i) \cong \bigoplus_{j \in J} (M'_j, \rho'_j),$$

where every (M_i, ρ_i) and every (M'_j, ρ'_j) is indecomposable. Then there exists a bijection

$$\alpha : I \longrightarrow J$$

such that, for every $i \in I$,

$$(M_i, \rho_i) \cong (M'_{\alpha(i)}, \rho'_{\alpha(i)}).$$

Proof. We regard k-linear representations of G as modules over the group algebra kG. More precisely, a representation

$$\rho : G \longrightarrow \mathrm{GL}(M)$$

determines a left kG-module structure on M by

$$\left(\sum_{g\in G} a_g g\right) m := \sum_{g\in G} a_g\, \rho(g)(m), \qquad a_g \in k,\ m \in M.$$

Conversely, every left kG-module gives a k-linear representation of G. Hence $\mathrm{Rep}(G, \mathrm{vect}_k)$ is equivalent to the category of finite-dimensional left kG-modules.

Since G is finite, the group algebra kG is finite-dimensional over k. Therefore every finite-dimensional kG-module has finite composition length.

Applying the Krull–Schmidt theorem for finite-length modules to the kG-module M, we obtain a decomposition

$$M \cong M_1 \oplus \cdots \oplus M_r$$

into indecomposable kG-modules. Translating this back into representations gives a decomposition

$$(M, \rho) \cong (M_1, \rho_1) \oplus \cdots \oplus (M_r, \rho_r)$$

into indecomposable representations. This proves (1).

For (2), suppose that

$$(M, \rho) \cong \bigoplus_{i\in I} (M_i, \rho_i) \cong \bigoplus_{j\in J} (M'_j, \rho'_j),$$

where all summands are indecomposable. Since M is finite-dimensional, both index sets I and J are finite.

Passing again to kG-modules, this gives two decompositions of the same finite-length kG-module M into indecomposable direct summands:

$$M \cong \bigoplus_{i\in I} M_i \cong \bigoplus_{j\in J} M'_j.$$

By the Krull–Schmidt uniqueness theorem for finite-length modules, there exists a bijection

$$\alpha : I \longrightarrow J$$

such that

$$M_i \cong M'_{\alpha(i)}$$

as kG-modules for every $i \in I$.

Translating the kG-module isomorphisms back into k-linear isomorphisms, we obtain

$$(M_i, \rho_i) \cong (M'_{\alpha(i)}, \rho'_{\alpha(i)})$$

as k-linear representations of G. This proves the desired uniqueness. $\square$

Remark 11.3.23. *The theorem says that a finite-dimensional representation can be decomposed into indecomposable pieces, and that these pieces are unique up to isomorphism and reordering.*

11.4 Linearization, regular representation, and group algebra

Let k be a field and let (Ω, ρ) be a set-representation of G. Thus

$$\rho : G \longrightarrow \mathrm{Sym}(\Omega)$$

is a group homomorphism.

Denote by

$$k\Omega$$

the k-vector space with basis Ω. That is, every element of $k\Omega$ is a finite k-linear combination

$$\sum_{\omega \in \Omega} a_\omega \omega, \qquad a_\omega \in k.$$

The action of G on Ω induces a k-linear representation

$$k\rho : G \longrightarrow \mathrm{GL}(k\Omega)$$

defined on basis elements by

$$(k\rho)(g)(\omega) = \rho(g)(\omega) = g \cdot \omega, \qquad g \in G,\ \omega \in \Omega,$$

and extended k-linearly.

Lemma 11.4.1. *The map*

$$k\rho : G \longrightarrow \mathrm{GL}(k\Omega)$$

is a group homomorphism.

Proof. For each $g \in G$, the map $\rho(g) : \Omega \to \Omega$ is a bijection. Hence its k-linear extension

$$(k\rho)(g) : k\Omega \to k\Omega$$

is an invertible k-linear map.

For $g, h \in G$ and $\omega \in \Omega$, we have

$$(k\rho)(gh)(\omega) = (gh) \cdot \omega = g \cdot (h \cdot \omega) = (k\rho)(g)\big((k\rho)(h)(\omega)\big).$$

Since the basis elements $\omega \in \Omega$ span $k\Omega$, it follows that

$$(k\rho)(gh) = (k\rho)(g) \circ (k\rho)(h).$$

Also,

$$(k\rho)(1_G) = \mathrm{id}_{k\Omega} .$$

Therefore $k\rho$ is a group homomorphism. $\square$

Definition 11.4.2 (Linearization of a G-set)**.** *The k-linear representation*

$$(k\Omega, k\rho)$$

is called the linearization of the G-set (Ω, ρ).

Let

$$f : (\Omega, \rho) \longrightarrow (\Lambda, \sigma)$$

be a morphism of G-sets. Thus $f : \Omega \to \Lambda$ is a map satisfying

$$f(g \cdot \omega) = g \cdot f(\omega)$$

for all $g \in G$ and $\omega \in \Omega$.

Define a k-linear map

$$kf : k\Omega \longrightarrow k\Lambda$$

by

$$kf(\omega) = f(\omega), \qquad \omega \in \Omega,$$

and extend k-linearly.

Then kf is a morphism of k-linear representations. Indeed, for $g \in G$ and $\omega \in \Omega$,

$$kf\big((k\rho)(g)(\omega)\big) = kf(g \cdot \omega) = f(g \cdot \omega) = g \cdot f(\omega) = (k\sigma)(g)(kf(\omega)).$$

Hence

$$kf \circ (k\rho)(g) = (k\sigma)(g) \circ kf.$$

Therefore the construction

$$(\Omega, \rho) \longmapsto (k\Omega, k\rho)$$

induces a functor

$$k[-] : \mathrm{Rep}(G, \mathrm{Set}) \longrightarrow \mathrm{Rep}(G, \mathrm{vect}_k).$$

Let (Ω, ρ) and (Λ, σ) be two G-sets.

Proposition 11.4.3. *There is a natural isomorphism of k-linear representations*

$$k(\Omega \sqcup \Lambda) \cong k\Omega \oplus k\Lambda.$$

Proof. The disjoint union $\Omega \sqcup \Lambda$ has basis elements coming either from Ω or from Λ. Hence the assignment

$$\omega \longmapsto (\omega, 0), \qquad \lambda \longmapsto (0, \lambda)$$

induces a k-linear isomorphism

$$k(\Omega \sqcup \Lambda) \cong k\Omega \oplus k\Lambda.$$

This isomorphism is G-equivariant because G acts on each summand by the corresponding action. $\square$

Proposition 11.4.4. *There is a natural isomorphism of k-linear representations*

$$k(\Omega \times \Lambda) \cong k\Omega \otimes_k k\Lambda.$$

Proof. Define

$$\Phi : k(\Omega \times \Lambda) \longrightarrow k\Omega \otimes_k k\Lambda$$

on basis elements by

$$\Phi(\omega, \lambda) = \omega \otimes \lambda.$$

Since the elements (ω, λ) form a basis of $k(\Omega \times \Lambda)$, this defines a k-linear map.

The elements

$$\omega \otimes \lambda, \qquad \omega \in \Omega, \ \lambda \in \Lambda,$$

form a basis of $k\Omega \otimes_k k\Lambda$, so Φ is a k-linear isomorphism.

It is G-equivariant because

$$\Phi\big(g \cdot (\omega, \lambda)\big) = \Phi(g \cdot \omega, g \cdot \lambda) = (g \cdot \omega) \otimes (g \cdot \lambda),$$

while

$$g \cdot \Phi(\omega, \lambda) = g \cdot (\omega \otimes \lambda) = (g \cdot \omega) \otimes (g \cdot \lambda).$$

Hence

$$k(\Omega \times \Lambda) \cong k\Omega \otimes_k k\Lambda$$

as k-linear representations of G. $\square$

Remark 11.4.5. *Thus linearization transforms disjoint unions of G-sets into direct sums of k-linear representations, and transforms products of G-sets into tensor products of k-linear representations.*

Remark 11.4.6. *Even if (Ω, ρ) is transitive as a G-set and*

$$|\Omega| > 1,$$

the linearized representation

$$(k\Omega, k\rho)$$

need not be irreducible.

Indeed, define

$$S_\Omega := \sum_{\omega \in \Omega} \omega \in k\Omega.$$

Since Ω is a basis of $k\Omega$, we have

$$S_\Omega \neq 0.$$

For every $g \in G$,

$$g \cdot S_\Omega = \sum_{\omega \in \Omega} g \cdot \omega.$$

Since $g : \Omega \to \Omega$ is a bijection, the set

$$\{g \cdot \omega \mid \omega \in \Omega\}$$

is again Ω. Hence

$$g \cdot S_\Omega = S_\Omega.$$

Therefore the line

$$kS_\Omega$$

is G-stable. If $|\Omega| > 1$, this is a nonzero proper subrepresentation of $k\Omega$. Hence $k\Omega$ is not irreducible.

Definition 11.4.7 (Left regular representation)**.** *Let G act on itself by left multiplication:*

$$g \cdot x := gx, \qquad g, x \in G.$$

The linearization of this G-set is the representation

$$kG$$

with basis G, where G acts by

$$g \cdot x = gx.$$

This representation is called the left regular representation of G.

Let

$$S_G := \sum_{g \in G} g \in kG.$$

Then the line

$$kS_G$$

is G-stable, since

$$h \cdot S_G = \sum_{g \in G} hg = \sum_{g \in G} g = S_G$$

for every $h \in G$.

Lemma 11.4.8. *Assume that k has characteristic $p > 0$ and that*

$$p \mid |G|.$$

Then the G-stable line

$$kS_G \subseteq kG$$

has no G-stable complement in the left regular representation kG.

Proof. Suppose, for contradiction, that there exists a G-stable subspace $V' \subseteq kG$ such that

$$kG = kS_G \oplus V'.$$

Let

$$\pi : kG \longrightarrow kS_G$$

be the projection with kernel V'. Thus

$$\pi|_{kS_G} = \mathrm{id}_{kS_G} .$$

Since both kS_G and V' are G-stable, the projection π is G-equivariant:

$$\pi(gx) = g\pi(x), \qquad g \in G, \ x \in kG.$$

Write

$$\pi(1) = \lambda S_G$$

for some $\lambda \in k$. Then, for every $g \in G$,

$$\pi(g) = \pi(g \cdot 1) = g \cdot \pi(1) = g \cdot (\lambda S_G) = \lambda S_G.$$

Therefore

$$\pi(S_G) = \pi\left(\sum_{g \in G} g\right) = \sum_{g \in G} \pi(g) = \sum_{g \in G} \lambda S_G = |G| \lambda S_G.$$

Since $\mathrm{char}(k) = p$ and $p \mid |G|$, we have

$$|G| = 0 \qquad \text{in } k.$$

Hence

$$\pi(S_G) = 0.$$

But $S_G \in kS_G$ and π restricts to the identity on kS_G, so

$$\pi(S_G) = S_G.$$

This is a contradiction, because $S_G \neq 0$. Therefore kS_G has no G-stable complement in kG. $\square$

Let k be a field, and let

$$(M, \rho), \qquad (N, \sigma)$$

be two k-linear representations of a group G.

The group G acts on the k-vector space

$$\mathrm{Hom}_k(M, N)$$

by

$$g \cdot \alpha := \sigma(g) \circ \alpha \circ \rho(g)^{-1}, \qquad g \in G, \quad \alpha \in \mathrm{Hom}_k(M, N).$$

Equivalently, the following diagram commutes:

$$\begin{tikzcd} M \arrow[r, "\alpha"] \arrow[d, "\rho(g)"'] & N \arrow[d, "\sigma(g)"] \\ M \arrow[r, "g\cdot\alpha"'] & N. \end{tikzcd}$$

Lemma 11.4.9. *The formula*

$$g \cdot \alpha = \sigma(g) \circ \alpha \circ \rho(g)^{-1}$$

defines a k-linear representation of G on $\mathrm{Hom}_k(M, N)$*.*

Proof. For each $g \in G$, the map

$$\alpha \longmapsto \sigma(g) \circ \alpha \circ \rho(g)^{-1}$$

is k-linear. Moreover, for $g, h \in G$,

$$\begin{aligned} g \cdot (h \cdot \alpha) &= \sigma(g) \circ \big(\sigma(h) \circ \alpha \circ \rho(h)^{-1}\big) \circ \rho(g)^{-1} \\ &= \sigma(gh) \circ \alpha \circ \big(\rho(h)^{-1} \circ \rho(g)^{-1}\big) \\ &= \sigma(gh) \circ \alpha \circ \rho(gh)^{-1} \\ &= (gh) \cdot \alpha. \end{aligned}$$

Also $1_G \cdot \alpha = \alpha$. Hence this is a representation of G. □

Taking $N = k$ with the trivial representation, we obtain a natural representation on the dual vector space

$$M^* := \mathrm{Hom}_k(M, k).$$

Definition 11.4.10 (Contragredient representation)**.** *Let (M, ρ) be a k-linear representation of G. Its contragredient representation is the representation on*

$$M^* = \mathrm{Hom}_k(M, k)$$

defined by

$$(g \cdot \alpha)(m) = \alpha\big(\rho(g)^{-1} m\big), \qquad g \in G, \ \alpha \in M^*, \ m \in M.$$

Equivalently,

$$g \cdot \alpha = \alpha \circ \rho(g)^{-1}.$$

The tensor product

$$M \otimes_k N$$

also carries a natural G-representation.

Definition 11.4.11 (Tensor product representation)**.** *The tensor product of* (M, ρ) *and* (N, σ) *is*

$$(M, \rho) \otimes (N, \sigma) := (M \otimes_k N, \rho \otimes \sigma),$$

where

$$(\rho \otimes \sigma)(g)(m \otimes n) := \rho(g)m \otimes \sigma(g)n.$$

In action notation,

$$g \cdot (m \otimes n) = (g \cdot m) \otimes (g \cdot n).$$

A representation of degree 1 is the same as a group homomorphism

$$\rho : G \longrightarrow k^{\times}.$$

Indeed, if $M = k$, then

$$\mathrm{GL}(M) = \mathrm{GL}_1(k) \cong k^{\times}.$$

Definition 11.4.12. *A one-dimensional representation of* G *over* k *is a pair*

$$(k, \rho),$$

where

$$\rho : G \longrightarrow k^{\times}$$

is a group homomorphism.

Remark 11.4.13. *Since* $k^{\times}$ *is abelian, every homomorphism*

$$\rho : G \longrightarrow k^{\times}$$

kills the commutator subgroup $[G, G]$*. Hence* ρ *factors uniquely through the abelianization:*

$$G/[G, G] \longrightarrow k^{\times}.$$

Therefore one-dimensional representations of G *over* k *are naturally the same as group homomorphisms*

$$G/[G, G] \longrightarrow k^{\times}.$$

Example 11.4.14 (One-dimensional representations of D_8)**.** *Let*

$$G = D_8 = \langle \sigma, \tau \mid \sigma^2 = \tau^2 = (\sigma\tau)^4 = 1 \rangle.$$

Then

$$[G, G] = Z(G) = \langle (\sigma\tau)^2 \rangle,$$

and hence

$$G/[G, G] \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Therefore, over a field k with $\mathrm{char}(k) \neq 2$, the group G has four one-dimensional representations:

	σ	τ
ρ_1	1	1
ρ_2	1	-1
ρ_3	-1	1
ρ_4	-1	-1

Let (M, ρ) be a k-linear representation of G, and let

$$(k, \sigma)$$

be a one-dimensional representation of G. Then

$$M \otimes_k k \cong M$$

as k-vector spaces. Under this identification, the tensor product representation

$$(M, \rho) \otimes (k, \sigma)$$

corresponds to the representation

$$(M, \rho \otimes \sigma),$$

where

$$(\rho \otimes \sigma)(g) = \sigma(g)\rho(g).$$

Lemma 11.4.15. *Let (S, ρ) be an irreducible k-linear representation of G, and let (k, σ) be a one-dimensional representation of G. Then*

$$(S, \rho) \otimes (k, \sigma)$$

is irreducible.

Proof. Under the identification $S \otimes_k k \cong S$, the tensor product representation is given by

$$g \longmapsto \sigma(g)\rho(g).$$

Let $W \subseteq S$ be a nonzero subspace stable under this twisted action. For $w \in W$ and $g \in G$, we have

$$\sigma(g)\rho(g)w \in W.$$

Since $\sigma(g) \in k^\times$, multiplication by $\sigma(g)$ is invertible. Hence

$$\rho(g)w \in W.$$

Thus W is stable under the original action ρ. Since (S, ρ) is irreducible, we must have

$$W = S.$$

Therefore the twisted representation is irreducible. □

Let G be a finite group and k a field.

Definition 11.4.16 (Group algebra)**.** *The group algebra kG is the k-vector space with basis G:*

$$kG = \left\{ \sum_{g \in G} x_g g \;\middle|\; x_g \in k \right\}.$$

Multiplication is defined by extending the group multiplication k-bilinearly. Thus, if

$$x = \sum_{s \in G} x_s s, \qquad y = \sum_{t \in G} y_t t,$$

then

$$xy = \sum_{s,t \in G} x_s y_t \, st.$$

Equivalently,

$$xy = \sum_{g \in G} \left(\sum_{s \in G} x_s y_{s^{-1}g} \right) g.$$

Proposition 11.4.17. *The group algebra kG is a k-algebra.*

Proof. The multiplication on kG is defined by k-bilinear extension of the multiplication in G. Since multiplication in G is associative, the induced multiplication on kG is associative. The identity element is $1_G \in G \subseteq kG$. Hence kG is a unital associative k-algebra. □

Theorem 11.4.18. *The category of k-linear representations of G is naturally equivalent to the category of left kG-modules:*

$$\mathrm{Rep}(G, \mathrm{vect}_k) \simeq kG\text{-}\mathrm{mod}\,.$$

Proof. Let (M, ρ) be a k-linear representation of G. Define an action of kG on M by

$$\left(\sum_{g \in G} a_g g\right) m := \sum_{g \in G} a_g \, \rho(g)(m).$$

This makes M into a left kG-module.

Conversely, if M is a left kG-module, then each $g \in G \subseteq kG$ acts on M by a k-linear automorphism, since $g^{-1} \in G$ acts as its inverse. Hence we obtain a group homomorphism

$$\rho : G \longrightarrow \mathrm{GL}(M), \qquad \rho(g)(m) = gm.$$

This gives a k-linear representation of G.

These two constructions are inverse to each other, and morphisms are exactly the k-linear maps commuting with the G-action, equivalently the kG-module homomorphisms. Therefore the two categories are naturally equivalent. $\square$

Chapter 12

Lecture 12

12.1 Group algebra, class functions, and symmetric algebras

Definition 12.1.1 (Group algebra)**.** *Let G be a finite group, and let R be a commutative unital ring. The group algebra RG is the free R-module with basis G, whose multiplication is determined by the group multiplication of G.*

Thus every element of RG can be written uniquely in the form

$$x = \sum_{s \in G} x_s s, \qquad y = \sum_{t \in G} y_t t,$$

where $x_s, y_t \in R$. Their product is given by

$$xy = \sum_{s,t \in G} x_s y_t st = \sum_{g \in G} \left(\sum_{\substack{s,t \in G \\ st = g}} x_s y_t \right) g.$$

From now on, let k be a field, and consider the group algebra kG.

Definition 12.1.2. *Define a k-linear form*

$$\pi : kG \longrightarrow k$$

by

$$\pi(g) = \begin{cases} 1, & g = 1, \\ 0, & g \neq 1, \end{cases}$$

for all $g \in G$. Equivalently, $\pi(x)$ is the coefficient of the identity element $1 \in G$ in $x \in kG$.

Lemma 12.1.3. *Let $x, y \in kG$. Then*

$$\pi(xy) = \pi(yx).$$

Proof. Write

$$x = \sum_{s \in G} x_s s, \qquad y = \sum_{t \in G} y_t t.$$

Then the coefficient of $1 \in G$ in xy is

$$\pi(xy) = \sum_{\substack{s,t \in G \\ st=1}} x_s y_t = \sum_{s \in G} x_s y_{s^{-1}}.$$

Similarly,

$$\pi(yx) = \sum_{\substack{t,s \in G \\ ts=1}} y_t x_s = \sum_{t \in G} y_t x_{t^{-1}}.$$

After replacing t by s^{-1}, we get

$$\pi(yx) = \sum_{s \in G} y_{s^{-1}} x_s = \sum_{s \in G} x_s y_{s^{-1}} = \pi(xy),$$

because k is commutative. □

Lemma 12.1.4. *The map*

$$\psi : kG \longrightarrow \mathrm{Hom}_k(kG, k), \qquad x \longmapsto \big(y \mapsto \pi(xy)\big),$$

is a k-linear isomorphism.

Proof. It is clear that ψ is k-linear.

Let $f \in \mathrm{Hom}_k(kG, k)$. We claim that the inverse image of f under ψ is

$$\psi^{-1}(f) = \sum_{g \in G} f(g^{-1}) g.$$

Indeed, set

$$x = \sum_{g \in G} f(g^{-1}) g.$$

For any $h \in G$, we have

$$\psi(x)(h) = \pi(xh) = \pi\left(\sum_{g \in G} f(g^{-1}) gh\right).$$

The coefficient of 1 occurs precisely when $gh = 1$, that is, $g = h^{-1}$. Therefore

$$\psi(x)(h) = f\big((h^{-1})^{-1}\big) = f(h).$$

Since G is a basis of kG, this proves $\psi(x) = f$. Hence ψ is surjective. The same formula also shows uniqueness of x, so ψ is injective. Therefore ψ is a k-linear isomorphism. □

Definition 12.1.5 (Class function)**.** *A function*

$$f : G \longrightarrow k$$

is called a class function if

$$f(gxg^{-1}) = f(x)$$

for all $g, x \in G$.

Remark 12.1.6. *If $f : G \to k$ is a class function, then*

$$f(xy) = f(yx)$$

for all $x, y \in G$. Indeed,

$$xy = x(yx)x^{-1},$$

so xy and yx are conjugate.

Extending f linearly, we obtain a k-linear map

$$f : kG \longrightarrow k.$$

Thus class functions on G may be regarded as special linear forms on kG.

Remark 12.1.7. *There is a natural identification*

$$\{\, \textit{functions } G \to k \,\} \cong \mathrm{Hom}_k(kG, k) = (kG)^*.$$

Indeed, since G is a k-basis of kG, a linear form on kG is uniquely determined by its values on the basis elements $g \in G$.

Definition 12.1.8 (Symmetric algebra)**.** *Let A be a finite-dimensional k-algebra. We say that A is a* symmetric algebra *if there exists a k-bilinear form*

$$\langle -, - \rangle : A \times A \longrightarrow k$$

such that:

1. *$\langle -, - \rangle$ is nondegenerate;*
2. *$\langle a, b \rangle = \langle b, a \rangle$ for all $a, b \in A$;*
3. *$\langle ab, c \rangle = \langle a, bc \rangle$ for all $a, b, c \in A$.*

Definition 12.1.9 (Equivalent definition)**.** *A finite-dimensional k-algebra A is called symmetric if*

$$A \cong A^*$$

as A-bimodules, where

$$A^* = \mathrm{Hom}_k(A, k).$$

Proposition 12.1.10 (Equivalent definitions of symmetric algebras)**.** *Let A be a finite-dimensional unital associative k-algebra. Put*

$$A^* = \mathrm{Hom}_k(A, k),$$

and give A^* the A-bimodule structure

$$(a \cdot f \cdot b)(x) = f(bxa), \qquad a, b, x \in A, \ f \in A^*.$$

Then the following are equivalent:

(i) *There exists a nondegenerate k-bilinear form*

$$\langle -, - \rangle : A \times A \longrightarrow k$$

such that

$$\langle a, b \rangle = \langle b, a \rangle$$

for all $a, b \in A$, and

$$\langle ab, c \rangle = \langle a, bc \rangle$$

for all $a, b, c \in A$.

(ii) *There is an isomorphism*

$$A \cong A^*$$

as A-bimodules.

Proof. Assume first that $\langle -, - \rangle$ is a nondegenerate symmetric associative bilinear form on A. Define

$$\Phi : A \longrightarrow A^*, \qquad \Phi(a)(x) = \langle x, a \rangle.$$

Since the bilinear form is nondegenerate and A is finite-dimensional, Φ is a k-linear isomorphism.

We claim that Φ is an A-bimodule homomorphism. Let $a, c, d, x \in A$. Then

$$\begin{aligned}(c \cdot \Phi(a) \cdot d)(x) &= \Phi(a)(dxc) \\ &= \langle dxc, a \rangle \\ &= \langle dx, ca \rangle \\ &= \langle d, xca \rangle \\ &= \langle xca, d \rangle \\ &= \langle xc, ad \rangle \\ &= \langle x, cad \rangle \\ &= \Phi(cad)(x).\end{aligned}$$

Therefore

$$\Phi(cad) = c \cdot \Phi(a) \cdot d.$$

Hence $\Phi : A \to A^*$ is an isomorphism of A-bimodules.

Conversely, assume that there is an A-bimodule isomorphism

$$\Phi : A \longrightarrow A^*.$$

Define a k-bilinear form on A by

$$\langle x, y \rangle = \Phi(y)(x).$$

Since Φ is a k-linear isomorphism, this bilinear form is nondegenerate.

We first prove associativity. For $a, b, c \in A$, since Φ is a left A-module homomorphism, we have

$$\Phi(bc) = b \cdot \Phi(c).$$

Thus

$$\begin{aligned}\langle a, bc \rangle &= \Phi(bc)(a) \\ &= (b \cdot \Phi(c))(a) \\ &= \Phi(c)(ab) \\ &= \langle ab, c \rangle.\end{aligned}$$

Hence

$$\langle ab, c \rangle = \langle a, bc \rangle.$$

It remains to prove symmetry. Let

$$\lambda = \Phi(1) \in A^*.$$

Since Φ is an A-bimodule homomorphism, for every $y \in A$ we have

$$\Phi(y) = \Phi(y \cdot 1) = y \cdot \Phi(1) = y \cdot \lambda$$

and also

$$\Phi(y) = \Phi(1 \cdot y) = \Phi(1) \cdot y = \lambda \cdot y.$$

Therefore, for all $x, y \in A$,

$$\Phi(y)(x) = (y \cdot \lambda)(x) = \lambda(xy),$$

and also

$$\Phi(y)(x) = (\lambda \cdot y)(x) = \lambda(yx).$$

Thus

$$\lambda(xy) = \lambda(yx)$$

for all $x, y \in A$. Hence

$$\begin{aligned}
\langle x, y \rangle &= \Phi(y)(x) \\
&= \lambda(xy) \\
&= \lambda(yx) \\
&= \Phi(x)(y) \\
&= \langle y, x \rangle.
\end{aligned}$$

So the bilinear form is symmetric.

Therefore the existence of a nondegenerate symmetric associative bilinear form on A is equivalent to the existence of an A-bimodule isomorphism

$$A \cong A^*.$$

□

Corollary 12.1.11. *The group algebra kG is a symmetric algebra.*

Proof. The bilinear form

$$\langle x, y \rangle = \pi(xy)$$

is symmetric by the first lemma:

$$\langle x, y \rangle = \pi(xy) = \pi(yx) = \langle y, x \rangle.$$

The second lemma says precisely that the induced map

$$kG \longrightarrow (kG)^*, \qquad x \longmapsto \langle x, - \rangle,$$

is a k-linear isomorphism. Therefore the bilinear form is nondegenerate, and hence kG is a symmetric algebra. □

Let

$$F(G, k)$$

denote the k-vector space of all functions $G \to k$. We identify $F(G, k)$ with $(kG)^* =$

$\mathrm{Hom}_k(kG, k)$ by extending functions linearly from G to kG.

Let

$$\mathrm{CF}(G, k)$$

denote the subspace of class functions on G, that is,

$$\mathrm{CF}(G, k) = \{\, f : G \to k \mid f(hgh^{-1}) = f(g) \text{ for all } g, h \in G \,\}.$$

Recall that for $f \in F(G, k)$, we define

$$f^\circ = \sum_{g \in G} f(g^{-1})g \in kG.$$

Lemma 12.1.12. *Let $a \in kG$ and $f \in F(G, k)$. Define $a \cdot f \in F(G, k)$ by*

$$(a \cdot f)(x) = f(xa), \qquad x \in kG.$$

Then

$$(a \cdot f)^\circ = af^\circ.$$

Proof. Write

$$a = \sum_{h \in G} a_h h.$$

Then

$$\begin{aligned} (a \cdot f)^\circ &= \sum_{g \in G} (a \cdot f)(g^{-1})g \\ &= \sum_{g \in G} f(g^{-1}a)g \\ &= \sum_{g \in G} \sum_{h \in G} a_h f(g^{-1}h)g. \end{aligned}$$

On the other hand,

$$\begin{aligned} af^\circ &= \left(\sum_{h \in G} a_h h \right) \left(\sum_{t \in G} f(t^{-1})t \right) \\ &= \sum_{h, t \in G} a_h f(t^{-1}) ht. \end{aligned}$$

Putting $g = ht$, equivalently $t = h^{-1}g$, we get

$$t^{-1} = g^{-1}h.$$

Hence

$$af^\circ = \sum_{g \in G} \sum_{h \in G} a_h f(g^{-1}h)g = (a \cdot f)^\circ.$$

$\square$

Let

$$\mathrm{Cl}(G)$$

denote the set of conjugacy classes of G. For each $C \in \mathrm{Cl}(G)$, define the characteristic function

$$\gamma_C : G \longrightarrow k$$

by

$$\gamma_C(g) = \begin{cases} 1, & g \in C, \\ 0, & g \notin C. \end{cases}$$

Also define the class sum

$$\widehat{C} = \sum_{g \in C} g \in kG.$$

For a class function $f \in \mathrm{CF}(G, k)$, if $g_C \in C$ is a chosen representative for each conjugacy class C, then

$$f = \sum_{C \in \mathrm{Cl}(G)} f(g_C)\gamma_C.$$

Moreover,

$$f^\circ = \sum_{C \in \mathrm{Cl}(G)} f(g_C^{-1})\widehat{C}.$$

Lemma 12.1.13. *Let $Z(kG)$ denote the center of the group algebra kG, that is,*

$$Z(kG) = \{ x \in kG \mid xy = yx \text{ for all } y \in kG \}.$$

Then:

1. *The family*

$$\{ \gamma_C \mid C \in \mathrm{Cl}(G) \}$$

   is a basis of $\mathrm{CF}(G, k)$.

2. *The family*

$$\{ \widehat{C} \mid C \in \mathrm{Cl}(G) \}$$

   is a basis of $Z(kG)$.

3. *Hence*

$$\dim_k \mathrm{CF}(G, k) = |\,\mathrm{Cl}(G)|.$$

4. *The map*

$$F(G, k) \longrightarrow kG, \qquad f \longmapsto f^\circ,$$

   restricts to a k-linear isomorphism

$$\mathrm{CF}(G, k) \xrightarrow{\sim} Z(kG).$$

   In particular,

$$\dim_k Z(kG) = |\,\mathrm{Cl}(G)|.$$

Proof. First, every class function is constant on each conjugacy class. Therefore, if $g_C \in C$ is a chosen representative for each $C \in \mathrm{Cl}(G)$, then every $f \in \mathrm{CF}(G,k)$ can be written uniquely as

$$f = \sum_{C \in \mathrm{Cl}(G)} f(g_C)\gamma_C.$$

Thus

$$\{\, \gamma_C \mid C \in \mathrm{Cl}(G) \,\}$$

is a basis of $\mathrm{CF}(G,k)$. In particular,

$$\dim_k \mathrm{CF}(G,k) = |\,\mathrm{Cl}(G)|.$$

Next, we prove that the class sums form a basis of the center. Let

$$x = \sum_{g \in G} x_g g \in kG.$$

Then $x \in Z(kG)$ if and only if

$$hxh^{-1} = x$$

for every $h \in G$. But

$$hxh^{-1} = \sum_{g \in G} x_g hgh^{-1}.$$

Therefore $hxh^{-1} = x$ for all $h \in G$ if and only if the coefficient x_g is constant on conjugacy classes. Hence every central element has a unique expression of the form

$$x = \sum_{C \in \mathrm{Cl}(G)} a_C \widehat{C}, \qquad a_C \in k.$$

Thus

$$\{\, \widehat{C} \mid C \in \mathrm{Cl}(G) \,\}$$

is a basis of $Z(kG)$.

Finally, for $C \in \mathrm{Cl}(G)$, we have

$$\gamma_C^\circ = \sum_{g \in G} \gamma_C(g^{-1})g = \sum_{g^{-1} \in C} g = \sum_{g \in C^{-1}} g = \widehat{C^{-1}}.$$

Since the map

$$C \longmapsto C^{-1}$$

is a permutation of the set of conjugacy classes, the map

$$f \longmapsto f^\circ$$

sends the basis

$$\{\, \gamma_C \mid C \in \mathrm{Cl}(G) \,\}$$

of $\mathrm{CF}(G,k)$ onto the basis

$$\{\,\widehat{C} \mid C \in \mathrm{Cl}(G)\,\}$$

of $Z(kG)$. Therefore it restricts to a k-linear isomorphism

$$\mathrm{CF}(G,k) \xrightarrow{\sim} Z(kG).$$

Consequently,

$$\dim_k Z(kG) = |\,\mathrm{Cl}(G)|.$$

$\square$

12.2 Schur's lemma, and Maschke's theorem

Let G be a finite group and let k be a field. A k-representation of G is a pair (M,ρ), where M is a k-vector space and

$$\rho : G \longrightarrow \mathrm{GL}(M)$$

is a group homomorphism.

Proposition 12.2.1. *Let (M,ρ) be a k-representation of G. Then ρ extends uniquely by k-linearity to a unital k-algebra homomorphism*

$$\widetilde{\rho} : kG \longrightarrow \mathrm{End}_k(M).$$

Explicitly, if

$$x = \sum_{g\in G} a_g g \in kG,$$

then

$$\widetilde{\rho}(x) = \sum_{g\in G} a_g \rho(g).$$

Proof. Since kG is the free k-vector space with basis G, the map $\rho : G \to \mathrm{GL}(M) \subseteq \mathrm{End}_k(M)$ extends uniquely to a k-linear map

$$\widetilde{\rho} : kG \to \mathrm{End}_k(M).$$

We need to check that $\widetilde{\rho}$ is multiplicative.

Let

$$x = \sum_{g\in G} a_g g, \qquad y = \sum_{h\in G} b_h h.$$

Then

$$xy = \sum_{g,h\in G} a_g b_h gh.$$

Therefore

$$\begin{aligned}
\widetilde{\rho}(xy) &= \sum_{g,h\in G} a_g b_h \rho(gh) \\
&= \sum_{g,h\in G} a_g b_h \rho(g)\rho(h) \\
&= \left(\sum_{g\in G} a_g \rho(g)\right)\left(\sum_{h\in G} b_h \rho(h)\right) \\
&= \widetilde{\rho}(x)\widetilde{\rho}(y).
\end{aligned}$$

Moreover,

$$\widetilde{\rho}(1_G) = \rho(1_G) = \mathrm{id}_M .$$

Hence $\widetilde{\rho}$ is a unital k-algebra homomorphism. □

Conversely, suppose we are given a unital k-algebra homomorphism

$$\rho : kG \longrightarrow \mathrm{End}_k(M).$$

Then its restriction to $G \subseteq kG$ gives a group homomorphism

$$G \longrightarrow \mathrm{GL}(M).$$

Indeed, for $g \in G$, the inverse of g in kG is g^{-1}, so

$$\rho(g)\rho(g^{-1}) = \rho(gg^{-1}) = \rho(1) = \mathrm{id}_M .$$

Thus $\rho(g) \in \mathrm{GL}(M)$.

Proposition 12.2.2. *Giving a k-representation of G on M is equivalent to giving a left kG-module structure on M.*

Proof. Let

$$\rho : kG \longrightarrow \mathrm{End}_k(M)$$

be the algebra homomorphism associated to a representation. Define

$$kG \times M \longrightarrow M, \qquad (x, m) \longmapsto x \cdot m := \rho(x)(m).$$

Since ρ is a unital algebra homomorphism, we have

$$(xy) \cdot m = x \cdot (y \cdot m), \qquad 1 \cdot m = m.$$

Thus M becomes a left kG-module.

Conversely, if M is a left kG-module, then every $g \in G$ acts on M by an invertible k-linear map, with inverse given by the action of g^{-1}. Hence we get a group homomorphism

$$G \longrightarrow \mathrm{GL}(M), \qquad g \longmapsto (m \mapsto g \cdot m).$$

These two constructions are inverse to each other. $\square$

Therefore one has an equivalence of categories

$$\mathrm{Rep}_k(G) \simeq kG\text{-}\mathrm{Mod}\,.$$

Under this equivalence, morphisms of representations are precisely kG-module homomorphisms.

Definition 12.2.3 (Regular representation)**.** *The* left regular representation *of G is the representation on kG given by left multiplication:*

$$\rho : G \longrightarrow \mathrm{GL}(kG), \qquad \rho(g)(x) = gx.$$

Equivalently, it is the kG-module kG acting on itself by left multiplication.

The associated algebra homomorphism is

$$\rho : kG \longrightarrow \mathrm{End}_k(kG), \qquad \rho(a)(x) = ax.$$

Lemma 12.2.4. *The algebra homomorphism*

$$\rho : kG \longrightarrow \mathrm{End}_k(kG)$$

associated with the regular representation is injective. In particular, the regular representation is faithful.

Proof. Suppose $a \in kG$ satisfies

$$\rho(a) = 0.$$

Then $\rho(a)$ is the zero endomorphism of kG. Applying it to the identity element $1 \in kG$, we get

$$0 = \rho(a)(1) = a \cdot 1 = a.$$

Therefore $a = 0$, and hence ρ is injective. $\square$

Lemma 12.2.5 (Schur's lemma)**.** *Let (S, ρ) and (S', ρ') be two irreducible k representations of G. Equivalently, let S and S' be simple left kG-modules. Then:*

1. *If S and S' are not isomorphic, then*

$$\mathrm{Hom}_{kG}(S, S') = 0.$$

2. *The endomorphism algebra*

$$\mathrm{End}_{kG}(S) = \mathrm{Hom}_{kG}(S, S)$$

is a division k-algebra.

3. *If $k = \overline{k}$, then*

$$\operatorname{End}_{kG}(S) = k \operatorname{id}_S .$$

Proof. Let

$$f : S \longrightarrow S'$$

be a kG-module homomorphism. Then $\ker f$ is a submodule of S, and $\operatorname{Im} f$ is a submodule of S'. Since S and S' are simple, we have

$$\ker f = 0 \quad \text{or} \quad \ker f = S,$$

and

$$\operatorname{Im} f = 0 \quad \text{or} \quad \operatorname{Im} f = S'.$$

If $f \neq 0$, then $\ker f \neq S$ and $\operatorname{Im} f \neq 0$. Hence

$$\ker f = 0, \qquad \operatorname{Im} f = S'.$$

Thus every nonzero kG-module homomorphism

$$f : S \to S'$$

is an isomorphism. Therefore, if $S \ncong S'$, no nonzero homomorphism can exist, and so

$$\operatorname{Hom}_{kG}(S, S') = 0.$$

Taking $S' = S$, we see that every nonzero element of $\operatorname{End}_{kG}(S)$ is an automorphism of S. Hence

$$\operatorname{End}_{kG}(S)$$

is a division k-algebra.

Finally, assume that $k = \overline{k}$. Let

$$f \in \operatorname{End}_{kG}(S).$$

Since S is finite-dimensional over the algebraically closed field k, the k-linear map $f : S \to S$ has an eigenvalue $\lambda \in k$. Thus

$$f - \lambda \operatorname{id}_S$$

is not invertible as a k-linear map. But it is still a kG-module endomorphism of S. Since $\operatorname{End}_{kG}(S)$ is a division algebra, every nonzero element is invertible. Therefore

$$f - \lambda \operatorname{id}_S = 0.$$

Hence

$$f = \lambda \operatorname{id}_S .$$

So

$$\mathrm{End}_{kG}(S) = k\,\mathrm{id}_S\,.$$

$\square$

Corollary 12.2.6. *Let $k = \overline{k}$, and let*

$$\rho : G \longrightarrow \mathrm{GL}_n(k)$$

be an irreducible representation of G. If $M \in \mathrm{GL}_n(k)$ satisfies

$$M^{-1}\rho(g)M = \rho(g)$$

for all $g \in G$, then M is a scalar matrix.

Proof. The condition

$$M^{-1}\rho(g)M = \rho(g)$$

is equivalent to

$$\rho(g)M = M\rho(g)$$

for all $g \in G$. Thus M is an endomorphism of the representation (k^n, ρ), that is,

$$M \in \mathrm{End}_{kG}(k^n).$$

Since the representation is irreducible and k is algebraically closed, Schur's lemma gives

$$\mathrm{End}_{kG}(k^n) = k\,\mathrm{id}_{k^n}\,.$$

Therefore

$$M = \lambda I_n$$

for some $\lambda \in k$. $\square$

Theorem 12.2.7. *If*

$$\mathrm{char}\, k \mid |G|,$$

then kG is not semisimple.

Proof. Let

$$\varepsilon : kG \longrightarrow k$$

be the augmentation map, defined by

$$\varepsilon\left(\sum_{g\in G} a_g g\right) = \sum_{g\in G} a_g.$$

This is a homomorphism of left kG-modules, where k is the trivial kG-module.

Assume, for contradiction, that kG is semisimple. Then the surjection

$$\varepsilon : kG \longrightarrow k$$

splits as a kG-module homomorphism. Hence there exists a kG-module homomorphism

$$s : k \longrightarrow kG$$

such that

$$\varepsilon \circ s = \mathrm{id}_k .$$

Let

$$x = s(1) \in kG.$$

Since s is a kG-module homomorphism and k is the trivial module, for every $h \in G$ we have

$$hx = x.$$

Thus x is fixed by left multiplication by every element of G.

Write

$$x = \sum_{g \in G} a_g g.$$

The condition $hx = x$ for all $h \in G$ implies that all coefficients a_g are equal. Therefore

$$x = a \sum_{g \in G} g$$

for some $a \in k$. Hence

$$\varepsilon(x) = a|G|.$$

But $\operatorname{char} k \mid |G|$, so $|G| = 0$ in k. Thus

$$\varepsilon(x) = 0.$$

On the other hand,

$$\varepsilon(x) = \varepsilon(s(1)) = 1,$$

a contradiction. Therefore kG is not semisimple. $\square$

Theorem 12.2.8 (Maschke's theorem)**.** *If*

$$\operatorname{char} k \nmid |G|,$$

then kG is semisimple.

Proof. Let V be a left kG-module, and let

$$W \subseteq V$$

be a kG-submodule. We prove that W has a kG-submodule complement in V.

Let

$$\iota : W \hookrightarrow V$$

be the inclusion map. Since W is a kG-submodule, ι is a kG-module homomorphism.

As k-vector spaces, choose a complement W' of W in V, so that

$$V = W \oplus W'.$$

Let

$$p : V \longrightarrow W$$

be the k-linear projection onto W with respect to this decomposition. Thus

$$p \circ \iota = \mathrm{id}_W .$$

The map p need not be a kG-module homomorphism. We now average it over the group.

Let

$$\rho : G \longrightarrow \mathrm{GL}(V)$$

be the representation corresponding to the kG-module V. Define

$$\pi : V \longrightarrow W$$

by

$$\pi(v) = \frac{1}{|G|} \sum_{g \in G} \rho(g)^{-1} p(\rho(g)v).$$

This is well-defined because $|G| \neq 0$ in k.

We claim that π is a kG-module homomorphism. Let $h \in G$. Then

$$\begin{aligned}
\rho(h)^{-1} \pi \rho(h) &= \rho(h)^{-1} \left(\frac{1}{|G|} \sum_{g \in G} \rho(g)^{-1} p \rho(g) \right) \rho(h) \\
&= \frac{1}{|G|} \sum_{g \in G} \rho(h)^{-1} \rho(g)^{-1} p \rho(g) \rho(h) \\
&= \frac{1}{|G|} \sum_{g \in G} \rho(gh)^{-1} p \rho(gh).
\end{aligned}$$

Since $g \mapsto gh$ is a permutation of G, we get

$$\rho(h)^{-1} \pi \rho(h) = \pi.$$

Equivalently,

$$\pi \rho(h) = \rho(h) \pi$$

for all $h \in G$. Hence π is a kG-module homomorphism.

Next, for $w \in W$, since W is a kG-submodule, we have

$$\rho(g)w \in W$$

for all $g \in G$. Therefore

$$p(\rho(g)w) = \rho(g)w.$$

Thus

$$\begin{aligned}
\pi(w) &= \frac{1}{|G|}\sum_{g\in G}\rho(g)^{-1}p(\rho(g)w) \\
&= \frac{1}{|G|}\sum_{g\in G}\rho(g)^{-1}\rho(g)w \\
&= \frac{1}{|G|}\sum_{g\in G} w \\
&= w.
\end{aligned}$$

Hence

$$\pi \circ \iota = \mathrm{id}_W .$$

Now set

$$U = \ker \pi.$$

Since π is a kG-module homomorphism, U is a kG-submodule of V. We claim that

$$V = W \oplus U.$$

For every $v \in V$, we have

$$v = \pi(v) + \big(v - \pi(v)\big).$$

Here $\pi(v) \in W$, and

$$\pi\big(v - \pi(v)\big) = \pi(v) - \pi^2(v) = \pi(v) - \pi(v) = 0,$$

because π is the identity on W. Hence

$$v - \pi(v) \in U.$$

Thus $V = W + U$.

Finally, if $w \in W \cap U$, then

$$\pi(w) = w$$

because π is the identity on W, while

$$\pi(w) = 0$$

because $w \in U = \ker \pi$. Hence $w = 0$. Therefore

$$W \cap U = 0.$$

So

$$V = W \oplus U$$

as kG-modules.

Thus every kG-submodule of every kG-module has a complement. Therefore kG is semisimple. $\square$

12.3 Splitting fields and the structure of group algebras

Definition 12.3.1. *Let A be a finite-dimensional k-algebra. We say that A is* split *over k if for every simple left A-module S, one has*

$$\operatorname{End}_A(S) = k\,\mathrm{id}_S\,.$$

Equivalently, every simple A-module has endomorphism ring exactly k.

If $A = kG$, then we say that k is a splitting field *for G if kG is split over k.*

Theorem 12.3.2 (Finite-dimensional density theorem)**.** *Let A be a k-algebra, and let S be a finite-dimensional simple left A-module. Put*

$$D := \operatorname{End}_A(S)^{\mathrm{op}}.$$

We regard S as a right D-vector space by

$$s \cdot \varphi := \varphi(s), \qquad s \in S,\ \varphi \in \operatorname{End}_A(S).$$

Let

$$I = \operatorname{Ann}_A(S) = \{\, a \in A \mid as = 0 \text{ for all } s \in S \,\}.$$

Then the natural algebra homomorphism

$$\overline{\rho} : A/I \longrightarrow \operatorname{End}_D(S), \qquad a + I \longmapsto L_a,$$

where

$$L_a : S \longrightarrow S, \qquad s \longmapsto as,$$

is an isomorphism.

Proof. By Schur's lemma,

$$\operatorname{End}_A(S)$$

is a division ring. Hence

$$D = \operatorname{End}_A(S)^{\mathrm{op}}$$

is also a division ring, and S is a right vector space over D.

For every $a \in A$, the map

$$L_a : S \longrightarrow S, \qquad s \longmapsto as,$$

is right D-linear. Indeed, for $s \in S$ and $\varphi \in \mathrm{End}_A(S)$, we have

$$L_a(s \cdot \varphi) = L_a(\varphi(s)) = a\varphi(s) = \varphi(as) = L_a(s) \cdot \varphi,$$

because φ is A-linear. Thus we get an algebra homomorphism

$$\rho : A \longrightarrow \mathrm{End}_D(S), \qquad a \longmapsto L_a.$$

Its kernel is precisely

$$\mathrm{Ker}(\rho) = \{\, a \in A \mid L_a = 0 \,\} = \{\, a \in A \mid as = 0 \text{ for all } s \in S \,\} = \mathrm{Ann}_A(S) = I.$$

Therefore ρ induces an injective homomorphism

$$\overline{\rho} : A/I \hookrightarrow \mathrm{End}_D(S).$$

It remains to prove that $\overline{\rho}$ is surjective.

We first prove the following density claim. Let

$$x_1, \ldots, x_m \in S$$

be right D-linearly independent, and let

$$y_1, \ldots, y_m \in S$$

be arbitrary. Then there exists $a \in A$ such that

$$ax_i = y_i \qquad (1 \leq i \leq m).$$

To prove the claim, consider the A-submodule

$$N := \{\, (ax_1, \ldots, ax_m) \in S^m \mid a \in A \,\} \subseteq S^m.$$

We claim that

$$N = S^m.$$

Suppose not. Since S^m is a finite direct sum of copies of the simple module S, it is semisimple. Hence the proper submodule N is contained in the kernel of some nonzero A-module homomorphism

$$F : S^m \longrightarrow S.$$

Every such homomorphism has the form

$$F(z_1, \ldots, z_m) = \varphi_1(z_1) + \cdots + \varphi_m(z_m),$$

where

$$\varphi_i \in \mathrm{End}_A(S),$$

and not all φ_i are zero.

Since $N \subseteq \mathrm{Ker}(F)$, for every $a \in A$ we have

$$0 = F(ax_1, \ldots, ax_m) = \sum_{i=1}^{m} \varphi_i(ax_i).$$

Because each φ_i is A-linear,

$$\sum_{i=1}^{m} \varphi_i(ax_i) = \sum_{i=1}^{m} a\varphi_i(x_i) = a\left(\sum_{i=1}^{m} \varphi_i(x_i)\right).$$

Taking $a = 1$, we obtain

$$\sum_{i=1}^{m} \varphi_i(x_i) = 0.$$

In terms of the right D-module structure on S, this is

$$\sum_{i=1}^{m} x_i \cdot \varphi_i = 0.$$

Since $x_1, \ldots, x_m$ are right D-linearly independent, it follows that

$$\varphi_1 = \cdots = \varphi_m = 0,$$

contradicting the fact that $F \neq 0$. Therefore

$$N = S^m.$$

Hence for any $(y_1, \ldots, y_m) \in S^m$, there exists $a \in A$ such that

$$(ax_1, \ldots, ax_m) = (y_1, \ldots, y_m).$$

This proves the density claim.

Now let

$$T \in \mathrm{End}_D(S).$$

Since S is finite-dimensional over k, it is also finite-dimensional as a right D-vector space. Choose a right D-basis

$$s_1, \ldots, s_n$$

of S. Applying the density claim to

$$x_i = s_i, \qquad y_i = T(s_i),$$

there exists $a \in A$ such that

$$as_i = T(s_i) \qquad (1 \leq i \leq n).$$

Since both L_a and T are right D-linear and they agree on the right D-basis $s_1, \dots, s_n$, we get

$$L_a = T.$$

Thus every element of $\mathrm{End}_D(S)$ lies in the image of ρ. Therefore $\overline{\rho}$ is surjective.

Hence

$$A/I \cong \mathrm{End}_D(S).$$

□

Remark 12.3.3. *If one writes*

$$D = \mathrm{End}_A(S)$$

instead of $D = \mathrm{End}_A(S)^{\mathrm{op}}$, then the same statement is obtained by remembering that S is naturally a right $\mathrm{End}_A(S)^{\mathrm{op}}$-vector space via

$$s \cdot \varphi = \varphi(s).$$

In the split case $D = k$, this distinction disappears.

Proposition 12.3.4. *Let A be a finite-dimensional k-algebra, and let S be a simple left A-module. Then the following are equivalent:*

1. $$\mathrm{End}_A(S) = k\,\mathrm{id}_S\,.$$

2. *For every field extension K/k, the scalar extension*

$$S_K := K \otimes_k S$$

is a simple left A_K-module, where

$$A_K := K \otimes_k A.$$

Proof. Let

$$D = \mathrm{End}_A(S).$$

By Schur's lemma, D is a division algebra over k.

Assume first that

$$D = k.$$

Let

$$I = \mathrm{Ann}_A(S).$$

By the density theorem, since S is a finite-dimensional simple A-module, the natural map

$$A/I \longrightarrow \mathrm{End}_D(S)$$

is an isomorphism. Since $D = k$, this becomes

$$A/I \cong \mathrm{End}_k(S).$$

After extending scalars from k to K, we get

$$A_K/(K \otimes_k I) \cong \mathrm{End}_K(K \otimes_k S).$$

Thus the action of A_K on $S_K = K \otimes_k S$ contains the full matrix algebra

$$\mathrm{End}_K(S_K).$$

Hence every nonzero A_K-submodule of S_K is stable under the full endomorphism algebra of S_K, and therefore must be all of S_K. Thus S_K is simple.

Conversely, assume that for every field extension K/k, the A_K-module S_K is simple. Since A is finite-dimensional and S is finite-dimensional, base change gives

$$K \otimes_k \mathrm{End}_A(S) \cong \mathrm{End}_{A_K}(S_K).$$

Take $K = \overline{k}$. Then $S_{\overline{k}}$ is simple by assumption, so Schur's lemma implies that

$$\mathrm{End}_{A_{\overline{k}}}(S_{\overline{k}})$$

is a division algebra. Therefore

$$\overline{k} \otimes_k D$$

is a finite-dimensional division algebra over the algebraically closed field $\overline{k}$.

But any finite-dimensional division algebra over an algebraically closed field is just the field itself. Hence

$$\overline{k} \otimes_k D \cong \overline{k}.$$

Taking dimensions over $\overline{k}$, this implies

$$\dim_k D = 1.$$

Therefore

$$D = k,$$

that is,

$$\mathrm{End}_A(S) = k\,\mathrm{id}_S\,.$$

$\square$

Remark 12.3.5. *If $k = \overline{k}$, then Schur's lemma implies that k is a splitting field for G. Indeed, for every simple kG-module S,*

$$\mathrm{End}_{kG}(S) = k\,\mathrm{id}_S\,.$$

Theorem 12.3.6 (Brauer)**.** *Let G be a finite group. If k contains a primitive $|G|$-th root of unity, then k is a splitting field for G.*

Proof. Put
$$n = |G|.$$

Since k contains a primitive n-th root of unity, we have

$$\operatorname{char} k \nmid n.$$

Hence, by Maschke's theorem, kG is semisimple.

Let $\overline{k}$ be an algebraic closure of k. It is enough to prove that every simple $\overline{k}G$-module is realizable over k. Indeed, if every simple $\overline{k}G$-module has a k-form, then for every simple kG-module S, the scalar extension
$$\overline{k} \otimes_k S$$
is again simple. Equivalently,
$$\mathrm{End}_{kG}(S) = k,$$
so k is a splitting field for G.

Let V be a simple $\overline{k}G$-module, and let

$$\chi : G \longrightarrow \overline{k}$$

be its character. For each $g \in G$, the order of g divides n, so

$$g^n = 1.$$

Therefore the linear operator representing g on V satisfies

$$X^n - 1 = 0.$$

Since k contains all n-th roots of unity and $\operatorname{char} k \nmid n$, the polynomial $X^n - 1$ splits over k with distinct roots. Hence all eigenvalues of g on V lie in k. It follows that

$$\chi(g) \in k$$

for all $g \in G$. Thus every irreducible character of G over $\overline{k}$ is k-valued.

We now use Brauer's induction theorem. It says that every irreducible character χ of G is an integral linear combination of characters induced from linear characters of subgroups.

More precisely,

$$\chi = \sum_i a_i \operatorname{Ind}_{H_i}^G(\lambda_i), \qquad a_i \in \mathbb{Z},$$

where each $H_i \leq G$, and each λ_i is a one-dimensional character of H_i.

Since every element of H_i has order dividing n, the values of λ_i are n-th roots of unity. Hence

$$\lambda_i(H_i) \subseteq k.$$

Therefore each λ_i is afforded by a one-dimensional kH_i-module. Consequently,

$$\operatorname{Ind}_{H_i}^G(\lambda_i)$$

is afforded by a kG-module.

Thus χ is an integral linear combination of characters afforded by kG-modules.

Now recall the standard Schur-index criterion: if χ is an irreducible $\overline{k}$-character which is k-valued, then the Schur index $m_k(\chi)$ is the greatest common divisor of the multiplicities with which χ occurs in scalar extensions of kG-modules. In particular, if χ is an integral linear combination of kG-characters, then

$$m_k(\chi) \mid 1.$$

Hence

$$m_k(\chi) = 1.$$

Therefore V is realizable over k. That is, there exists a simple kG-module S such that

$$\overline{k} \otimes_k S \cong V.$$

Since this holds for every simple $\overline{k}G$-module V, every simple kG-module remains simple after extension of scalars to $\overline{k}$. Equivalently,

$$\operatorname{End}_{kG}(S) = k$$

for every simple kG-module S. Hence k is a splitting field for G. □

Proposition 12.3.7. *Assume that*

$$\operatorname{char} k = p$$

and that G is a finite p-group. Then the only simple kG-module is the trivial module k.

Proof. Let S be a simple kG-module. We show that S contains a nonzero G-fixed vector.

Choose

$$0 \neq x \in S.$$

Consider the finite subset
$$X = \{\, gx \mid g \in G \,\} \subseteq S.$$
Let S_0 be the subgroup of the additive group of S generated by X. Since $\operatorname{char} k = p$, the additive group of S has exponent p. Hence S_0 is a finite abelian p-group.

Moreover, S_0 is stable under the action of G. Indeed, if $h \in G$ and $gx \in X$, then
$$h(gx) = (hg)x \in X.$$
Thus G acts on the finite p-group S_0.

We now use the standard fact that if a finite p-group acts on a finite p-group, then the set of fixed points has cardinality congruent to the whole set modulo p. Equivalently,
$$|S_0^G| \equiv |S_0| \pmod{p}.$$
Since S_0 is a finite p-group, we have
$$|S_0| \equiv 0 \pmod{p}.$$
Therefore
$$|S_0^G| \equiv 0 \pmod{p}.$$
But $0 \in S_0^G$, so S_0^G is not empty. Since its cardinality is divisible by p, it must contain an element different from 0. Hence there exists
$$0 \neq y \in S_0$$
such that
$$gy = y \qquad \text{for all } g \in G.$$
Now consider the k-subspace
$$ky \subseteq S.$$
Since $gy = y$ for all $g \in G$, the subspace ky is stable under the action of G. Hence ky is a nonzero kG-submodule of S.

Because S is simple, we must have
$$S = ky.$$
Therefore S is one-dimensional, and G acts trivially on S. Hence
$$S \cong k$$
as kG-modules, where k is the trivial kG-module. □

Remark 12.3.8. *Thus, when* $\operatorname{char} k = p$ *and* G *is a finite* p*-group, the group algebra* kG *has only one simple module up to isomorphism, namely the trivial module.*

Theorem 12.3.9. *Assume that*

$$\operatorname{char} k \nmid |G|,$$

or equivalently, that $|G|$ is invertible in k. Then kG is semisimple. Hence, as a ring,

$$kG \cong \bigoplus_{i=1}^{r} M_{n_i}(D_i),$$

where each D_i is a finite-dimensional division algebra over k.

Proof. By Maschke's theorem, kG is semisimple. Since kG is finite-dimensional over k, it is an Artinian semisimple ring. Therefore the Artin–Wedderburn theorem implies that

$$kG \cong \bigoplus_{i=1}^{r} M_{n_i}(D_i),$$

where each D_i is a division algebra finite-dimensional over k. □

Theorem 12.3.10. *Assume further that k is a splitting field for G. In particular, this holds if $k = \overline{k}$. Let*

$$S_1, \dots, S_r$$

be a complete set of pairwise non-isomorphic simple kG-modules, and set

$$d_i = \dim_k S_i.$$

Then, as a left kG-module,

$$kG \cong S_1^{\oplus d_1} \oplus \cdots \oplus S_r^{\oplus d_r}.$$

Consequently,

$$|G| = d_1^2 + \cdots + d_r^2.$$

Moreover, as a k-algebra,

$$kG \cong M_{d_1}(k) \oplus \cdots \oplus M_{d_r}(k).$$

Proof. Since kG is semisimple, every simple module is projective and every kG-module is semisimple. Thus the regular module decomposes as

$$kG \cong S_1^{\oplus m_1} \oplus \cdots \oplus S_r^{\oplus m_r}$$

for some multiplicities m_i.

Because k is a splitting field, Schur's lemma gives

$$\operatorname{End}_{kG}(S_i) = k.$$

The Artin–Wedderburn decomposition therefore has the form

$$kG \cong M_{d_1}(k) \oplus \cdots \oplus M_{d_r}(k),$$

where $d_i = \dim_k S_i$. As a left module over $M_{d_i}(k)$, the algebra $M_{d_i}(k)$ is a direct sum of d_i copies of its natural simple module k^{d_i}. Hence

$$kG \cong S_1^{\oplus d_1} \oplus \cdots \oplus S_r^{\oplus d_r}$$

as a left kG-module.

Taking k-dimensions gives

$$|G| = \dim_k kG = \sum_{i=1}^{r} d_i \dim_k S_i = \sum_{i=1}^{r} d_i^2.$$

□

12.4 Characters and orthogonality relations

Let now

$$k = \mathbb{C}.$$

Let (V, ρ) be a finite-dimensional complex representation of G, so

$$\rho : G \longrightarrow \mathrm{GL}(V).$$

After choosing a basis of V, this may be written as

$$\rho : G \longrightarrow \mathrm{GL}_d(\mathbb{C}), \qquad d = \dim_{\mathbb{C}} V.$$

Definition 12.4.1 (Character)**.** *Given a complex representation (V, ρ) of G, its character is the function*

$$\chi_V : G \longrightarrow \mathbb{C}$$

defined by

$$\chi_V(g) = \operatorname{tr} \rho(g).$$

Proposition 12.4.2. *Let V and W be finite-dimensional complex representations of G. Then:*

1. *χ_V is a class function.*

2. *For every $g \in G$,*

$$\chi_V(g^{-1}) = \overline{\chi_V(g)}.$$

3. *If $V \cong W$ as $\mathbb{C}G$-modules, then*

$$\chi_V = \chi_W.$$

4. *We have*

$$\chi_{V \oplus W} = \chi_V + \chi_W, \qquad \chi_{V \otimes W} = \chi_V \chi_W.$$

5. *For the dual representation V^*,*

$$\chi_{V^*}(g) = \chi_V(g^{-1}) = \overline{\chi_V(g)}.$$

Hence

$$\chi_{V^*} = \overline{\chi_V}.$$

6. *For the representation $\operatorname{Hom}_{\mathbb{C}}(V, W)$,*

$$\chi_{\operatorname{Hom}_{\mathbb{C}}(V,W)}(g) = \chi_V(g^{-1})\chi_W(g).$$

Equivalently,

$$\chi_{\operatorname{Hom}_{\mathbb{C}}(V,W)} = \overline{\chi_V}\chi_W.$$

7. *One has*

$$\chi_V(1) = \dim_{\mathbb{C}} V.$$

Moreover,

$$|\chi_V(g)| \leq \dim_{\mathbb{C}} V.$$

Equality holds if and only if g acts on V as a scalar. In particular,

$$\chi_V(g) = \dim_{\mathbb{C}} V$$

if and only if g acts trivially on V.

Proof. For (1), let $h, g \in G$. Then

$$\rho(hgh^{-1}) = \rho(h)\rho(g)\rho(h)^{-1}.$$

Since trace is invariant under conjugation,

$$\chi_V(hgh^{-1}) = \operatorname{tr}\rho(hgh^{-1}) = \operatorname{tr}\rho(g) = \chi_V(g).$$

Thus χ_V is a class function.

For (2), since g has finite order, $\rho(g)$ also has finite order. Hence its eigenvalues are roots of unity. If the eigenvalues of $\rho(g)$ are

$$\lambda_1, \ldots, \lambda_d,$$

then the eigenvalues of $\rho(g^{-1}) = \rho(g)^{-1}$ are

$$\lambda_1^{-1}, \ldots, \lambda_d^{-1}.$$

Since each λ_i is a root of unity, we have

$$\lambda_i^{-1} = \overline{\lambda_i}.$$

Therefore

$$\chi_V(g^{-1}) = \sum_{i=1}^{d} \lambda_i^{-1} = \sum_{i=1}^{d} \overline{\lambda_i} = \overline{\chi_V(g)}.$$

For (3), if $V \cong W$, then after choosing compatible bases, the matrices of $\rho_V(g)$ and $\rho_W(g)$ are conjugate for every $g \in G$. Hence their traces are equal.

For (4), the matrix of g on $V \oplus W$ is block diagonal:

$$\rho_{V\oplus W}(g) = \begin{pmatrix} \rho_V(g) & 0 \\ 0 & \rho_W(g) \end{pmatrix}.$$

Therefore

$$\chi_{V\oplus W}(g) = \chi_V(g) + \chi_W(g).$$

Also,

$$\rho_{V\otimes W}(g) = \rho_V(g) \otimes \rho_W(g),$$

so

$$\mathrm{tr}(\rho_V(g) \otimes \rho_W(g)) = \mathrm{tr}\,\rho_V(g)\,\mathrm{tr}\,\rho_W(g).$$

Thus

$$\chi_{V\otimes W}(g) = \chi_V(g)\chi_W(g).$$

For (5), the action of g on V^* is dual to the action of g^{-1} on V. Hence

$$\chi_{V^*}(g) = \mathrm{tr}\,\rho_V(g^{-1}) = \chi_V(g^{-1}) = \overline{\chi_V(g)}.$$

For (6), we use the natural identification

$$\mathrm{Hom}_{\mathbb{C}}(V, W) \cong V^* \otimes W.$$

Therefore, by (4) and (5),

$$\chi_{\mathrm{Hom}_{\mathbb{C}}(V,W)} = \chi_{V^*}\chi_W = \overline{\chi_V}\chi_W.$$

Equivalently, for every $g \in G$,

$$\chi_{\mathrm{Hom}_{\mathbb{C}}(V,W)}(g) = \chi_V(g^{-1})\chi_W(g).$$

Finally, for (7), we have

$$\rho(1) = \mathrm{id}_V,$$

so

$$\chi_V(1) = \operatorname{tr}(\mathrm{id}_V) = \dim_{\mathbb{C}} V.$$

Let the eigenvalues of $\rho(g)$ be

$$\lambda_1, \dots, \lambda_d.$$

Since g has finite order, each λ_i is a root of unity, hence

$$|\lambda_i| = 1.$$

Therefore

$$|\chi_V(g)| = |\lambda_1 + \cdots + \lambda_d| \leq |\lambda_1| + \cdots + |\lambda_d| = d.$$

Equality holds if and only if all λ_i have the same argument, that is, if and only if all λ_i are equal. Since $\rho(g)$ is diagonalizable, this is equivalent to saying that $\rho(g)$ is a scalar operator.

Moreover,

$$\chi_V(g) = d$$

if and only if all eigenvalues of $\rho(g)$ are equal to 1, which is equivalent to

$$\rho(g) = \mathrm{id}_V.$$

Thus g acts trivially on V. □

For simplicity, write

$$\mathrm{CF}(G) = \mathrm{CF}(G, \mathbb{C}).$$

Thus $\mathrm{CF}(G)$ is the $\mathbb{C}$-vector space of complex class functions on G.

Definition 12.4.3. *For $f_1, f_2 \in \mathrm{CF}(G)$, define*

$$\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}.$$

Since f_1 and f_2 are class functions, the above sum may be written as a sum over conjugacy classes. Let $\mathrm{Cl}(G)$ denote the set of conjugacy classes of G. For each $C \in \mathrm{Cl}(G)$, choose a representative $g_C \in C$. Then

$$\begin{aligned} \langle f_1, f_2 \rangle &= \frac{1}{|G|} \sum_{C \in \mathrm{Cl}(G)} |C| f_1(g_C) \overline{f_2(g_C)} \\ &= \frac{1}{|G|} \sum_{C \in \mathrm{Cl}(G)} \frac{|G|}{|C_G(g_C)|} f_1(g_C) \overline{f_2(g_C)} \\ &= \sum_{C \in \mathrm{Cl}(G)} \frac{1}{|C_G(g_C)|} f_1(g_C) \overline{f_2(g_C)}. \end{aligned}$$

Here

$$C_G(g_C) = \{\, h \in G \mid h g_C = g_C h \,\}$$

is the centralizer of g_C in G, and we used the orbit-stabilizer formula

$$|C| = \frac{|G|}{|C_G(g_C)|}.$$

Proposition 12.4.4. *The pairing*

$$\langle -, - \rangle : \mathrm{CF}(G) \times \mathrm{CF}(G) \longrightarrow \mathbb{C}$$

defined by

$$\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}$$

is a Hermitian inner product on the complex vector space $\mathrm{CF}(G)$.

Moreover, if f_1 and f_2 are characters of complex representations of G, then

$$\langle f_1, f_2 \rangle \in \mathbb{R}.$$

Proof. It is clear that the pairing is linear in the first variable and conjugate linear in the second variable. Also,

$$\begin{aligned} \overline{\langle f_1, f_2 \rangle} &= \overline{\frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}} \\ &= \frac{1}{|G|} \sum_{g \in G} \overline{f_1(g)} f_2(g) \\ &= \langle f_2, f_1 \rangle. \end{aligned}$$

Finally,

$$\langle f, f \rangle = \frac{1}{|G|} \sum_{g \in G} |f(g)|^2 \geq 0,$$

and equality holds if and only if $f(g) = 0$ for all $g \in G$, that is, $f = 0$. Thus $\langle -, - \rangle$ is a Hermitian inner product.

Now suppose that f_1 and f_2 are characters. For every $g \in G$, one has

$$\overline{f_i(g)} = f_i(g^{-1}), \qquad i = 1, 2.$$

Hence

$$\begin{aligned} \langle f_1, f_2 \rangle &= \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)} \\ &= \frac{1}{|G|} \sum_{g \in G} f_1(g) f_2(g^{-1}) \\ &= \frac{1}{|G|} \sum_{g \in G} f_1(g^{-1}) f_2(g) \\ &= \frac{1}{|G|} \sum_{g \in G} \overline{f_1(g)} f_2(g) \\ &= \langle f_2, f_1 \rangle. \end{aligned}$$

But Hermitian symmetry gives

$$\langle f_2, f_1\rangle = \overline{\langle f_1, f_2\rangle}.$$

Therefore

$$\langle f_1, f_2\rangle = \overline{\langle f_1, f_2\rangle},$$

and hence

$$\langle f_1, f_2\rangle \in \mathbb{R}.$$

$\square$

Lemma 12.4.5. *Let V be a finite-dimensional $\mathbb{C}G$-module. Define*

$$V^G = \{\, v \in V \mid gv = v \text{ for all } g \in G\,\}.$$

Then V^G is a $\mathbb{C}G$-submodule of V, and

$$\dim_{\mathbb{C}} V^G = \langle \chi_V, 1_G\rangle = \frac{1}{|G|}\sum_{g\in G}\chi_V(g),$$

where 1_G denotes the character of the trivial representation.

Proof. It is clear that V^G is a complex subspace of V. Moreover, if $v \in V^G$, then for every $h \in G$,

$$hv = v.$$

Thus V^G is stable under the action of G, and the induced action on V^G is trivial. Hence V^G is a $\mathbb{C}G$-submodule of V.

Define a linear map

$$\pi : V \longrightarrow V$$

by

$$\pi(v) = \left(\frac{1}{|G|}\sum_{g\in G} g\right) v = \frac{1}{|G|}\sum_{g\in G} gv.$$

We claim that

$$\operatorname{Im}\pi = V^G.$$

First, for $h \in G$, we have

$$\begin{aligned}
h\pi(v) &= h\left(\frac{1}{|G|}\sum_{g\in G} gv\right)\\
&= \frac{1}{|G|}\sum_{g\in G} hgv\\
&= \frac{1}{|G|}\sum_{g\in G} gv\\
&= \pi(v),
\end{aligned}$$

because $g \mapsto hg$ is a permutation of G. Hence

$$\operatorname{Im} \pi \subseteq V^G.$$

Conversely, if $v \in V^G$, then

$$\pi(v) = \frac{1}{|G|} \sum_{g \in G} gv = \frac{1}{|G|} \sum_{g \in G} v = v.$$

Thus $V^G \subseteq \operatorname{Im} \pi$. Therefore

$$\operatorname{Im} \pi = V^G.$$

Also,

$$\pi^2 = \left(\frac{1}{|G|} \sum_{g \in G} g \right) \left(\frac{1}{|G|} \sum_{h \in G} h \right) = \frac{1}{|G|^2} \sum_{g,h \in G} gh.$$

For each $x \in G$, there are exactly $|G|$ pairs $(g, h) \in G \times G$ such that $gh = x$. Hence

$$\pi^2 = \frac{1}{|G|} \sum_{x \in G} x = \pi.$$

Thus π is a projection onto V^G. Therefore

$$V = \ker \pi \oplus \operatorname{Im} \pi, \qquad \pi|_{\operatorname{Im} \pi} = \mathrm{id}_{\operatorname{Im} \pi}.$$

It follows that

$$\operatorname{tr}(\pi) = \dim_{\mathbb{C}} \operatorname{Im} \pi = \dim_{\mathbb{C}} V^G.$$

On the other hand,

$$\begin{aligned}
\operatorname{tr}(\pi) &= \operatorname{tr} \left(\frac{1}{|G|} \sum_{g \in G} \rho(g) \right) \\
&= \frac{1}{|G|} \sum_{g \in G} \operatorname{tr}(\rho(g)) \\
&= \frac{1}{|G|} \sum_{g \in G} \chi_V(g).
\end{aligned}$$

Since $1_G(g) = 1$ for all $g \in G$, we have

$$\frac{1}{|G|} \sum_{g \in G} \chi_V(g) = \langle \chi_V, 1_G \rangle.$$

Therefore

$$\dim_{\mathbb{C}} V^G = \langle \chi_V, 1_G \rangle.$$

$\square$

Lemma 12.4.6. *Let k be any field, and let V, W be kG-modules. Give* $\mathrm{Hom}_k(V, W)$ *the G-action defined by*

$$(g\varphi)(v) = g\varphi(g^{-1}v), \qquad g \in G,\ \varphi \in \mathrm{Hom}_k(V, W),\ v \in V.$$

Then

$$\mathrm{Hom}_k(V, W)^G = \mathrm{Hom}_{kG}(V, W).$$

Proof. Let $\varphi \in \mathrm{Hom}_k(V, W)$. Then

$$\varphi \in \mathrm{Hom}_k(V, W)^G$$

if and only if

$$g\varphi = \varphi$$

for all $g \in G$. By definition of the action, this means

$$g\varphi(g^{-1}v) = \varphi(v)$$

for all $g \in G$ and $v \in V$. Replacing v by gv, this is equivalent to

$$g\varphi(v) = \varphi(gv)$$

for all $g \in G$ and $v \in V$. This is precisely the condition that φ be a kG-module homomorphism. Hence

$$\mathrm{Hom}_k(V, W)^G = \mathrm{Hom}_{kG}(V, W).$$

□

Theorem 12.4.7 (Row orthogonality)**.** *Let V and W be simple $\mathbb{C}G$-modules. Then*

$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1, & V \cong W, \\ 0, & V \ncong W. \end{cases}$$

Proof. By Schur's lemma,

$$\dim_{\mathbb{C}} \mathrm{Hom}_{\mathbb{C}G}(V, W) = \begin{cases} 1, & V \cong W, \\ 0, & V \ncong W. \end{cases}$$

On the other hand, by the previous lemma,

$$\mathrm{Hom}_{\mathbb{C}G}(V, W) = \mathrm{Hom}_{\mathbb{C}}(V, W)^G.$$

Therefore

$$\begin{aligned} \dim_{\mathbb{C}} \mathrm{Hom}_{\mathbb{C}G}(V, W) &= \dim_{\mathbb{C}} \mathrm{Hom}_{\mathbb{C}}(V, W)^G \\ &= \left\langle \chi_{\mathrm{Hom}_{\mathbb{C}}(V,W)}, 1_G \right\rangle. \end{aligned}$$

But

$$\mathrm{Hom}_{\mathbb{C}}(V,W)\cong V^*\otimes W$$

as $\mathbb{C}G$-modules. Hence

$$\chi_{\mathrm{Hom}_{\mathbb{C}}(V,W)}=\chi_{V^*}\chi_W=\overline{\chi_V}\chi_W.$$

Thus

$$\begin{aligned}\dim_{\mathbb{C}}\mathrm{Hom}_{\mathbb{C}G}(V,W)&=\frac{1}{|G|}\sum_{g\in G}\overline{\chi_V(g)}\chi_W(g)\\&=\langle\chi_W,\chi_V\rangle.\end{aligned}$$

Since χ_V and χ_W are characters, the previous proposition gives

$$\langle\chi_W,\chi_V\rangle=\langle\chi_V,\chi_W\rangle.$$

Therefore

$$\langle\chi_V,\chi_W\rangle=\dim_{\mathbb{C}}\mathrm{Hom}_{\mathbb{C}G}(V,W).$$

By Schur's lemma, this dimension is 1 if $V\cong W$, and 0 otherwise. Hence

$$\langle\chi_V,\chi_W\rangle=\begin{cases}1, & V\cong W,\\ 0, & V\not\cong W.\end{cases}$$

□

Definition 12.4.8. *The set of irreducible characters of G is denoted by*

$$\mathrm{Irr}(G).$$

Thus

$$\mathrm{Irr}(G)=\{\,\chi_S\mid S\ \textit{is a simple}\ \mathbb{C}G\textit{-module}\,\}.$$

Equivalently, $\mathrm{Irr}(G)$ *is the set of characters afforded by simple complex representations of G.*

Corollary 12.4.9. *The set* $\mathrm{Irr}(G)$ *is $\mathbb{C}$-linearly independent in* $\mathrm{CF}(G)$*.*

Proof. Suppose that

$$\sum_{\chi\in\mathrm{Irr}(G)}a_\chi\chi=0,\qquad a_\chi\in\mathbb{C}.$$

Let $\psi\in\mathrm{Irr}(G)$. Taking the inner product with ψ, and using the row orthogonality relations, we get

$$0=\left\langle\sum_{\chi\in\mathrm{Irr}(G)}a_\chi\chi,\psi\right\rangle=\sum_{\chi\in\mathrm{Irr}(G)}a_\chi\langle\chi,\psi\rangle=a_\chi.$$

Since this holds for every $\psi\in\mathrm{Irr}(G)$, all coefficients are zero. Hence $\mathrm{Irr}(G)$ is $\mathbb{C}$-linearly independent. □

Corollary 12.4.10. *The set* $\mathrm{Irr}(G)$ *is a* $\mathbb{C}$*-basis of* $\mathrm{CF}(G)$*.*

Proof. Let
$$\mathrm{Irr}(G) = \{\chi_1, \dots, \chi_r\},$$
where χ_i is afforded by the simple $\mathbb{C}G$-module S_i. Since $\mathbb{C}$ is a splitting field for G, the Artin–Wedderburn decomposition gives

$$\mathbb{C}G \cong M_{d_1}(\mathbb{C}) \oplus \cdots \oplus M_{d_r}(\mathbb{C}),$$

where
$$d_i = \dim_{\mathbb{C}} S_i.$$

Taking centers, we get

$$Z(\mathbb{C}G) \cong Z(M_{d_1}(\mathbb{C})) \oplus \cdots \oplus Z(M_{d_r}(\mathbb{C})).$$

Since
$$Z(M_{d_i}(\mathbb{C})) = \mathbb{C}I_{d_i},$$
we have
$$\dim_{\mathbb{C}} Z(\mathbb{C}G) = r.$$

On the other hand, we have already seen that

$$\mathrm{CF}(G) \cong Z(\mathbb{C}G)$$

as $\mathbb{C}$-vector spaces. Therefore

$$\dim_{\mathbb{C}} \mathrm{CF}(G) = r = |\,\mathrm{Irr}(G)|.$$

Since $\mathrm{Irr}(G)$ is linearly independent and has cardinality equal to $\dim_{\mathbb{C}} \mathrm{CF}(G)$, it is a basis of $\mathrm{CF}(G)$. ☐

Corollary 12.4.11. *The number of irreducible complex characters of* G *is equal to the number of conjugacy classes of* G*. That is,*

$$|\,\mathrm{Irr}(G)| = |\,\mathrm{Cl}(G)|.$$

Proof. Since $\mathrm{Irr}(G)$ is a basis of $\mathrm{CF}(G)$, we have

$$|\,\mathrm{Irr}(G)| = \dim_{\mathbb{C}} \mathrm{CF}(G).$$

But $\mathrm{CF}(G)$ has a basis given by the characteristic functions of the conjugacy classes. Hence

$$\dim_{\mathbb{C}} \mathrm{CF}(G) = |\,\mathrm{Cl}(G)|.$$

Therefore

$$|\operatorname{Irr}(G)| = |\operatorname{Cl}(G)|.$$

$\square$

Lemma 12.4.12. *One has*

$$|G| = \sum_{\chi \in \operatorname{Irr}(G)} \chi(1)^2.$$

Proof. Let

$$S_1, \ldots, S_r$$

be a complete set of pairwise non-isomorphic simple $\mathbb{C}G$-modules, and let

$$\chi_i = \chi_{S_i}, \qquad d_i = \dim_{\mathbb{C}} S_i.$$

As a left $\mathbb{C}G$-module, the regular module decomposes as

$$\mathbb{C}G \cong S_1^{\oplus d_1} \oplus \cdots \oplus S_r^{\oplus d_r}.$$

Taking dimensions gives

$$|G| = \dim_{\mathbb{C}} \mathbb{C}G = \sum_{i=1}^{r} d_i \dim_{\mathbb{C}} S_i = \sum_{i=1}^{r} d_i^2.$$

Since

$$d_i = \chi_i(1),$$

we obtain

$$|G| = \sum_{i=1}^{r} \chi_i(1)^2 = \sum_{\chi \in \operatorname{Irr}(G)} \chi(1)^2.$$

$\square$

Corollary 12.4.13. *Let V be a finite-dimensional $\mathbb{C}G$-module. Then*

$$\langle \chi_V, \chi_V \rangle = 1$$

if and only if V is simple.

Proof. Since $\mathbb{C}G$ is semisimple, we may write

$$V \cong S_1^{\oplus n_1} \oplus \cdots \oplus S_r^{\oplus n_r},$$

where $S_1, \ldots, S_r$ are pairwise non-isomorphic simple $\mathbb{C}G$-modules and $n_i \geq 0$. Then

$$\chi_V = n_1 \chi_{S_1} + \cdots + n_r \chi_{S_r}.$$

By row orthogonality,

$$\langle \chi_V, \chi_V \rangle = n_1^2 + \cdots + n_r^2.$$

Therefore

$$\langle \chi_V, \chi_V \rangle = 1$$

if and only if exactly one of the n_i's is equal to 1, and all the others are 0. This is equivalent to saying that V is simple. □

Lemma 12.4.14. *Let V and W be finite-dimensional $\mathbb{C}G$-modules. Suppose*

$$V \cong S_1^{\oplus n_1} \oplus \cdots \oplus S_r^{\oplus n_r},$$

and

$$W \cong S_1^{\oplus m_1} \oplus \cdots \oplus S_r^{\oplus m_r},$$

where $S_1, \ldots, S_r$ are pairwise non-isomorphic simple $\mathbb{C}G$-modules and $n_i, m_i \geq 0$. Then

$$\langle \chi_V, \chi_W \rangle = n_1 m_1 + \cdots + n_r m_r.$$

Proof. We have

$$\chi_V = \sum_{i=1}^{r} n_i \chi_{S_i}, \qquad \chi_W = \sum_{i=1}^{r} m_i \chi_{S_i}.$$

Therefore, by bilinearity of the inner product and row orthogonality,

$$\begin{aligned} \langle \chi_V, \chi_W \rangle &= \left\langle \sum_{i=1}^{r} n_i \chi_{S_i}, \sum_{j=1}^{r} m_j \chi_{S_j} \right\rangle \\ &= \sum_{i,j} n_i m_j \langle \chi_{S_i}, \chi_{S_j} \rangle \\ &= \sum_{i=1}^{r} n_i m_i. \end{aligned}$$

□

Corollary 12.4.15. *Let V and W be finite-dimensional $\mathbb{C}G$-modules. Then*

$$\chi_V = \chi_W$$

if and only if

$$V \cong W.$$

Proof. If $V \cong W$, then clearly $\chi_V = \chi_W$.

Conversely, write

$$V \cong S_1^{\oplus n_1} \oplus \cdots \oplus S_r^{\oplus n_r},$$

and

$$W \cong S_1^{\oplus m_1} \oplus \cdots \oplus S_r^{\oplus m_r}.$$

If $\chi_V = \chi_W$, then for each i,

$$n_i = \langle \chi_V, \chi_{S_i} \rangle = \langle \chi_W, \chi_{S_i} \rangle = m_i.$$

Hence V and W have the same simple direct summands with the same multiplicities. Therefore

$$V \cong W.$$

□

Corollary 12.4.16. *Let V be a finite-dimensional $\mathbb{C}G$-module, and suppose*

$$V \cong S_1^{\oplus n_1} \oplus \cdots \oplus S_r^{\oplus n_r},$$

where $S_1, \dots, S_r$ are pairwise non-isomorphic simple $\mathbb{C}G$-modules. Then

$$n_i = \langle \chi_V, \chi_{S_i} \rangle$$

for every i.

Proof. Since

$$\chi_V = \sum_{j=1}^{r} n_j \chi_{S_j},$$

row orthogonality gives

$$\langle \chi_V, \chi_{S_i} \rangle = \sum_{j=1}^{r} n_j \langle \chi_{S_j}, \chi_{S_i} \rangle = n_i.$$

□

Let

$$\mathrm{Irr}(G) = \{\chi_1, \dots, \chi_r\},$$

and let

$$\mathrm{Cl}(G) = \{C_1, \dots, C_r\}$$

be the set of conjugacy classes of G. Choose a representative

$$g_j \in C_j$$

for each j. The character table of G is the matrix

$$X = (\chi_i(g_j))_{1 \le i,j \le r}.$$

Theorem 12.4.17 (Column orthogonality relations)**.** *With the notation above, for*

$1 \le j, \ell \le r$, one has

$$\sum_{i=1}^{r} \chi_i(g_j)\overline{\chi_i(g_\ell)} = \begin{cases} |C_G(g_j)|, & j = \ell, \\ 0, & j \neq \ell. \end{cases}$$

Equivalently, for arbitrary $g, h \in G$,

$$\sum_{\chi \in \mathrm{Irr}(G)} \chi(g)\overline{\chi(h)} = \begin{cases} |C_G(g)|, & g \text{ and } h \text{ are conjugate in } G, \\ 0, & g \text{ and } h \text{ are not conjugate in } G. \end{cases}$$

Proof. Let

$$D = \frac{1}{|G|} \operatorname{diag}(|C_1|, \ldots, |C_r|).$$

The row orthogonality relations say that

$$\langle \chi_i, \chi_\ell \rangle = \delta_{i\ell}.$$

Writing the inner product as a sum over conjugacy classes gives

$$\frac{1}{|G|} \sum_{j=1}^{r} |C_j| \chi_i(g_j)\overline{\chi_\ell(g_j)} = \delta_{i\ell}.$$

In matrix form, this is

$$XD\overline{X}^t = I_r.$$

Hence X is invertible, and

$$D\overline{X}^t = X^{-1}.$$

Multiplying on the right by X, we get

$$D\overline{X}^t X = I_r.$$

Therefore

$$\overline{X}^t X = D^{-1}.$$

But

$$D^{-1} = \operatorname{diag}\left(\frac{|G|}{|C_1|}, \ldots, \frac{|G|}{|C_r|}\right).$$

By the orbit-stabilizer formula for the conjugation action,

$$|C_j| = \frac{|G|}{|C_G(g_j)|}.$$

Hence

$$D^{-1} = \operatorname{diag}\left(|C_G(g_1)|, \ldots, |C_G(g_r)|\right).$$

The (j, ℓ)-entry of $\overline{X}^t X$ is

$$\sum_{i=1}^{r} \overline{\chi_i(g_j)}\chi_i(g_\ell).$$

Thus

$$\sum_{i=1}^{r} \overline{\chi_i(g_j)}\chi_i(g_\ell) = \begin{cases} |C_G(g_j)|, & j = \ell, \\ 0, & j \neq \ell. \end{cases}$$

Taking complex conjugates gives

$$\sum_{i=1}^{r} \chi_i(g_j)\overline{\chi_i(g_\ell)} = \begin{cases} |C_G(g_j)|, & j = \ell, \\ 0, & j \neq \ell. \end{cases}$$

Finally, if $g, h \in G$ are arbitrary, then their character values depend only on their conjugacy classes. Therefore the formula above is equivalent to

$$\sum_{\chi \in \mathrm{Irr}(G)} \chi(g)\overline{\chi(h)} = \begin{cases} |C_G(g)|, & g \text{ and } h \text{ are conjugate in } G, \\ 0, & g \text{ and } h \text{ are not conjugate in } G. \end{cases}$$

□

12.5 Examples: Computing Character Tables by Orthogonality Relations

In this section we illustrate how to compute character tables using the row and column orthogonality relations.

Recall that if

$$\mathrm{Irr}(G) = \{\chi_1, \ldots, \chi_r\}$$

is the set of irreducible complex characters of G, and if

$$C_1, \ldots, C_r$$

are the conjugacy classes of G, with representatives $g_1, \ldots, g_r$, then the row orthogonality relations say that

$$\langle \chi_i, \chi_j \rangle = \frac{1}{|G|} \sum_{\ell=1}^{r} |C_\ell| \chi_i(g_\ell)\overline{\chi_j(g_\ell)} = \delta_{ij}.$$

The column orthogonality relations say that

$$\sum_{\chi \in \mathrm{Irr}(G)} \chi(g_i)\overline{\chi(g_j)} = \begin{cases} |C_G(g_i)|, & g_i \text{ and } g_j \text{ are conjugate}, \\ 0, & g_i \text{ and } g_j \text{ are not conjugate}. \end{cases}$$

In particular, for a representative g_i,

$$\sum_{\chi \in \mathrm{Irr}(G)} |\chi(g_i)|^2 = |C_G(g_i)|.$$

Example 12.5.1 (The character table of S_3). *The conjugacy classes of S_3 are represented by*

$$1, \qquad (12), \qquad (123).$$

Their sizes and centralizer orders are

representative	1	(12)	(123)
class size	1	3	2
$\lvert C_{S_3}(g)\rvert$	6	2	3

Hence S_3 has exactly three irreducible complex characters.

There are two one-dimensional characters:

$$\chi_{\mathrm{triv}} = (1,1,1), \qquad \chi_{\mathrm{sgn}} = (1,-1,1).$$

If the remaining irreducible character has degree d, then

$$1^2 + 1^2 + d^2 = |S_3| = 6,$$

so

$$d = 2.$$

Write the remaining character as

$$\chi = (2, a, b),$$

where $a = \chi((12))$ and $b = \chi((123))$.

We first use the column orthogonality relation. For the transposition class,

$$|\chi_{\mathrm{triv}}((12))|^2 + |\chi_{\mathrm{sgn}}((12))|^2 + |\chi((12))|^2 = |C_{S_3}((12))|.$$

Thus

$$1^2 + (-1)^2 + |a|^2 = 2,$$

and hence

$$a = 0.$$

For the 3-cycle class,

$$1^2 + 1^2 + |b|^2 = 3,$$

so

$$|b| = 1.$$

To determine the sign of b, use row orthogonality with the trivial character:

$$0 = \langle \chi, \chi_{\mathrm{triv}} \rangle = \frac{1}{6}\big(1 \cdot 2 + 3 \cdot a + 2 \cdot b\big).$$

Since $a = 0$, this becomes

$$2 + 2b = 0,$$

hence

$$b = -1.$$

Therefore the character table of S_3 is

S_3	1	(12)	(123)
class size	1	3	2
χ_{triv}	1	1	1
χ_{sgn}	1	-1	1
χ	2	0	-1

Example 12.5.2 (The character table of D_8)**.** *Let*

$$D_8 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle$$

be the dihedral group of order 8. Its conjugacy classes are represented by

$$1, \qquad r^2, \qquad r, \qquad s, \qquad rs.$$

More explicitly,

$$\{1\}, \qquad \{r^2\}, \qquad \{r, r^3\}, \qquad \{s, r^2 s\}, \qquad \{rs, r^3 s\}.$$

Their sizes and centralizer orders are

representative	1	r^2	r	s	rs
class size	1	1	2	2	2
$\lvert C_{D_8}(g) \rvert$	8	8	4	4	4

Hence D_8 has exactly five irreducible complex characters.

Since

$$D_8/[D_8, D_8] \cong C_2 \times C_2,$$

there are four one-dimensional characters. They are determined by the possible values of r and s, both of which must be ± 1:

	1	r^2	r	s	rs
χ_{++}	1	1	1	1	1
χ_{+-}	1	1	1	-1	-1
χ_{-+}	1	1	-1	1	-1
χ_{--}	1	1	-1	-1	1

Let the remaining irreducible character be

$$\chi = (2, a, b, c, d),$$

where the entries correspond to the classes

$$1, \quad r^2, \quad r, \quad s, \quad rs.$$

Its degree is 2*, because*

$$1^2 + 1^2 + 1^2 + 1^2 + \chi(1)^2 = |D_8| = 8.$$

We now use column orthogonality. For the class of r*,*

$$1^2 + 1^2 + (-1)^2 + (-1)^2 + |b|^2 = |C_{D_8}(r)| = 4.$$

Thus

$$b = 0.$$

Similarly, for the two reflection classes,

$$c = 0, \qquad d = 0.$$

For the central element r^2*, column orthogonality gives*

$$1^2 + 1^2 + 1^2 + 1^2 + |a|^2 = |C_{D_8}(r^2)| = 8.$$

Hence

$$|a| = 2.$$

To determine a*, use row orthogonality with the trivial character:*

$$0 = \langle \chi, \chi_{++} \rangle = \frac{1}{8}\big(1 \cdot 2 + 1 \cdot a + 2 \cdot b + 2 \cdot c + 2 \cdot d\big).$$

Since $b = c = d = 0$*, this gives*

$$2 + a = 0,$$

so

$$a = -2.$$

Therefore the character table of D_8 is

D_8	1	r^2	r	s	rs
class size	1	1	2	2	2
χ_{++}	1	1	1	1	1
χ_{+-}	1	1	1	-1	-1
χ_{-+}	1	1	-1	1	-1
χ_{--}	1	1	-1	-1	1
χ	2	-2	0	0	0

Example 12.5.3 (The character table of S_4)**.** *The conjugacy classes of S_4 are determined by cycle type. We take representatives*

$$1, \qquad (12), \qquad (12)(34), \qquad (123), \qquad (1234).$$

The class sizes and centralizer orders are

representative	1	(12)	(12)(34)	(123)	(1234)
class size	1	6	3	8	6
$\lvert C_{S_4}(g)\rvert$	24	4	8	3	4

Hence S_4 has exactly five irreducible complex characters.

We already know two one-dimensional characters:

$$\chi_{\mathrm{triv}} = (1, 1, 1, 1, 1),$$

and

$$\chi_{\mathrm{sgn}} = (1, -1, 1, 1, -1).$$

Next consider the permutation representation of S_4 on

$$\mathbb{C}^4.$$

Its character is the number of fixed points of a permutation. Hence its values are

$$(4, 2, 0, 1, 0).$$

Removing the one-dimensional trivial subrepresentation, we get the standard representation V_{std}, whose character is

$$\chi_{\mathrm{std}} = (4, 2, 0, 1, 0) - (1, 1, 1, 1, 1) = (3, 1, -1, 0, -1).$$

We check that this character is irreducible:

$$\langle \chi_{\mathrm{std}}, \chi_{\mathrm{std}} \rangle = \frac{1}{24}\left(1 \cdot 3^2 + 6 \cdot 1^2 + 3 \cdot (-1)^2 + 8 \cdot 0^2 + 6 \cdot (-1)^2\right) = 1.$$

Therefore χ_{std} is irreducible.

Twisting by the sign character gives another irreducible character:

$$\chi_{\text{sgn}}\chi_{\text{std}} = (3, -1, -1, 0, 1).$$

So far we have four irreducible characters, of degrees

$$1, \quad 1, \quad 3, \quad 3.$$

Let the remaining irreducible character be

$$\chi = (2, a, b, c, d),$$

where the entries correspond to

$$1, \quad (12), \quad (12)(34), \quad (123), \quad (1234).$$

Its degree is 2*, since*

$$1^2 + 1^2 + 3^2 + 3^2 + \chi(1)^2 = |S_4| = 24.$$

We first use column orthogonality. For the transposition class (12)*,*

$$1^2 + (-1)^2 + 1^2 + (-1)^2 + |a|^2 = |C_{S_4}((12))| = 4.$$

Therefore

$$a = 0.$$

For the 4*-cycle class* (1234)*,*

$$1^2 + (-1)^2 + (-1)^2 + 1^2 + |d|^2 = |C_{S_4}((1234))| = 4,$$

so

$$d = 0.$$

For the double transposition class (12)(34)*,*

$$1^2 + 1^2 + (-1)^2 + (-1)^2 + |b|^2 = |C_{S_4}((12)(34))| = 8,$$

hence

$$|b| = 2.$$

For the 3*-cycle class* (123)*,*

$$1^2 + 1^2 + 0^2 + 0^2 + |c|^2 = |C_{S_4}((123))| = 3,$$

hence

$$|c| = 1.$$

Column orthogonality has determined

$$a = 0, \qquad d = 0, \qquad |b| = 2, \qquad |c| = 1.$$

To determine the signs of b *and* c*, we now use row orthogonality.*

Orthogonality with χ_{std} *gives*

$$0 = \langle \chi, \chi_{\mathrm{std}} \rangle = \frac{1}{24} \left(1 \cdot 2 \cdot 3 + 6 \cdot a \cdot 1 + 3 \cdot b \cdot (-1) + 8 \cdot c \cdot 0 + 6 \cdot d \cdot (-1) \right).$$

Since $a = d = 0$*, this becomes*

$$6 - 3b = 0.$$

Hence

$$b = 2.$$

Orthogonality with the trivial character gives

$$0 = \langle \chi, \chi_{\mathrm{triv}} \rangle = \frac{1}{24} \left(1 \cdot 2 + 6 \cdot a + 3 \cdot b + 8 \cdot c + 6 \cdot d \right).$$

Using $a = d = 0$ *and* $b = 2$*, we obtain*

$$2 + 6 + 8c = 0.$$

Therefore

$$c = -1.$$

Thus the character table of S_4 *is*

S_4	1	(12)	(12)(34)	(123)	(1234)
class size	1	6	3	8	6
χ_{triv}	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_{std}	3	1	-1	0	-1
$\chi_{\mathrm{sgn}}\chi_{\mathrm{std}}$	3	-1	-1	0	1
χ	2	0	2	-1	0

Chapter 13

Lecture 13

13.1 Commutator Subspaces in Characteristic p

Let A be a finite-dimensional associative k-algebra, where

$$\operatorname{char} k = p > 0.$$

Put

$$S(A) = \operatorname{span}_k\{\, ab - ba \mid a, b \in A \,\}.$$

When the algebra A is clear from the context, we simply write $S = S(A)$.

For each integer $m \geq 0$, define

$$T_m(A) = \{\, r \in A \mid r^{p^m} \in S(A) \,\}.$$

We also define

$$T(A) = \bigcup_{m \geq 0} T_m(A) = \{\, r \in A \mid r^{p^m} \in S(A) \text{ for some } m \geq 0 \,\}.$$

Lemma 13.1.1. *For every integer $e \geq 0$, every $a, b \in A$, and every $\lambda, \mu \in k$, one has*

$$(\lambda a + \mu b)^{p^e} \equiv \lambda^{p^e} a^{p^e} + \mu^{p^e} b^{p^e} \pmod{S(A)}.$$

More generally, for $x_1, \ldots, x_t \in A$ and $\lambda_1, \ldots, \lambda_t \in k$,

$$\left(\sum_{i=1}^{t} \lambda_i x_i\right)^{p^e} \equiv \sum_{i=1}^{t} \lambda_i^{p^e} x_i^{p^e} \pmod{S(A)}.$$

Proof. The case $e = 0$ is clear. Assume $e \geq 1$, and put

$$q = p^e.$$

Let $x_1, \ldots, x_q \in A$, and set

$$w = x_1 x_2 \cdots x_q.$$

A cyclic permutation of w differs from w by an element of $S(A)$. Indeed,

$$x_1 x_2 \cdots x_q - x_2 \cdots x_q x_1 = x_1(x_2 \cdots x_q) - (x_2 \cdots x_q)x_1 \in S(A).$$

Hence every word of length q is congruent modulo $S(A)$ to each of its cyclic permutations.

Now expand

$$(\lambda a + \mu b)^q.$$

The terms are words of length q in the two letters a and b. The two constant words contribute

$$\lambda^q a^q + \mu^q b^q.$$

Every other word is a mixed word. Since $q = p^e$, the cyclic orbit of a mixed word has size divisible by p: its size is a divisor of q, and it is not 1. All words in such an orbit are congruent modulo $S(A)$, and their scalar coefficients are the same. Therefore the sum of the terms in each mixed orbit is congruent modulo $S(A)$ to

$$|\mathcal{O}| \, w = 0,$$

because $\operatorname{char} k = p$. Thus

$$(\lambda a + \mu b)^q \equiv \lambda^q a^q + \mu^q b^q \pmod{S(A)}.$$

The formula for a finite sum follows by induction on the number of summands. □

Lemma 13.1.2. *For every fixed integer $m \geq 0$, the subset*

$$T_m(A) = \{ \, r \in A \mid r^{p^m} \in S(A) \, \}$$

is a k-subspace of A, and

$$S(A) \subseteq T_m(A).$$

Proof. Put

$$q = p^m.$$

We first prove that $S(A) \subseteq T_m(A)$. Let $a, b \in A$, and set

$$c = ab - ba.$$

By the previous lemma,

$$c^q = (ab - ba)^q \equiv (ab)^q - (ba)^q \pmod{S(A)}.$$

Here we use the fact that, in characteristic p,

$$(-1)^q = -1.$$

Moreover,

$$(ab)^q - (ba)^q = a\bigl(b(ab)^{q-1}\bigr) - \bigl(b(ab)^{q-1}\bigr)a \in S(A).$$

Hence

$$(ab - ba)^q \in S(A).$$

Now let $s \in S(A)$. Since $S(A)$ is spanned by commutators, we may write

$$s = \sum_i \lambda_i c_i,$$

where each c_i is a commutator. By the previous lemma,

$$s^q = \left(\sum_i \lambda_i c_i\right)^q \equiv \sum_i \lambda_i^q c_i^q \pmod{S(A)}.$$

Each $c_i^q \in S(A)$, so the right-hand side lies in $S(A)$. Therefore

$$s^q \in S(A).$$

Hence $s \in T_m(A)$, and so

$$S(A) \subseteq T_m(A).$$

It remains to prove that $T_m(A)$ is a k-subspace. Let $x, y \in T_m(A)$, and let $\lambda, \mu \in k$. Then

$$x^q, y^q \in S(A).$$

Again by the previous lemma,

$$(\lambda x + \mu y)^q \equiv \lambda^q x^q + \mu^q y^q \pmod{S(A)}.$$

The right-hand side lies in $S(A)$, and hence

$$(\lambda x + \mu y)^q \in S(A).$$

Therefore

$$\lambda x + \mu y \in T_m(A).$$

Thus $T_m(A)$ is a k-subspace of A. □

Corollary 13.1.3. *The subset*

$$T(A) = \{\, r \in A \mid r^{p^m} \in S(A) \text{ for some } m \geq 0 \,\}$$

is a k-subspace of A, and

$$S(A) \subseteq T(A).$$

Proof. For every $m \geq 0$, the previous lemma gives

$$S(A) \subseteq T_m(A).$$

Moreover,

$$T_m(A) \subseteq T_{m+1}(A).$$

Indeed, if $x \in T_m(A)$, then $x^{p^m} \in S(A)$. Since $S(A) \subseteq T_1(A)$, we have

$$\left(x^{p^m}\right)^p = x^{p^{m+1}} \in S(A).$$

Hence $x \in T_{m+1}(A)$.

Therefore

$$T(A) = \bigcup_{m \geq 0} T_m(A)$$

is the union of an increasing chain of k-subspaces, and hence is itself a k-subspace. Also,

$$S(A) \subseteq T(A).$$

□

Lemma 13.1.4. *Let*

$$A = M_n(k).$$

Then

$$S(A) = T(A) = \{\, X \in M_n(k) \mid \operatorname{tr}(X) = 0 \,\}.$$

Proof. Put

$$V = \{\, X \in M_n(k) \mid \operatorname{tr}(X) = 0 \,\}.$$

Since

$$\operatorname{tr}(XY - YX) = 0$$

for all $X, Y \in M_n(k)$, we have

$$S(A) \subseteq V.$$

Conversely, let E_{ij} denote the standard matrix units. If $i \neq j$, then

$$E_{ij} = E_{ii}E_{ij} - E_{ij}E_{ii} \in S(A).$$

Also, for $i \neq j$,

$$E_{ii} - E_{jj} = E_{ij}E_{ji} - E_{ji}E_{ij} \in S(A).$$

The trace-zero matrices are spanned by the off-diagonal matrix units E_{ij} with $i \neq j$, together with the diagonal differences $E_{ii} - E_{jj}$. Hence

$$V \subseteq S(A).$$

Therefore

$$S(A) = V.$$

By the previous corollary,

$$S(A) \subseteq T(A) \subseteq A.$$

Since $S(A) = V$ has codimension 1 in $A = M_n(k)$, we have either

$$T(A) = S(A) \qquad \text{or} \qquad T(A) = A.$$

But

$$E_{11}^{p^m} = E_{11} \qquad \text{for every } m \geq 0,$$

and

$$\operatorname{tr}(E_{11}) = 1 \neq 0.$$

Hence

$$E_{11}^{p^m} \notin S(A) \qquad \text{for every } m \geq 0.$$

Therefore $E_{11} \notin T(A)$, so $T(A) \neq A$. It follows that

$$T(A) = S(A).$$

Consequently,

$$S(A) = T(A) = \{\, X \in M_n(k) \mid \operatorname{tr}(X) = 0 \,\}.$$

□

Proposition 13.1.5. *Let A be a finite-dimensional k-algebra, where*

$$\operatorname{char} k = p > 0.$$

Put

$$S(A) = \operatorname{span}_k\{\, ab - ba \mid a, b \in A \,\},$$

and

$$T(A) = \{\, a \in A \mid a^{p^m} \in S(A) \text{ for some } m \geq 0 \,\}.$$

Assume that k is a splitting field for A. Then

$$\#\{\, \text{non-isomorphic simple } A\text{-modules} \,\} = \dim_k A - \dim_k T(A).$$

Proof. Let

$$J = J(A)$$

be the Jacobson radical of A, and set

$$\overline{A} = A/J.$$

Since A is finite-dimensional, J is nilpotent. Hence for every $x \in J$, there exists $m \geq 0$ such that

$$x^{p^m} = 0.$$

Since $0 \in S(A)$, it follows that

$$J \subseteq T(A).$$

Let

$$\pi : A \longrightarrow \overline{A} = A/J$$

be the natural quotient map. We claim that

$$\pi(T(A)) = T(\overline{A}).$$

First, since

$$\pi(S(A)) = S(\overline{A}),$$

if $a \in T(A)$, then for some $m \geq 0$,

$$a^{p^m} \in S(A).$$

Therefore

$$\pi(a)^{p^m} = \pi(a^{p^m}) \in S(\overline{A}),$$

and hence

$$\pi(a) \in T(\overline{A}).$$

Thus

$$\pi(T(A)) \subseteq T(\overline{A}).$$

Conversely, suppose that

$$\overline{a} = \pi(a) \in T(\overline{A}).$$

Then for some $m \geq 0$,

$$\overline{a}^{p^m} \in S(\overline{A}).$$

Since $\pi(S(A)) = S(\overline{A})$, there exist $s \in S(A)$ and $j \in J$ such that

$$a^{p^m} = s + j.$$

Choose $\ell \geq 0$ sufficiently large such that

$$j^{p^\ell} = 0.$$

By the congruence above,

$$(s + j)^{p^\ell} \equiv s^{p^\ell} + j^{p^\ell} \pmod{S(A)}.$$

Therefore

$$a^{p^{m+\ell}} = (a^{p^m})^{p^\ell} = (s + j)^{p^\ell} \equiv s^{p^\ell} \pmod{S(A)}.$$

Since $s \in S(A)$, the lemma on $T_\ell(A)$ gives

$$s^{p^\ell} \in S(A).$$

Hence

$$a^{p^{m+\ell}} \in S(A).$$

Therefore

$$a \in T(A).$$

This proves

$$\pi(T(A)) = T(\overline{A}).$$

Since $J \subseteq T(A)$, we get

$$A/T(A) \cong \overline{A}/T(\overline{A}).$$

Hence

$$\dim_k A - \dim_k T(A) = \dim_k \overline{A} - \dim_k T(\overline{A}).$$

Now k is a splitting field for A, so the semisimple algebra

$$\overline{A} = A/J(A)$$

is split semisimple. Thus by the Artin–Wedderburn theorem,

$$\overline{A} \cong M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k),$$

where

$$r = \#\{\,\text{non-isomorphic simple } A\text{-modules}\,\}.$$

The simple A-modules and the simple $\overline{A}$-modules are the same, because $J(A)$ annihilates every simple A-module.

For a finite direct sum of algebras, commutators and p^m-powers are computed componentwise. Hence

$$S(\overline{A}) = S(M_{n_1}(k)) \oplus \cdots \oplus S(M_{n_r}(k)).$$

Also,

$$T(\overline{A}) = T(M_{n_1}(k)) \oplus \cdots \oplus T(M_{n_r}(k)).$$

Indeed, if an element belongs componentwise to the right-hand side, then each component satisfies the defining condition for some exponent p^{m_i}; taking a common exponent p^m with $m \geq \max_i m_i$ shows that the whole element lies in $T(\overline{A})$.

By the previous matrix algebra lemma,

$$T(M_{n_i}(k)) = \{\, X \in M_{n_i}(k) \mid \operatorname{tr}(X) = 0 \,\}.$$

Therefore each quotient

$$M_{n_i}(k)/T(M_{n_i}(k))$$

is one-dimensional over k. Hence

$$\dim_k \overline{A} - \dim_k T(\overline{A}) = \sum_{i=1}^{r} \dim_k\big(M_{n_i}(k)/T(M_{n_i}(k))\big) = r.$$

Consequently,

$$\dim_k A - \dim_k T(A) = r = \#\{\,\text{non-isomorphic simple } A\text{-modules}\,\}.$$

□

13.2 Brauer's Theorem and Modular Systems

Definition 13.2.1 (p-regular and p-singular elements)**.** *Let G be a finite group and let p be a prime number. An element $g \in G$ is called p-**regular** if*

$$p \nmid \operatorname{ord}(g).$$

*It is called p-**singular** in the sense of these notes if*

$$\operatorname{ord}(g) = p^e$$

for some integer $e \geq 0$. Equivalently, g is a p-element.

Lemma 13.2.2 (p-regular and p-singular decomposition)**.** *Let G be a finite group. For every $x \in G$, there exist unique elements $s, t \in G$ such that*

$$x = st, \qquad st = ts,$$

where s is p-regular and t is p-singular.

Proof. Let

$$\operatorname{ord}(x) = p^a m, \qquad \gcd(p, m) = 1.$$

Since $\gcd(p^a, m) = 1$, choose integers $r, q \in \mathbb{Z}$ such that

$$rm + qp^a = 1.$$

Define

$$t = x^{rm}, \qquad s = x^{qp^a}.$$

Then $s, t \in \langle x \rangle$, so $st = ts$, and

$$st = x^{rm+qp^a} = x.$$

Moreover, the order of t divides p^a, so t is p-singular, while the order of s divides m, so s is p-regular.

For uniqueness, suppose

$$x = s_1 t_1 = s_2 t_2$$

are two such decompositions, with s_i p-regular, t_i p-singular, and $s_i t_i = t_i s_i$. Since s_i and t_i have coprime orders, both s_i and t_i are powers of x. Thus they lie in the cyclic group $\langle x \rangle$. Inside the finite cyclic group $\langle x \rangle$, the p-part and the p'-part of an element are unique. Hence

$$s_1 = s_2, \qquad t_1 = t_2.$$

□

Lemma 13.2.3 (Commutator subspace of a group algebra)**.** *Let G be a finite group and let k be a field. Put*

$$S(kG) = \operatorname{span}_k\{\, ab - ba \mid a, b \in kG \,\}.$$

Then

$$S(kG) = \left\{ \sum_{g \in G} \lambda_g g \;\middle|\; \sum_{g \in C} \lambda_g = 0 \text{ for every conjugacy class } C \subseteq G \right\}.$$

Proof. First observe that if $g, h \in G$, then

$$hgh^{-1} - g = h(gh^{-1}) - (gh^{-1})h \in S(kG).$$

Hence differences of conjugate group elements lie in $S(kG)$.

Let

$$\alpha = \sum_{g \in G} \lambda_g g \in S(kG).$$

Since each commutator $ab - ba$ has total coefficient 0 on every conjugacy class, it follows that

$$\sum_{g \in C} \lambda_g = 0$$

for every conjugacy class C.

Conversely, suppose

$$\alpha = \sum_{g \in G} \lambda_g g$$

satisfies

$$\sum_{g \in C} \lambda_g = 0$$

for every conjugacy class C. For each conjugacy class C, choose a representative $c_C \in C$. Then

$$\sum_{g \in C} \lambda_g g = \sum_{g \in C} \lambda_g (g - c_C),$$

because $\sum_{g \in C} \lambda_g = 0$. Since each $g - c_C$ is a difference of conjugate elements, it lies in

$S(kG)$. Therefore

$$\sum_{g \in C} \lambda_g g \in S(kG)$$

for every conjugacy class C, and hence

$$\alpha \in S(kG).$$

This proves the claim. $\square$

Theorem 13.2.4 (Brauer)**.** *Let G be a finite group, and let k be a splitting field for G with*

$$\operatorname{char} k = p > 0.$$

Then the number of non-isomorphic simple kG-modules is equal to the number of conjugacy classes of p-regular elements of G.

Proof. Put

$$A = kG.$$

As before, define

$$S(A) = \operatorname{span}_k\{\, ab - ba \mid a, b \in A \,\},$$

and

$$T(A) = \{\, a \in A \mid a^{p^m} \in S(A) \text{ for some } m \geq 0 \,\}.$$

By the previous proposition,

$$\#\{\, \text{non-isomorphic simple } kG\text{-modules} \,\} = \dim_k kG - \dim_k T(A).$$

Thus it remains to show that

$$\dim_k kG - \dim_k T(A)$$

is the number of conjugacy classes of p-regular elements.

Let

$$\{x_i\}_{i \in I}$$

be a set of representatives of the conjugacy classes of p-regular elements of G. We claim that

$$\{\, x_i + T(A) \mid i \in I \,\}$$

is a k-basis of $kG/T(A)$.

First we prove that these elements span $kG/T(A)$. Let $x \in G$. By the p-regular and p-singular decomposition, write

$$x = st, \qquad st = ts,$$

where s is p-regular and t is p-singular. Since t has p-power order, for sufficiently large n

we have

$$t^{p^n} = 1.$$

Hence, in characteristic p,

$$(x-s)^{p^n} = \big(s(t-1)\big)^{p^n} = s^{p^n}(t-1)^{p^n} = s^{p^n}(t^{p^n}-1) = 0.$$

Therefore

$$x - s \in T(A).$$

Since s is p-regular, it is conjugate to some x_i. Hence

$$s - x_i \in S(A) \subseteq T(A).$$

Therefore

$$x + T(A) = x_i + T(A).$$

Since the group elements form a k-basis of kG, this proves that the classes $x_i + T(A)$ span $kG/T(A)$.

It remains to prove linear independence. Suppose

$$\sum_{i \in I} \lambda_i x_i \in T(A).$$

Then, by the definition of $T(A)$, there exists $m \geq 0$ such that

$$\left(\sum_{i \in I} \lambda_i x_i\right)^{p^m} \in S(A).$$

Enlarging m if necessary, we may assume that

$$x_i^{p^m} = x_i \qquad \text{for all } i \in I.$$

Indeed, the orders of the x_i are prime to p, so one can choose m sufficiently large such that

$$p^m \equiv 1 \pmod{\operatorname{ord}(x_i)}$$

for all i.

Using the congruence

$$\left(\sum_i \lambda_i x_i\right)^{p^m} \equiv \sum_i \lambda_i^{p^m} x_i^{p^m} \pmod{S(A)},$$

we get

$$\sum_{i \in I} \lambda_i^{p^m} x_i \in S(A).$$

By the preceding lemma, the sum of coefficients on each conjugacy class must be zero. Since the x_i lie in distinct conjugacy classes, this gives

$$\lambda_i^{p^m} = 0 \qquad \text{for all } i.$$

Hence

$$\lambda_i = 0 \qquad \text{for all } i.$$

Therefore the classes $x_i + T(A)$ are linearly independent.

Consequently,

$$\dim_k kG/T(A) = |I|,$$

which is precisely the number of conjugacy classes of p-regular elements of G. Hence

$$\#\{\,\text{non-isomorphic simple } kG\text{-modules}\,\} = |I|.$$

$\square$

13.3 p-Modular Systems

Definition 13.3.1 (p-modular system)**.** *Let G be a finite group and let p be a prime number. A p-**modular system** for G is a triple*

$$(F, R, k)$$

satisfying the following conditions:

1. *F is a field of characteristic 0, equipped with a complete discrete valuation.*
2. *R is the valuation ring of F, with maximal ideal*

$$J(R) = (\pi).$$

3. *The residue field is*

$$k = R/(\pi),$$

and

$$\operatorname{char} k = p.$$

*If both F and k are splitting fields for G, then (F, R, k) is called a **splitting** p-**modular system** for G.*

Lemma 13.3.2. *Let R be a complete discrete valuation ring with maximal ideal*

$$J(R) = (\pi),$$

and residue field

$$k = R/(\pi).$$

Let G be a finite group. Then:

1. *If S is a simple RG-module, then*
$$\pi S = 0.$$

2. *The simple RG-modules are exactly the simple kG-modules, via the quotient homomorphism*
$$RG \longrightarrow kG.$$

3. *If U is an RG-module, then*
$$\pi U \subseteq \mathrm{Rad}(U).$$
In particular,
$$\pi RG \subseteq J(RG).$$

4. *For every RG-module U,*
$$\mathrm{Rad}(U)/\pi U = \mathrm{Rad}(U/\pi U).$$

Proof. First we prove that
$$\pi RG \subseteq J(RG).$$

Let $z \in \pi RG$. Since RG is finite over the complete discrete valuation ring R, it is complete with respect to the π-adic topology. Hence the geometric series
$$1 + z + z^2 + \cdots$$
converges in RG, and gives an inverse of $1 - z$. Thus $1 - z$ is a unit. More generally, for every $y \in RG$, one has $yz \in \pi RG$, so
$$1 - yz$$
is a unit. By the standard characterization of the Jacobson radical, this implies
$$z \in J(RG).$$

Hence
$$\pi RG \subseteq J(RG).$$

Now let S be a simple RG-module. Since the Jacobson radical annihilates every simple module, we have
$$J(RG)S = 0.$$

Since $\pi RG \subseteq J(RG)$, it follows that
$$\pi S = 0.$$

This proves (1).

Since every simple RG-module is annihilated by π, it factors through

$$RG/\pi RG \cong kG.$$

Therefore every simple RG-module is a simple kG-module. Conversely, every simple kG-module becomes a simple RG-module by restriction along the quotient map

$$RG \longrightarrow kG.$$

This proves (2).

Let U be an RG-module. If M is a maximal submodule of U, then U/M is simple. By (1),

$$\pi(U/M) = 0.$$

Hence

$$\pi U \subseteq M.$$

Taking the intersection over all maximal submodules M of U, we obtain

$$\pi U \subseteq \mathrm{Rad}(U).$$

Applying this to the regular module $U = RG$, we get

$$\pi RG \subseteq J(RG).$$

This proves (3).

Finally, since every maximal submodule of U contains πU, the maximal submodules of $U/\pi U$ are precisely the quotients

$$M/\pi U,$$

where M runs through the maximal submodules of U. Therefore

$$\mathrm{Rad}(U/\pi U) = \bigcap_{M \text{ maximal in } U} M/\pi U = \left(\bigcap_{M \text{ maximal in } U} M \right) /\pi U = \mathrm{Rad}(U)/\pi U.$$

This proves (4). $\square$

Corollary 13.3.3. *Let P and Q be finitely generated projective RG-modules. Then*

$$P \cong Q \quad \textit{as } RG\textit{-modules}$$

if and only if

$$P/\pi P \cong Q/\pi Q \quad \textit{as } kG\textit{-modules.}$$

Proof. The forward implication is immediate by reducing modulo π.

Conversely, suppose that

$$\overline{f} : P/\pi P \longrightarrow Q/\pi Q$$

is an isomorphism of kG-modules. Let

$$q_P : P \longrightarrow P/\pi P, \qquad q_Q : Q \longrightarrow Q/\pi Q$$

be the natural quotient maps.

Since P is projective and q_Q is surjective, the map

$$\overline{f} \circ q_P : P \longrightarrow Q/\pi Q$$

lifts to an RG-homomorphism

$$f : P \longrightarrow Q$$

such that

$$q_Q \circ f = \overline{f} \circ q_P.$$

Similarly, the inverse

$$\overline{f}^{-1} : Q/\pi Q \longrightarrow P/\pi P$$

lifts to an RG-homomorphism

$$g : Q \longrightarrow P.$$

Then the induced maps modulo π satisfy

$$\overline{gf} = \overline{g}\,\overline{f} = \mathrm{id}_{P/\pi P},$$

and similarly

$$\overline{fg} = \mathrm{id}_{Q/\pi Q}\,.$$

Hence

$$(gf - \mathrm{id}_P)(P) \subseteq \pi P,$$

and

$$(fg - \mathrm{id}_Q)(Q) \subseteq \pi Q.$$

Therefore the endomorphisms gf and fg are surjective by Nakayama's lemma, since

$$\pi RG \subseteq J(RG).$$

Since P and Q are finitely generated over the finite R-algebra RG, they are finite R-modules. In particular, they are Noetherian. Hence a surjective endomorphism of P or Q is injective. Thus gf and fg are automorphisms.

From gf being an automorphism, f is injective. From fg being an automorphism, f is surjective. Therefore f is an isomorphism:

$$P \cong Q.$$

□

13.4 RG-Lattices and Lifting of Projectives

Definition 13.4.1 (RG-lattice)**.** *Let R be a commutative ring and let G be a finite group. An RG-module L is called an RG-**lattice** if L is finitely generated as an RG-module and is free as an R-module.*

Proposition 13.4.2 (Lifting projective covers from kG to RG)**.** *Let R be a complete discrete valuation ring with maximal ideal*

$$J(R) = (\pi),$$

and let

$$k = R/(\pi).$$

Let G be a finite group. Then the following hold.

1. *For every simple kG-module S, there exists an indecomposable projective RG-module $\widehat{P}_S$ such that*
$$\widehat{P}_S = RG\widehat{e}_S,$$
where $\widehat{e}_S \in RG$ is a primitive idempotent satisfying
$$\widehat{e}_S S \neq 0.$$
Moreover, $\widehat{P}_S$ is a projective cover of S as an RG-module.

2. *If P_S denotes the projective cover of S as a kG-module, then*
$$\widehat{P}_S/\pi\widehat{P}_S \cong P_S.$$
Moreover, the canonical epimorphism
$$\widehat{P}_S \longrightarrow \widehat{P}_S/\pi\widehat{P}_S \cong P_S$$
is a projective cover of P_S as an RG-module. In particular, S is the unique simple quotient of $\widehat{P}_S$.

   Consequently, for simple kG-modules S and T,
$$\widehat{P}_S \cong \widehat{P}_T \quad \Longleftrightarrow \quad S \cong T.$$

3. *Every finitely generated RG-module has a projective cover.*

4. *Every finitely generated indecomposable projective RG-module is isomorphic to $\widehat{P}_S$ for some simple kG-module S.*

5. *Every finitely generated projective kG-module can be lifted to a projective RG-lattice. Such a lift is unique up to isomorphism.*

Proof. Let
$$\overline{RG} = RG/\pi RG \cong kG.$$
Since R is complete, idempotents lift modulo the ideal πRG. Moreover,
$$\pi RG \subseteq J(RG),$$
so primitive idempotents lift to primitive idempotents.

Let S be a simple kG-module. Choose a primitive idempotent $e_S \in kG$ such that
$$P_S = kGe_S$$
is the projective cover of S, and
$$e_S S \neq 0.$$
Lift e_S to a primitive idempotent
$$\widehat{e}_S \in RG.$$
Define
$$\widehat{P}_S := RG\widehat{e}_S.$$
Since $\widehat{e}_S$ is an idempotent, $RG\widehat{e}_S$ is a direct summand of the regular module RG, hence is projective. Since $\widehat{e}_S$ is primitive, $RG\widehat{e}_S$ is indecomposable.

Reducing modulo π, we get
$$\widehat{P}_S/\pi\widehat{P}_S \cong (RG/\pi RG)(\widehat{e}_S + \pi RG) \cong kGe_S = P_S.$$
Thus
$$\widehat{P}_S/\pi\widehat{P}_S \cong P_S.$$

Since
$$\pi\widehat{P}_S \subseteq \mathrm{Rad}(\widehat{P}_S),$$
the canonical epimorphism
$$\widehat{P}_S \longrightarrow \widehat{P}_S/\pi\widehat{P}_S$$
is essential. Hence
$$\widehat{P}_S \longrightarrow P_S$$
is a projective cover of P_S as an RG-module.

Composing with the projective cover
$$P_S \longrightarrow S$$
in kG-modules, we obtain an essential epimorphism
$$\widehat{P}_S \longrightarrow S.$$
Therefore $\widehat{P}_S$ is a projective cover of S as an RG-module.

Moreover,

$$\operatorname{Top}(\widehat{P}_S) = \widehat{P}_S / \operatorname{Rad}(\widehat{P}_S) \cong P_S / \operatorname{Rad}(P_S) \cong S.$$

Hence S is the unique simple quotient of $\widehat{P}_S$. It follows that for simple kG-modules S and T,

$$\widehat{P}_S \cong \widehat{P}_T \iff \operatorname{Top}(\widehat{P}_S) \cong \operatorname{Top}(\widehat{P}_T) \iff S \cong T.$$

Now let U be a finitely generated RG-module. Since $RG/J(RG)$ is a semisimple Artinian ring, the top

$$\operatorname{Top}(U) = U / \operatorname{Rad}(U)$$

is a semisimple RG-module. Hence

$$\operatorname{Top}(U) \cong S_1 \oplus \cdots \oplus S_m$$

for some simple kG-modules S_i. Then

$$\widehat{P}_{S_1} \oplus \cdots \oplus \widehat{P}_{S_m} \longrightarrow S_1 \oplus \cdots \oplus S_m \cong \operatorname{Top}(U)$$

is a projective cover. Since the canonical map

$$U \longrightarrow \operatorname{Top}(U)$$

is an essential epimorphism, lifting along it gives an essential epimorphism from a projective module onto U. Thus U has a projective cover.

Let P be a finitely generated indecomposable projective RG-module. The canonical epimorphism

$$P \longrightarrow \operatorname{Top}(P)$$

is a projective cover. Since $\operatorname{Top}(P)$ is semisimple, write

$$\operatorname{Top}(P) \cong S_1 \oplus \cdots \oplus S_m.$$

Then the projective cover of $\operatorname{Top}(P)$ is

$$\widehat{P}_{S_1} \oplus \cdots \oplus \widehat{P}_{S_m}.$$

By uniqueness of projective covers,

$$P \cong \widehat{P}_{S_1} \oplus \cdots \oplus \widehat{P}_{S_m}.$$

Since P is indecomposable, we must have $m = 1$. Hence

$$P \cong \widehat{P}_S$$

for some simple kG-module S.

Finally, let Q be a finitely generated projective kG-module. Since kG is Artinian, Q

decomposes as a finite direct sum of indecomposable projective kG-modules:

$$Q \cong \bigoplus_S P_S^{\oplus m_S}.$$

Define

$$\widehat{Q} := \bigoplus_S \widehat{P}_S^{\oplus m_S}.$$

Then $\widehat{Q}$ is a projective RG-lattice and

$$\widehat{Q}/\pi\widehat{Q} \cong \bigoplus_S P_S^{\oplus m_S} \cong Q.$$

Thus Q lifts to a projective RG-lattice. The uniqueness follows from the isomorphism criterion below. $\square$

Corollary 13.4.3. *Let L be an RG-lattice. Then L is projective as an RG-module if and only if*

$$L/\pi L$$

is projective as a kG-module.

Proof. If L is projective over RG, then L is a direct summand of a finite free RG-module. Reducing modulo π, we see that $L/\pi L$ is a direct summand of a finite free kG-module. Hence $L/\pi L$ is projective over kG.

Conversely, assume that $L/\pi L$ is projective over kG. Let

$$P \longrightarrow L$$

be a projective cover of L as an RG-module. Reducing modulo π, we obtain an essential epimorphism

$$P/\pi P \longrightarrow L/\pi L.$$

Since $L/\pi L$ is projective, this essential epimorphism must be an isomorphism.

Let K be the kernel of $P \to L$. Since P and L are free over the DVR R, reduction modulo π gives

$$K/\pi K = 0.$$

By Nakayama's lemma, $K = 0$. Hence

$$P \cong L,$$

and therefore L is projective. $\square$

Corollary 13.4.4. *Let L_1 and L_2 be finitely generated projective RG-modules. Then*

$$L_1 \cong L_2 \quad \Longleftrightarrow \quad L_1/\pi L_1 \cong L_2/\pi L_2$$

as kG-modules.

Proof. The forward implication is immediate by reduction modulo π.

Conversely, suppose

$$L_1/\pi L_1 \cong L_2/\pi L_2$$

as kG-modules. Since L_i is projective over RG, the natural map

$$L_i \longrightarrow L_i/\pi L_i$$

is a projective cover. Therefore L_1 and L_2 are projective covers of isomorphic RG-modules. By uniqueness of projective covers,

$$L_1 \cong L_2.$$

□

13.5 Full Lattices

Definition 13.5.1 (Full RG-lattice). *Let R be a principal ideal domain, and let $F = \mathrm{Frac}(R)$ be its fraction field. Let G be a finite group, and let U be a finite-dimensional FG-module.*

*A **full RG-lattice** in U is an RG-submodule*

$$L \subseteq U$$

such that:

1. *L is finitely generated and free as an R-module;*
2. *$FL = U$.*

Equivalently, L contains an F-basis of U, or equivalently, the natural map

$$F \otimes_R L \longrightarrow U, \qquad a \otimes \ell \longmapsto a\ell,$$

is an isomorphism of FG-modules.

Remark 13.5.2 (Realizing a lattice inside its scalar extension). *Let R be a principal ideal domain with fraction field F, and let L be an RG-lattice. Since L is free as an R-module, it is torsion-free. Therefore the natural map*

$$L \longrightarrow F \otimes_R L, \qquad \ell \longmapsto 1 \otimes \ell,$$

is injective. Hence we may regard L as an RG-submodule of the FG-module

$$F \otimes_R L.$$

In this sense, the FG-module $F \otimes_R L$ has a form whose structure constants are defined over R.

Lemma 13.5.3. *Let R be a principal ideal domain with fraction field F, and let U be a finite-dimensional F-vector space. Let $L \subseteq U$ be a finitely generated R-submodule containing an F-basis of U. Then L is a full R-lattice in U.*

If, moreover, U is an FG-module and L is G-stable, then L is a full RG-lattice in U.

Proof. Since $L \subseteq U$, the R-module L is torsion-free. Since R is a principal ideal domain and L is finitely generated, L is free as an R-module. Choose an R-basis

$$x_1, \dots, x_m$$

of L.

We claim that $x_1, \dots, x_m$ are F-linearly independent. Suppose

$$\lambda_1 x_1 + \cdots + \lambda_m x_m = 0, \qquad \lambda_i \in F.$$

Write

$$\lambda_i = \frac{a_i}{b_i}, \qquad a_i, b_i \in R, \quad b_i \neq 0.$$

Let

$$b = b_1 b_2 \cdots b_m.$$

Multiplying the relation by b, we get

$$b\lambda_1 x_1 + \cdots + b\lambda_m x_m = 0,$$

where each $b\lambda_i \in R$. Since $x_1, \dots, x_m$ are R-linearly independent, we get

$$b\lambda_i = 0$$

for every i. Since $b \neq 0$, it follows that

$$\lambda_i = 0$$

for every i. Hence $x_1, \dots, x_m$ are F-linearly independent.

Since L contains an F-basis of U, we have

$$FL = U.$$

Therefore $x_1, \dots, x_m$ also span U over F. Hence

$$x_1, \dots, x_m$$

form an F-basis of U. Thus

$$F \otimes_R L \cong U,$$

and L is a full R-lattice in U.

If U is an FG-module and L is G-stable, then L is an RG-submodule of U. Hence L is a full RG-lattice in U. $\square$

Proposition 13.5.4 (Uniqueness of the lift of an indecomposable projective)**.** *Let R be a complete discrete valuation ring with maximal ideal*

$$(\pi),$$

residue field

$$k = R/(\pi),$$

and let G be a finite group.

Let S be a simple kG-module, and let

$$P_S$$

be its projective cover as a kG-module. Let

$$\widehat{P}_S$$

be the projective RG-lattice lifting P_S, so that

$$\widehat{P}_S/\pi\widehat{P}_S \cong P_S.$$

If L is an RG-lattice such that

$$L/\pi L \cong P_S$$

as kG-modules, then

$$L \cong \widehat{P}_S$$

as RG-modules.

Proof. Since

$$L/\pi L \cong P_S$$

is projective as a kG-module, the projectivity criterion for lattices implies that L is projective as an RG-module.

Consider the natural epimorphism

$$L \longrightarrow L/\pi L \cong P_S \longrightarrow S.$$

Since $\widehat{P}_S$ is projective and $\widehat{P}_S \longrightarrow S$ is the projective cover of S, this map factors through

L. Hence there is an RG-homomorphism

$$f : \widehat{P}_S \longrightarrow L$$

such that the diagram

$$\begin{tikzcd} \widehat{P}_S \arrow[r, "f"] \arrow[dr] & L \arrow[d] \\ & S \end{tikzcd}$$

commutes.

We claim that f is surjective. Since

$$\pi L \subseteq \operatorname{Rad}(L),$$

and

$$\operatorname{Rad}(L)/\pi L = \operatorname{Rad}(L/\pi L),$$

we have

$$\operatorname{Top}(L) = L/\operatorname{Rad}(L) \cong (L/\pi L)/\operatorname{Rad}(L/\pi L) \cong P_S/\operatorname{Rad}(P_S) \cong S.$$

Similarly,

$$\operatorname{Top}(\widehat{P}_S) \cong S.$$

The morphism f induces a nonzero map

$$\operatorname{Top}(\widehat{P}_S) \longrightarrow \operatorname{Top}(L),$$

hence an isomorphism, since both sides are isomorphic to the simple module S. Therefore

$$f(\widehat{P}_S) + \operatorname{Rad}(L) = L.$$

By Nakayama's lemma,

$$f(\widehat{P}_S) = L.$$

Thus f is surjective.

It remains to show that f is injective. Since $\widehat{P}_S$ and L are R-free lattices, we have

$$\operatorname{rank}_R \widehat{P}_S = \dim_k(\widehat{P}_S/\pi\widehat{P}_S) = \dim_k P_S,$$

and

$$\operatorname{rank}_R L = \dim_k(L/\pi L) = \dim_k P_S.$$

Hence

$$\operatorname{rank}_R \widehat{P}_S = \operatorname{rank}_R L.$$

Since $f : \widehat{P}_S \to L$ is a surjective homomorphism between finite free R-modules of the same rank, its kernel has R-rank 0. But $\ker f$ is a submodule of the free R-module $\widehat{P}_S$, hence is torsion-free. Therefore

$$\ker f = 0.$$

Thus f is an isomorphism. $\square$

Corollary 13.5.5 (Uniqueness of projective lifts)**.** *Let Q be a finitely generated projective kG-module. If L_1 and L_2 are projective RG-lattices such that*

$$L_1/\pi L_1 \cong Q, \qquad L_2/\pi L_2 \cong Q,$$

then

$$L_1 \cong L_2$$

as RG-modules.

Proof. Decompose Q into indecomposable projective kG-modules:

$$Q \cong \bigoplus_S P_S^{\oplus m_S},$$

where S runs through the isomorphism classes of simple kG-modules. By the preceding proposition, each indecomposable summand P_S has a unique projective RG-lattice lift, namely $\widehat{P}_S$. Hence any projective RG-lattice lifting Q is isomorphic to

$$\bigoplus_S \widehat{P}_S^{\oplus m_S}.$$

Therefore

$$L_1 \cong L_2.$$

$\square$

Corollary 13.5.6 (Existence of full lattices)**.** *Let R be a principal ideal domain with fraction field F, and let G be a finite group. If U is a finite-dimensional FG-module, then U contains a full RG-lattice.*

Proof. Let

$$u_1, \ldots, u_n$$

be an F-basis of U. Define

$$L = \sum_{i=1}^{n} \sum_{g \in G} R\, g u_i.$$

Since G is finite, L is finitely generated as an R-module. It is also stable under the action of G, hence is an RG-submodule of U.

Moreover, L contains the F-basis $u_1, \ldots, u_n$, because $1 \in G$. Therefore, by the previous lemma, L is a full RG-lattice in U. $\square$

Chapter 14

Lecture 14

14.1 Brauer–Nesbitt and Decomposition Numbers

Recall a p-modular system (F, R, k), where R is a complete discrete valuation ring with uniformizer π, fraction field F, and residue field $k = R/(\pi)$.

Theorem 14.1.1 (Brauer–Nesbitt)**.** *Let (F, R, k) be a p-modular system, and let G be a finite group. Let U be a finitely generated FG-module. Let L_1, L_2 be full RG-lattices in U. Then*

$$L_1/\pi L_1 \qquad \textit{and} \qquad L_2/\pi L_2$$

have the same composition factors, with the same multiplicities, as kG-modules.

Proof. First observe that $L_1 + L_2$ is also a full RG-lattice in U. Thus, replacing L_1, L_2 if necessary, we may assume

$$L_1 \subseteq L_2.$$

As R-modules, L_1 and L_2 are free of the same rank. Hence L_2/L_1 is a torsion R-module. Thus L_2/L_1 has a finite composition series as an R-module, and hence we may reduce to the case where L_1 is a maximal RG-submodule of L_2.

Then L_2/L_1 is simple. Since L_2/L_1 is torsion over the DVR R, we have

$$\pi L_2 \subseteq L_1.$$

Thus

$$\pi L_1 \subseteq \pi L_2 \subseteq L_1 \subseteq L_2.$$

It remains to show that

$$L_2/L_1 \cong \pi L_2/\pi L_1.$$

Indeed, define

$$\varphi : L_2 \longrightarrow \pi L_2/\pi L_1, \qquad x \longmapsto \pi x + \pi L_1.$$

This is a surjective RG-homomorphism, and

$$\ker(\varphi) = L_1.$$

Therefore

$$L_2/L_1 \cong \pi L_2/\pi L_1.$$

Using the chains

$$\pi L_1 \subseteq \pi L_2 \subseteq L_1$$

and

$$\pi L_2 \subseteq L_1 \subseteq L_2,$$

the above isomorphism shows that $L_1/\pi L_1$ and $L_2/\pi L_2$ have the same composition factors with the same multiplicities. $\square$

Definition 14.1.2 (Decomposition matrix)**.** *Assume (F, R, k) is splitting for G. The decomposition matrix $D = (d_{TS})$ is the matrix whose rows are indexed by the simple FG-modules T, and whose columns are indexed by the simple kG-modules S. Its entries are*

$$d_{TS} = [T_0/\pi T_0 : S],$$

where T_0 is a full RG-lattice in T, and S is a simple kG-module.

The integer d_{TS} is called a decomposition number.

14.2 Grothendieck Groups and the Cartan Triangle

Assume (F, R, k) is splitting for G, and assume R is complete.

Define $G_0(FG)$ to be the free abelian group with basis the isomorphism classes of simple FG-modules. Similarly, define $G_0(kG)$ to be the free abelian group with basis the isomorphism classes of simple kG-modules.

Define $K_0(kG)$ to be the free abelian group with basis the isomorphism classes of indecomposable projective kG-modules.

If T is a simple FG-module, write $[T]$ for the corresponding basis element of $G_0(FG)$. If S is a simple kG-module, write $[S]$ for the corresponding basis element of $G_0(kG)$. If P is an indecomposable projective kG-module, write $[P]$ for the corresponding basis element of $K_0(kG)$.

If U is a kG-module with composition factors

$$S_1, \ldots, S_r$$

with multiplicities

$$n_1, \ldots, n_r,$$

we write

$$[U] = n_1[S_1] + \cdots + n_r[S_r] \in G_0(kG).$$

Similarly, for an FG-module V, if

$$V \cong T_1^{n_1} \oplus \cdots \oplus T_s^{n_s},$$

where the T_i are simple FG-modules, we write

$$[V] = n_1[T_1] + \cdots + n_s[T_s] \in G_0(FG).$$

If P is a projective kG-module and

$$P \cong P_1^{n_1} \oplus \cdots \oplus P_s^{n_s},$$

where the P_i are indecomposable projective kG-modules, then

$$[P] = n_1[P_1] + \cdots + n_s[P_s] \in K_0(kG).$$

Consider the following Cartan triangle:

$$\begin{tikzcd} & G_0(FG) \arrow[dr, "d"] & \\ K_0(kG) \arrow[ur, "e"] \arrow[rr, "c"'] & & G_0(kG). \end{tikzcd}$$

If P_S is an indecomposable projective kG-module, then

$$e([P_S]) = \left[F \otimes_R \widehat{P}_S\right] \in G_0(FG),$$

where $\widehat{P}_S$ is an RG-lattice lifting P_S. Recall that $\widehat{P}_S$ is unique up to isomorphism.

If V is a simple FG-module containing a full RG-lattice V_0, define

$$d([V]) = [V_0/\pi V_0] \in G_0(kG).$$

With respect to the bases above, the matrix of d is the transpose of the decomposition matrix D.

Let P_S be the indecomposable projective kG-module which is the projective cover of a simple kG-module S. Define

$$c([P_S]) = [P_S].$$

Thus $c : K_0(kG) \to G_0(kG)$ sends a projective module to its class as a kG-module. If

$$[P_T] = \sum_S C_{ST}[S],$$

where S, T run through simple kG-modules, then the matrix of c is the Cartan matrix

$$C = (C_{ST}).$$

Theorem 14.2.1. *With the notation above,*

$$c = d \circ e.$$

Proof. Let P_S be an indecomposable projective kG-module. Then

$$c([P_S]) = [P_S],$$

while

$$e([P_S]) = \left[F \otimes_R \widehat{P}_S\right].$$

Choose the full RG-lattice $\widehat{P}_S$ in $F \otimes_R \widehat{P}_S$. Reducing modulo π, we get

$$d\left(\left[F \otimes_R \widehat{P}_S\right]\right) = [P_S].$$

Hence

$$(d \circ e)([P_S]) = [P_S] = c([P_S]).$$

Since the classes $[P_S]$ form a basis of $K_0(kG)$, it follows that

$$c = d \circ e.$$

□

14.3 Hom-Lattices

Proposition 14.3.1. *Let (F, R, k) be a p-modular system. Let U, V be FG-modules containing full RG-lattices U_0, V_0, respectively.*

1. *The R-module*

$$\mathrm{Hom}_{RG}(U_0, V_0)$$

is a full RG-lattice in

$$\mathrm{Hom}_{FG}(U, V).$$

2. *If U_0 is a projective RG-lattice, then*

$$\mathrm{Hom}_{RG}(U_0, V_0)/\pi\mathrm{Hom}_{RG}(U_0, V_0) \cong \mathrm{Hom}_{RG}(U_0, V_0/\pi V_0)$$

and

$$\mathrm{Hom}_{RG}(U_0, V_0/\pi V_0) \cong \mathrm{Hom}_{kG}(U_0/\pi U_0, V_0/\pi V_0).$$

Proof. First, $\mathrm{Hom}_{RG}(U_0, V_0)$ may be viewed as a subset of $\mathrm{Hom}_{FG}(U, V)$. Choose R-bases

$$u_1, \ldots, u_r \qquad \text{and} \qquad v_1, \ldots, v_s$$

of U_0 and V_0, respectively. These are also F-bases of U and V. Any RG-homomorphism

$U_0 \to V_0$ is represented by a matrix over R, and therefore determines an FG-homomorphism $U \to V$.

We have

$$\mathrm{Hom}_{RG}(U_0, V_0) \subseteq \mathrm{Hom}_R(U_0, V_0) \cong R^{rs}.$$

Since R is a PID, this submodule is R-free.

It remains to show fullness. Let

$$\phi : U \to V$$

be an FG-homomorphism. Write

$$\phi(u_i) = \sum_j \lambda_{ji} v_j, \qquad \lambda_{ji} \in F.$$

Choose $a \in R \setminus \{0\}$ such that

$$a\lambda_{ji} \in R$$

for all i, j. Then

$$a\phi : U_0 \to V_0$$

is an RG-homomorphism. Therefore

$$\mathrm{Hom}_{RG}(U_0, V_0)$$

spans $\mathrm{Hom}_{FG}(U, V)$ over F, and hence is a full lattice.

Now assume U_0 is projective. Then

$$\mathrm{Hom}_{RG}(U_0, \pi V_0) = \pi \mathrm{Hom}_{RG}(U_0, V_0).$$

Consider the natural map

$$\mathrm{Hom}_{RG}(U_0, V_0) \longrightarrow \mathrm{Hom}_{RG}(U_0, V_0/\pi V_0).$$

Since U_0 is projective, this map is surjective. Its kernel is

$$\mathrm{Hom}_{RG}(U_0, \pi V_0) = \pi \mathrm{Hom}_{RG}(U_0, V_0).$$

Hence

$$\mathrm{Hom}_{RG}(U_0, V_0)/\pi \mathrm{Hom}_{RG}(U_0, V_0) \cong \mathrm{Hom}_{RG}(U_0, V_0/\pi V_0).$$

For the second isomorphism, every morphism

$$U_0 \to V_0/\pi V_0$$

contains πU_0 in its kernel, and therefore factors uniquely through

$$U_0/\pi U_0.$$

Thus
$$\operatorname{Hom}_{RG}(U_0, V_0/\pi V_0) \cong \operatorname{Hom}_{kG}(U_0/\pi U_0, V_0/\pi V_0).$$

□

Corollary 14.3.2. *Suppose U_0, V_0 are full RG-lattices in FG-modules U, V, respectively. Assume U_0 is projective. Then*

$$\dim_F \operatorname{Hom}_{FG}(U, V) = \dim_k \operatorname{Hom}_{kG}(U_0/\pi U_0, V_0/\pi V_0).$$

Proof. Both sides are equal to the R-rank of $\operatorname{Hom}_{RG}(U_0, V_0)$. □

14.4 Brauer Reciprocity and the Cartan Matrix

Theorem 14.4.1 (Brauer reciprocity)**.** *Assume (F, R, k) is splitting for G, and R is complete. Let S be a simple kG-module, and let T be a simple FG-module containing a full RG-lattice T_0.*

Let P_S be the projective cover of S, and let $\widehat{P}_S$ be its lift to an RG-lattice. Then the multiplicity of T in
$$F \otimes_R \widehat{P}_S$$
is equal to the multiplicity of S as a composition factor of

$$T_0/\pi T_0.$$

Equivalently,
$$\left[F \otimes_R \widehat{P}_S : T\right] = [T_0/\pi T_0 : S]\,.$$

Proof. By the preceding corollary,

$$\dim_F \operatorname{Hom}_{FG}\left(F \otimes_R \widehat{P}_S, T\right) = \dim_k \operatorname{Hom}_{kG}\left(P_S, T_0/\pi T_0\right).$$

The left-hand side equals

$$\dim_F \operatorname{End}_{FG}(T) \cdot \left[F \otimes_R \widehat{P}_S : T\right].$$

The right-hand side equals

$$\dim_k \operatorname{End}_{kG}(S) \cdot [T_0/\pi T_0 : S]\,.$$

Since (F, R, k) is splitting, we have

$$\dim_F \operatorname{End}_{FG}(T) = 1, \qquad \dim_k \operatorname{End}_{kG}(S) = 1.$$

Therefore
$$\left[F \otimes_R \widehat{P}_S : T\right] = [T_0/\pi T_0 : S]\,.$$

□

With respect to the standard bases, the matrix of e is D, and the matrix of d is D^T, where D is the decomposition matrix.

Corollary 14.4.2. *Assume (F, R, k) is splitting for G, and R is complete. Then the Cartan matrix satisfies*

$$C = D^T D.$$

In particular, C is symmetric.

Proof. Since

$$c = d \circ e,$$

and the matrices of d and e are D^T and D, respectively, the matrix of c is

$$C = D^T D.$$

□

14.5 Integral Elements

Let S be a commutative ring with 1, and let $R \subseteq S$ be a subring with $1_R = 1_S$.

Definition 14.5.1. *An element $s \in S$ is called integral over R if there exists a monic polynomial*

$$f(x) \in R[x]$$

such that

$$f(s) = 0.$$

Theorem 14.5.2. *For $s \in S$, the following are equivalent:*

1. *s is integral over R.*
2. *$R[s]$ is contained in some finitely generated R-submodule $M \subseteq S$ such that*

$$sM \subseteq M.$$

Theorem 14.5.3. *The elements of S which are integral over R form a subring of S.*

Theorem 14.5.4. *The algebraic integers in $\mathbb{Q}$ are precisely the ordinary integers:*

$$\{x \in \mathbb{Q} \mid x \text{ is integral over } \mathbb{Z}\} = \mathbb{Z}.$$

Theorem 14.5.5. *Let G be a finite group, let $g \in G$, and let χ be a complex character* *of G. Then*

$$\chi(g)$$

is an algebraic integer.

Remark 14.5.6. *Recall that an element of $\mathbb{C}$ is called an algebraic integer if it is integral over $\mathbb{Z}$.*

14.6 Central Elements in Group Rings

Consider the group rings

$$\mathbb{C}G \qquad \text{and} \qquad \mathbb{Z}G.$$

We identify

$$\mathbb{Z} \cdot 1_G \subseteq Z(\mathbb{Z}G).$$

Proposition 14.6.1. *The center $Z(\mathbb{Z}G)$ is integral over $\mathbb{Z}$.*

Proof. Let $C_1, \ldots, C_r$ be the conjugacy classes of G, and define the class sums

$$\overline{C}_i = \sum_{g \in C_i} g.$$

Then

$$Z(\mathbb{Z}G) = \bigoplus_{i=1}^{r} \mathbb{Z}\,\overline{C}_i.$$

Thus $Z(\mathbb{Z}G)$ is a finitely generated abelian group, hence a finitely generated $\mathbb{Z}$-module. Therefore every element of $Z(\mathbb{Z}G)$ is integral over $\mathbb{Z}$. $\square$

Corollary 14.6.2. *Let $C_1, \ldots, C_r$ be the conjugacy classes of G, and let*

$$\lambda_1, \ldots, \lambda_r$$

be algebraic integers in $\mathbb{C}$. Then

$$\sum_{i=1}^{r} \lambda_i \overline{C}_i$$

is integral over $\mathbb{Z}$.

Let

$$\rho_1, \ldots, \rho_r$$

be the simple complex representations of G, with degrees

$$d_1, \ldots, d_r,$$

and characters

$$\chi_1, \ldots, \chi_r.$$

Thus

$$\rho_i : G \to \mathrm{GL}_{d_i}(\mathbb{C}).$$

Extending linearly gives

$$\rho_i : \mathbb{C}G \to M_{d_i}(\mathbb{C}).$$

Under the Artin–Wedderburn decomposition

$$\mathbb{C}G \cong \bigoplus_{i=1}^{r} M_{d_i}(\mathbb{C}),$$

the map ρ_i is the projection onto the i-th factor.

Proposition 14.6.3. *Fix i, and let $x \in Z(\mathbb{C}G)$. Then*

$$\rho_i(x) = \lambda I_{d_i}$$

for some $\lambda \in \mathbb{C}$. In fact, if

$$x = \sum_{g \in G} a_g g,$$

then

$$\lambda = \frac{1}{d_i}\,\mathrm{tr}(\rho_i(x)) = \frac{1}{d_i}\sum_{g \in G} a_g \chi_i(g).$$

Proof. Since $x \in Z(\mathbb{C}G)$, for every $y \in \mathbb{C}G$,

$$xy = yx.$$

Therefore, for every i,

$$\rho_i(x)\rho_i(y) = \rho_i(y)\rho_i(x).$$

By Schur's lemma, $\rho_i(x)$ is a scalar matrix:

$$\rho_i(x) = \lambda I_{d_i}.$$

Taking traces gives

$$\lambda = \frac{1}{d_i}\,\mathrm{tr}(\rho_i(x)).$$

If

$$x = \sum_{g \in G} a_g g,$$

then

$$\mathrm{tr}(\rho_i(x)) = \mathrm{tr}\left(\sum_{g \in G} a_g \rho_i(g)\right) = \sum_{g \in G} a_g\,\mathrm{tr}(\rho_i(g)) = \sum_{g \in G} a_g \chi_i(g).$$

Hence

$$\lambda = \frac{1}{d_i}\sum_{g \in G} a_g \chi_i(g).$$

□

14.7 Degrees of Irreducible Complex Representations

Theorem 14.7.1. *Let G be a finite group, and let*

$$\rho_i : G \to \mathrm{GL}_{d_i}(\mathbb{C})$$

be a simple complex representation of degree d_i. Then

$$d_i \mid |G|.$$

Proof. Let χ_i be the character of ρ_i. Define

$$x = \sum_{g \in G} \overline{\chi_i(g)}\, g \in \mathbb{C}G.$$

Since

$$\overline{\chi_i(g)} = \chi_i(g^{-1}),$$

the coefficients are constant on conjugacy classes, so $x \in Z(\mathbb{C}G)$.

By the previous proposition,

$$\rho_i(x) = \frac{1}{d_i}\left(\sum_{g \in G} \overline{\chi_i(g)}\chi_i(g)\right) I_{d_i}.$$

By the orthogonality relations,

$$\sum_{g \in G} \overline{\chi_i(g)}\chi_i(g) = |G|.$$

Therefore

$$\rho_i(x) = \frac{|G|}{d_i} I_{d_i}.$$

For every $g \in G$, the number $\chi_i(g)$ is an algebraic integer. Hence

$$\overline{\chi_i(g)} = \chi_i(g^{-1})$$

is also an algebraic integer. Therefore

$$x = \sum_{C \in \mathrm{Cl}(G)} \overline{\chi_i(g_C)} \left(\sum_{g \in C} g\right)$$

is integral over $\mathbb{Z}$, where g_C is any representative of the conjugacy class C.

Thus $\rho_i(x)$ is integral over $\mathbb{Z}$. Hence the scalar

$$\frac{|G|}{d_i}$$

is integral over $\mathbb{Z}$. But

$$\frac{|G|}{d_i} \in \mathbb{Q}.$$

Since the only rational algebraic integers are the ordinary integers, we get

$$\frac{|G|}{d_i} \in \mathbb{Z}.$$

Therefore

$$d_i \mid |G|.$$

□

14.8 Character Table Examples

We recall several standard facts about irreducible characters.

- Linear characters of G are the irreducible characters of the abelianization

$$G/[G,G].$$

- The number of irreducible characters equals the number of conjugacy classes:

$$|\operatorname{Irr}(G)| = |\operatorname{Cl}(G)|.$$

- The sum of the squares of the degrees is the order of the group:

$$|G| = \sum_{\chi \in \operatorname{Irr}(G)} \chi(1)^2.$$

- For every $\chi \in \operatorname{Irr}(G)$,

$$\chi(1) \mid |G|.$$

- The irreducible characters satisfy the usual orthogonality relations.

Example 14.8.1 (The character table of S_3)**.** *For S_3, we have*

$$S_3/[S_3,S_3] = S_3/A_3 \cong C_2.$$

Thus S_3 has two linear characters. Since

$$|S_3| = 6$$

and the irreducible degrees satisfy

$$6 = \sum_{\chi \in \operatorname{Irr}(S_3)} \chi(1)^2,$$

the remaining irreducible character has degree 2.

The conjugacy classes are

$$(1), \qquad (12), \qquad (123).$$

The character table is

	(1)	(12)	(123)
1	1	1	1
ε	1	-1	1
χ	2	0	-1

Example 14.8.2 (The character table of D_8)**.** *Let*

$$D_8 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle$$

be the dihedral group of order 8.

One standard ordering of the conjugacy classes is

$$C_1 = \{1\}, \qquad C_2 = \{s, r^2 s\}, \qquad C_3 = \{rs, r^3 s\},$$

$$C_4 = \{r, r^3\}, \qquad C_5 = \{r^2\}.$$

The character table is

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	-1	1	-1	1
χ_3	1	1	-1	-1	1
χ_4	1	-1	-1	1	1
χ_5	2	0	0	0	-2

Chapter 15

Homework

15.1 Homework Week 1

Exercise 15.1.1. *Let R be a ring, and let M be an R-module. Suppose that S is a ring and*

$$\theta : S \to R$$

is a ring homomorphism satisfying $\theta(1_S) = 1_R$. Let S act on M by

$$sx = \theta(s)x,$$

where $x \in M$ and $s \in S$. Prove that M is an S-module.

Proof. This action is well-defined because $\forall s \in S$, $\theta(s) \in R$ and $\theta(s)x \in M$ for all $x \in M$. We need to verify the module axioms, for all $s, t \in S$ and $x, y \in M$:

- $s(x + y) = \theta(s)(x + y) = \theta(s)x + \theta(s)y = sx + sy$
- $(s + t)x = \theta(s + t)x = \theta(s)x + \theta(t)x = sx + tx$
- $(st)x = \theta(st)x = \theta(s)\theta(t)x = \theta(s)(\theta(t)x) = s(tx)$.
- $1_S x = \theta(1_S)x = 1_R x = x$

Hence M is an S-module. $\square$

Exercise 15.1.2. *Let R be a ring, and let M and N be two right R-modules. Let* $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ *be the set of group homomorphisms from M to N (as abelian groups). The action of R on* $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ *is given as follows: for any $a \in R$ and $\varphi \in$* $\mathrm{Hom}_{\mathbb{Z}}(M, N)$*, define*

$$(\varphi a)(x) = \varphi(xa), \qquad \forall x \in M.$$

Prove that $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ *is a left R-module.*

Proof. This action is well defined: for every $r \in R$ and every $\varphi \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$, the map

$$\varphi r \colon M \to N, \qquad (\varphi r)(x) = \varphi(xr),$$

is again a group homomorphism.

We now verify the module axioms. Let $r_1, r_2 \in R$, let $\varphi, \psi \in \operatorname{Hom}_{\mathbb{Z}}(M, N)$, and let $x \in M$. Then

$$\begin{aligned}
((\varphi+\psi)r_1)(x) &= (\varphi+\psi)(xr_1) \\
&= \varphi(xr_1) + \psi(xr_1) \\
&= (\varphi r_1)(x) + (\psi r_1)(x), \\
(\varphi(r_1+r_2))(x) &= \varphi(x(r_1+r_2)) \\
&= \varphi(xr_1 + xr_2) \\
&= \varphi(xr_1) + \varphi(xr_2) \\
&= (\varphi r_1)(x) + (\varphi r_2)(x), \\
((\varphi r_1)r_2)(x) &= (\varphi r_1)(xr_2) \\
&= \varphi((xr_2)r_1) \\
&= \varphi(x(r_2 r_1)) \\
&= (\varphi(r_2 r_1))(x), \\
(\varphi 1_R)(x) &= \varphi(x 1_R) \\
&= \varphi(x).
\end{aligned}$$

Hence $\operatorname{Hom}_{\mathbb{Z}}(M, N)$ is a left R-module. □

Exercise 15.1.3. *Let R be a ring, and let M be an R-module. Denote*

$$\operatorname{Ann}_R M = \{r \in R \mid rm = 0,\ \forall m \in M\}.$$

We call M a faithful R-module if $\operatorname{Ann}_R M = \{0\}$. Define the action of the quotient ring $R/\operatorname{Ann}_R M$ on M by

$$\overline{r}\,x = rx,$$

where $x \in M$, $r \in R$, and $\overline{r} = r + \operatorname{Ann}_R M$. Prove that M is a faithful $R/\operatorname{Ann}_R M$-module.

Proof. First we show that the action is well-defined. Suppose

$$\bar{r} = \bar{r}' \in R/\operatorname{Ann}_R M.$$

Then $r - r' \in \operatorname{Ann}_R M$, so for every $x \in M$,

$$(r - r')x = 0.$$

Hence $rx = r'x$, so $\bar{r}\,x$ does not depend on the choice of representative.

The module axioms follow immediately from those of the R-module structure on M.

Now we prove faithfulness. Suppose $\bar{r} \in R/\operatorname{Ann}_R M$ acts trivially on M, that is,

$$\bar{r}\,x = 0 \qquad \forall x \in M.$$

By definition, this means

$$rx = 0 \qquad \forall x \in M.$$

Therefore $r \in \operatorname{Ann}_R M$, so $\bar{r} = \bar{0}$ in the quotient ring. Hence

$$\operatorname{Ann}_{R/\operatorname{Ann}_R M}(M) = \{\bar{0}\},$$

and M is faithful as an $R/\operatorname{Ann}_R M$-module. $\square$

Exercise 15.1.4. *Let $\varphi : M \to M$ be a homomorphism of R-modules such that*

$$\varphi\varphi = \varphi.$$

Prove that

$$M = \ker\varphi \oplus \operatorname{Im}\varphi.$$

Proof. Let $x \in M$. Then

$$x = \varphi(x) + (x - \varphi(x)),$$

where $\varphi(x) \in \operatorname{Im}\varphi$ and $x - \varphi(x) \in \ker\varphi$. Hence

$$M = \operatorname{Im}\varphi + \ker\varphi.$$

It remains to show that

$$\ker\varphi \cap \operatorname{Im}\varphi = \{0\}.$$

Suppose $x \in \ker\varphi \cap \operatorname{Im}\varphi$. Then $x = \varphi(y)$ for some $y \in M$ and $\varphi(x) = 0$. Thus

$$0 = \varphi(x) = \varphi(\varphi(y)) = \varphi(y) = x,$$

so $x = 0$. Hence $\ker\varphi \cap \operatorname{Im}\varphi = \{0\}$. $\square$

Exercise 15.1.5. *Show that any R-module is a homomorphism image of some free R-module.*

Proof. Let M be a left R-module. Consider the free R-module

$$F = \bigoplus_{x \in M} Re_x$$

with basis $\{e_x\}_{x \in M}$ indexed by the underlying set of M. Define an R-module homomorphism

$$\pi : F \to M$$

by

$$\pi(e_x) = x \qquad (x \in M),$$

and extend R-linearly. Thus

$$\pi\Big(\sum_{i=1}^{n} r_i e_{x_i}\Big) = \sum_{i=1}^{n} r_i x_i.$$

This is well-defined and R-linear.

For any $m \in M$, we have

$$\pi(e_m) = m,$$

so π is surjective. Therefore M is a homomorphic image of the free module F. Equivalently,

$$M \cong F/\ker \pi.$$

□

Exercise 15.1.6. *Let $\mathbb{Q}$ be the field of rational numbers. Is the additive group $(\mathbb{Q}, +)$ a free $\mathbb{Z}$-module?*

Proof. Claim: any two non-zero element of $\mathbb{Q}$ are linearly dependent over $\mathbb{Z}$. Hence if $\mathbb{Q}$ is a free $\mathbb{Z}$-module, then it must have a basis of cardinality 1.

Proof of claim: let $0 \neq x = \frac{a}{b}$ and $0 \neq y = \frac{c}{d}$ be two elements of $\mathbb{Q}$. Then $dcx - aby = 0$. If $a = 0$ or $c = 0$, then $x = 0$ or $y = 0$, which contradicts the assumption. Hence $a, c \neq 0$ which means dc, ab are two non-zero integer coefficients. Hence x and y are linearly dependent over $\mathbb{Z}$. And if $\mathbb{Q}$ is a free $\mathbb{Z}$-module, then it must have a basis of cardinality 1.

But if $\mathbb{Q}$ is a cyclic $\mathbb{Z}$-module, then it has only one generator. Suppose it is $z = \frac{e}{f}$. Then $\mathbb{Q} = \{n\frac{e}{f} \mid n \in \mathbb{Z}\}$. But this is not possible because we can easily pick $\frac{1}{2f} \notin \mathbb{Q}$, which is absurd.

Therefore $\mathbb{Q}$ is not a free $\mathbb{Z}$-module. □

15.2 Homework Week 2

Exercise 15.2.1. *Let R be a commutative ring, and suppose that φ is an endomorphism of the free R-module R^m. Prove that if φ is surjective, then φ is injective. Consider that: if φ is injective, is φ surjective?*

Proof. Let $\{e_1, \ldots, e_m\}$ be the standard basis of R^m, and let $A \in M_m(R)$ be the matrix of φ with respect to this basis. Since φ is surjective, for each j there exists $v_j \in R^m$ with $\varphi(v_j) = e_j$. Write $v_j = \sum_{i=1}^m b_{ij} e_i$ and let $B = (b_{ij}) \in M_m(R)$. Then $AB = I_m$, so

$$\det(A)\det(B) = \det(AB) = 1.$$

Hence $\det(A)$ is a unit in R. It follows that A is invertible (with $A^{-1} = \det(A)^{-1}\operatorname{adj}(A)$), so φ is an isomorphism. In particular φ is injective.

Counterexample for the converse. The converse is false in general. Let $R = \mathbb{Z}$ and $m = 1$, so $R^m = \mathbb{Z}$. Define $\varphi \colon \mathbb{Z} \to \mathbb{Z}$ by $\varphi(n) = 2n$. Then φ is injective but not surjective; for example, $1 \notin \operatorname{Im} \varphi$. □

Exercise 15.2.2. *Let $\varphi : M \to N$ be a homomorphism of R-modules. Prove that:*

1. *φ is a monomorphism if and only if for any R-module T and any two R-module homomorphisms $\xi, \eta : T \to M$, $\varphi\xi = \varphi\eta$ implies $\xi = \eta$;*

2. *φ is an epimorphism if and only if for any R-module S and any two R-module homomorphisms $\rho, \theta : N \to S$, $\rho\varphi = \theta\varphi$ implies $\rho = \theta$.*

Proof. 1. ($\Rightarrow$) Suppose φ is injective and $\varphi\xi = \varphi\eta$ for $\xi, \eta : T \to M$. Then for every $t \in T$,

$$\varphi(\xi(t)) = \varphi(\eta(t)),$$

so $\xi(t) - \eta(t) \in \ker\varphi = \{0\}$. Hence $\xi(t) = \eta(t)$ for all $t \in T$, and therefore $\xi = \eta$.

($\Leftarrow$) We show $\ker\varphi = \{0\}$. Let $T = \ker\varphi$, let $\xi : T \hookrightarrow M$ be the inclusion, and let $\eta = 0 : T \to M$ be the zero map. Then $\varphi\xi(t) = \varphi(t) = 0 = \varphi\eta(t)$ for all $t \in T$, so $\varphi\xi = \varphi\eta$. By hypothesis $\xi = \eta$, which means $t = \xi(t) = \eta(t) = 0$ for all $t \in T$. Hence $\ker\varphi = \{0\}$, so φ is injective.

2. ($\Rightarrow$) Suppose φ is surjective and $\rho\varphi = \theta\varphi$. For any $n \in N$, pick $m \in M$ with $\varphi(m) = n$. Then

$$\rho(n) = \rho(\varphi(m)) = \theta(\varphi(m)) = \theta(n),$$

so $\rho = \theta$.

($\Leftarrow$) We show φ is surjective. Let $S = N/\operatorname{Im}\varphi$, let $\rho : N \to S$ be the canonical projection, and let $\theta = 0 : N \to S$ be the zero map. For any $m \in M$, we have

$$\rho(\varphi(m)) = \overline{\varphi(m)} = \bar{0} = \theta(\varphi(m)),$$

so $\rho\varphi = \theta\varphi$. By hypothesis $\rho = \theta$, i.e. the projection $N \to N/\operatorname{Im}\varphi$ is the zero map. This means $N = \operatorname{Im}\varphi$, so φ is surjective.

□

Exercise 15.2.3. *Let M and M_i $(1 \leq i \leq n)$ be R-modules. Show that $M \simeq \bigoplus_{i=1}^n M_i$ if and only if for every $i \in \{1, \dots, n\}$, there exists an $\varphi_i \in \operatorname{End}_R(M)$ such that $\operatorname{Im}\varphi_i \simeq M_i$, $\varphi_i\varphi_j = 0$ for $i \neq j$, $\varphi_i^2 = \varphi_i$, and $\varphi_1 + \varphi_2 + \cdots + \varphi_n = \mathrm{id}_M$.*

Proof. ($\rightarrow$) Suppose $f : M \xrightarrow{\sim} \bigoplus_{i=1}^n M_i$. Let $\iota_i : M_i \hookrightarrow \bigoplus_{j=1}^n M_j$ be the canonical inclusion and $\pi_i : \bigoplus_{j=1}^n M_j \twoheadrightarrow M_i$ the canonical projection. Define

$$\varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f \in \operatorname{End}_R(M).$$

Then:

- $\varphi_i^2 = f^{-1}\iota_i\pi_i f f^{-1}\iota_i\pi_i f = f^{-1}\iota_i(\pi_i\iota_i)\pi_i f = f^{-1}\iota_i\pi_i f = \varphi_i$, since $\pi_i\iota_i = \mathrm{id}_{M_i}$.
- For $i \neq j$:$\varphi_i\varphi_j = f^{-1}\iota_i(\pi_i\iota_j)\pi_j f = 0$, since $\pi_i\iota_j = 0$ when $i \neq j$.
- $\sum_{i=1}^n \varphi_i = f^{-1}\left(\sum_{i=1}^n \iota_i\pi_i\right) f = f^{-1} \circ \mathrm{id} \circ f = \mathrm{id}_M$, since $\sum_i \iota_i\pi_i = \mathrm{id}_{\bigoplus M_j}$.
- $\operatorname{Im}\varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f(M) \simeq M_i$.

($\Leftarrow$) Suppose such φ_i exist. Define $g : M \to \bigoplus_{i=1}^n \operatorname{Im}\varphi_i$ by $g(x) = (\varphi_1(x), \dots, \varphi_n(x))$.

Surjective: for any $(y_1, \ldots, y_n) \in \bigoplus \operatorname{Im} \varphi_i$, let $x = y_1 + \cdots + y_n \in M$. Then

$$\varphi_j(x) = \sum_{i=1}^{n} \varphi_j(y_i) = \sum_{i=1}^{n} \varphi_j(\varphi_i(x_i)) = \varphi_j(\varphi_j(x_j)) = \varphi_j^2(x_j) = \varphi_j(x_j) = y_j,$$

where we used $\varphi_j \varphi_i = 0$ for $j \neq i$ and $\varphi_j^2 = \varphi_j$. Hence $g(x) = (y_1, \ldots, y_n)$.

Injective: if $g(x) = 0$, then $\varphi_i(x) = 0$ for all i, so $x = \mathrm{id}_M(x) = \sum_i \varphi_i(x) = 0$.

Since $\operatorname{Im} \varphi_i \simeq M_i$, we conclude $M \simeq \bigoplus_{i=1}^n M_i$. □

Exercise 15.2.4.

1. *Prove* $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/(m,n)\mathbb{Z}$, *where* $m, n \in \mathbb{Z}$.
2. *Let* A *be an abelian group, prove* $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.
3. *Let* A, B *be two finitely generated abelian groups. What is* $A \otimes_{\mathbb{Z}} B$*?*

Proof.

1. We use the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$. Tensoring with $\mathbb{Z}/m\mathbb{Z}$ over $\mathbb{Z}$ gives the right-exact sequence

$$\mathbb{Z}/m\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \to 0.$$

   Hence $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq (\mathbb{Z}/m\mathbb{Z})/n(\mathbb{Z}/m\mathbb{Z})$. Now $n(\mathbb{Z}/m\mathbb{Z}) = (n\mathbb{Z} + m\mathbb{Z})/m\mathbb{Z}$. Since $n\mathbb{Z} + m\mathbb{Z} = (m,n)\mathbb{Z}$, we get

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \frac{\mathbb{Z}/m\mathbb{Z}}{(m,n)\mathbb{Z}/m\mathbb{Z}} \simeq \mathbb{Z}/(m,n)\mathbb{Z}.$$

2. Similarly, tensor the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$ with A:

$$A \xrightarrow{\cdot n} A \to A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \to 0.$$

   By right-exactness, $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.

3. By the fundamental theorem of finitely generated abelian groups,

$$A \simeq \mathbb{Z}^r \oplus \bigoplus_{i=1}^{s} \mathbb{Z}/d_i\mathbb{Z}, \qquad B \simeq \mathbb{Z}^t \oplus \bigoplus_{j=1}^{u} \mathbb{Z}/e_j\mathbb{Z}.$$

   Since tensor product distributes over direct sums, using (1), (2), and $\mathbb{Z} \otimes_{\mathbb{Z}} M \simeq M$, we get

$$A \otimes_{\mathbb{Z}} B \simeq \mathbb{Z}^{rt} \oplus \bigoplus_{j=1}^{u} (\mathbb{Z}/e_j\mathbb{Z})^r \oplus \bigoplus_{i=1}^{s} (\mathbb{Z}/d_i\mathbb{Z})^t \oplus \bigoplus_{i=1}^{s} \bigoplus_{j=1}^{u} \mathbb{Z}/(d_i, e_j)\mathbb{Z}.$$

□

Exercise 15.2.5. *Let $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ be the unique monomorphism. Prove that*

$$\mathrm{id} \otimes \varphi : \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$$

is a zero homomorphism.

Proof. The unique monomorphism $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ is given by $\varphi(\bar{1}) = \bar{2}$. By the previous exercise, $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$.

It suffices to check on the generator $\bar{1} \otimes \bar{1}$ of $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$:

$$(\mathrm{id} \otimes \varphi)(\bar{1} \otimes \bar{1}) = \bar{1} \otimes \varphi(\bar{1}) = \bar{1} \otimes \bar{2} = \bar{1} \otimes 2 \cdot \bar{1} = 2 \cdot (\bar{1} \otimes \bar{1}) = (2 \cdot \bar{1}) \otimes \bar{1} = \bar{0} \otimes \bar{1} = 0,$$

where we used $2 \cdot \bar{1} = \bar{0}$ in $\mathbb{Z}/2\mathbb{Z}$. Since the generator maps to 0, the map $\mathrm{id} \otimes \varphi$ is the zero homomorphism.

This also shows that tensoring a monomorphism need not yield a monomorphism, i.e. the functor $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} -$ is not left-exact. ☐

Exercise 15.2.6. *Let V_1 and V_2 be vector spaces over a field F, with $\dim_F V_1 = n$ and $\dim_F V_2 = m$. Let $\{\epsilon_1, \dots, \epsilon_n\}$ and $\{\eta_1, \dots, \eta_m\}$ be bases of V_1 and V_2, respectively. Suppose that $\mathscr{A} \colon V_1 \to V_1$ and $\mathscr{B} \colon V_2 \to V_2$ are F-linear maps, represented by the matrices $A = (a_{ij})_{n\times n}$ and $B = (b_{ij})_{m\times m}$ with respect to these bases. Prove that*

$$\{\epsilon_i \otimes \eta_j \mid 1 \le i \le n,\, 1 \le j \le m\}$$

is an F-basis of $V_1 \otimes_F V_2$. Determine the matrix representing the F-linear map $\mathscr{A} \otimes \mathscr{B}$ under this basis.

Proof. **Step 1.** The set $\{\epsilon_i \otimes \eta_j\}$ is a basis.

Spanning. Every element of $V_1 \otimes_F V_2$ is a finite sum of simple tensors $v \otimes w$. Writing $v = \sum_i a_i \epsilon_i$ and $w = \sum_j b_j \eta_j$, by bilinearity of the tensor product we get

$$v \otimes w = \sum_{i,j} a_i b_j \, (\epsilon_i \otimes \eta_j).$$

Hence $\{\epsilon_i \otimes \eta_j\}$ spans $V_1 \otimes_F V_2$.

Linear independence. Suppose $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$. For each pair (p, q), let $f_p \in V_1^*$ and $g_q \in V_2^*$ be the dual basis functionals:$f_p(\epsilon_i) = \delta_{pi}$ and $g_q(\eta_j) = \delta_{qj}$. The bilinear map $(v, w) \mapsto f_p(v)g_q(w)$ from $V_1 \times V_2$ to F induces (by the universal property of the tensor product) an F-linear map $\Phi_{pq} : V_1 \otimes_F V_2 \to F$ with $\Phi_{pq}(\epsilon_i \otimes \eta_j) = \delta_{pi}\delta_{qj}$. Applying Φ_{pq} to the relation $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$ yields $c_{pq} = 0$. Since (p, q) was arbitrary, all $c_{ij} = 0$.

Therefore $\{\epsilon_i \otimes \eta_j : 1 \le i \le n,\, 1 \le j \le m\}$ is an F-basis of $V_1 \otimes_F V_2$, and $\dim_F(V_1 \otimes_F V_2) = nm$.

Step 2: matrix of $\mathscr{A} \otimes \mathscr{B}$.

Order the basis lexicographically:$\epsilon_1 \otimes \eta_1,\ \epsilon_1 \otimes \eta_2,\ \dots,\ \epsilon_1 \otimes \eta_m,\ \epsilon_2 \otimes \eta_1,\ \dots,\ \epsilon_n \otimes \eta_m$. Since

$\mathscr{A}(\epsilon_i) = \sum_{k=1}^n a_{ki}\epsilon_k$ and $\mathscr{B}(\eta_j) = \sum_{l=1}^m b_{lj}\eta_l$, we have

$$(\mathscr{A} \otimes \mathscr{B})(\epsilon_i \otimes \eta_j) = \mathscr{A}(\epsilon_i) \otimes \mathscr{B}(\eta_j) = \sum_{k=1}^n \sum_{l=1}^m a_{ki}b_{lj} \, (\epsilon_k \otimes \eta_l).$$

The entry in row (k, l) and column (i, j) of the representing matrix is $a_{ki}b_{lj}$. Written as an $nm \times nm$ block matrix, this is the **Kronecker product**

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{pmatrix}.$$

□

15.3 Homework Week 3

Exercise 15.3.1. *Let M, N, $(M_i)_{i\in I}$, and $(N_j)_{j\in J}$ be R-modules. Prove the following isomorphisms of abelian groups:*

1.

$$\mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i,\, N\right) \cong \prod_{i\in I} \mathrm{Hom}_R(M_i, N).$$

2.

$$\mathrm{Hom}_R\left(M,\, \prod_{j\in J} N_j\right) \cong \prod_{j\in J} \mathrm{Hom}_R(M, N_j).$$

Proof. We prove the two statements separately.

(1) Define

$$\Phi : \mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i,\, N\right) \longrightarrow \prod_{i\in I} \mathrm{Hom}_R(M_i, N)$$

by

$$\Phi(f) = (f \circ \iota_i)_{i\in I},$$

where

$$\iota_i : M_i \to \bigoplus_{i\in I} M_i$$

is the canonical inclusion.

We show that Φ is an isomorphism.

First, Φ is a group homomorphism, since composition is compatible with addition:

$$\Phi(f + g) = \big((f + g) \circ \iota_i\big)_{i\in I} = \big(f \circ \iota_i + g \circ \iota_i\big)_{i\in I} = \Phi(f) + \Phi(g).$$

Now let $(f_i)_{i\in I} \in \prod_{i\in I} \mathrm{Hom}_R(M_i, N)$. Define

$$f : \bigoplus_{i\in I} M_i \to N$$

by

$$f\big((m_i)_{i\in I}\big) = \sum_{i\in I} f_i(m_i).$$

This is well-defined, because an element of $\bigoplus_{i\in I} M_i$ has only finitely many nonzero components, so the above sum is finite. It is immediate that f is R-linear, and

$$f \circ \iota_i = f_i \qquad \text{for all } i \in I.$$

Hence $\Phi(f) = (f_i)_{i\in I}$, so Φ is surjective.

To prove injectivity, suppose $\Phi(f) = \Phi(g)$. Then

$$f \circ \iota_i = g \circ \iota_i \qquad \text{for all } i \in I.$$

Take any $x = (m_i)_{i\in I} \in \bigoplus_{i\in I} M_i$. Since x has finite support, we may write

$$x = \sum_{i\in I} \iota_i(m_i),$$

with only finitely many nonzero terms. Therefore,

$$f(x) = \sum_{i\in I} f(\iota_i(m_i)) = \sum_{i\in I} g(\iota_i(m_i)) = g(x).$$

So $f = g$, and Φ is injective.

Thus Φ is an isomorphism:

$$\mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i,\, N\right) \cong \prod_{i\in I} \mathrm{Hom}_R(M_i, N).$$

(2) Let

$$\pi_j : \prod_{j\in J} N_j \to N_j$$

be the canonical projection. Define

$$\Psi : \mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) \longrightarrow \prod_{j\in J} \mathrm{Hom}_R(M, N_j)$$

by

$$\Psi(f) = (\pi_j \circ f)_{j\in J}.$$

Again, Ψ is a group homomorphism.

We show that Ψ is an isomorphism. Given a family

$$(g_j)_{j\in J} \in \prod_{j\in J} \mathrm{Hom}_R(M, N_j),$$

define

$$g : M \to \prod_{j\in J} N_j$$

by

$$g(m) = (g_j(m))_{j\in J}.$$

Since each g_j is R-linear, g is also R-linear. Moreover,

$$\pi_j \circ g = g_j \qquad \text{for all } j \in J,$$

so

$$\Psi(g) = (g_j)_{j\in J}.$$

Hence Ψ is surjective.

For injectivity, suppose $\Psi(f) = \Psi(g)$. Then

$$\pi_j \circ f = \pi_j \circ g \qquad \text{for all } j \in J.$$

Thus for every $m \in M$ and every $j \in J$,

$$\pi_j(f(m)) = \pi_j(g(m)).$$

Since two elements of $\prod_{j\in J} N_j$ are equal if and only if all their components are equal, it follows that

$$f(m) = g(m) \qquad \text{for all } m \in M.$$

Hence $f = g$, and Ψ is injective.

Therefore

$$\mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) \cong \prod_{j\in J} \mathrm{Hom}_R(M, N_j).$$

This completes the proof. $\square$

Exercise 15.3.2 (Five Lemma)**.** *Suppose that the following diagram of homomorphisms of R-modules*

$$\begin{tikzcd}
M_1 \arrow[r] \arrow[d, "\varphi_1"'] & M_2 \arrow[r] \arrow[d, "\varphi_2"'] & M_3 \arrow[r] \arrow[d, "\varphi_3"'] & M_4 \arrow[r] \arrow[d, "\varphi_4"'] & M_5 \arrow[d, "\varphi_5"'] \\
N_1 \arrow[r] & N_2 \arrow[r] & N_3 \arrow[r] & N_4 \arrow[r] & N_5
\end{tikzcd}$$

is commutative and has exact rows. Prove that:

1. *If φ_1 is surjective, and φ_2 and φ_4 are injective, then φ_3 is injective.*
2. *If φ_5 is injective, and φ_2 and φ_4 are surjective, then φ_3 is surjective.*

Proof. We prove the two assertions separately.

(1) Assume that φ_1 is surjective and that φ_2, φ_4 are injective. We show that φ_3 is injective.

Let $x \in M_3$ and suppose that

$$\varphi_3(x) = 0.$$

We must prove that $x = 0$.

Let $\alpha_3 : M_3 \to M_4$ and $\beta_3 : N_3 \to N_4$ denote the third horizontal maps. By commutativity,

$$\varphi_4(\alpha_3(x)) = \beta_3(\varphi_3(x)) = \beta_3(0) = 0.$$

Since φ_4 is injective, it follows that $\alpha_3(x) = 0$. Exactness of the top row at M_3 yields

$$x = \alpha_2(y)$$

for some $y \in M_2$, where $\alpha_2 : M_2 \to M_3$ is the second horizontal map.

Again by commutativity,

$$0 = \varphi_3(x) = \varphi_3(\alpha_2(y)) = \beta_2(\varphi_2(y)),$$

where $\beta_2 : N_2 \to N_3$ is the second horizontal map. Hence $\varphi_2(y) \in \ker(\beta_2)$. By exactness of the bottom row at N_2,

$$\ker(\beta_2) = \mathrm{Im}(\beta_1),$$

so there exists $z \in N_1$ such that

$$\beta_1(z) = \varphi_2(y),$$

where $\beta_1 : N_1 \to N_2$ is the first horizontal map.

Since φ_1 is surjective, there exists $w \in M_1$ such that

$$\varphi_1(w) = z.$$

Using commutativity once more,

$$\varphi_2(\alpha_1(w)) = \beta_1(\varphi_1(w)) = \beta_1(z) = \varphi_2(y),$$

where $\alpha_1 : M_1 \to M_2$ is the first horizontal map. Because φ_2 is injective, we get

$$\alpha_1(w) = y.$$

Therefore

$$x = \alpha_2(y) = \alpha_2(\alpha_1(w)) = 0,$$

since the composition of two consecutive maps in an exact sequence is zero. Thus $\ker(\varphi_3) = 0$, so φ_3 is injective.

(2) Assume that φ_5 is injective and that φ_2, φ_4 are surjective. We show that φ_3 is surjective.

Let $n \in N_3$. We must find $m \in M_3$ such that

$$\varphi_3(m) = n.$$

Let $\beta_3 : N_3 \to N_4$ and $\alpha_3 : M_3 \to M_4$ denote the third horizontal maps. Since φ_4 is surjective, there exists $u \in M_4$ such that

$$\varphi_4(u) = \beta_3(n).$$

Let $\alpha_4 : M_4 \to M_5$ and $\beta_4 : N_4 \to N_5$ be the next horizontal maps. Then

$$\varphi_5(\alpha_4(u)) = \beta_4(\varphi_4(u)) = \beta_4(\beta_3(n)) = 0.$$

Because φ_5 is injective, it follows that $\alpha_4(u) = 0$. Exactness of the top row at M_4 shows that there exists $m \in M_3$ such that

$$\alpha_3(m) = u.$$

Consider

$$n - \varphi_3(m) \in N_3.$$

By commutativity,

$$\beta_3\big(n - \varphi_3(m)\big) = \beta_3(n) - \beta_3(\varphi_3(m)) = \beta_3(n) - \varphi_4(\alpha_3(m)) = \beta_3(n) - \varphi_4(u) = 0.$$

Hence $n - \varphi_3(m) \in \ker(\beta_3)$. By exactness of the bottom row at N_3,

$$\ker(\beta_3) = \operatorname{Im}(\beta_2),$$

so there exists $v \in N_2$ such that

$$\beta_2(v) = n - \varphi_3(m),$$

where $\beta_2 : N_2 \to N_3$ is the second horizontal map.

Since φ_2 is surjective, there exists $y \in M_2$ such that

$$\varphi_2(y) = v.$$

Then

$$\varphi_3(\alpha_2(y)) = \beta_2(\varphi_2(y)) = \beta_2(v) = n - \varphi_3(m),$$

where $\alpha_2 : M_2 \to M_3$ is the second horizontal map. Therefore

$$n = \varphi_3(m) + \varphi_3(\alpha_2(y)) = \varphi_3\big(m + \alpha_2(y)\big).$$

Thus n lies in the image of φ_3, proving that φ_3 is surjective. □

Exercise 15.3.3 (Snake Lemma)**.** *Suppose that the following diagram of homomorphisms* *of R-modules*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \longrightarrow & 0 \\ & & \Big\downarrow{\scriptstyle \varphi'} & & \Big\downarrow{\scriptstyle \varphi} & & \Big\downarrow{\scriptstyle \varphi''} & & \\ 0 & \longrightarrow & N' & \xrightarrow{\mu} & N & \xrightarrow{\nu} & N'' & \longrightarrow & 0 \end{array}$$

is commutative and has exact rows. Prove that

$$0 \longrightarrow \ker\varphi' \xrightarrow{\alpha'} \ker\varphi \xrightarrow{\beta'} \ker\varphi'' \xrightarrow{\delta} \operatorname{coker}\varphi' \xrightarrow{\mu'} \operatorname{coker}\varphi \xrightarrow{\nu'} \operatorname{coker}\varphi'' \longrightarrow 0$$

is a long exact sequence, where $\alpha', \beta', \mu', \nu'$ are the homomorphisms induced by α, β, μ, ν, respectively.

Proof. Since the rows are exact, α and μ are injective, while β and ν are surjective.

We first define the connecting homomorphism

$$\delta : \ker\varphi'' \to \operatorname{coker}\varphi'.$$

Let $x \in \ker\varphi'' \subseteq M''$. Since $\beta : M \to M''$ is surjective, choose $y \in M$ such that

$$\beta(y) = x.$$

By commutativity,

$$\nu(\varphi(y)) = \varphi''(\beta(y)) = \varphi''(x) = 0.$$

Hence $\varphi(y) \in \ker\nu = \operatorname{Im}\mu$. Therefore there exists $z \in N'$ such that

$$\mu(z) = \varphi(y).$$

Define

$$\delta(x) = \overline{z} = z + \operatorname{Im}\varphi' \in \operatorname{coker}\varphi'.$$

We must check that this is well-defined. First, suppose $y_1, y_2 \in M$ both satisfy

$$\beta(y_1) = \beta(y_2) = x.$$

Then $y_1 - y_2 \in \ker\beta = \operatorname{Im}\alpha$, so there exists $m' \in M'$ such that

$$y_1 - y_2 = \alpha(m').$$

If $\mu(z_i) = \varphi(y_i)$ for $i = 1, 2$, then

$$\mu(z_1 - z_2) = \varphi(y_1) - \varphi(y_2) = \varphi(\alpha(m')) = \mu(\varphi'(m')),$$

where we used commutativity. Since μ is injective, it follows that

$$z_1 - z_2 = \varphi'(m') \in \operatorname{Im}\varphi'.$$

Thus $\overline{z_1} = \overline{z_2}$ in $\operatorname{coker}\varphi'$, so $\delta(x)$ is independent of the choice of y.

Next, if $z, z' \in N'$ both satisfy

$$\mu(z) = \mu(z') = \varphi(y),$$

then $\mu(z - z') = 0$, and since μ is injective, $z = z'$. Hence $\delta(x)$ is also independent of the choice of z.

It is straightforward to verify that δ is a homomorphism.

Now we prove exactness.

Exactness at $\ker\varphi'$. Since α induces α', we have

$$\alpha'(\ker\varphi') \subseteq \ker\varphi.$$

Because α is injective, so is α'. Hence

$$\ker\alpha' = 0.$$

Therefore the sequence is exact at $\ker\varphi'$.

Exactness at $\ker\varphi$. We show that

$$\operatorname{Im}\alpha' = \ker\beta'.$$

Take $x' \in \ker\varphi'$. Then

$$\beta'(\alpha'(x')) = \beta(\alpha(x')) = 0,$$

so $\operatorname{Im}\alpha' \subseteq \ker\beta'$.

Conversely, let $y \in \ker\varphi$ satisfy $\beta'(y) = 0$. Then $\beta(y) = 0$, hence by exactness of the top row there exists $x' \in M'$ such that

$$\alpha(x') = y.$$

Since $\varphi(y) = 0$, we get

$$0 = \varphi(\alpha(x')) = \mu(\varphi'(x')).$$

As μ is injective, $\varphi'(x') = 0$, so $x' \in \ker\varphi'$. Thus

$$y = \alpha'(x'),$$

proving $\ker\beta' \subseteq \operatorname{Im}\alpha'$.

Exactness at $\ker\varphi''$. We show that

$$\operatorname{Im}\beta' = \ker\delta.$$

First let $y \in \ker\varphi$. Then $\beta(y) \in \ker\varphi''$, since

$$\varphi''(\beta(y)) = \nu(\varphi(y)) = 0.$$

Thus $\beta'(y) \in \ker \varphi''$. To compute $\delta(\beta'(y))$, choose the lift y itself. Then $\varphi(y) = 0$, so we may take $z = 0$, and hence
$$\delta(\beta'(y)) = \overline{0} = 0.$$
Therefore $\operatorname{Im} \beta' \subseteq \ker \delta$.

Conversely, let $x \in \ker \varphi''$ and suppose $\delta(x) = 0$. Choose $y \in M$ with
$$\beta(y) = x,$$
and choose $z \in N'$ such that
$$\mu(z) = \varphi(y).$$
Since $\delta(x) = \overline{z} = 0$ in $\operatorname{coker} \varphi'$, there exists $x' \in M'$ such that
$$z = \varphi'(x').$$
Hence
$$\varphi(y - \alpha(x')) = \varphi(y) - \varphi(\alpha(x')) = \mu(z) - \mu(\varphi'(x')) = 0.$$
So $y - \alpha(x') \in \ker \varphi$, and
$$\beta'(y - \alpha(x')) = \beta(y - \alpha(x')) = \beta(y) = x,$$
because $\beta\alpha = 0$. Thus $x \in \operatorname{Im} \beta'$, proving
$$\ker \delta \subseteq \operatorname{Im} \beta'.$$

Exactness at $\operatorname{coker} \varphi'$**.** We show that
$$\operatorname{Im} \delta = \ker \mu'.$$

Let $x \in \ker \varphi''$, and write
$$\delta(x) = \overline{z} \in \operatorname{coker} \varphi'$$
as above, with $\mu(z) = \varphi(y)$. Then
$$\mu'(\delta(x)) = \overline{\mu(z)} = \overline{\varphi(y)} = 0$$
in $\operatorname{coker} \varphi$. Hence $\operatorname{Im} \delta \subseteq \ker \mu'$.

Conversely, let $\overline{z} \in \operatorname{coker} \varphi'$ satisfy
$$\mu'(\overline{z}) = 0.$$
This means that $\mu(z) \in \operatorname{Im} \varphi$, so there exists $y \in M$ such that
$$\varphi(y) = \mu(z).$$

Applying ν, we get
$$\varphi''(\beta(y)) = \nu(\varphi(y)) = \nu(\mu(z)) = 0.$$
Thus $\beta(y) \in \ker \varphi''$. By construction of δ,
$$\delta(\beta(y)) = \overline{z}.$$
So $\overline{z} \in \operatorname{Im} \delta$, proving
$$\ker \mu' \subseteq \operatorname{Im} \delta.$$

Exactness at $\operatorname{coker} \varphi$**.** We show that
$$\operatorname{Im} \mu' = \ker \nu'.$$

For $\overline{z} \in \operatorname{coker} \varphi'$, we have
$$\nu'(\mu'(\overline{z})) = \overline{\nu(\mu(z))} = \overline{0} = 0,$$
so $\operatorname{Im} \mu' \subseteq \ker \nu'$.

Conversely, let $\overline{n} \in \operatorname{coker} \varphi$ satisfy
$$\nu'(\overline{n}) = 0.$$
Then $\overline{\nu(n)} = 0$ in $\operatorname{coker} \varphi''$, so there exists $m'' \in M''$ such that
$$\varphi''(m'') = \nu(n).$$
Since β is surjective, choose $m \in M$ such that
$$\beta(m) = m''.$$
Then
$$\nu(n - \varphi(m)) = \nu(n) - \varphi''(\beta(m)) = \nu(n) - \varphi''(m'') = 0.$$
Hence
$$n - \varphi(m) \in \ker \nu = \operatorname{Im} \mu,$$
so there exists $z \in N'$ such that
$$\mu(z) = n - \varphi(m).$$
Therefore in $\operatorname{coker} \varphi$,
$$\overline{n} = \overline{\mu(z)} = \mu'(\overline{z}),$$
showing that $\overline{n} \in \operatorname{Im} \mu'$. Thus
$$\ker \nu' \subseteq \operatorname{Im} \mu'.$$

Exactness at $\operatorname{coker} \varphi''$ **and termination.** Since $\nu : N \to N''$ is surjective, the induced map

$$\nu' : \operatorname{coker}\varphi \to \operatorname{coker}\varphi''$$

is surjective as well: for any $\overline{n''} \in \operatorname{coker}\varphi''$, choose $n \in N$ such that

$$\nu(n) = n'',$$

then

$$\nu'(\overline{n}) = \overline{n''}.$$

Hence

$$\operatorname{Im}\nu' = \operatorname{coker}\varphi'',$$

which is equivalent to exactness at $\operatorname{coker}\varphi''$ and the final 0.

Collecting all the above, we obtain the exact sequence

$$0 \longrightarrow \ker\varphi' \xrightarrow{\alpha'} \ker\varphi \xrightarrow{\beta'} \ker\varphi'' \xrightarrow{\delta} \operatorname{coker}\varphi' \xrightarrow{\mu'} \operatorname{coker}\varphi \xrightarrow{\nu'} \operatorname{coker}\varphi'' \longrightarrow 0.$$

□

15.4 Homework Week 4

Exercise 15.4.1. *Let J be an R-module. Show that the following are equivalent.*

1. *J is an injective module.*
2. *For any R-module monomorphism $\varphi\colon J \to M$, there exists an R-module homomorphism $\psi\colon M \to J$ such that $\psi\varphi = \mathrm{id}_J$.*
3. *If J is a submodule of some R-module M, then J is a direct summand of M.*

Proof. We prove (1) ⇒ (2) ⇒ (3) ⇒ (1).

(1) ⇒ (2). Assume that J is injective, and let $\varphi\colon J \to M$ be a monomorphism. Apply the defining property of injectivity to the monomorphism φ and the map $\mathrm{id}_J\colon J \to J$:

$$\begin{tikzcd} J \arrow[r, hook, "\varphi"] \arrow[d, "\mathrm{id}_J"'] & M \arrow[dl, dashed, "\exists\psi"] \\ J & \end{tikzcd}$$

Then there exists an R-module homomorphism $\psi\colon M \to J$ such that

$$\psi \circ \varphi = \mathrm{id}_J\,.$$

(2) ⇒ (3). Assume (2). If $J \subseteq M$, let $\iota\colon J \hookrightarrow M$ be the inclusion map. By (2), there exists $\psi\colon M \to J$ such that

$$\psi \circ \iota = \mathrm{id}_J\,.$$

We claim that

$$M = J \oplus \ker \psi.$$

Indeed, for any $m \in M$,

$$m = \iota(\psi(m)) + \big(m - \iota(\psi(m))\big),$$

and

$$\psi\big(m - \iota(\psi(m))\big) = \psi(m) - \psi(\iota(\psi(m))) = \psi(m) - \psi(m) = 0,$$

so $m - \iota(\psi(m)) \in \ker \psi$. Hence $M = J + \ker \psi$. If $x \in J \cap \ker \psi$, then

$$x = (\psi \circ \iota)(x) = \psi(x) = 0,$$

so $J \cap \ker \psi = 0$. Therefore $M = J \oplus \ker \psi$, and J is a direct summand of M.

(3) ⇒ (1). Assume (3). To prove that J is injective, let

$$i\colon A \hookrightarrow B$$

be a monomorphism of R-modules, and let

$$f\colon A \to J$$

be an R-module homomorphism. We must show that f extends to a map $g\colon B \to J$.

Consider the quotient module

$$P = (B \oplus J)/N, \qquad N = \{(i(a), -f(a)) \mid a \in A\}.$$

Define maps

$$\eta\colon J \to P, \qquad \eta(j) = [(0, j)],$$

and

$$\beta\colon B \to P, \qquad \beta(b) = [(b, 0)].$$

We first show that η is injective. If $\eta(j) = 0$, then $(0, j) \in N$, so for some $a \in A$,

$$(0, j) = (i(a), -f(a)).$$

Hence $i(a) = 0$. Since i is injective, $a = 0$, and therefore $j = -f(0) = 0$. Thus η is a monomorphism, so we may identify J with the submodule $\eta(J) \subseteq P$.

By (3), this submodule is a direct summand of P. Therefore there exists an R-module homomorphism

$$s\colon P \to J$$

such that

$$s \circ \eta = \mathrm{id}_J .$$

Now define

$$g = s \circ \beta\colon B \to J.$$

For any $a \in A$, we have in P:
$$[(i(a), 0)] = [(0, f(a))],$$
because
$$(i(a), 0) - (0, f(a)) = (i(a), -f(a)) \in N.$$
Hence
$$g(i(a)) = s([(i(a), 0)]) = s([(0, f(a))]) = (s \circ \eta)(f(a)) = f(a).$$
So $g \circ i = f$, and f extends to B. Therefore J is injective. $\square$

Exercise 15.4.2. *Suppose that $(M_i)_{i \in I}$ are R-modules. Prove that*
$$\bigoplus_{i \in I} M_i$$
is a flat R-module if and only if each M_i is a flat R-module.

Proof. Set
$$M = \bigoplus_{i \in I} M_i.$$

First assume that each M_i is flat. Let
$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$
be a short exact sequence of R-modules. Since each M_i is flat, for every $i \in I$ the sequence
$$0 \longrightarrow M_i \otimes_R A \longrightarrow M_i \otimes_R B \longrightarrow M_i \otimes_R C \longrightarrow 0$$
is exact. Taking direct sums over $i \in I$, we obtain an exact sequence
$$0 \longrightarrow \bigoplus_{i \in I}(M_i \otimes_R A) \longrightarrow \bigoplus_{i \in I}(M_i \otimes_R B) \longrightarrow \bigoplus_{i \in I}(M_i \otimes_R C) \longrightarrow 0.$$
Using the canonical isomorphisms
$$\left(\bigoplus_{i \in I} M_i\right) \otimes_R X \cong \bigoplus_{i \in I}(M_i \otimes_R X) \qquad (X = A, B, C),$$
we get
$$0 \longrightarrow M \otimes_R A \longrightarrow M \otimes_R B \longrightarrow M \otimes_R C \longrightarrow 0.$$
Hence M is flat.

Conversely, assume that M is flat. Fix $i \in I$, and write
$$M = M_i \oplus N_i, \qquad N_i = \bigoplus_{j \neq i} M_j.$$
Let
$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be a short exact sequence of R-modules. Since M is flat, the sequence

$$0 \longrightarrow M \otimes_R A \longrightarrow M \otimes_R B \longrightarrow M \otimes_R C \longrightarrow 0$$

is exact. Using

$$M \otimes_R X \cong (M_i \otimes_R X) \oplus (N_i \otimes_R X) \qquad (X = A, B, C),$$

this becomes

$$0 \longrightarrow (M_i \otimes_R A) \oplus (N_i \otimes_R A) \longrightarrow (M_i \otimes_R B) \oplus (N_i \otimes_R B) \longrightarrow (M_i \otimes_R C) \oplus (N_i \otimes_R C) \longrightarrow 0.$$

Since kernels and images of direct sum maps decompose componentwise, it follows that

$$0 \longrightarrow M_i \otimes_R A \longrightarrow M_i \otimes_R B \longrightarrow M_i \otimes_R C \longrightarrow 0$$

is exact. Therefore M_i is flat. As i was arbitrary, each M_i is flat. $\square$

Exercise 15.4.3. *Prove that the following three conditions on an R-module V are equivalent.*

1. (Ascending chain condition.) *For every infinite chain of R-submodules of V,*

$$V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n \subseteq \cdots,$$

there is a number m such that

$$V_m = V_{m+1} = \cdots.$$

2. *Any non-empty set of submodules of V contains a maximal element with respect to inclusion.*

3. *Every R-submodule of V is finitely generated.*

Proof. We prove (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1).

(1) $\Rightarrow$ (2)**.** Assume that (2) fails. Then there exists a non-empty set $\mathcal{S}$ of submodules of V having no maximal element. Choose $V_1 \in \mathcal{S}$. Since V_1 is not maximal in $\mathcal{S}$, there exists $V_2 \in \mathcal{S}$ such that

$$V_1 \subsetneq V_2.$$

Inductively, having chosen $V_n \in \mathcal{S}$, since V_n is not maximal there exists $V_{n+1} \in \mathcal{S}$ with

$$V_n \subsetneq V_{n+1}.$$

Thus we obtain a strictly ascending chain

$$V_1 \subsetneq V_2 \subsetneq \cdots,$$

which contradicts (1). Hence every non-empty set of submodules contains a maximal element.

(2) $\Rightarrow$ (3). Let U be a submodule of V. Consider the set

$$\mathcal{T} = \{N \leq U \mid N \text{ is finitely generated}\}.$$

This set is non-empty, since $0 \in \mathcal{T}$. By (2), $\mathcal{T}$ has a maximal element, say N.

We claim that $N = U$. If not, choose $u \in U \setminus N$. Then

$$N + Ru$$

is a finitely generated submodule of U properly containing N, contradicting the maximality of N in $\mathcal{T}$. Therefore $U = N$, so U is finitely generated. Hence every submodule of V is finitely generated.

(3) $\Rightarrow$ (1). Let

$$V_1 \subseteq V_2 \subseteq \cdots$$

be an ascending chain of submodules of V. Set

$$U = \bigcup_{n \geq 1} V_n.$$

Since the chain is ascending, U is a submodule of V. By (3), U is finitely generated, say

$$U = Ru_1 + \cdots + Ru_r.$$

For each j, choose n_j such that $u_j \in V_{n_j}$, and let

$$m = \max\{n_1, \ldots, n_r\}.$$

Then $u_1, \ldots, u_r \in V_m$, so $U \subseteq V_m$. Since always $V_m \subseteq U$, we get $U = V_m$. Therefore for all $n \geq m$,

$$V_n = U = V_m,$$

so the chain stabilizes. Hence (1) holds. $\square$

Exercise 15.4.4. *Let W be an R-submodule of V. Prove that V is Noetherian if and only if both W and V/W are Noetherian.*

Proof. We use the characterization that an R-module is Noetherian if and only if every submodule is finitely generated.

Assume first that V is Noetherian. Then every submodule of V is finitely generated.

Since every submodule of W is also a submodule of V, it follows that every submodule of W is finitely generated. Hence W is Noetherian.

Now let $\overline{U}$ be a submodule of V/W, and let

$$\pi \colon V \to V/W$$

be the quotient map. Then $U := \pi^{-1}(\overline{U})$ is a submodule of V, so U is finitely generated, say

$$U = Rv_1 + \cdots + Rv_n.$$

Applying π, we get

$$\overline{U} = R(v_1 + W) + \cdots + R(v_n + W).$$

Thus every submodule of V/W is finitely generated, and therefore V/W is Noetherian.

Conversely, assume that both W and V/W are Noetherian. Let U be any submodule of V. We show that U is finitely generated.

First,

$$U \cap W$$

is a submodule of W, so it is finitely generated. Choose generators

$$U \cap W = Rw_1 + \cdots + Rw_m.$$

Next, since

$$(U + W)/W$$

is a submodule of V/W, it is finitely generated as well. Choose elements $u_1, \ldots, u_n \in U$ such that

$$(U + W)/W = R(u_1 + W) + \cdots + R(u_n + W).$$

Now let $u \in U$. Since $u + W \in (U + W)/W$, there exist $a_1, \ldots, a_n \in R$ such that

$$u + W = \sum_{i=1}^{n} a_i(u_i + W).$$

Therefore

$$u - \sum_{i=1}^{n} a_i u_i \in W.$$

Because also $u, u_i \in U$, we have

$$u - \sum_{i=1}^{n} a_i u_i \in U \cap W.$$

So there exist $b_1, \ldots, b_m \in R$ such that

$$u - \sum_{i=1}^{n} a_i u_i = \sum_{j=1}^{m} b_j w_j.$$

Hence

$$u = \sum_{i=1}^{n} a_i u_i + \sum_{j=1}^{m} b_j w_j.$$

This shows that U is generated by

$$u_1, \ldots, u_n, w_1, \ldots, w_m.$$

Thus every submodule of V is finitely generated, so V is Noetherian. $\square$

Exercise 15.4.5. *Let V be a Noetherian R-module and let $f\colon V \to V$ be an epimorphism. Show that f is an R-isomorphism.*

Proof. Since f is an epimorphism, it is surjective. It therefore suffices to prove that f is injective.

For each $n \geq 1$, set

$$K_n = \ker(f^n).$$

Then

$$K_1 \subseteq K_2 \subseteq K_3 \subseteq \cdots,$$

because if $f^n(x) = 0$, then

$$f^{n+1}(x) = f(f^n(x)) = 0.$$

Since V is Noetherian, this ascending chain stabilizes. Hence there exists $m \geq 1$ such that

$$K_m = K_{m+1}.$$

We claim that $K_m = 0$. Let $x \in K_m$. Because f is surjective, f^m is also surjective, there exists $y \in V$ such that

$$f^m(y) = x.$$

Now

$$f^{2m}(y) = f^m(f^m(y)) = f^m(x) = 0,$$

so $y \in K_{2m} = K_m$. Therefore

$$x = f^m(y) = 0.$$

Thus $K_m = 0$.

Since

$$\ker(f) = K_1 \subseteq K_m = 0,$$

we obtain $\ker(f) = 0$. So f is injective. Being both surjective and injective, f is an isomorphism. $\square$

15.5 Homework Week 5

Exercise 15.5.1. *Are the following statements correct?*

(1) Any submodule of a semisimple module is also semisimple.

(2) Any semisimple Noetherian module is Artinian.

Proof. **(1)** True. Let V be a semisimple module, so

$$V = \bigoplus_{i \in I} S_i$$

where each S_i is simple. Let $W \leq V$ be a submodule. We show that W is a sum of simple submodules, hence semisimple.

Consider the set $\mathcal{S}$ of all subsets $J \subseteq I$ such that the sum

$$W + \bigoplus_{j \in J} S_j$$

is direct, i.e.

$$W \cap \bigoplus_{j \in J} S_j = 0.$$

By Zorn's lemma, there exists a maximal such subset J_0. Set

$$U := W \oplus \bigoplus_{j \in J_0} S_j.$$

For any $i \in I \setminus J_0$, the maximality of J_0 forces $U \cap S_i \neq 0$. Since S_i is simple, we have $S_i \subseteq U$. Therefore $S_i \subseteq U$ for every $i \in I$, so $V = U$. Hence

$$V = W \oplus \bigoplus_{j \in J_0} S_j,$$

and

$$W \cong V \Big/ \bigoplus_{j \in J_0} S_j \cong \bigoplus_{i \in I \setminus J_0} S_i,$$

which is semisimple.

(2) True. Let V be a semisimple Noetherian module. Since V is semisimple,

$$V = \bigoplus_{i \in I} S_i$$

Since V is Noetherian, it is finitely generated. Let $v_1, \ldots, v_n$ generate V. For each k, because $V = \bigoplus_{i \in I} S_i$ is a direct sum, the element v_k has only finitely many nonzero components, so there exists a finite subset $I_k \subseteq I$ such that

$$v_k \in \bigoplus_{i \in I_k} S_i.$$

Hence

$$V = \langle v_1, \ldots, v_n \rangle \subseteq \bigoplus_{i \in I_1 \cup \cdots \cup I_n} S_i.$$

Since the reverse inclusion is obvious, we get

$$V = \bigoplus_{i \in I_1 \cup \cdots \cup I_n} S_i,$$

and therefore V is a finite direct sum of simple modules. Each S_i is simple, hence both Noetherian and Artinian. A finite direct sum of Artinian modules is Artinian, so V is Artinian. $\square$

Exercise 15.5.2. *Let V be an R-module with a composition series. Prove that the following conditions are equivalent.*

(1) V is semisimple.

(2) Every irreducible submodule of V is a direct summand of V.

(3) Every maximal submodule of V is a direct summand of V.

Proof. We prove
$$(1) \implies (2) \implies (3) \implies (1).$$

(1) $\implies$ (2). If V is semisimple, then every submodule of V is a direct summand. In particular, every irreducible submodule of V is a direct summand.

(2) $\implies$ (3). Let M be a maximal submodule of V. Then V/M is simple.

Since M is a maximal submodule of V, the quotient V/M is simple. Now let S be a submodule of V such that $S \nsubseteq M$. Then $S + M \neq M$, so

$$(S + M)/M \neq 0.$$

By the second isomorphism theorem,

$$S/(S \cap M) \cong (S + M)/M.$$

Since $(S + M)/M$ is a nonzero submodule of the simple module V/M, it follows that $(S + M)/M$ is simple. Hence $S/(S \cap M)$ is simple.

Choose $S \subseteq V$ minimal subject to $S \nsubseteq M$. Then $S \cap M$ is a maximal submodule of S, and by minimality of S, the quotient $S/(S \cap M)$ is simple. But $S \cap M$ is a proper submodule of S, so by minimality of S, it must be 0. Thus S is simple.

Therefore S is an irreducible submodule of V with $S \nsubseteq M$. By (2), S is a direct summand of V, say
$$V = S \oplus T.$$

Since $S \nsubseteq M$ and S is simple, we have $S \cap M = 0$. Also M is maximal and $S \nsubseteq M$, so

$$M + S = V.$$

Hence
$$V = M \oplus S,$$

and therefore M is a direct summand of V.

(3) $\Longrightarrow$ (1). Let

$$V = V_0 \supsetneq V_1 \supsetneq \cdots \supsetneq V_n = 0$$

be a composition series of V. We argue by induction on n, the composition length of V.

If $n = 1$, then V is simple, hence semisimple.

Assume $n > 1$, and suppose the statement holds for modules of composition length $< n$. Since V/V_1 is simple, V_1 is a maximal submodule of V. By (3), V_1 is a direct summand of V; thus

$$V = V_1 \oplus S$$

for some simple submodule $S \cong V/V_1$.

Now V_1 has composition length $n-1$. We claim that every maximal submodule of V_1 is a direct summand of V_1. Indeed, let N be a maximal submodule of V_1. Then

$$V/N \cong (V_1/N) \oplus S,$$

and since V_1/N is simple, $N \oplus S$ is a maximal submodule of V. By (3), $N \oplus S$ is a direct summand of V, say

$$V = (N \oplus S) \oplus T.$$

Because

$$V = V_1 \oplus S,$$

intersecting with V_1 gives

$$V_1 = N \oplus T.$$

So N is a direct summand of V_1.

Hence V_1 satisfies condition (3), and by the induction hypothesis V_1 is semisimple. Therefore

$$V = V_1 \oplus S$$

is semisimple as well.

This proves (3) $\Longrightarrow$ (1), and therefore the three conditions are equivalent. □

Exercise 15.5.3. *Prove that*

$$\mathrm{Soc}(U \oplus V) = \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Proof. We first show

$$\mathrm{Soc}(U) \oplus \mathrm{Soc}(V) \subseteq \mathrm{Soc}(U \oplus V).$$

Indeed, $\mathrm{Soc}(U)$ and $\mathrm{Soc}(V)$ are semisimple, so $\mathrm{Soc}(U) \oplus \mathrm{Soc}(V)$ is a semisimple submodule of $U \oplus V$. Hence it is contained in $\mathrm{Soc}(U \oplus V)$.

For the reverse inclusion, let S be a simple submodule of $U \oplus V$, and let

$$\pi_U : U \oplus V \to U, \qquad \pi_V : U \oplus V \to V$$

be the canonical projections. Consider the restrictions

$$\pi_U|_S : S \to U, \qquad \pi_V|_S : S \to V.$$

Since S is simple, the image of any homomorphism defined on S is either 0 or simple. Therefore $\pi_U(S)$ is either 0 or a simple submodule of U, and similarly $\pi_V(S)$ is either 0 or a simple submodule of V. Hence

$$\pi_U(S) \subseteq \mathrm{Soc}(U), \qquad \pi_V(S) \subseteq \mathrm{Soc}(V).$$

Now for any $s \in S$, we can write

$$s = (\pi_U(s), \pi_V(s)).$$

Since $\pi_U(s) \in \mathrm{Soc}(U)$ and $\pi_V(s) \in \mathrm{Soc}(V)$, it follows that

$$s \in \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Thus

$$S \subseteq \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Since $\mathrm{Soc}(U \oplus V)$ is the sum of all simple submodules of $U \oplus V$, we obtain

$$\mathrm{Soc}(U \oplus V) \subseteq \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Combining the two inclusions, we conclude that

$$\mathrm{Soc}(U \oplus V) = \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

□

Exercise 15.5.4. *Let R be an arbitrary ring. Then every two-sided ideal of the matrix ring $M_n(R)$ can be expressed in the form $M_n(I)$, where I is a two-sided ideal of R. In particular, R is simple if and only if $M_n(R)$ is simple.*

Hint. For an ideal L of $M_n(R)$, let I be the set of $(1,1)$-entries of L. Show that I is an ideal of R and $L = M_n(I)$.

Proof. Let E_{ij} denote the matrix with 1 in position (i, j) and 0 elsewhere. For a two-sided ideal L of $M_n(R)$, define

$$I := \{a \in R \mid aE_{11} \in L\}.$$

We first note that for any $A = (a_{kl}) \in L$, we have

$$E_{1k} \cdot A \cdot E_{l1} = a_{kl}\, E_{11} \in L,$$

so every entry of every matrix in L belongs to I. In particular,

$$I = \{a_{kl} \mid A = (a_{kl}) \in L,\ 1 \le k, l \le n\}.$$

I **is a two-sided ideal of** R**.** If $a, b \in I$, then $aE_{11}, bE_{11} \in L$, so $(a-b)E_{11} = aE_{11} - bE_{11} \in L$, hence $a - b \in I$. For $r \in R$, we have $(rE_{11}) \cdot (aE_{11}) = raE_{11} \in L$ and $(aE_{11}) \cdot (rE_{11}) = arE_{11} \in L$, so $ra, ar \in I$. Thus I is a two-sided ideal of R.

$L = M_n(I)$**.** Since every entry of every matrix in L lies in I, we have $L \subseteq M_n(I)$. Conversely, let $aE_{ij} \in M_n(I)$ with $a \in I$. Then $aE_{11} \in L$, and

$$aE_{ij} = E_{i1} \cdot (aE_{11}) \cdot E_{1j} \in L.$$

Since every matrix in $M_n(I)$ is a sum of such elements aE_{ij}, we get $M_n(I) \subseteq L$. Therefore $L = M_n(I)$.

R **is simple** $\Longleftrightarrow$ $M_n(R)$ **is simple.** The map $I \mapsto M_n(I)$ is a bijection from the set of two-sided ideals of R to the set of two-sided ideals of $M_n(R)$. Hence R has no nontrivial two-sided ideals if and only if $M_n(R)$ has no nontrivial two-sided ideals. $\square$

Exercise 15.5.5. *Let W be a submodule of a semisimple module V. Show that W and V/W are also semisimple.*

Proof. Since V is semisimple, write $V = \bigoplus_{i \in I} S_i$ with each S_i simple.

W **is semisimple.** By Zorn's lemma, choose a maximal subset $J \subseteq I$ such that $W \cap \bigoplus_{j \in J} S_j = 0$. Set $U = W \oplus \bigoplus_{j \in J} S_j$. For any $i \in I \setminus J$, the maximality of J gives $U \cap S_i \neq 0$, and since S_i is simple, $S_i \subseteq U$. Therefore $V \subseteq U$, so $V = U = W \oplus \bigoplus_{j \in J} S_j$. Thus

$$W \cong V \Big/ \bigoplus_{j \in J} S_j \cong \bigoplus_{i \in I \setminus J} S_i,$$

which is a direct sum of simple modules, hence semisimple.

V/W **is semisimple.** From the decomposition $V = W \oplus \bigoplus_{j \in J} S_j$, we obtain

$$V/W \cong \bigoplus_{j \in J} S_j,$$

which is a direct sum of simple modules, hence semisimple. $\square$

15.6 Homework Week 6

Exercise 15.6.1. *Let D be a division ring and n a positive integer.*

(a) Show that the natural left $M_n(D)$-module, consisting of column vectors of length n with entries in D, is a simple module.

(b) Show that $M_n(D)$ is semisimple and has up to isomorphism only one simple module.

(c) Show that every ring of the form

$$M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$$

is semisimple, where D_i are division rings.

(d) *Show that $M_n(D)$ is a simple ring, namely, one in which the only 2-sided ideals are the zero ideal and the whole ring.*

Proof. Let $R = M_n(D)$ and let $V = D^n$ be the natural left R-module of column vectors.

(a) We show that V is simple. Let $0 \neq W \leq V$, and choose

$$v = (v_1, \dots, v_n)^t \in W, \qquad v \neq 0.$$

Then $v_j \neq 0$ for some j. For each $i \in \{1, \dots, n\}$, let $A_i \in R$ be the matrix whose (i, j)-entry is v_j^{-1} and whose other entries are 0. Then

$$A_i v = e_i,$$

where e_i is the i-th standard basis vector in D^n. Since W is an R-submodule and $v \in W$, we get $e_i \in W$ for all i. Hence

$$W = D^n = V.$$

Therefore V is a simple left R-module.

(b) Let $E_{11}, \dots, E_{nn}$ be the standard diagonal matrix units. Then

$$1 = E_{11} + \cdots + E_{nn},$$

and hence

$$R = RE_{11} \oplus \cdots \oplus RE_{nn}.$$

Indeed, RE_{ii} is precisely the set of matrices whose only possibly nonzero column is the i-th column, so every matrix is uniquely the sum of its columns.

For each i, define

$$\phi_i : RE_{ii} \longrightarrow V$$

by sending a matrix in RE_{ii} to its i-th column. This is an isomorphism of left R-modules. By part (a), V is simple, so each RE_{ii} is simple. Thus ${}_RR$ is a direct sum of simple submodules, and therefore R is semisimple.

Now let S be a simple left R-module. Pick $0 \neq s \in S$. Then the map

$$\psi : R \longrightarrow S, \qquad A \mapsto As$$

is a nonzero R-homomorphism, hence surjective since S is simple. Because R is semisimple, every quotient of R is semisimple. Since S is simple, it must be isomorphic to one of the simple direct summands RE_{ii}. Therefore

$$S \cong RE_{ii} \cong V$$

for some i. Hence, up to isomorphism, R has only one simple left module.

(c) Let

$$A = M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r).$$

As a left A-module over itself,

$$_AA = {}_AM_{n_1}(D_1) \oplus \cdots \oplus {}_AM_{n_r}(D_r).$$

Each summand is semisimple by part (b), and a finite direct sum of semisimple modules is semisimple. Hence A is a semisimple ring.

(d) Let I be a nonzero two-sided ideal of R. Choose $0 \neq A = (a_{pq}) \in I$, and pick indices i, j such that $a_{ij} \neq 0$. For arbitrary r, s, let E_{uv} denote the standard matrix units. Then

$$E_{ri}AE_{js} = a_{ij}E_{rs} \in I.$$

Since $a_{ij} \in D^\times$, we also have

$$(a_{ij}^{-1}E_{rr})(a_{ij}E_{rs}) = E_{rs} \in I.$$

Thus every matrix unit E_{rs} belongs to I. Therefore

$$1 = E_{11} + \cdots + E_{nn} \in I,$$

so $I = R$. Hence the only two-sided ideals of R are 0 and R, i.e. R is a simple ring.

$\square$

Exercise 15.6.2. *Let U be an A-module for a semisimple finite-dimensional algebra A (over a field). Show that if $\operatorname{End}_A(U)$ is a division ring then U is simple.*

Proof. Since A is a semisimple finite-dimensional algebra over a field, it is a semisimple Artinian ring. Hence every left A-module is semisimple. In particular, U is a semisimple A-module.

We claim that U must be simple. Suppose not. Then U is not simple, so because U is semisimple, there exist nonzero submodules $U_1, U_2 \leq U$ such that

$$U = U_1 \oplus U_2.$$

Let

$$\pi_1 : U \to U_1$$

be the projection onto the first summand. Since U_1 is an A-submodule, π_1 is an A-module endomorphism of U, so

$$\pi_1 \in \operatorname{End}_A(U).$$

Moreover,

$$\pi_1^2 = \pi_1,$$

so π_1 is an idempotent.

Now $\pi_1 \neq 0$ because $U_1 \neq 0$, and $\pi_1 \neq 1_U$ because $U_2 \neq 0$. Thus π_1 is a nontrivial idempotent in $\mathrm{End}_A(U)$.

But a division ring has no idempotents other than 0 and 1: indeed, if $e^2 = e$ and $e \neq 0$, then multiplying by e^{-1} gives $e = 1$. This contradicts the assumption that $\mathrm{End}_A(U)$ is a division ring.

Therefore U must be simple. □

Exercise 15.6.3. *Let A be a ring. Then A is a semisimple Artinian ring if and only if every A-module is projective.*

Proof. Here all modules are left A-modules.

(⇒)

Assume that A is a semisimple Artinian ring. Then every A-module is semisimple.

Let M be any A-module. To show that M is projective, consider an arbitrary short exact sequence

$$0 \longrightarrow K \longrightarrow N \xrightarrow{\pi} M \longrightarrow 0.$$

Since N is semisimple, the submodule K is a direct summand of N. Thus there exists a submodule $N' \leq N$ such that

$$N = K \oplus N'.$$

Now $\pi|_{N'} : N' \to M$ is surjective, and

$$\ker(\pi|_{N'}) = N' \cap K = 0.$$

Hence $\pi|_{N'}$ is an isomorphism. Therefore the above short exact sequence splits. Since every short exact sequence ending in M splits, M is projective.

Since M was arbitrary, every A-module is projective.

(⇐)

Assume that every A-module is projective. Let M be any A-module and let $N \leq M$ be a submodule. Then the quotient M/N is projective, so the short exact sequence

$$0 \longrightarrow N \longrightarrow M \longrightarrow M/N \longrightarrow 0$$

splits. Hence N is a direct summand of M.

Therefore every submodule of every A-module is a direct summand. It follows that every A-module is semisimple. In particular, the left regular module ${}_AA$ is semisimple.

Now A is cyclic as a left A-module, generated by 1. Since ${}_AA$ is semisimple, it is a sum of simple submodules:

$$A = \sum_{\lambda \in \Lambda} S_\lambda.$$

Because $1 \in A$, we may write

$$1 = s_1 + \cdots + s_n \qquad (s_i \in S_{\lambda_i}).$$

Then for any $a \in A$,

$$a = a \cdot 1 = as_1 + \cdots + as_n \in S_{\lambda_1} + \cdots + S_{\lambda_n}.$$

Thus

$$A = S_{\lambda_1} + \cdots + S_{\lambda_n}.$$

Since A is semisimple, this finite sum is in fact a direct sum (after deleting repetitions if necessary):

$$A \cong S_1 \oplus \cdots \oplus S_m$$

with each S_i simple.

Hence ${}_AA$ is a finite direct sum of simple modules, so it is Artinian. Therefore A is a semisimple Artinian ring.

This proves that A is a semisimple Artinian ring if and only if every A-module is projective. $\square$

Exercise 15.6.4. *Let k be a field. Suppose the group algebra kG is semisimple, and the simple kG-modules are $S_1, \dots, S_r$ with degrees $d_i = \dim_k S_i$. Show that*

$$\sum_{i=1}^{r} d_i^2 \geq |G|$$

with equality if and only if $\operatorname{End}_{kG}(S_i) = k$ for all i.

Proof. Let

$$A = kG, \qquad E_i = \operatorname{End}_A(S_i) \quad (1 \leq i \leq r).$$

Since A is a semisimple finite-dimensional k-algebra, by the Wedderburn–Artin theorem there exist positive integers $n_1, \dots, n_r$ such that

$$A \cong \bigoplus_{i=1}^{r} M_{n_i}(E_i^{\mathrm{op}}),$$

where $S_1, \dots, S_r$ form a complete set of representatives of the isomorphism classes of simple A-modules.

For each i, the simple module S_i is naturally a right E_i-vector space of dimension n_i. Hence

$$d_i = \dim_k S_i = n_i \dim_k E_i.$$

Taking k-dimensions in the Wedderburn decomposition gives

$$|G| = \dim_k(kG) = \dim_k A = \sum_{i=1}^{r} \dim_k M_{n_i}(E_i^{\mathrm{op}}) = \sum_{i=1}^{r} n_i^2 \dim_k E_i.$$

Using $d_i = n_i \dim_k E_i$, we obtain

$$|G| = \sum_{i=1}^{r} n_i^2 \dim_k E_i = \sum_{i=1}^{r} \frac{d_i^2}{\dim_k E_i}.$$

Now each E_i is a division k-algebra, so $\dim_k E_i \geq 1$. Therefore

$$|G| = \sum_{i=1}^{r} \frac{d_i^2}{\dim_k E_i} \leq \sum_{i=1}^{r} d_i^2.$$

Thus

$$\sum_{i=1}^{r} d_i^2 \geq |G|.$$

It remains to determine when equality holds. Since each $d_i > 0$ and each $\dim_k E_i \geq 1$, we have

$$\sum_{i=1}^{r} \frac{d_i^2}{\dim_k E_i} = \sum_{i=1}^{r} d_i^2$$

if and only if

$$\dim_k E_i = 1 \qquad \text{for all } i.$$

But $E_i = \mathrm{End}_A(S_i)$ is a k-algebra, so $\dim_k E_i = 1$ is equivalent to

$$E_i = k.$$

Hence equality holds if and only if

$$\mathrm{End}_{kG}(S_i) = k \qquad \text{for all } i.$$

This proves the claim. □

Exercise 15.6.5. *Suppose that we have R-module homomorphisms*

$$U \xrightarrow{f} V \xrightarrow{g} W.$$

Show that: if gf is an essential epimorphism and f is an epimorphism, then both f and g are essential epimorphisms.

Proof. Assume that

$$U \xrightarrow{f} V \xrightarrow{g} W$$

are R-module homomorphisms such that gf is an essential epimorphism and f is an epimorphism.

We must show that both f and g are essential epimorphisms.

Step 1: f is an essential epimorphism.

Since f is already assumed to be an epimorphism, it remains to show that it is essential.

Let $U' \leq U$ be a submodule such that

$$f(U') = V.$$

Then applying g, we get

$$(gf)(U') = g(f(U')) = g(V) = W,$$

because gf is an epimorphism. Thus U' maps onto W under gf.

But gf is an *essential* epimorphism, so no proper submodule of U can map surjectively onto W. Hence we must have

$$U' = U.$$

Therefore f is an essential epimorphism.

Step 2: g is an essential epimorphism.

First, g is an epimorphism: indeed, since gf is surjective and f is surjective, for any $w \in W$ there exists $u \in U$ such that

$$g(f(u)) = w.$$

Thus $w \in g(V)$, so $g(V) = W$.

Now let $V' \leq V$ be a submodule such that

$$g(V') = W.$$

We claim that $V' = V$.

Consider the inverse image

$$f^{-1}(V') = \{u \in U \mid f(u) \in V'\}.$$

This is a submodule of U, and

$$(gf)\big(f^{-1}(V')\big) = g(V') = W.$$

Since gf is an essential epimorphism, it follows that

$$f^{-1}(V') = U.$$

Applying f to both sides gives

$$V = f(U) = f\big(f^{-1}(V')\big) \subseteq V'.$$

Hence $V' = V$.

Therefore no proper submodule of V maps surjectively onto W under g, so g is an essential epimorphism.

We conclude that both f and g are essential epimorphisms. $\square$

15.7 Homework Week 7

Exercise 15.7.1. *Let V be an A-module.*

(a) Show that an endomorphism

$$e\colon V \to V$$

is a projection onto an A-submodule W if and only if e is idempotent as an element of $\mathrm{End}_A(V)$.

(b) Show that direct sum decompositions

$$V = W_1 \oplus W_2$$

as A-modules are in bijection with idempotent decompositions

$$1 = e + f$$

in $\mathrm{End}_A(V)$.

(c) Show that $e \in \mathrm{End}_A(V)$ is primitive if and only if $e(V)$ is indecomposable.

(d) Suppose that V is semisimple with finitely many simple summands and let

$$e_1, e_2 \in \mathrm{End}_A(V)$$

be idempotents. Show that

$$e_1(V) \cong e_2(V)$$

as A-modules if and only if e_1 and e_2 are conjugate by an invertible element of $\mathrm{End}_A(V)$.

(e) Let k be a field. Show that all primitive idempotent elements in $M_n(k)$ are conjugate under the action of the unit group $GL_n(k)$, which consists of all the invertible matrices in $M_n(k)$.

Proof. (a) Suppose first that $e\colon V \to V$ is a projection onto an A-submodule W. Then $e(V) = W$ and $e(w) = w$ for every $w \in W$. Hence for any $v \in V$,

$$e^2(v) = e(e(v)) = e(v),$$

because $e(v) \in W$. Thus $e^2 = e$, so e is idempotent.

Conversely, suppose that $e^2 = e$. Set

$$W := e(V).$$

Since e is A-linear, W is an A-submodule of V. Moreover, if $w = e(v) \in W$, then

$$e(w) = e^2(v) = e(v) = w.$$

Thus e fixes W pointwise and maps V onto W, so e is a projection onto W.

In fact, one also has

$$V = e(V) \oplus \ker(e),$$

since for any $v \in V$,

$$v = e(v) + (v - e(v)), \qquad e(v - e(v)) = e(v) - e^2(v) = 0,$$

and if $x \in e(V) \cap \ker(e)$, then $x = e(v)$ for some v, hence

$$x = e(x) = 0.$$

(b) Suppose first that

$$V = W_1 \oplus W_2.$$

Define

$$e(w_1 + w_2) = w_1, \qquad f(w_1 + w_2) = w_2 \qquad (w_i \in W_i).$$

Then $e, f \in \mathrm{End}_A(V)$, and e and f are the projections onto W_1 and W_2, respectively. By part (a), both are idempotent, and clearly

$$e + f = 1.$$

Also,

$$ef = fe = 0.$$

Conversely, suppose

$$1 = e + f$$

with $e^2 = e$ and $f^2 = f$. Then

$$e = e(e + f) = e + ef,$$

so $ef = 0$. Similarly,

$$f = (e + f)f = ef + f,$$

so again $ef = 0$, and likewise

$$fe = 0.$$

By part (a), e and f are projections onto $e(V)$ and $f(V)$, respectively. For every $v \in V$,

$$v = (e + f)(v) = e(v) + f(v),$$

so

$$V = e(V) + f(V).$$

If $x \in e(V) \cap f(V)$, say $x = e(v) = f(w)$, then

$$x = e(x) = ef(w) = 0.$$

Hence
$$V = e(V) \oplus f(V).$$

These two constructions are inverse to each other, so direct sum decompositions

$$V = W_1 \oplus W_2$$

are in bijection with idempotent decompositions

$$1 = e + f$$

in $\mathrm{End}_A(V)$.

(c) We show that e is primitive if and only if $e(V)$ is indecomposable.

First suppose that e is not primitive. Then there exist nonzero idempotents $e_1, e_2 \in \mathrm{End}_A(V)$ such that
$$e = e_1 + e_2, \qquad e_1e_2 = e_2e_1 = 0.$$

Since
$$ee_i = (e_1 + e_2)e_i = e_i,$$

we have $e_i(V) \subseteq e(V)$ for $i = 1, 2$. Moreover, for any $v \in V$,

$$e(v) = e_1(v) + e_2(v),$$

so
$$e(V) = e_1(V) + e_2(V).$$

If $x \in e_1(V) \cap e_2(V)$, say $x = e_1(v) = e_2(w)$, then

$$x = e_1(x) = e_1e_2(w) = 0.$$

Hence
$$e(V) = e_1(V) \oplus e_2(V).$$

Since e_1 and e_2 are nonzero, both $e_1(V)$ and $e_2(V)$ are nonzero. Thus $e(V)$ is decomposable.

Conversely, suppose that $e(V)$ is decomposable:

$$e(V) = U_1 \oplus U_2$$

with $U_1, U_2 \neq 0$. Let
$$p_i \colon e(V) \to e(V)$$

be the projection onto U_i ($i = 1, 2$). Define

$$\widetilde{e}_i := p_i \circ e \in \mathrm{End}_A(V).$$

Since e acts as the identity on $e(V)$, we have $e \circ p_i = p_i$, and therefore

$$\widetilde{e}_i^2 = p_i e p_i e = p_i p_i e = \widetilde{e}_i.$$

Also,

$$\widetilde{e}_1 \widetilde{e}_2 = p_1 e p_2 e = p_1 p_2 e = 0, \qquad \widetilde{e}_2 \widetilde{e}_1 = 0,$$

and

$$\widetilde{e}_1 + \widetilde{e}_2 = (p_1 + p_2)e = e.$$

Since $U_i \neq 0$, each $\widetilde{e}_i \neq 0$. Thus e is not primitive.

Hence e is primitive if and only if $e(V)$ is indecomposable.

(d) Assume first that $e_2 = u e_1 u^{-1}$ for some invertible $u \in \mathrm{End}_A(V)$. Then for any $x = e_1(v) \in e_1(V)$,

$$u(x) = u e_1(v) = e_2 u(v) \in e_2(V),$$

so $u(e_1(V)) \subseteq e_2(V)$. Conversely, if $y = e_2(w) \in e_2(V)$, then

$$y = u e_1 u^{-1}(w) \in u(e_1(V)).$$

Hence

$$u(e_1(V)) = e_2(V),$$

and therefore the restriction

$$u|_{e_1(V)} \colon e_1(V) \to e_2(V)$$

is an isomorphism of A-modules. So

$$e_1(V) \cong e_2(V).$$

Conversely, suppose

$$e_1(V) \cong e_2(V).$$

Since V is semisimple with finitely many simple summands, we may write

$$V \cong \bigoplus_{j=1}^{r} S_j^{n_j},$$

where $S_1, \ldots, S_r$ are pairwise nonisomorphic simple A-modules. Because $e_i(V)$ is a submodule of a semisimple module, it is semisimple, so

$$e_i(V) \cong \bigoplus_{j=1}^{r} S_j^{m_j}$$

for some integers m_j, and the m_j are the same for $i = 1, 2$ since $e_1(V) \cong e_2(V)$.

By part (a),

$$V = e_i(V) \oplus \ker(e_i).$$

Therefore

$$\ker(e_i) \cong \bigoplus_{j=1}^{r} S_j^{n_j - m_j},$$

and hence

$$\ker(e_1) \cong \ker(e_2).$$

Choose isomorphisms

$$\alpha\colon e_1(V) \xrightarrow{\sim} e_2(V), \qquad \beta\colon \ker(e_1) \xrightarrow{\sim} \ker(e_2).$$

Using the decompositions

$$V = e_1(V) \oplus \ker(e_1) \qquad \text{and} \qquad V = e_2(V) \oplus \ker(e_2),$$

define

$$u\colon V \to V, \qquad u(x+y) = \alpha(x) + \beta(y) \quad \bigl(x \in e_1(V),\ y \in \ker(e_1)\bigr).$$

Then u is an invertible A-endomorphism of V. For $x \in e_1(V)$ and $y \in \ker(e_1)$,

$$ue_1(x+y) = u(x) = \alpha(x),$$

while

$$e_2u(x+y) = e_2\bigl(\alpha(x) + \beta(y)\bigr) = \alpha(x),$$

because e_2 is the identity on $e_2(V)$ and zero on $\ker(e_2)$. Hence

$$ue_1 = e_2u,$$

so

$$e_2 = ue_1u^{-1}.$$

Thus e_1 and e_2 are conjugate by an invertible element of $\mathrm{End}_A(V)$.

(e) Let $V = k^n$. Then

$$M_n(k) \cong \mathrm{End}_k(V).$$

Let $e \in M_n(k)$ be a primitive idempotent. By part (c), $e(V)$ is indecomposable as a k-vector space. But a nonzero k-vector space is indecomposable if and only if it has dimension 1: indeed, if $\dim_k W \geq 2$, then any nonzero proper subspace has a complement, so W is decomposable.

Therefore

$$\dim_k e(V) = 1.$$

Since e is idempotent, part (a) gives

$$V = e(V) \oplus \ker(e).$$

Choose a basis $v_1, \ldots, v_n$ of V such that v_1 spans $e(V)$ and $v_2, \ldots, v_n$ is a basis of $\ker(e)$. Then

$$e(v_1) = v_1, \qquad e(v_i) = 0 \quad (i \geq 2).$$

Hence the matrix of e with respect to this basis is

$$E_{11} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Therefore e is conjugate to E_{11} by some element of $GL_n(k)$. It follows that any two primitive idempotents in $M_n(k)$ are conjugate to each other.

□

Exercise 15.7.2. *Prove: No idempotent e of A is contained in $J(A)$. [Hint: $e(1-e) = 0$.]*

Proof. Assume that $e \in J(A)$ and that e is idempotent, so

$$e^2 = e.$$

Since $e \in J(A)$, a standard characterization of the Jacobson radical implies that

$$1 - e$$

is a unit in A.

On the other hand,

$$e(1-e) = e - e^2 = 0.$$

Multiplying this equality on the right by $(1-e)^{-1}$, we obtain

$$e = 0.$$

Therefore any idempotent contained in $J(A)$ must be zero. In particular, no nonzero idempotent of A lies in $J(A)$. □

Exercise 15.7.3. *Prove: Let A be an Artinian ring with Artinian center $Z(A)$. Then*

$$J(Z(A)) = J(A) \cap Z(A).$$

Proof. Let

$$B := Z(A).$$

We prove the two inclusions separately.

First, we show that

$$J(B) \subseteq J(A) \cap B.$$

Take $z \in J(B)$. Since B is Artinian, its Jacobson radical is nilpotent, so there exists $m \geq 1$ such that

$$J(B)^m = 0.$$

In particular,

$$z^m = 0.$$

Now let $x \in A$. Because $z \in Z(A)$, we have $xz = zx$, hence

$$(xz)^m = x^m z^m = 0.$$

So xz is nilpotent, and therefore

$$1 - xz$$

is invertible in A, with inverse

$$1 + xz + \cdots + (xz)^{m-1}.$$

By the standard characterization of the Jacobson radical, this implies

$$z \in J(A).$$

Since also $z \in B$, we get

$$z \in J(A) \cap B.$$

Thus

$$J(B) \subseteq J(A) \cap B.$$

Next, we show that

$$J(A) \cap B \subseteq J(B).$$

Take $z \in J(A) \cap B$. Since A is Artinian, $J(A)$ is nilpotent, so there exists $n \geq 1$ such that

$$J(A)^n = 0.$$

Hence

$$z^n = 0.$$

Now let $c \in B$. Since B is commutative, we have

$$(cz)^n = c^n z^n = 0.$$

Therefore cz is nilpotent in B, and so

$$1 - cz$$

is invertible in B, with inverse

$$1 + cz + \cdots + (cz)^{n-1}.$$

Again by the standard characterization of the Jacobson radical, this shows that

$$z \in J(B).$$

Hence

$$J(A) \cap B \subseteq J(B).$$

Combining the two inclusions, we conclude that

$$J(Z(A)) = J(B) = J(A) \cap B.$$

□

Exercise 15.7.4. *Prove: Let*

$$A = A_1 \oplus \cdots \oplus A_n$$

be a direct sum of 2*-sided ideals. Then*

(i)

$$Z(A) = Z(A_1) \oplus Z(A_2) \oplus \cdots \oplus Z(A_n).$$

(ii)

$$J(A) = J(A_1) \oplus J(A_2) \oplus \cdots \oplus J(A_n).$$

Proof. For $i \neq j$, since A_i and A_j are two-sided ideals and the sum is direct,

$$A_i A_j \subseteq A_i \cap A_j = 0.$$

Write

$$1_A = e_1 + \cdots + e_n, \qquad e_i \in A_i.$$

Then e_i is the identity element of A_i. Indeed, if $a_i \in A_i$, then all mixed products vanish, so

$$a_i = 1_A a_i = e_i a_i, \qquad a_i = a_i 1_A = a_i e_i.$$

Thus each A_i is a unital ring, with identity e_i. Moreover multiplication in A is coordinatewise with respect to the decomposition

$$A = A_1 \oplus \cdots \oplus A_n.$$

Hence

$$\Phi\colon A \longrightarrow A_1 \times \cdots \times A_n, \qquad a_1 + \cdots + a_n \longmapsto (a_1, \ldots, a_n)$$

is an isomorphism of rings. Since n is finite, we identify

$$A \cong A_1 \times \cdots \times A_n.$$

(i) First we prove

$$Z(A) \subseteq Z(A_1) \oplus \cdots \oplus Z(A_n).$$

Let $z \in Z(A)$, and write

$$z = z_1 + \cdots + z_n, \qquad z_i \in A_i.$$

Fix i, and let $x_i \in A_i$. Since z is central in A,

$$zx_i = x_i z.$$

But all mixed products vanish, so

$$zx_i = z_i x_i, \qquad x_i z = x_i z_i.$$

Hence $z_i x_i = x_i z_i$ for every $x_i \in A_i$, and therefore

$$z_i \in Z(A_i).$$

Since this holds for every i, we get

$$z \in Z(A_1) \oplus \cdots \oplus Z(A_n).$$

Conversely, suppose

$$z = z_1 + \cdots + z_n, \qquad z_i \in Z(A_i).$$

Let

$$a = a_1 + \cdots + a_n, \qquad a_i \in A_i.$$

Again using $A_i A_j = 0$ for $i \neq j$, we have

$$za = \sum_{i=1}^{n} z_i a_i, \qquad az = \sum_{i=1}^{n} a_i z_i.$$

Since $z_i \in Z(A_i)$, each $z_i a_i = a_i z_i$. Hence $za = az$, so $z \in Z(A)$. Therefore

$$Z(A) = Z(A_1) \oplus \cdots \oplus Z(A_n).$$

(ii) First we prove

$$J(A_1) \oplus \cdots \oplus J(A_n) \subseteq J(A).$$

Let

$$x = x_1 + \cdots + x_n, \qquad x_i \in J(A_i).$$

Take any

$$a = a_1 + \cdots + a_n \in A, \qquad a_i \in A_i.$$

Since $J(A_i)$ is an ideal of A_i, we have $a_i x_i \in J(A_i)$. Hence $e_i - a_i x_i$ is invertible in A_i. Choose $u_i \in A_i$ such that

$$u_i(e_i - a_i x_i) = e_i.$$

Put $u = u_1 + \cdots + u_n$. Since multiplication is coordinatewise,

$$u(1_A - ax) = \sum_{i=1}^{n} u_i(e_i - a_i x_i) = \sum_{i=1}^{n} e_i = 1_A.$$

Therefore $1_A - ax$ has a left inverse for every $a \in A$. By the Jacobson radical criterion,

$$x \in J(A).$$

Thus

$$J(A_1) \oplus \cdots \oplus J(A_n) \subseteq J(A).$$

Conversely, we prove

$$J(A) \subseteq J(A_1) \oplus \cdots \oplus J(A_n).$$

Let

$$x = x_1 + \cdots + x_n \in J(A), \qquad x_i \in A_i.$$

Fix i, and let $a_i \in A_i$. Since $x \in J(A)$, the element

$$1_A - a_i x = 1_A - a_i x_i$$

has a left inverse in A. Thus there exists $v = v_1 + \cdots + v_n \in A$, with $v_j \in A_j$, such that

$$v(1_A - a_i x_i) = 1_A.$$

Taking the A_i-component of this equality gives

$$v_i(e_i - a_i x_i) = e_i.$$

Hence $e_i - a_i x_i$ has a left inverse in A_i for every $a_i \in A_i$. By the Jacobson radical criterion applied inside A_i,

$$x_i \in J(A_i).$$

Since i was arbitrary,

$$x \in J(A_1) \oplus \cdots \oplus J(A_n).$$

Combining the two inclusions, we conclude that

$$J(A) = J(A_1) \oplus \cdots \oplus J(A_n).$$

□

Exercise 15.7.5. *Let*

$$\phi\colon U \to V$$

be a homomorphism of A-modules. Show that

$$\phi(\operatorname{Soc} U) \subseteq \operatorname{Soc} V,$$

and that if ϕ is an isomorphism then ϕ restricts to an isomorphism

$$\operatorname{Soc} U \to \operatorname{Soc} V.$$

Proof. Recall that

$$\operatorname{Soc}(U) = \sum \{\, S \leq U \mid S \text{ is a simple } A\text{-submodule} \,\}.$$

We first show that

$$\phi(\operatorname{Soc} U) \subseteq \operatorname{Soc} V.$$

Let $S \leq U$ be a simple submodule. Then $\phi(S)$ is a submodule of V, and since S is simple, the restriction

$$\phi|_S\colon S \to \phi(S)$$

is either the zero map or an isomorphism. Hence $\phi(S)$ is either 0 or a simple submodule of V. In either case,

$$\phi(S) \subseteq \operatorname{Soc} V.$$

Since $\operatorname{Soc}(U)$ is the sum of all simple submodules $S \leq U$, we get

$$\phi(\operatorname{Soc} U) = \sum_{S \text{ simple } \leq U} \phi(S) \subseteq \operatorname{Soc} V.$$

Now assume that ϕ is an isomorphism. Then $\phi^{-1}\colon V \to U$ is also an A-module homomorphism. Applying the first part to ϕ^{-1}, we obtain

$$\phi^{-1}(\operatorname{Soc} V) \subseteq \operatorname{Soc} U.$$

Applying ϕ to both sides gives

$$\operatorname{Soc} V \subseteq \phi(\operatorname{Soc} U).$$

Together with the already proved inclusion

$$\phi(\operatorname{Soc} U) \subseteq \operatorname{Soc} V,$$

we conclude that

$$\phi(\operatorname{Soc} U) = \operatorname{Soc} V.$$

Therefore the restriction

$$\phi|_{\operatorname{Soc} U} \colon \operatorname{Soc} U \to \operatorname{Soc} V$$

is surjective. It is also injective because ϕ itself is injective. Hence

$$\phi|_{\operatorname{Soc} U} \colon \operatorname{Soc} U \xrightarrow{\sim} \operatorname{Soc} V$$

is an isomorphism of A-modules. □

Exercise 15.7.6. *If*

$$\varphi \colon A \to A'$$

is an epimorphism of rings, then

$$\varphi(J(A)) \subseteq J(A').$$

Proof. Let $a \in J(A)$. We show that $\varphi(a) \in J(A')$.

We use the standard characterization of the Jacobson radical:

$$x \in J(R) \iff \text{for every } r \in R,\ \ 1 - rx \text{ has a left inverse in } R.$$

So let $y' \in A'$. Since $\varphi \colon A \to A'$ is surjective, there exists $y \in A$ such that

$$\varphi(y) = y'.$$

Because $a \in J(A)$, the element

$$1 - ya$$

has a left inverse in A. Thus there exists $b \in A$ such that

$$b(1 - ya) = 1.$$

Apply φ to this equality. Since φ is a ring homomorphism, we get

$$\varphi(b)\,\varphi(1 - ya) = \varphi(1) = 1,$$

that is,

$$\varphi(b)\bigl(1 - \varphi(y)\varphi(a)\bigr) = 1.$$

Using $\varphi(y) = y'$, this becomes

$$\varphi(b)(1 - y'\varphi(a)) = 1.$$

Hence $1 - y'\varphi(a)$ has a left inverse in A'.

Since $y' \in A'$ was arbitrary, the above characterization implies that

$$\varphi(a) \in J(A').$$

Therefore

$$\varphi(J(A)) \subseteq J(A').$$

$\square$

15.8 Homework Week 8

Exercise 15.8.1. *Let A be a finite-dimensional algebra over a field k. Suppose that V is a finitely generated A-module, and that a certain simple A-module S occurs as a composition factor of V with multiplicity 1. Suppose that there exist nonzero homomorphisms*

$$S \longrightarrow V \qquad \text{and} \qquad V \longrightarrow S.$$

Prove that S is a direct summand of V.

Proof. Since A is finite-dimensional over k and V is finitely generated over A, the module V is finite-dimensional over k. Hence V has finite length.

Let

$$\iota : S \longrightarrow V, \qquad \rho : V \longrightarrow S$$

be nonzero A-homomorphisms. Since S is simple, ι is injective and ρ is surjective.

We claim that

$$\rho \circ \iota \neq 0.$$

Suppose otherwise that $\rho \circ \iota = 0$. Then

$$\iota(S) \subseteq \ker \rho.$$

Thus S occurs as a composition factor of $\ker \rho$. Since ρ is surjective, we have a short exact sequence

$$0 \longrightarrow \ker \rho \longrightarrow V \longrightarrow S \longrightarrow 0.$$

By additivity of composition multiplicities in short exact sequences,

$$[V : S] = [\ker \rho : S] + [S : S] \geq 1 + 1 = 2,$$

contradicting the assumption that S occurs in V with multiplicity 1. Therefore

$$\rho \circ \iota \neq 0.$$

By Schur's lemma, the nonzero endomorphism

$$\rho \circ \iota : S \longrightarrow S$$

is an isomorphism. Define

$$\sigma = \iota \circ (\rho \circ \iota)^{-1} : S \longrightarrow V.$$

Then

$$\rho \circ \sigma = \rho \circ \iota \circ (\rho \circ \iota)^{-1} = \mathrm{id}_S .$$

Hence $\rho : V \to S$ splits. Therefore

$$V = \ker \rho \oplus \sigma(S),$$

and since $\sigma(S) \cong S$, the simple module S is a direct summand of V. $\square$

Exercise 15.8.2. *Suppose that U, V, W are modules over a finite-dimensional algebra A over a field, and let*

$$p_W : P_W \longrightarrow W$$

be a projective cover of W. Assume that all modules involved are finite-dimensional.

(a) Show that a homomorphism

$$\alpha : U \longrightarrow W$$

is an essential epimorphism if and only if there is a surjective homomorphism

$$h : P_W \longrightarrow U$$

such that the composite

$$P_W \xrightarrow{h} U \xrightarrow{\alpha} W$$

is a projective cover of W. In this situation, show that

$$h : P_W \longrightarrow U$$

must be a projective cover of U.

(b) Prove the following "extension and converse" to Nakayama's lemma: let V be any submodule of U. Then the canonical epimorphism

$$q : U \longrightarrow U/V$$

is an essential epimorphism if and only if

$$V \subseteq \operatorname{Rad} U.$$

Proof. **(a)** Recall that an epimorphism $f : X \to Y$ is essential if whenever $X_0 \le X$ satisfies $f(X_0) = Y$, one must have $X_0 = X$.

First suppose that

$$\alpha : U \longrightarrow W$$

is an essential epimorphism. Since P_W is projective and α is surjective, the map $p_W : P_W \to W$ lifts through α. Thus there exists an A-homomorphism

$$h : P_W \longrightarrow U$$

such that

$$\alpha \circ h = p_W.$$

Because p_W is surjective, we have

$$\alpha(h(P_W)) = W.$$

Since α is essential, this forces

$$h(P_W) = U.$$

Hence h is surjective. Moreover, the composite

$$P_W \xrightarrow{h} U \xrightarrow{\alpha} W$$

is precisely p_W, hence is a projective cover of W.

Conversely, suppose that there exists a surjective homomorphism

$$h : P_W \longrightarrow U$$

such that

$$\alpha \circ h : P_W \longrightarrow W$$

is a projective cover of W. In particular, $\alpha \circ h$ is an essential epimorphism. Since h is surjective, α is also surjective.

Let $U_0 \leq U$ be a submodule such that

$$\alpha(U_0) = W.$$

Then

$$(\alpha \circ h)\big(h^{-1}(U_0)\big) = W.$$

Since $\alpha \circ h$ is essential, we must have

$$h^{-1}(U_0) = P_W.$$

Applying h, and using the surjectivity of h, we get

$$U = h(P_W) \subseteq U_0.$$

Hence $U_0 = U$. Therefore $\alpha : U \to W$ is an essential epimorphism.

Finally, we show that in this situation

$$h : P_W \longrightarrow U$$

is a projective cover of U. We already know that P_W is projective and that h is surjective. It remains to show that h is essential.

Let $X \leq P_W$ be a submodule such that

$$h(X) = U.$$

Then

$$(\alpha \circ h)(X) = \alpha(U) = W.$$

Since $\alpha \circ h$ is essential, we obtain

$$X = P_W.$$

Thus h is an essential epimorphism. Therefore $h : P_W \to U$ is a projective cover of U.

(b) Let

$$q : U \longrightarrow U/V$$

be the canonical epimorphism.

First suppose that q is essential. We prove that

$$V \subseteq \operatorname{Rad} U.$$

If not, then there exists a maximal submodule M of U such that

$$V \nsubseteq M.$$

Since M is maximal and $V \nsubseteq M$, we have

$$M + V = U.$$

Therefore

$$q(M) = U/V.$$

But $M \neq U$, contradicting the essentiality of q. Hence

$$V \subseteq \operatorname{Rad} U.$$

Conversely, suppose that

$$V \subseteq \operatorname{Rad} U.$$

Let $X \leq U$ be a submodule such that

$$q(X) = U/V.$$

This is equivalent to

$$X + V = U.$$

We show that $X = U$. Suppose, for contradiction, that $X \neq U$. Since U is finite-dimensional, X is contained in some maximal submodule M of U. Since

$$V \subseteq \operatorname{Rad} U \subseteq M,$$

we get

$$U = X + V \subseteq M,$$

which is impossible because $M \neq U$. Therefore $X = U$, and hence q is an essential epimorphism.

Thus

$$U \longrightarrow U/V$$

is an essential epimorphism if and only if

$$V \subseteq \operatorname{Rad} U.$$

$\square$

Exercise 15.8.3. *Let v be a valuation of a field K. Prove the following properties:*

(i) $v(\pm 1) = 0$.

(ii) $v(a) = v(-a)$.

(iii) $v(a^{-1}) = -v(a)$.

(iv) If $v(b) < v(a)$, then

$$v(a+b) = v(b).$$

Proof. We use the defining properties of a valuation:

$$v(xy) = v(x) + v(y), \qquad v(x+y) \geq \min\{v(x), v(y)\}.$$

(i) Since $1 \cdot 1 = 1$, we have

$$v(1) = v(1 \cdot 1) = v(1) + v(1).$$

Hence

$$v(1) = 0.$$

Also,

$$(-1)^2 = 1,$$

so

$$2v(-1) = v((-1)^2) = v(1) = 0.$$

Therefore

$$v(-1) = 0.$$

Thus

$$v(\pm 1) = 0.$$

(ii) For $a \in K^{\times}$,

$$-a = (-1)a.$$

Therefore

$$v(-a) = v((-1)a) = v(-1) + v(a) = v(a).$$

(iii) For $a \in K^\times$, we have
$$aa^{-1} = 1.$$
Thus
$$v(a) + v(a^{-1}) = v(aa^{-1}) = v(1) = 0.$$
Hence
$$v(a^{-1}) = -v(a).$$

(iv) Assume that $a, b \in K^\times$ and
$$v(b) < v(a).$$
First, $a + b \neq 0$; otherwise $a = -b$, and by part **(ii)** we would have
$$v(a) = v(-b) = v(b),$$
contradicting $v(b) < v(a)$.

By the valuation inequality,
$$v(a+b) \geq \min\{v(a), v(b)\} = v(b).$$
We now prove the reverse inequality. Since
$$b = (a+b) - a,$$
we have
$$v(b) = v((a+b) - a) \geq \min\{v(a+b), v(-a)\} = \min\{v(a+b), v(a)\}.$$
If $v(a+b) > v(b)$, then both $v(a+b)$ and $v(a)$ are strictly larger than $v(b)$, so
$$\min\{v(a+b), v(a)\} > v(b),$$
contradicting the previous inequality. Therefore
$$v(a+b) \leq v(b).$$
Combining this with
$$v(a+b) \geq v(b),$$
we obtain
$$v(a+b) = v(b).$$

□

15.9 Homework Midterm

Exercise 15.9.1. *Let R be a ring with 1.*

1. *(i) Let*

$$h : U \longrightarrow V$$

be a monomorphism of R-modules. Then the following two conditions are equivalent.

(a) A homomorphism

$$k : V \longrightarrow Y$$

must be a monomorphism whenever the composite map

$$U \xrightarrow{h} V \xrightarrow{k} Y$$

is a monomorphism.

(b) If $0 \neq W \subset V$, then

$$h(U) \cap W \neq 0.$$

The monomorphism h satisfying the above conditions is said to be essential. *If there is an injective R-module Q with an essential monomorphism*

$$h : V \longrightarrow Q,$$

then the diagram

$$V \xrightarrow{h} Q$$

is called an injective hull *of V.*

(ii) Prove that any R-module V has an injective hull.

2. *(i) Prove that the following holds for an idempotent $e \in R$:*

$$J(eRe) = eJ(R)e = eRe \cap J(R).$$

(ii) Prove that

$$J(M_n(R)) = M_n(J(R)).$$

Here $M_n(R)$ means the matrix ring.

3. *Prove that a submodule of a finitely generated free module over a PID is free.*

Proof of 1(i). We identify U with its image $h(U) \subseteq V$.

First assume (a). Suppose that there exists a nonzero submodule $W \subseteq V$ such that

$$h(U) \cap W = 0.$$

Let

$$q : V \longrightarrow V/W$$

be the canonical quotient map. Since $h(U) \cap W = 0$, the composite

$$U \xrightarrow{h} V \xrightarrow{q} V/W$$

is a monomorphism. Hence by (a), q must be a monomorphism. But

$$\ker q = W \neq 0,$$

a contradiction. Therefore for every nonzero submodule $W \subseteq V$, one has

$$h(U) \cap W \neq 0.$$

Thus (b) holds.

Conversely, assume (b). Let

$$k : V \longrightarrow Y$$

be an R-homomorphism such that kh is a monomorphism. We prove that k is a monomorphism.

Let

$$W = \ker k.$$

If $W \neq 0$, then by (b),

$$h(U) \cap W \neq 0.$$

Choose $0 \neq h(u) \in h(U) \cap W$. Then

$$k(h(u)) = 0.$$

Since kh is a monomorphism, this implies $u = 0$, hence $h(u) = 0$, a contradiction. Therefore $W = 0$, so k is a monomorphism.

Hence (a) and (b) are equivalent. □

Definition 15.9.2 (Essential extension)**.** *Let $V \subseteq E$ be an inclusion of R-modules. We say that V is* essential *in E, or that E is an* essential extension *of V, if for every nonzero submodule $W \subseteq E$, one has*

$$W \cap V \neq 0.$$

Equivalently, every nonzero submodule of E must meet V nontrivially.

Remark 15.9.3. *Intuitively, the condition $V \subseteq E$ being essential means that E contains no nonzero submodule which is completely disjoint from V. Thus E is obtained from V in a way that does not create an independent "new part" away from V.*

Lemma 15.9.4 (Existence of a maximal essential extension inside a fixed ambient module)**.** *Let V be an R-module, and suppose that V is embedded into an R-module*

I:

$$V \subseteq I.$$

Consider the set

$$\mathcal{S} = \{\, E \mid V \subseteq E \subseteq I,\ \textit{and } V \subseteq E \textit{ is an essential extension} \,\}.$$

Then $\mathcal{S}$ has a maximal element.

Proof. First, $\mathcal{S} \neq \varnothing$. Indeed, $V \in \mathcal{S}$, because $V \subseteq V$ is clearly essential: if $0 \neq W \subseteq V$, then

$$W \cap V = W \neq 0.$$

We partially order $\mathcal{S}$ by inclusion. To apply Zorn's lemma, we show that every chain in $\mathcal{S}$ has an upper bound in $\mathcal{S}$.

Let

$$\{E_\alpha\}_{\alpha \in A}$$

be a chain in $\mathcal{S}$. This means that for any $\alpha, \beta \in A$, either

$$E_\alpha \subseteq E_\beta \quad \text{or} \quad E_\beta \subseteq E_\alpha.$$

Define

$$E = \bigcup_{\alpha \in A} E_\alpha.$$

Because the family $\{E_\alpha\}_{\alpha \in A}$ is a chain, the union E is again an R-submodule of I. Clearly,

$$V \subseteq E \subseteq I.$$

It remains to prove that $V \subseteq E$ is essential. Let

$$0 \neq W \subseteq E$$

be a nonzero submodule. Choose a nonzero element

$$0 \neq w \in W.$$

Since $E = \bigcup_{\alpha \in A} E_\alpha$, there exists some $\alpha \in A$ such that

$$w \in E_\alpha.$$

Hence

$$Rw \subseteq E_\alpha.$$

Because $w \neq 0$ and R has identity, we have

$$Rw \neq 0.$$

Since $E_\alpha \in \mathcal{S}$, the inclusion $V \subseteq E_\alpha$ is essential. Therefore

$$Rw \cap V \neq 0.$$

But $Rw \subseteq W$, so

$$W \cap V \neq 0.$$

Thus $V \subseteq E$ is essential, and hence

$$E \in \mathcal{S}.$$

Therefore every chain in $\mathcal{S}$ has an upper bound in $\mathcal{S}$. By Zorn's lemma, $\mathcal{S}$ has a maximal element. Denote it by E. □

Remark 15.9.5. *The maximality of E means the following: if E' is another submodule of I such that*

$$E \subseteq E' \subseteq I$$

and $V \subseteq E'$ is still essential, then necessarily

$$E' = E.$$

Equivalently, there is no strictly larger submodule E' with

$$E \subsetneq E' \subseteq I$$

such that $V \subseteq E'$ remains an essential extension.

Remark 15.9.6. *In the proof of the existence of injective hulls, one usually first uses the standard fact that every R-module V can be embedded into some injective module I. Then the above lemma gives a maximal essential extension*

$$V \subseteq E \subseteq I.$$

The remaining step is to prove that this maximal essential extension E is itself injective. Once this is done, the inclusion

$$V \hookrightarrow E$$

is an injective hull of V.

Lemma 15.9.7 (Baer's criterion)**.** *Let R be a ring with identity, and let E be a left R-module. Then E is injective if and only if for every left ideal $\mathfrak{a} \subseteq R$, every R-homomorphism*

$$f : \mathfrak{a} \longrightarrow E$$

can be extended to an R-homomorphism

$$\widetilde{f} : R \longrightarrow E.$$

Equivalently, for every left ideal $\mathfrak{a} \subseteq R$*, every diagram*

$$\begin{tikzcd} \mathfrak{a} \arrow[r, "f"] \arrow[d, hook] & E \\ R \arrow[ur, dashed, "\widetilde{f}"'] & \end{tikzcd}$$

can be completed by some $\widetilde{f} : R \to E$*.*

Proof. First suppose that E is injective. Let $\mathfrak{a} \subseteq R$ be a left ideal, and let

$$f : \mathfrak{a} \longrightarrow E$$

be an R-homomorphism. Since $\mathfrak{a} \hookrightarrow R$ is a monomorphism of left R-modules, the injectivity of E implies that f extends to an R-homomorphism

$$\widetilde{f} : R \longrightarrow E.$$

Thus the stated condition holds.

Conversely, assume that every homomorphism from a left ideal of R to E can be extended to R. We prove that E is injective.

Let $A \subseteq B$ be an inclusion of left R-modules, and let

$$f : A \longrightarrow E$$

be an R-homomorphism. We will prove that f extends to B.

Consider the set $\mathcal{S}$ of pairs (C, g), where

$$A \subseteq C \subseteq B$$

is a submodule and

$$g : C \longrightarrow E$$

is an R-homomorphism satisfying

$$g|_A = f.$$

We partially order $\mathcal{S}$ by declaring

$$(C, g) \leq (C', g')$$

if

$$C \subseteq C' \quad \text{and} \quad g'|_C = g.$$

The set $\mathcal{S}$ is nonempty, since $(A, f) \in \mathcal{S}$. If

$$\{(C_i, g_i)\}_{i \in I}$$

is a chain in $\mathcal{S}$, then

$$C = \bigcup_{i \in I} C_i$$

is a submodule of B, and we may define

$$g : C \longrightarrow E$$

by

$$g(c) = g_i(c)$$

whenever $c \in C_i$. This is well-defined because the pairs form a chain. Hence (C, g) is an upper bound of the chain. By Zorn's lemma, $\mathcal{S}$ has a maximal element, say

$$(C, g).$$

We claim that $C = B$. Suppose not. Choose

$$b \in B \setminus C.$$

Define

$$\mathfrak{a} = \{\, r \in R \mid rb \in C \,\}.$$

Then $\mathfrak{a}$ is a left ideal of R. Indeed, if $r, s \in \mathfrak{a}$, then

$$(r + s)b = rb + sb \in C,$$

and if $t \in R$, then

$$(tr)b = t(rb) \in C.$$

Define an R-homomorphism

$$h : \mathfrak{a} \longrightarrow E$$

by

$$h(r) = g(rb).$$

This is well-defined and R-linear because g is R-linear.

By the assumed condition, h extends to an R-homomorphism

$$\widetilde{h} : R \longrightarrow E.$$

Let

$$e = \widetilde{h}(1).$$

Then for every $r \in R$,

$$\widetilde{h}(r) = re,$$

and in particular, for every $r \in \mathfrak{a}$,

$$g(rb) = h(r) = \widetilde{h}(r) = re.$$

Now define

$$g' : C + Rb \longrightarrow E$$

by

$$g'(c + rb) = g(c) + re, \qquad c \in C,\ r \in R.$$

We check that g' is well-defined. Suppose

$$c + rb = c' + r'b.$$

Then

$$(r - r')b = c' - c \in C,$$

so $r - r' \in \mathfrak{a}$. Hence

$$g((r - r')b) = (r - r')e.$$

But

$$g((r - r')b) = g(c' - c) = g(c') - g(c).$$

Therefore

$$g(c') - g(c) = (r - r')e,$$

which implies

$$g(c) + re = g(c') + r'e.$$

Thus g' is well-defined. It is straightforward to check that g' is an R-homomorphism.

Moreover, g' extends g, since

$$g'(c) = g'(c + 0b) = g(c).$$

Thus

$$(C, g) < (C + Rb, g')$$

unless $C + Rb = C$. But $b \notin C$, so $C + Rb \neq C$. This contradicts the maximality of (C, g).

Therefore $C = B$. Hence $f : A \to E$ extends to a homomorphism $B \to E$. This proves that E is injective. □

Proof of 1(ii). It remains to prove that E is injective. We use Baer's criterion. Suppose, for contradiction, that E is not injective. Then there exists a left ideal

$$I \subseteq R$$

and an R-homomorphism

$$f : I \longrightarrow E$$

which cannot be extended to a homomorphism $R \to E$.

Construct an R-module

$$P = (E \oplus R)/N,$$

where
$$N = \{(f(a), -a) \mid a \in I\}.$$
The natural map
$$E \longrightarrow P, \qquad e \longmapsto (e, 0) + N$$
is injective. Indeed, if $(e, 0) \in N$, then
$$(e, 0) = (f(a), -a)$$
for some $a \in I$, hence $a = 0$ and $e = f(0) = 0$.

Choose, by Zorn's lemma, a submodule $C \subseteq P$ maximal subject to
$$C \cap E = 0.$$
Then the image of E in P/C is essential. Indeed, if D/C is a nonzero submodule of P/C, then $D \supsetneq C$. By maximality of C, we must have
$$D \cap E \neq 0.$$
Hence
$$(D/C) \cap E \neq 0.$$
Thus $E \subseteq P/C$ is an essential extension.

Since $V \subseteq E$ is essential and $E \subseteq P/C$ is essential, by transitivity $V \subseteq P/C$ is essential. But E was chosen to be a maximal essential extension of V. Therefore
$$P/C = E.$$

Let
$$x = (0, 1) + N + C \in P/C.$$
Since $P/C = E$, there exists $e \in E$ such that
$$x = e.$$
For every $a \in I$, we have in P/C
$$ax = (0, a) + N + C = (f(a), 0) + N + C = f(a).$$
On the other hand, since $x = e$, we also have
$$ax = ae.$$
Therefore
$$f(a) = ae \qquad \text{for all } a \in I.$$

Hence f extends to an R-homomorphism

$$R \longrightarrow E, \qquad r \longmapsto re,$$

contradicting the choice of f.

Thus E is injective. Since $V \subseteq E$ is an essential monomorphism and E is injective, the inclusion

$$V \hookrightarrow E$$

is an injective hull of V. □

Proof of 2(i). We prove

$$J(eRe) = eJ(R)e = eRe \cap J(R).$$

First observe that

$$eJ(R)e \subseteq eRe \cap J(R),$$

because $J(R)$ is a two-sided ideal of R. Conversely, if

$$x \in eRe \cap J(R),$$

then $x = exe$, so

$$x \in eJ(R)e.$$

Hence

$$eJ(R)e = eRe \cap J(R).$$

It remains to prove

$$J(eRe) = eJ(R)e.$$

First let $a \in eJ(R)e$. Then $a \in eRe$. For every $y \in eRe$, we have

$$ya \in J(R).$$

By the usual characterization of the Jacobson radical, $1 - ya$ has a left inverse in R. Thus there exists $u \in R$ such that

$$u(1 - ya) = 1.$$

Multiplying by e on both sides gives

$$eue(e - ya) = e.$$

Hence $e - ya$ has a left inverse in the ring eRe, whose identity is e. Therefore

$$a \in J(eRe).$$

Thus

$$eJ(R)e \subseteq J(eRe).$$

Conversely, let

$$a \in J(eRe).$$

We show that $a \in J(R)$. It is enough to show that a annihilates every simple left R-module.

Let S be a simple left R-module. Then eS is either 0 or a simple left eRe-module. Indeed, if $0 \neq T \subseteq eS$ is an eRe-submodule and $0 \neq t \in T$, then $Rt = S$, and hence

$$eS = eRt = eRe\,t \subseteq T.$$

Therefore $T = eS$.

Since $a \in J(eRe)$, it annihilates every simple eRe-module. Hence

$$a(eS) = 0.$$

But $a \in eRe$, so $a = ae$. Therefore, for every $s \in S$,

$$as = aes = a(es) = 0.$$

Thus a annihilates every simple left R-module, and hence

$$a \in J(R).$$

Since also $a \in eRe$, we get

$$a \in eRe \cap J(R) = eJ(R)e.$$

Therefore

$$J(eRe) \subseteq eJ(R)e.$$

Combining the two inclusions gives

$$J(eRe) = eJ(R)e = eRe \cap J(R).$$

□

Proof of 2(ii). Let

$$S = M_n(R).$$

We prove

$$J(S) = M_n(J(R)).$$

Let E_{ij} denote the standard matrix units. Applying part $2(i)$ to the idempotent $E_{11} \in S$, we obtain

$$J(E_{11}SE_{11}) = E_{11}J(S)E_{11}.$$

But

$$E_{11}SE_{11} \cong R,$$

so

$$E_{11}J(S)E_{11} \cong J(R).$$

Now take $A = (a_{ij}) \in J(S)$. Then

$$E_{1i}AE_{j1} \in E_{11}J(S)E_{11}.$$

But

$$E_{1i}AE_{j1} = a_{ij}E_{11}.$$

Therefore $a_{ij} \in J(R)$ for all i, j. Hence

$$J(S) \subseteq M_n(J(R)).$$

Conversely, we show that

$$M_n(J(R)) \subseteq J(S).$$

It is enough to prove that for every $a \in J(R)$ and every i, j,

$$aE_{ij} \in J(S).$$

Let $X = (x_{pq}) \in S$. We must show that

$$I_n - X(aE_{ij})$$

has a left inverse in S.

Set

$$u = \begin{pmatrix} x_{1i}a \\ x_{2i}a \\ \vdots \\ x_{ni}a \end{pmatrix}, \qquad v^T = \begin{pmatrix} 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \end{pmatrix},$$

where the 1 occurs in the j-th position. Then

$$X(aE_{ij}) = uv^T,$$

and

$$v^T u = x_{ji}a.$$

Since $a \in J(R)$, the element

$$1 - x_{ji}a$$

has a left inverse in R. Hence there exists $c \in R$ such that

$$c(1 - x_{ji}a) = 1.$$

Equivalently,

$$c - c(v^T u) = 1.$$

Now compute:

$$(I_n + ucv^T)(I_n - uv^T) = I_n + u\big(c - 1 - c(v^T u)\big)v^T = I_n.$$

Thus $I_n - X(aE_{ij})$ has a left inverse for every $X \in S$. Therefore

$$aE_{ij} \in J(S).$$

Since $M_n(J(R))$ is generated additively by such elements aE_{ij}, we get

$$M_n(J(R)) \subseteq J(S).$$

Therefore

$$J(M_n(R)) = M_n(J(R)).$$

□

Proof of 3. Let R be a PID, and let F be a finitely generated free R-module. Then

$$F \cong R^n$$

for some $n \geq 0$. We prove by induction on n that every submodule of R^n is free.

For $n = 0$, the claim is trivial. For $n = 1$, every submodule of R is an ideal of R. Since R is a PID, every ideal is principal. Hence every submodule of R is free.

Assume now that $n \geq 2$, and suppose the result is true for R^{n-1}. Let

$$N \subseteq R^n$$

be a submodule. Consider the projection onto the first coordinate:

$$\pi : R^n \longrightarrow R.$$

Then $\pi(N)$ is an ideal of R. Since R is a PID, there exists $d \in R$ such that

$$\pi(N) = (d).$$

Let

$$K = N \cap \ker \pi.$$

Since

$$\ker \pi \cong R^{n-1},$$

the induction hypothesis implies that K is free.

We have an exact sequence

$$0 \longrightarrow K \longrightarrow N \xrightarrow{\pi} \pi(N) \longrightarrow 0.$$

Since

$$\pi(N) = (d)$$

is a principal ideal, it is a free R-module of rank 1 if $d \neq 0$, and it is the zero module if

$d = 0$. In either case, $\pi(N)$ is projective. Hence the short exact sequence splits. Therefore

$$N \cong K \oplus \pi(N).$$

Since both K and $\pi(N)$ are free, their direct sum is free. Hence N is free.

By induction, every submodule of a finitely generated free module over a PID is free. $\square$

15.10 Homework Week 9

Exercise 15.10.1. *Let φ be a multiplicative valuation of a field K. Then the following hold:*

1. *The sequence (a_n) is a Cauchy sequence if and only if*

$$\lim_{n\to\infty} \varphi(a_{n+1} - a_n) = 0.$$

2. *If*

$$\lim_{n\to\infty} a_n = a \neq 0,$$

then

$$\varphi(a_n) = \varphi(a)$$

for all sufficiently large n.

Proof. We use the metric on K induced by φ, namely

$$d(x, y) = \varphi(x - y).$$

1. Suppose first that (a_n) is a Cauchy sequence. Then, for every $\varepsilon > 0$, there exists N such that for all $m, n \geq N$,

$$\varphi(a_m - a_n) < \varepsilon.$$

Taking $m = n + 1$, we get

$$\varphi(a_{n+1} - a_n) < \varepsilon$$

for all $n \geq N$. Hence

$$\lim_{n\to\infty} \varphi(a_{n+1} - a_n) = 0.$$

Conversely, suppose that

$$\lim_{n\to\infty} \varphi(a_{n+1} - a_n) = 0.$$

We show that (a_n) is Cauchy. Let $\varepsilon > 0$ be given. Then there exists N such that for all $k \geq N$,

$$\varphi(a_{k+1} - a_k) < \varepsilon.$$

If $m > n \geq N$, then

$$a_m - a_n = (a_m - a_{m-1}) + (a_{m-1} - a_{m-2}) + \cdots + (a_{n+1} - a_n).$$

Since φ is non-archimedean, we have

$$\varphi(a_m - a_n) \leq \max_{n \leq k \leq m-1} \varphi(a_{k+1} - a_k) < \varepsilon.$$

Therefore (a_n) is a Cauchy sequence.

2. Suppose that

$$\lim_{n\to\infty} a_n = a \neq 0.$$

Then

$$\lim_{n\to\infty} \varphi(a_n - a) = 0.$$

Since $a \neq 0$, we have $\varphi(a) > 0$. Hence there exists N such that for all $n \geq N$,

$$\varphi(a_n - a) < \varphi(a).$$

For such n, we write

$$a_n = a + (a_n - a).$$

By the strong triangle inequality,

$$\varphi(a_n) \leq \max\{\varphi(a), \varphi(a_n - a)\} = \varphi(a).$$

On the other hand,

$$a = a_n - (a_n - a),$$

so

$$\varphi(a) \leq \max\{\varphi(a_n), \varphi(a_n - a)\}.$$

Since $\varphi(a_n - a) < \varphi(a)$, the maximum on the right-hand side must be $\varphi(a_n)$. Thus

$$\varphi(a) \leq \varphi(a_n).$$

Combining the two inequalities gives

$$\varphi(a_n) = \varphi(a)$$

for all $n \geq N$.

$\square$

Exercise 15.10.2. *Let φ be a multiplicative valuation of a field K. Let V be a finite dimensional K-space with basis $u_1, \ldots, u_n$. For*

$$v = a_1u_1 + \cdots + a_nu_n \in V,$$

define $\|v\|_0$ by

$$\|v\|_0 = \max\{\varphi(a_i) \mid 1 \leq i \leq n\}.$$

Then this is a norm on V. Moreover, the following hold concerning the metric induced by this norm:

1. *Let*

$$v_r = a_{r1}u_1 + \cdots + a_{rn}u_n \in V$$

be given for $r = 1, 2, \ldots$. Then the sequence (v_r) is a Cauchy sequence if and only if the sequence $(a_{ri})_r$ in K is a Cauchy sequence for every i.

Also,

$$\lim_{r\to\infty} v_r = \sum_{i=1}^{n} a_i u_i$$

if and only if

$$\lim_{r\to\infty} a_{ri} = a_i$$

for every i.

2. *If K is complete, then so is V.*

Proof. First we prove that $\|\cdot\|_0$ is a norm on V.

Let

$$v = a_1u_1 + \cdots + a_nu_n.$$

Then

$$\|v\|_0 = \max\{\varphi(a_i) \mid 1 \le i \le n\}.$$

Since $\varphi(a_i) \ge 0$ for every i, we have

$$\|v\|_0 \ge 0.$$

Moreover,

$$\|v\|_0 = 0$$

if and only if

$$\varphi(a_i) = 0$$

for every i, which is equivalent to

$$a_i = 0$$

for every i. Since $u_1, \ldots, u_n$ is a basis, this is equivalent to

$$v = 0.$$

Now let $\lambda \in K$. Then

$$\lambda v = \lambda a_1u_1 + \cdots + \lambda a_nu_n,$$

and therefore

$$\|\lambda v\|_0 = \max_{1\le i\le n} \varphi(\lambda a_i) = \max_{1\le i\le n} \varphi(\lambda)\varphi(a_i).$$

Hence

$$\|\lambda v\|_0 = \varphi(\lambda) \max_{1\le i\le n} \varphi(a_i) = \varphi(\lambda)\|v\|_0.$$

Finally, let

$$v = a_1u_1 + \cdots + a_nu_n, \qquad w = b_1u_1 + \cdots + b_nu_n.$$

Then

$$v + w = (a_1 + b_1)u_1 + \cdots + (a_n + b_n)u_n.$$

Thus

$$\|v + w\|_0 = \max_{1\le i\le n} \varphi(a_i + b_i).$$

Since φ is non-archimedean,

$$\varphi(a_i + b_i) \le \max\{\varphi(a_i), \varphi(b_i)\}$$

for every i. Hence

$$\|v + w\|_0 \le \max_{1\le i\le n} \max\{\varphi(a_i), \varphi(b_i)\}.$$

Therefore

$$\|v + w\|_0 \le \max\left\{\max_{1\le i\le n} \varphi(a_i), \max_{1\le i\le n} \varphi(b_i)\right\} = \max\{\|v\|_0, \|w\|_0\}.$$

So $\|\cdot\|_0$ is a non-archimedean norm on V.

1. Let

   $$v_r = a_{r1}u_1 + \cdots + a_{rn}u_n.$$

   For $r, s \ge 1$, we have

   $$v_r - v_s = (a_{r1} - a_{s1})u_1 + \cdots + (a_{rn} - a_{sn})u_n.$$

   Hence

   $$\|v_r - v_s\|_0 = \max_{1\le i\le n} \varphi(a_{ri} - a_{si}).$$

   Suppose first that (v_r) is Cauchy. Let i be fixed. Since

   $$\varphi(a_{ri} - a_{si}) \le \max_{1\le j\le n} \varphi(a_{rj} - a_{sj}) = \|v_r - v_s\|_0,$$

   it follows that $(a_{ri})_r$ is a Cauchy sequence in K.

   Conversely, suppose that for every i, the sequence $(a_{ri})_r$ is Cauchy in K. Let $\varepsilon > 0$ be given. For each i, there exists N_i such that for all $r, s \ge N_i$,

   $$\varphi(a_{ri} - a_{si}) < \varepsilon.$$

   Let

   $$N = \max\{N_1, \dots, N_n\}.$$

   Then for all $r, s \ge N$ and all i, we have

   $$\varphi(a_{ri} - a_{si}) < \varepsilon.$$

Therefore

$$\|v_r - v_s\|_0 = \max_{1 \le i \le n} \varphi(a_{ri} - a_{si}) < \varepsilon.$$

Hence (v_r) is Cauchy in V.

Next we prove the assertion about convergence. Let

$$v = \sum_{i=1}^{n} a_i u_i.$$

Then

$$v_r - v = (a_{r1} - a_1)u_1 + \cdots + (a_{rn} - a_n)u_n.$$

Hence

$$\|v_r - v\|_0 = \max_{1 \le i \le n} \varphi(a_{ri} - a_i).$$

Therefore

$$\lim_{r \to \infty} v_r = v$$

if and only if

$$\lim_{r \to \infty} \max_{1 \le i \le n} \varphi(a_{ri} - a_i) = 0.$$

Since there are only finitely many indices i, this is equivalent to

$$\lim_{r \to \infty} \varphi(a_{ri} - a_i) = 0$$

for every i, which is precisely

$$\lim_{r \to \infty} a_{ri} = a_i$$

for every i.

2. Suppose that K is complete. We show that V is complete.

Let (v_r) be a Cauchy sequence in V, where

$$v_r = a_{r1}u_1 + \cdots + a_{rn}u_n.$$

By part (1), for each i, the sequence $(a_{ri})_r$ is Cauchy in K. Since K is complete, there exists $a_i \in K$ such that

$$\lim_{r \to \infty} a_{ri} = a_i.$$

Define

$$v = \sum_{i=1}^{n} a_i u_i \in V.$$

Again by part (1), we get

$$\lim_{r \to \infty} v_r = v.$$

Thus every Cauchy sequence in V converges in V. Hence V is complete.

$\square$

15.11 Homework Week 10

Exercise 15.11.1. *Let R be a ring and let V be an R-module. Let I be an ideal of R. Suppose that d_I is defined on V and on R. Then:*

1. *Addition is continuous from $V \times V$ to V.*
2. *Multiplication is continuous from $R \times V$ to V.*
3. *If d_I is defined on the R-module W and $f \in \mathrm{Hom}_R(V, W)$, then f is continuous.*

Proof. Recall that, for an R-module M, the I-adic topology induced by d_I has a basis of neighbourhoods of $x \in M$ given by

$$x + I^n M, \qquad n \geq 1.$$

1. Let $(v_0, w_0) \in V \times V$. We show that addition is continuous at (v_0, w_0). Let

   $$v_0 + w_0 + I^n V$$

   be a neighbourhood of $v_0 + w_0$. If

   $$v - v_0 \in I^n V, \qquad w - w_0 \in I^n V,$$

   then
   $$(v + w) - (v_0 + w_0) = (v - v_0) + (w - w_0) \in I^n V.$$

   Hence
   $$v + w \in v_0 + w_0 + I^n V.$$

   Therefore addition $V \times V \to V$ is continuous.

2. Let $(r_0, v_0) \in R \times V$. We show that scalar multiplication is continuous at (r_0, v_0). Let

   $$r_0 v_0 + I^n V$$

   be a neighbourhood of $r_0 v_0$. Suppose

   $$r - r_0 \in I^n, \qquad v - v_0 \in I^n V.$$

   Then
   $$rv - r_0 v_0 = (r - r_0)v + r_0(v - v_0).$$

   Since $r - r_0 \in I^n$, we have
   $$(r - r_0)v \in I^n V.$$

   Also, because $I^n V$ is an R-submodule of V,

   $$r_0(v - v_0) \in I^n V.$$

Therefore

$$rv - r_0 v_0 \in I^n V.$$

Hence

$$rv \in r_0 v_0 + I^n V.$$

Thus multiplication $R \times V \to V$ is continuous.

3. Let $f \in \mathrm{Hom}_R(V, W)$. We first observe that

$$f(I^n V) \subseteq I^n W$$

for every $n \geq 1$. Indeed, if

$$x = \sum_{j=1}^{m} a_j v_j \in I^n V, \qquad a_j \in I^n,\ v_j \in V,$$

then

$$f(x) = f\left(\sum_{j=1}^{m} a_j v_j\right) = \sum_{j=1}^{m} a_j f(v_j) \in I^n W.$$

Now let $v_0 \in V$, and let

$$f(v_0) + I^n W$$

be a neighbourhood of $f(v_0)$. If

$$v - v_0 \in I^n V,$$

then

$$f(v) - f(v_0) = f(v - v_0) \in I^n W.$$

Hence

$$f(v) \in f(v_0) + I^n W.$$

Therefore f is continuous.

□

Exercise 15.11.2. *Let R be a commutative ring, and let A and B be R-algebras. Prove the following algebra isomorphisms:*

1.

$$M_n(A) \otimes_R B \cong M_n(A \otimes_R B).$$

2.

$$M_n(A) \otimes_R M_m(B) \cong M_{nm}(A \otimes_R B).$$

Proof. 1. Define a map

$$\Phi : M_n(A) \otimes_R B \longrightarrow M_n(A \otimes_R B)$$

on elementary tensors by

$$\Phi\big((a_{ij}) \otimes b\big) = (a_{ij} \otimes b).$$

This is well-defined and R-linear by the universal property of the tensor product.

We construct its inverse. Every element of $M_n(A \otimes_R B)$ can be written as a matrix whose (i, j)-entry is a finite sum of elementary tensors. Define

$$\Psi : M_n(A \otimes_R B) \longrightarrow M_n(A) \otimes_R B$$

by

$$\Psi\big((x_{ij})\big) = \sum_{i,j} \sum_{\ell} a_{ij\ell} E_{ij} \otimes b_{ij\ell},$$

whenever

$$x_{ij} = \sum_{\ell} a_{ij\ell} \otimes b_{ij\ell}.$$

This is well-defined because tensor product distributes over finite direct sums. It is immediate that

$$\Phi \circ \Psi = \mathrm{id}, \qquad \Psi \circ \Phi = \mathrm{id}\,.$$

Hence Φ is an R-module isomorphism.

It remains to check that Φ is an algebra homomorphism. Let $X = (x_{ij})$, $Y = (y_{ij}) \in M_n(A)$, and let $b, b' \in B$. Then

$$\Phi\big((X \otimes b)(Y \otimes b')\big) = \Phi(XY \otimes bb') = ((XY)_{ik} \otimes bb')_{i,k}.$$

On the other hand,

$$\Phi(X \otimes b)\Phi(Y \otimes b') = (x_{ij} \otimes b)_{i,j}(y_{jk} \otimes b')_{j,k},$$

whose (i, k)-entry is

$$\sum_j (x_{ij} \otimes b)(y_{jk} \otimes b') = \sum_j x_{ij} y_{jk} \otimes bb' = (XY)_{ik} \otimes bb'.$$

Thus

$$\Phi\big((X \otimes b)(Y \otimes b')\big) = \Phi(X \otimes b)\Phi(Y \otimes b').$$

Also,

$$\Phi(I_n \otimes 1_B) = I_n.$$

Therefore

$$M_n(A) \otimes_R B \cong M_n(A \otimes_R B)$$

as R-algebras.

2. Let E_{ij} denote the standard matrix units in $M_n(A)$, and let F_{kl} denote the standard matrix units in $M_m(B)$. We identify the rows and columns of $M_{nm}(A \otimes_R B)$ with

pairs

$$(i,k), \qquad 1 \le i \le n, \quad 1 \le k \le m.$$

Let $G_{(i,k),(j,l)}$ be the corresponding standard matrix unit in $M_{nm}(A \otimes_R B)$.

Define

$$\Theta : M_n(A) \otimes_R M_m(B) \longrightarrow M_{nm}(A \otimes_R B)$$

by

$$\Theta\big((aE_{ij}) \otimes (bF_{kl})\big) = (a \otimes b)G_{(i,k),(j,l)}.$$

Since

$$M_n(A) = \bigoplus_{i,j} AE_{ij}, \qquad M_m(B) = \bigoplus_{k,l} BF_{kl},$$

and tensor product distributes over finite direct sums, Θ is an R-module isomorphism.

We check multiplicativity on elementary generators. We have

$$(aE_{ij})(a'E_{i'j'}) = \delta_{j,i'}aa'E_{ij'},$$

and

$$(bF_{kl})(b'F_{k'l'}) = \delta_{l,k'}bb'F_{kl'}.$$

Therefore

$$\begin{aligned}&\Theta\left(\big((aE_{ij}) \otimes (bF_{kl})\big)\big((a'E_{i'j'}) \otimes (b'F_{k'l'})\big)\right)\\ &= \delta_{j,i'}\delta_{l,k'}(aa' \otimes bb')G_{(i,k),(j',l')}.\end{aligned}$$

On the other hand,

$$\begin{aligned}&\Theta\big((aE_{ij}) \otimes (bF_{kl})\big)\Theta\big((a'E_{i'j'}) \otimes (b'F_{k'l'})\big)\\ &= (a \otimes b)G_{(i,k),(j,l)}(a' \otimes b')G_{(i',k'),(j',l')}\\ &= \delta_{j,i'}\delta_{l,k'}(aa' \otimes bb')G_{(i,k),(j',l')}.\end{aligned}$$

Hence Θ preserves multiplication. It also sends the identity to the identity. Therefore

$$M_n(A) \otimes_R M_m(B) \cong M_{nm}(A \otimes_R B)$$

as R-algebras.

□

15.12 Homework Week 11

Exercise 15.12.1. *(i) Let (X,ρ) be a transitive* Set*-representation of G. For $x \in X$, let G_x denote the fixator of x in G. Then the map*

$$G/G_x \longrightarrow X, \qquad gG_x \longmapsto \rho(g)(x)$$

defines an isomorphism

$$(G/G_x, \rho_{G_x}) \xrightarrow{\sim} (X,\rho).$$

(ii) *For subgroups H and H' of G, the representations*

$$(G/H, \rho_H) \qquad \textit{and} \qquad (G/H', \rho_{H'})$$

are isomorphic if and only if H and H' are conjugate in G.

Proof. Write

$$g \cdot y := \rho(g)(y), \qquad g \in G,\ y \in X.$$

(i) Let

$$\Phi : G/G_x \longrightarrow X, \qquad gG_x \longmapsto g \cdot x.$$

We first show that Φ is well-defined. If $gG_x = g'G_x$, then $g' = gh$ for some $h \in G_x$. Hence

$$g' \cdot x = (gh) \cdot x = g \cdot (h \cdot x) = g \cdot x,$$

since $h \cdot x = x$.

The map Φ is surjective because the action is transitive. It is injective as well: if $\Phi(gG_x) = \Phi(g'G_x)$, then

$$g \cdot x = g' \cdot x.$$

Thus

$$(g'^{-1}g) \cdot x = x,$$

so $g'^{-1}g \in G_x$, and therefore $gG_x = g'G_x$.

Finally, Φ is G-equivariant. For $a \in G$,

$$\Phi\big(\rho_{G_x}(a)(gG_x)\big) = \Phi(agG_x) = ag \cdot x = \rho(a)(g \cdot x) = \rho(a)\big(\Phi(gG_x)\big).$$

Hence Φ is an isomorphism of G-sets.

(ii) Suppose first that

$$F : G/H \longrightarrow G/H'$$

is an isomorphism of G-sets. Write $F(H) = aH'$ for some $a \in G$. Since F is G-equivariant, it preserves stabilizers:

$$\mathrm{Stab}_G(H) = \mathrm{Stab}_G(F(H)).$$

But

$$\mathrm{Stab}_G(H) = H, \qquad \mathrm{Stab}_G(aH') = aH'a^{-1}.$$

Hence $H = aH'a^{-1}$, so H and H' are conjugate in G.

Conversely, suppose that H and H' are conjugate. Choose $a \in G$ such that

$$H = aH'a^{-1}.$$

Define

$$F : G/H \longrightarrow G/H', \qquad F(gH) = gaH'.$$

This is well-defined. Indeed, if $gH = g_1H$, then $g_1 = gh$ for some $h \in H$. Since $H = aH'a^{-1}$, we have $a^{-1}ha \in H'$. Therefore

$$g_1aH' = ghaH' = ga(a^{-1}ha)H' = gaH',$$

so $F(gH) = F(g_1H)$.

The map F is bijective, with inverse

$$F^{-1}(gH') = ga^{-1}H.$$

It is also G-equivariant, since for $t \in G$,

$$F(tgH) = tgaH' = tF(gH).$$

Hence

$$(G/H, \rho_H) \cong (G/H', \rho_{H'}).$$

□

Exercise 15.12.2. *Assume that G is the cyclic additive group $\mathbb{Z}/p\mathbb{Z}$, where p is a prime number, and let k be a field of characteristic p.*

Consider the morphism

$$G \longrightarrow \mathrm{GL}_2(k), \qquad \lambda \longmapsto \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}.$$

Prove that the line generated by the first standard basis vector of k^2 is G-stable but has no G-stable complement.

Proof. Let

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

be the standard basis of k^2. Denote the given representation by

$$\rho : G \longrightarrow \mathrm{GL}_2(k), \qquad \lambda \longmapsto \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}.$$

We identify $G = \mathbb{Z}/p\mathbb{Z}$ with the prime subfield $\mathbb{F}_p \subseteq k$.

Let

$$L := ke_1 \subseteq k^2.$$

For every $\lambda \in G$, we have

$$\rho(\lambda)e_1 = \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = e_1.$$

Hence $\rho(\lambda)L \subseteq L$ for every $\lambda \in G$, so L is G-stable.

We now prove that L has no G-stable complement. Suppose, for contradiction, that there exists a G-stable subspace $C \subseteq k^2$ such that

$$k^2 = L \oplus C.$$

Then C is one-dimensional and $C \neq L$. Hence C is generated by a vector of the form

$$v = ae_1 + e_2$$

for some $a \in k$.

Since C is G-stable, we must have

$$\rho(1)v \in C.$$

But

$$\rho(1)v = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ 1 \end{pmatrix} = \begin{pmatrix} a+1 \\ 1 \end{pmatrix} = v + e_1.$$

Since $C = kv$, there exists $\mu \in k$ such that

$$v + e_1 = \mu v.$$

Comparing the e_2-coordinates gives

$$1 = \mu.$$

Therefore

$$v + e_1 = v,$$

so $e_1 = 0$, a contradiction.

Thus no G-stable complement of $L = ke_1$ exists. $\square$

Exercise 15.12.3. *Let (M, ρ) and (N, σ) be two k-representations of G. Fix bases $(e_i)_{1 \leq i \leq m}$ and $(f_j)_{1 \leq j \leq n}$ of M and N, respectively. Then*

$$(e_i \otimes f_j)_{1 \leq i \leq m,\ 1 \leq j \leq n}$$

is a basis of $M \otimes_k N$. Denote by

$$(m, \rho), \qquad (n, \sigma), \qquad (mn, \rho \otimes \sigma)$$

the matrix representations associated with

$$(M, \rho), \qquad (N, \sigma), \qquad (M \otimes_k N, \rho \otimes \sigma),$$

respectively.

Determine the relation between $(\rho \otimes \sigma)(g)$ and $\rho(g), \sigma(g)$.

Proof. For $g \in G$, write the matrices of $\rho(g)$ and $\sigma(g)$ with respect to the chosen bases as

$$\rho(g) = (a_{\alpha i})_{1\le \alpha, i \le m}, \qquad \sigma(g) = (b_{\beta j})_{1 \le \beta, j \le n}.$$

Thus

$$\rho(g)e_i = \sum_{\alpha=1}^{m} a_{\alpha i} e_\alpha, \qquad \sigma(g) f_j = \sum_{\beta=1}^{n} b_{\beta j} f_\beta.$$

By definition of the tensor product representation, we have

$$(\rho \otimes \sigma)(g)(x \otimes y) = \rho(g)x \otimes \sigma(g)y$$

for all $x \in M$ and $y \in N$. Hence, for the basis element $e_i \otimes f_j$,

$$\begin{aligned}(\rho \otimes \sigma)(g)(e_i \otimes f_j) &= \rho(g)e_i \otimes \sigma(g) f_j \\ &= \left(\sum_{\alpha=1}^{m} a_{\alpha i} e_\alpha\right) \otimes \left(\sum_{\beta=1}^{n} b_{\beta j} f_\beta\right) \\ &= \sum_{\alpha=1}^{m} \sum_{\beta=1}^{n} a_{\alpha i} b_{\beta j}\, e_\alpha \otimes f_\beta.\end{aligned}$$

Therefore the coefficient of $e_\alpha \otimes f_\beta$ in the image of $e_i \otimes f_j$ is

$$a_{\alpha i} b_{\beta j}.$$

If the basis of $M \otimes_k N$ is ordered lexicographically as

$$e_1 \otimes f_1, \ldots, e_1 \otimes f_n, e_2 \otimes f_1, \ldots, e_2 \otimes f_n, \ldots, e_m \otimes f_1, \ldots, e_m \otimes f_n,$$

then the matrix of $(\rho \otimes \sigma)(g)$ is the Kronecker product

$$(\rho \otimes \sigma)(g) = \rho(g) \otimes \sigma(g).$$

Explicitly,

$$\rho(g) \otimes \sigma(g) = \begin{pmatrix} a_{11}\sigma(g) & a_{12}\sigma(g) & \cdots & a_{1m}\sigma(g) \\ a_{21}\sigma(g) & a_{22}\sigma(g) & \cdots & a_{2m}\sigma(g) \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}\sigma(g) & a_{m2}\sigma(g) & \cdots & a_{mm}\sigma(g) \end{pmatrix}.$$

Hence the tensor product representation is represented by the Kronecker product of the representing matrices:

$$(\rho \otimes \sigma)(g) = \rho(g) \otimes \sigma(g).$$

□

Exercise 15.12.4. *Let (M, ρ) be a k-representation of G. Fix a basis $(e_i)_{1 \le i \le m}$ of M. Let*

(M^*, ρ^*) be the contragredient representation of (M, ρ). Denote by

$$(m, \rho), \qquad (m, \rho^*)$$

the matrix representations associated with

$$(M, \rho), \qquad (M^*, \rho^*).$$

Determine the relation between $\rho(g)$ and $\rho^*(g)$.

Proof. Let

$$(e_1^*, \dots, e_m^*)$$

be the dual basis of M^*, so that

$$e_i^*(e_j) = \delta_{ij}.$$

Write the matrix of $\rho(g)$ with respect to the basis $(e_1, \dots, e_m)$ as

$$\rho(g) = A = (a_{ij})_{1 \le i,j \le m}.$$

Thus

$$\rho(g)e_j = \sum_{i=1}^{m} a_{ij} e_i.$$

By definition, the contragredient representation is given by

$$\rho^*(g)(\varphi) = \varphi \circ \rho(g^{-1}), \qquad \varphi \in M^*.$$

We want to compute the matrix of $\rho^*(g)$ with respect to the dual basis.

Let

$$\rho(g^{-1}) = A^{-1} = (b_{ij})_{1 \le i,j \le m}.$$

Then

$$\rho(g^{-1})e_j = \sum_{i=1}^{m} b_{ij} e_i.$$

For each dual basis vector e_ℓ^*, we have

$$\begin{aligned}
\rho^*(g)(e_\ell^*)(e_j) &= e_\ell^*(\rho(g^{-1})e_j) \\
&= e_\ell^* \left(\sum_{i=1}^{m} b_{ij} e_i \right) \\
&= b_{\ell j}.
\end{aligned}$$

Hence

$$\rho^*(g)(e_\ell^*) = \sum_{j=1}^{m} b_{\ell j} e_j^*.$$

Therefore the matrix of $\rho^*(g)$ is

$$(b_{\ell j})_{j,\ell} = (A^{-1})^t.$$

Consequently,
$$\rho^*(g) = \left(\rho(g)^{-1}\right)^t = \rho(g^{-1})^t.$$

Equivalently, the contragredient representation is represented by the inverse transpose matrix of the original representation:

$$\boxed{\rho^*(g) = \rho(g)^{-t}.}$$

$\square$

Exercise 15.12.5. *(i) Let G be a finite group acting on a finite set Ω. Let $\mathrm{Fix}^G(\Omega)$ denote the set of fixed points of Ω under G, and let $\Omega^\#$ denote the set of orbits of G on*

$$\Omega \setminus \mathrm{Fix}^G(\Omega).$$

Prove that
$$|\Omega| = |\mathrm{Fix}^G(\Omega)| + \sum_{\omega \in \Omega^\#} |\omega|.$$

From now on, we assume that p is a prime number and that G is a nontrivial p-group.

(ii) Assume now that $|\Omega|$ is a power of p different from 1. Prove that $|\mathrm{Fix}^G(\Omega)|$ is divisible by p.

(iii) Let k be a commutative field of characteristic p. Let M be a kG-module. Prove that

$$\mathrm{Fix}^G(M) \neq 0.$$

(HINT.

(a) Let $(e_i)_{i=1,\ldots,r}$ be a k-basis of M. Prove that the abelian group generated by

$$(ge_i)_{i=1,\ldots,r,\ g\in G}$$

is an $\mathbb{F}_pG$-module.

(b) Apply question (2) above to conclude.

)

(iv) Prove that, up to isomorphism, the trivial representation of G is the unique irreducible kG-module.

Proof. **(i).** The G-orbits form a partition of Ω. Moreover, an orbit has cardinality 1 if and only if it consists of a fixed point. Hence we have a disjoint union

$$\Omega = \mathrm{Fix}^G(\Omega) \sqcup \coprod_{\mathcal{O} \in \Omega^\#} \mathcal{O},$$

where $\Omega^{\#}$ denotes the set of G-orbits on $\Omega \setminus \operatorname{Fix}^G(\Omega)$. Taking cardinalities gives

$$|\Omega| = |\operatorname{Fix}^G(\Omega)| + \sum_{\mathcal{O} \in \Omega^{\#}} |\mathcal{O}|.$$

(ii). Assume now that G is a nontrivial p-group and that

$$|\Omega| = p^r$$

for some $r \geq 1$. Let $\mathcal{O} \in \Omega^{\#}$. Since $\mathcal{O}$ is not a fixed point orbit, we have $|\mathcal{O}| > 1$. By the orbit-stabilizer theorem,

$$|\mathcal{O}| = [G : G_\omega]$$

for any $\omega \in \mathcal{O}$. Since G is a p-group, the index $[G : G_\omega]$ is a power of p. Because it is greater than 1, it is divisible by p. Therefore every term $|\mathcal{O}|$ in the sum

$$|\Omega| = |\operatorname{Fix}^G(\Omega)| + \sum_{\mathcal{O} \in \Omega^{\#}} |\mathcal{O}|$$

is divisible by p. Since $|\Omega| = p^r$ is also divisible by p, it follows that

$$|\operatorname{Fix}^G(\Omega)|$$

is divisible by p.

(iii). Assume $M \neq 0$ is finite-dimensional over k, and let

$$(e_1, \ldots, e_r)$$

be a k-basis of M. Since $\operatorname{char}(k) = p$, the prime field $\mathbb{F}_p$ is contained in k. Define

$$M_0 := \sum_{g \in G} \sum_{i=1}^{r} \mathbb{F}_p \, g e_i \subseteq M.$$

Equivalently, M_0 is the abelian subgroup of M generated by the finitely many elements

$$g e_i, \qquad g \in G, \ 1 \leq i \leq r.$$

Since M has characteristic p as an additive group, this subgroup is an $\mathbb{F}_p$-vector space. It is finite-dimensional over $\mathbb{F}_p$, hence finite.

We show that M_0 is stable under the action of G. Let $h \in G$. Then

$$h(g e_i) = (hg) e_i,$$

and since $hg \in G$, the element $(hg)e_i$ is again one of the generators of M_0. Hence

$$h M_0 \subseteq M_0.$$

Therefore M_0 is an $\mathbb{F}_p G$-module.

Since $M \neq 0$, we have $M_0 \neq 0$. Thus, as a finite $\mathbb{F}_p$-vector space,

$$|M_0| = p^s$$

for some $s \geq 1$. We now apply part (ii) to the action of G on the finite set M_0. It follows that

$$|\operatorname{Fix}^G(M_0)|$$

is divisible by p. Since $0 \in \operatorname{Fix}^G(M_0)$, the fixed point set is nonempty. Being divisible by p, it must contain at least p elements, hence it contains a nonzero element. Therefore

$$\operatorname{Fix}^G(M_0) \neq 0.$$

Since $M_0 \subseteq M$, we obtain

$$\operatorname{Fix}^G(M) \neq 0.$$

(iv). Let S be an irreducible kG-module. By part (iii), we have

$$\operatorname{Fix}^G(S) \neq 0.$$

Choose

$$0 \neq v \in \operatorname{Fix}^G(S).$$

Then

$$gv = v \qquad \text{for all } g \in G.$$

Hence the one-dimensional subspace

$$kv \subseteq S$$

is G-stable. Indeed, for every $g \in G$ and every $\lambda \in k$,

$$g(\lambda v) = \lambda g v = \lambda v.$$

Therefore kv is a nonzero kG-submodule of S. Since S is irreducible, we must have

$$S = kv.$$

Thus S is one-dimensional, and G acts trivially on it. Hence S is isomorphic to the trivial representation.

Therefore, up to isomorphism, the trivial representation is the unique irreducible kG-module. $\square$

15.13 Homework Week 12

Exercise 15.13.1. *Write out a proof of Maschke's Theorem in the case of representations over $\mathbb{C}$ along the following lines.*

Given a representation

$$\rho : G \longrightarrow \mathrm{GL}(V),$$

where V is a vector space over $\mathbb{C}$, let

$$(\cdot, \cdot)$$

be any positive definite Hermitian form on V. Define a new form

$$(\cdot, \cdot)_1$$

on V by

$$(v, w)_1 = \frac{1}{|G|} \sum_{g \in G} (gv, gw).$$

Show that $(\cdot, \cdot)_1$ is a positive definite Hermitian form, preserved under the action of G; that is,

$$(v, w)_1 = (gv, gw)_1$$

always.

If W is a subrepresentation of V, show that

$$V = W \oplus W^{\perp}$$

as representations.

Proof. Throughout, write gv for $\rho(g)(v)$.

First we show that $(\cdot, \cdot)_1$ is Hermitian. Since each map

$$v \longmapsto gv$$

is linear, and since $(\cdot, \cdot)$ is Hermitian, the averaged form

$$(v, w)_1 = \frac{1}{|G|} \sum_{g \in G} (gv, gw)$$

is again sesquilinear and Hermitian.

Next we prove positive definiteness. If $v \neq 0$, then $gv \neq 0$ for every $g \in G$, because g acts invertibly on V. Hence

$$(gv, gv) > 0$$

for all $g \in G$. Therefore

$$(v, v)_1 = \frac{1}{|G|} \sum_{g \in G} (gv, gv) > 0.$$

Thus $(\cdot, \cdot)_1$ is positive definite.

Now let $h \in G$. Then

$$(hv, hw)_1 = \frac{1}{|G|} \sum_{g \in G} (ghv, ghw).$$

As g runs through G, so does gh. Hence, after the change of variable $t = gh$, we obtain

$$(hv, hw)_1 = \frac{1}{|G|} \sum_{t \in G} (tv, tw) = (v, w)_1.$$

Therefore $(\cdot, \cdot)_1$ is preserved by the action of G.

Let $W \subseteq V$ be a subrepresentation, and let $W^\perp$ denote the orthogonal complement of W with respect to $(\cdot, \cdot)_1$. Since $(\cdot, \cdot)_1$ is positive definite and V is finite-dimensional, we have a vector space direct sum

$$V = W \oplus W^\perp.$$

It remains to show that $W^\perp$ is G-stable.

Take $u \in W^\perp$, $h \in G$, and $w \in W$. Since W is a subrepresentation, we have

$$h^{-1}w \in W.$$

Using the G-invariance of $(\cdot, \cdot)_1$, we get

$$(hu, w)_1 = (hu, h(h^{-1}w))_1 = (u, h^{-1}w)_1 = 0,$$

because $u \in W^\perp$ and $h^{-1}w \in W$. Hence $hu \in W^\perp$.

Thus $W^\perp$ is a subrepresentation of V. Consequently,

$$V = W \oplus W^\perp$$

as representations. This proves Maschke's theorem over $\mathbb{C}$. $\square$

Exercise 15.13.2. *Prove the following theorem of Burnside: let G be a finite group, let k be an algebraically closed field in which $|G|$ is invertible, and let*

$$\rho : G \longrightarrow \mathrm{GL}(V)$$

be a representation over k. By taking a basis of V, write each endomorphism $\rho(g)$ as a matrix. Let

$$\dim V = n.$$

Show that the representation is simple if and only if there exist n^2 elements

$$g_1, \dots, g_{n^2} \in G$$

so that the matrices

$$\rho(g_1), \dots, \rho(g_{n^2})$$

are linearly independent, and that this happens if and only if the algebra homomorphism

$$kG \longrightarrow \mathrm{End}_k(V)$$

is surjective. Note that ρ itself is generally not surjective.

Proof. Let
$$\Phi : kG \longrightarrow \mathrm{End}_k(V)$$
be the algebra homomorphism induced by the representation, namely
$$\Phi\left(\sum_{g\in G} a_g g\right) = \sum_{g\in G} a_g \rho(g).$$
Set
$$A := \mathrm{Im}\,\Phi.$$
Then A is the k-subalgebra of $\mathrm{End}_k(V)$ spanned by the matrices
$$\rho(g), \qquad g \in G.$$

A subspace $W \subseteq V$ is G-stable if and only if it is stable under every element of A. Hence the representation V is simple as a representation of G if and only if V is a simple left A-module.

Since $|G|$ is invertible in k, Maschke's theorem implies that kG is semisimple. Therefore its quotient algebra A is also semisimple. Since k is algebraically closed and A is finite-dimensional semisimple, the Artin–Wedderburn theorem gives
$$A \cong M_{m_1}(k) \oplus \cdots \oplus M_{m_r}(k).$$

Suppose first that V is simple as an A-module. Since $A \subseteq \mathrm{End}_k(V)$, the action of A on V is faithful. But for the algebra
$$M_{m_1}(k) \oplus \cdots \oplus M_{m_r}(k),$$
a simple module is supported on exactly one simple matrix-algebra summand. If $r > 1$, then all other summands act by zero, contradicting faithfulness. Hence $r = 1$, and therefore
$$A \cong M_m(k)$$
for some m.

The unique simple left $M_m(k)$-module is k^m, so
$$\dim_k V = m.$$
Since $\dim_k V = n$, we have $m = n$. Hence
$$\dim_k A = m^2 = n^2.$$
But $A \subseteq \mathrm{End}_k(V)$, and
$$\dim_k \mathrm{End}_k(V) = n^2.$$

Therefore

$$A = \mathrm{End}_k(V).$$

Thus $\Phi : kG \to \mathrm{End}_k(V)$ is surjective.

Conversely, suppose that

$$\Phi : kG \longrightarrow \mathrm{End}_k(V)$$

is surjective. Then $A = \mathrm{End}_k(V)$. Let $0 \neq W \subseteq V$ be an A-submodule. Choose $0 \neq w \in W$. For any $v \in V$, there exists a linear endomorphism $T \in \mathrm{End}_k(V)$ such that

$$T(w) = v.$$

Since $A = \mathrm{End}_k(V)$, we have $T \in A$. Because W is stable under A, it follows that

$$v = T(w) \in W.$$

Thus $W = V$. Hence V is simple.

So the representation is simple if and only if

$$\Phi : kG \longrightarrow \mathrm{End}_k(V)$$

is surjective.

It remains to relate this to the existence of n^2 linearly independent matrices $\rho(g_i)$. Since

$$A = \mathrm{Span}_k\{\rho(g) \mid g \in G\},$$

there exist elements

$$g_1, \ldots, g_{n^2} \in G$$

such that

$$\rho(g_1), \ldots, \rho(g_{n^2})$$

are linearly independent if and only if

$$\dim_k A \geq n^2.$$

But $A \subseteq \mathrm{End}_k(V)$, and

$$\dim_k \mathrm{End}_k(V) = n^2.$$

Hence this happens if and only if

$$\dim_k A = n^2,$$

which is equivalent to

$$A = \mathrm{End}_k(V).$$

Therefore this happens if and only if Φ is surjective.

Combining the previous equivalences, the representation is simple if and only if there exist n^2 elements

$$g_1, \dots, g_{n^2} \in G$$

such that

$$\rho(g_1), \dots, \rho(g_{n^2})$$

are linearly independent, and this is equivalent to the surjectivity of

$$kG \longrightarrow \mathrm{End}_k(V).$$

$\square$

Exercise 15.13.3. *Let*

$$G = \langle x \mid x^n = 1 \rangle$$

be a cyclic group of order n, and let $\zeta_n \in \mathbb{C}$ be a primitive n-th root of unity. Then the simple complex characters of G are the n functions

$$\chi_r(x^s) = \zeta_n^{rs},$$

where

$$0 \leq r \leq n-1.$$

Proof. Let

$$G = \langle x \mid x^n = 1 \rangle.$$

For each $0 \leq r \leq n-1$, define a one-dimensional representation

$$\rho_r : G \longrightarrow \mathrm{GL}_1(\mathbb{C})$$

by

$$\rho_r(x) = \zeta_n^r.$$

Since $(\zeta_n^r)^n = 1$, this is a well-defined representation of G. Its character is given by

$$\chi_r(x^s) = \mathrm{tr}(\rho_r(x^s)) = \mathrm{tr}\big((\zeta_n^r)^s\big) = \zeta_n^{rs}.$$

Each ρ_r is one-dimensional, hence simple.

It remains to prove that every simple complex representation of G is one of these. Let

$$\rho : G \longrightarrow \mathrm{GL}(V)$$

be a simple complex representation, and write

$$T := \rho(x).$$

Since $x^n = 1$, we have

$$T^n = I.$$

Therefore the minimal polynomial of T divides

$$X^n - 1.$$

Over $\mathbb{C}$, the polynomial $X^n - 1$ splits into distinct linear factors:

$$X^n - 1 = \prod_{r=0}^{n-1} (X - \zeta_n^r).$$

Hence T is diagonalizable, and

$$V = \bigoplus_{\lambda^n = 1} V_\lambda,$$

where

$$V_\lambda = \{v \in V \mid Tv = \lambda v\}.$$

Each eigenspace V_λ is G-stable, because G is generated by x, and x acts on V_λ by the scalar λ. Since V is simple, exactly one eigenspace is nonzero. Thus

$$V = V_\lambda$$

for some n-th root of unity λ.

Moreover, if $\dim_{\mathbb{C}} V > 1$, then any one-dimensional subspace of V is stable under G, because x acts on all of V by the scalar λ. This would contradict the simplicity of V. Hence

$$\dim_{\mathbb{C}} V = 1.$$

Therefore V is a one-dimensional representation on which

$$x$$

acts by some n-th root of unity λ. Since ζ_n is primitive, there is a unique r with

$$0 \le r \le n - 1$$

such that

$$\lambda = \zeta_n^r.$$

Thus $V \cong \rho_r$, and its character is

$$\chi_r(x^s) = \zeta_n^{rs}.$$

Finally, the characters $\chi_0, \dots, \chi_{n-1}$ are distinct, since

$$\chi_r(x) = \zeta_n^r$$

are distinct for $0 \le r \le n - 1$. Therefore the simple complex characters of G are precisely

$$\chi_r(x^s) = \zeta_n^{rs}, \qquad 0 \le r \le n - 1.$$

□

15.14 Homework Week 13

Exercise 15.14.1. *Let G be a finite group and p a prime. Prove that each element $x \in G$ can be uniquely written as*

$$x = st,$$

where s is p-regular, t is p-singular, and $st = ts$. If $x_1 = s_1 t_1$ is such a decomposition of an element x_1 that is conjugate to x, prove that s is conjugate to s_1 and t is conjugate to t_1.

Proof. Fix $x \in G$. Since G is finite, x has finite order. Write

$$|x| = p^a m,$$

where $a \geq 0$ and $p \nmid m$. Because $\gcd(p^a, m) = 1$, there exist integers u, v such that

$$up^a + vm = 1.$$

Define

$$t := x^{vm}, \qquad s := x^{up^a}.$$

Then $s, t \in \langle x \rangle$, so $st = ts$, and

$$st = x^{up^a + vm} = x.$$

We check that t is p-singular and s is p-regular. Since

$$t^{p^a} = x^{vmp^a} = x^{v|x|} = 1,$$

the order of t divides p^a, hence $\operatorname{ord}(t) = p^k$ for some $k \leq a$. Thus t is p-singular. Likewise,

$$s^m = x^{up^a m} = x^{u|x|} = 1,$$

so $\operatorname{ord}(s) \mid m$. Since $p \nmid m$, we have $p \nmid \operatorname{ord}(s)$, and therefore s is p-regular. This proves existence.

For uniqueness, suppose

$$x = s_1 t_1 = s_2 t_2,$$

where each s_i is p-regular, each t_i is p-singular, and $s_i t_i = t_i s_i$. Since $p \nmid \operatorname{ord}(s_i)$ and $\operatorname{ord}(t_i)$ is a power of p, we have

$$\gcd\big(\operatorname{ord}(s_i), \operatorname{ord}(t_i)\big) = 1.$$

Hence

$$|x| = \operatorname{ord}(s_i) \operatorname{ord}(t_i).$$

The subgroup $\langle s_i, t_i \rangle$ is abelian and has order $\operatorname{ord}(s_i) \operatorname{ord}(t_i) = |x|$, so it equals $\langle x \rangle$. In particular, $s_i, t_i \in \langle x \rangle$.

The cyclic group $\langle x \rangle$ has order $|x| = p^a m$. In a cyclic group of this order, an element is p-regular if and only if its order divides m, and p-singular if and only if its order divides p^a. Under the identification

$$\langle x \rangle \cong \mathbb{Z}/|x|\mathbb{Z},$$

the Chinese remainder theorem gives a unique decomposition of each element into commuting p-regular and p-singular parts. Therefore $s_1 = s_2$ and $t_1 = t_2$.

Now suppose that x_1 is conjugate to x, say

$$x_1 = gxg^{-1},$$

and that $x = st$ and $x_1 = s_1 t_1$ are the corresponding decompositions. Define

$$s' := gsg^{-1}, \qquad t' := gtg^{-1}.$$

Conjugation preserves order, so s' is p-regular and t' is p-singular. Moreover,

$$s't' = g(st)g^{-1} = g(ts)g^{-1} = t's',$$

and

$$s't' = g(st)g^{-1} = gxg^{-1} = x_1.$$

By uniqueness of the decomposition of x_1, we have $s_1 = s'$ and $t_1 = t'$. Hence s is conjugate to s_1, and t is conjugate to t_1. □